Your Dreams Our Goal

POORNIMA UNIVERSITY

Member of Association of Indian Universities & Approved by UGC (Govt. of India) under 2(f) & 12(B)



FACULTY OF COMPUTER SCIENCE & ENGINEERING

PROGRAM: B.Tech.(CSE)

SCHEME & SYLLABUS BOOKLET

BATCH 2024-2028

SCHEME & SYLLABUS (2024-2028)

FACULTY OF COMPUTER SCIENCE & ENGINEERING

PROGRAM: B.Tech.(CSE)

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Disclaimer: The scheme, syllabus and other materials published in this booklet may be changed or modified as per the requirement after approval of competent authority. The decision taken by the management of Poornima University will be final and abiding to all.

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Vision

Our vision is to create knowledge based society with scientific temper, team spirit and dignity of labour to face global competitive challenges.

Mission

Our mission is to evolve and develop skill based systems for effective delivery of knowledge so as to equip young professionals with dedication and commitment to excellence in all spheres of life.

Quality Policy

To provide quality education through faculty development, updating of facilities and continual improvement for meeting norms laid down by the government, keeping the stakeholders satisfied. Poornima University has forged industrial alliances with Top MNC's worldwide which assures high educational standards, up to- date and forward-thinking curricula, and professional relevance. At Poornima University you will have a distinct advantage through exposure to the corporate standard environment through industry sponsored infrastructure and expert faculty. The University involves global industry leaders in many ways.

Knowledge Wheel

At Poornima, the academic atmosphere is a rare blend of modern technical as well as soft skills and traditional systems of learning processes.



About Program and Program Outcomes (PO):

Title of the Programme: Bachelor of Technology (B. Tech.)

Nature of the Programme: B. Tech. is four years full-time programme.

Program Outcomes (PO):

Engineering Graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and the need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcome (PSOs):

PSO1: The graduates are proficient in fundamental principles and methods of Computer Science, Mathematical and Scientific reasoning and are able to:

a) Apply fundamental concepts of integration, differentiation, vector calculus, probability & statistics, and discrete mathematics.

b) Design, create & evaluate algorithms and data structures appropriate to specific problems.

PSO2: The graduates possess in-depth knowledge of various components of hardware and system software. The students have thorough understanding:

a) Architecture of computer system and functionality of various units.

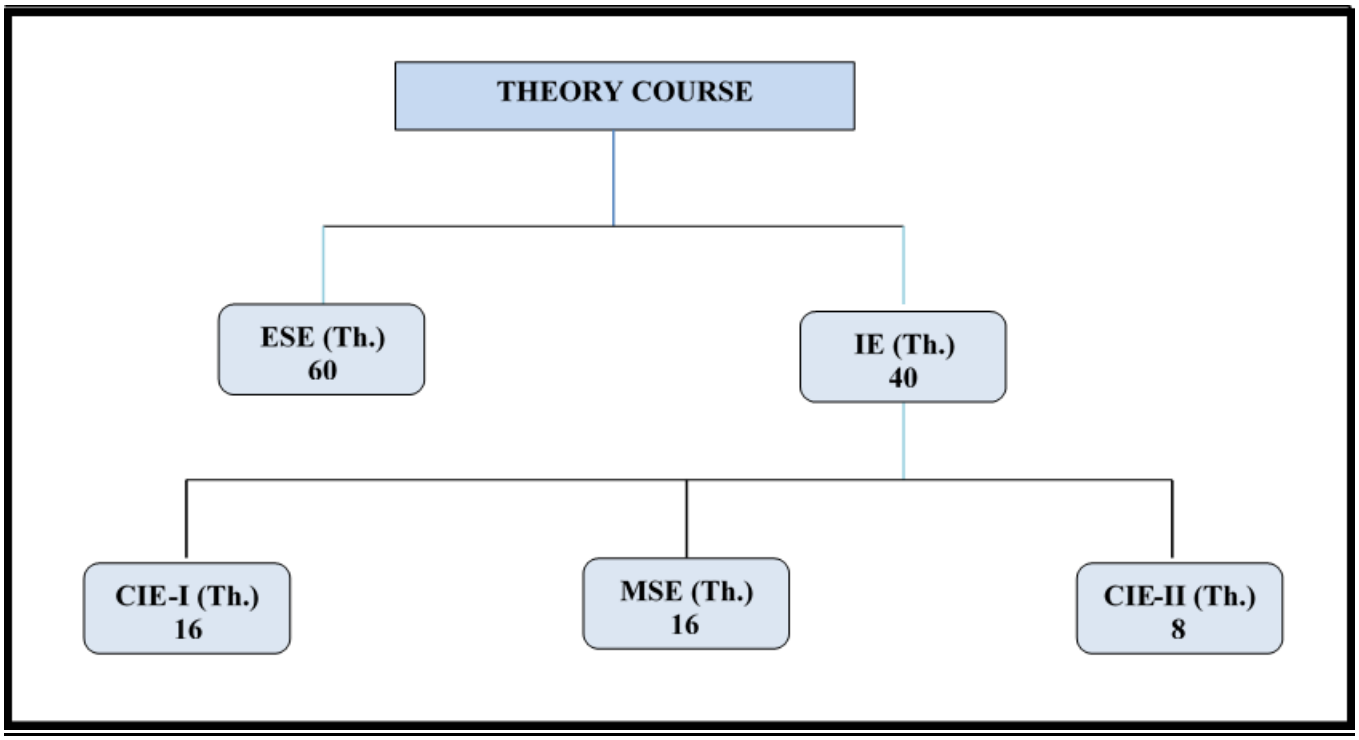
b) Role of operating system in managing the hardware units.

c) Computer networks and security related issues.

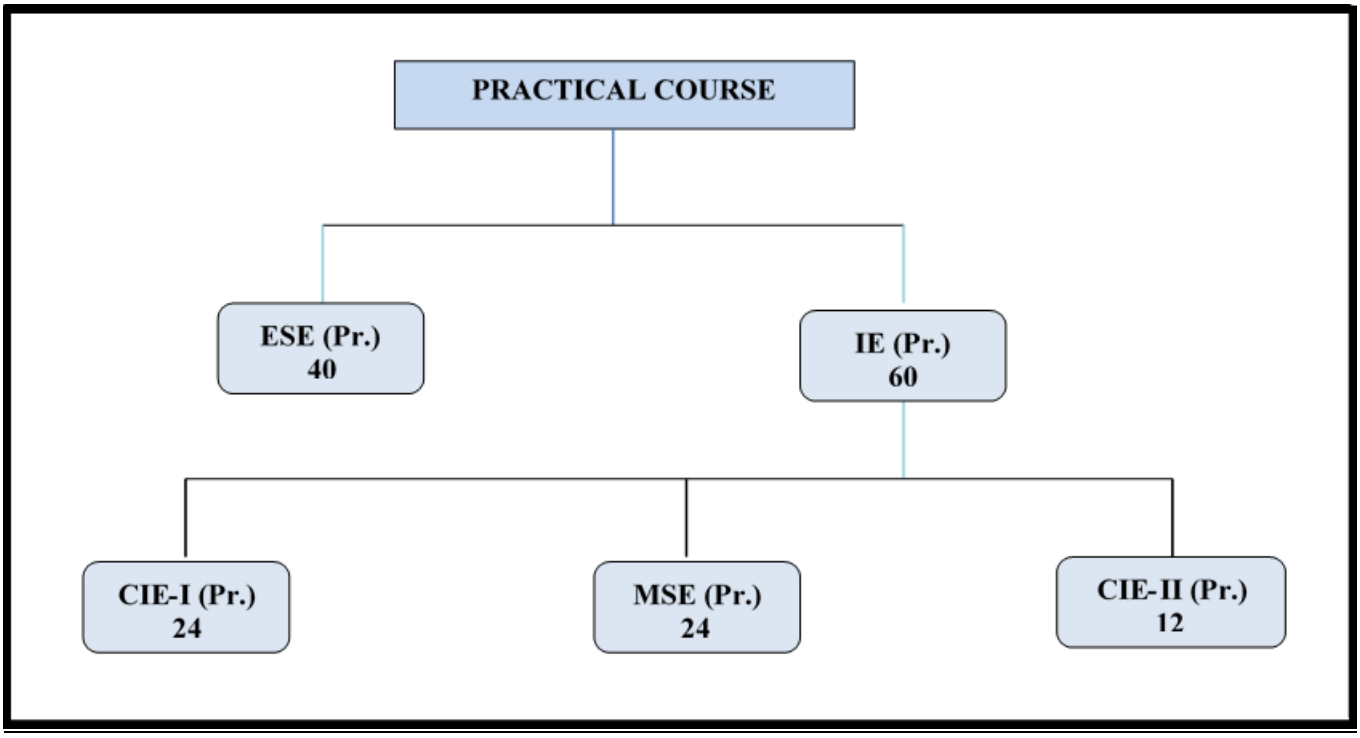
PSO3: The graduates are competent in object oriented programming languages and possess basic knowledge of several other programming languages.

Examination System :

Marks Distribution of Theory Course:



A. Marks Distribution of Practical Course :



Th.: Theory, Pr.: Practical, **ESE:** End Semester Examination, **MSE:** Mid Semester Examination, **CIE:** Continuous Internal Evaluation.

CO Wise Marks Distribution:

Exam Entity	Theory Subject		Practical/ Studio Subject	
	Maximum Marks	CO to be Covered	CO to be Covered	Maximum Marks
CIE-I	16 (8 + 8)	1 & 2	1 & 2	24 (12 + 12)
MSE	16 (8 + 8)	3 & 4	3 & 4	24 (12 + 12)
CIE-II (Activity/ Assignment)	8 (8)	5	5	12 (12)
ESE	60	-	-	40
TOTAL	100	-	-	100

Minimum Passing Percentage in All Exams:

S. No.	Program Name	Minimum Passing Percentage in		
		IE Component	ESE Component	Total Component
1	Course Work for PhD Registration	-	-	50%
2	B. Arch., FIRE Dept. (BBA, B. Com., MBA)	-	45%	50%
3	MBA, MCA, M.Des., M.Tech., M.Plan, MHA, MPH	-	40%	40%
4	B. Tech., B. Des., BVA, BCA, B.Sc., BBA, B.Com., B.A. & any other program	-	35%	35%

SGPA Calculation

$$SGPA = \frac{C_1G_1 + C_2G_2 + \ldots\ldots\ldots + C_nG_n}{C_1 + C_2 + \ldots\ldots + C_n}$$

$$SGPA = \frac{\sum_i C_i \times G_i}{\sum_i C_i}$$

where (as per teaching scheme & syllabus):

C_i is the number of credits of subject i,

G_i is the Grade Point for the subject I and i = 1 to n,

n = number of subjects in a course in the semester

CGPA Calculation

$$CGPA = \frac{C_1G_1 + C_2G_2 + \ldots\ldots\ldots + C_nG_n}{C_1 + C_2 + \ldots\ldots + C_n}$$

$$CGPA = \frac{\sum_i C_i \times G_i}{\sum_i C_i}$$

where (as per teaching scheme & syllabus):

C_i is the number of credits of subject i,

G_i is the Grade Point for the subject I and i = 1 to n,

n = number of subjects in a course of all the semesters up to which CGPA is computed

Grading Table:

Table-A
Applicable for B.Arch., FIRE Courses
(BBA, B.Com, MBA), & PhD. Course Work

Academic Performance	Grade	Grade Point	Marks Range (in %)
Outstanding	O	10	90≤ x ≤100
Excellent	A+	9	80≤ x <90
Very Good	A	8	70≤ x <80
Good	B+	7	60≤ x <70
Above Average	B	6	50≤ x <60
Fail	F	0	x <50
Absent	Ab	0	Absent

Table-B
Applicable for All Courses except Table-A

*** Not applicable for master programs**

Academic Performance	Grade	Grade Point	Marks Range (in %)
Outstanding	O	10	90≤ x ≤100
Excellent	A+	9	80≤ x <90
Very Good	A	8	70≤ x <80
Good	B+	7	60≤ x <70
Above Average	B	6	50≤ x <60
Average	C	5	40≤ x <50
Pass*	P	4	35≤ x <40
Fail	F	0	x <35
Absent	Ab	0	Absent

CGPA to percentage conversion rule:

Equivalent % of Marks in the Program = *CGPA* *10

Award of Class

CGPA	Percentage	Equivalent Division
7.50 ≤ CGPA	75% or more	First Division with Distinction
6.00 ≤ CGPA < 7.50	60% ≤ x <75%	First Division
5.00 ≤ CGPA < 6.00	50% ≤ x <60%	Second Division
4.00 ≤ CGPA < 5.00	40% ≤ x < 50%	Pass Class

Guidelines for Massive Open Online Courses (MOOCs)

(Session 2023-24)

Poornima University, in its never ending endeavor to equip students with best-of-class learning and knowledge, has undertaken to include MOOC courses as part of its credit scheme from session 2023-24 onwards. The objective behind this is to enable students to study courses designed by the best teachers in the country and to scale their knowledge base with the rest of learners from the nation. The MOOCs which are included under this scheme is can be chosen from SWAYAM-NPTEL.

1. Introduction of MOOCs: SWAYAM-NPTEL

About SWAYAM-NPTEL

NPTEL (National Programme on Technology Enhanced Learning), is a joint venture of the IITs and IISc, funded by the Ministry of Education (MoE) Government of India, and was launched in 2003. Initially started as a project to take quality education to all corners of the country, NPTEL now offers close to 600+ courses for certification every semester in about 22 disciplines.

Some highlights:

- Largest online repository in the world of courses in engineering, basic sciences and selected humanities and management subjects
- YouTube channel for NPTEL – most subscribed educational channel, 1.3 billion views and 40+ lakhs subscribers
- More than 56000 hours of video content, transcribed and subtitled
- Most accessed library of peer-reviewed educational content in the world
- Translation of more than 12000 hrs of English transcripts in regional Indian languages

NPTEL Online Certification:

The objective of enabling students obtain certificates for courses is to make students employable in the industry or pursue a suitable higher education programme. Through an online portal, 4, 8, or 12-week online courses, typically on topics relevant to students in all years of higher education along with basic core courses in sciences and humanities with exposure to relevant tools and technologies, are being offered. Enrolment to and learning from these courses is free. Following these online courses, an in-person, proctored certification exam is conducted and a certificate is provided through the participating institutions and industry, as applicable.

Some statistics regarding the open online courses since March 2014 till Dec 2021

Completed courses: 3496;

Enrollments across courses: 1.58 CRORE +

Number of exam registrations: 15.1 LAKH +

All the statistics pertaining to completed courses are available at https://beta.nptel.ac.in/courses.

All courses are completely free to enroll and learn from. The certification exam is optional and comes at a fee of Rs 1000/course exam.

2. MOOCs at Poornima University:

MOOCs envelops best in class teaching - learning processes along with meeting the requirements of various courses in terms of quality of teaching and evaluation system. To promote the MOOCs among students of Poornima University, it is decided to consider the credits earned through MOOCs.

(a) MOOCs as Credit Courses

(For this document, only those MOOCs will be considered which are available only on NPTEL platforms)

- Credit and Non-credit SWAYAM-NPTEL MOOCs can be opted by anyone, anytime, anywhere and in any language. However, prior-permission of the University Authorities is mandatory if the credits are to be transferred to regular degree.
- As Open Elective (for batches entered till 2022) / Multidisciplinary Courses (for batches admitted from 2023-24 onwards): Open Elective (for batches entered till 2022) courses were available at University level in offline mode till 2022-23 for which relevant booklets were published. From session 2023-24, Multidisciplinary Courses are introduces in lieu of open elective courses as per NEP 2020. These courses carries 02 credits. These category/type of courses (similar/different) are available as MOOC courses on SWAYAM-NPTEL platform which are being introduced from session 2023-24 onwards for all the students. The respective Deans / HODs shall provide all the information to all the students pertaining to MOOCs as per details given below:
    1. Deans / HODs shall prepare a list of up to 10 appropriate MOOC courses (From NPTEL Only) of Minimum 02/03 credits each, well in advance (at-least 15 days prior to commencement of semester) and take approval from the Office of Dean, Academics / Pro-President, PU.
    2. After approval, the respective Deans / HODs shall circulate a notice to all their respective students so that they can select any one course from the list, the credits (only 02) of which will be counted against Open Elective/ Multidisciplinary courses pertaining to that particular semester.
    3. The tutor of the class shall monitor the progress (assignments, feedback, any problem etc.) on weekly basis and report to Head/Dean and provide the academic support to students as per requirement.

(b) Important points related to MOOCs at Poornima University

- Only one MOOC shall be allowed in a particular semester for the purpose of credit transfer in the beginning.
- No attendance will be taken for MOOC courses.
- The method of assessments of MOOC such as assignments and examination are completely associated with that particular MOOC and no internal exam (IE component) will be conducted by the department as well as by the Examination Cell.
- The respective Dean / HOD must submit the detail of course i.e., code, name and credit of MOOC opted against that particular course in particular semester attached with highlighting in the related examination scheme of syllabus of that semester signed by BOS Convener / HoD and Dean of Faculty to the office of Pro-President before commencement of the classes.
- The center of examination for MOOCs will be finalized by SWAYAM-NPTEL. All the responsibility related to registration for MOOCs, timely submission of assignments, examinations etc. will be borne by the students only.
- NPTEL will award a certificate to all the students passing the examination.
- The list of registered students in MOOC along with name of course will be submitted to the Examination Cell by the Deans / HoDs before commencement of the classes.
- An ESE Exam of each said MOOC course will also be conducted by the University as per University norms.
- The award of marks/grading will be computed as given below:

Award of marks/grading	Remarks
1. 20% weightage taken from MOOC Certificate +80% weightage taken from ESE Exam of Poornima University **OR** 2. 100% of weightage taken from MOOC Certificate Note: The Higher Marks/Grades of the above two will be considered	The Certificate of MOOC to be Submitted as per date notified by COE, Poornima University

- Any student who would not be able to clear/pass the said course, will be required to appear as a back exam candidate of the University as per PU norms. Students who have not passed the MOOC exam are required to register and participate in the next semester for either the same subject or a similar subject (Ensuring at least 60% of the syllabus matches with the back subject and also approved by respective Dean) offered through NPTEL.
- The scorecard and related certificate of MOOC along with a consolidated list of students with marks of assignment and final exam will be submitted to the examination cell by the concerned Dean / HOD for further process. It is also recommended that alteration/changes/scaling in marks obtained by the students in any MOOC will not be considered.
- The exam registration fee of MOOC up to Max. INR 1000/- will be reimbursed to the student only after successful completion of the course in first attempt and submission of the fee receipt, score-card and certificate of the MOOC to the concerned department within stipulated time after declaration of the results.
- There will be no provision of re-evaluation of MOOC.

NOTE: This is to be noted that the procedure for getting approval from BOS, Faculty Board, Academic Council and BoM is to be followed as per regular process.

Attached Items:

Ability Enhancement Courses	Annexure-1
Value Added Course Booklet	Annexure-2

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BACHELOR OF TECHNOLOGY (B. TECH.)

(COMPUTER SCIENCE & ENGINEERING)

COMMON SCHEME & SYLLABUS

For

Cyber Security
Cloud & Full Stack Development
Artificial Intelligence & Data Science
Artificial Intelligence & Machine Learning

BATCH: 2024-28

POORNIMA UNIVERSITY, JAIPUR
Faculty of Computer Science and Engineering
Faculty of Engineering and Technology

Name of Program: B.Tech. CSE **Duration: 4 Years** **Total Credits: 172**

Teaching Scheme for Batch 2024-28

Semester-I

Course Code	Name of Course	Teaching Scheme			Marks Distribution			Credits
		Lecture (L)	Tutorial (T)	Practical (P)	IE	ESE	Total	
A.		**Major (Core Courses)**						
A.1	**Theory**							
BTXCSA1101/ BTXCSA1102	Engineering Physics / Engineering Chemistry	3	-	-	40	60	100	3
BTXCCE1103	Fundamental of Computer System	3	-	-	40	60	100	3
BTXCCV1104/ BTXCEE1105	Basics of Civil Engineering / Basics of Electrical and Electronics Engineering	3	-	-	40	60	100	3
BTXCSA1106/ BTXCME1107	Engineering Mathematics / Basic of Mechanical Engineering	3	-	-	40	60	100	3
A.2	**Practical**							
BTXCSA1201/ BTXCSA1202	Engineering Physics lab / Engineering Chemistry lab	-	-	2	60	40	100	1
BTXCCE1203	Programming in C Lab	-	-	2	60	40	100	1
BTXCCV1204/ BTXCEE1205	Computer Aided Design (CADD)/ Basics of Electrical and Electronics Engineering Lab	-	1	2	60	40	100	1
BTXCME1206/ BTXCME1207	Workshop Practice/ Engineering Graphics	-	1	2	60	40	100	1
BTXCCE1208	Exploratory Project-1	-	-	2	60	40	100	1
B.		**Minor Stream Courses/ Department Electives**						
B.1	**Theory**	-	-	-	-	-	-	-
B.2	**Practical**	-	-	-	-	-	-	-
C		**Multidisciplinary Courses**						
	-	-	-	-				
D		**Ability Enhancement Courses (AEC)**						
BUACHM1202	Foundation English	-	-	2	60	40	100	1
E		**Skill Enhancement Courses (SEC)**						
BULCSE1201	Skill Enhancement Generic Course-I	-	-	2	60	40	100	1
F		**Value Added Courses (VAC)**						
BUVCSA1102	Environmental Studies	2	-	-	40	60	100	2
G		**Summer Internship / Research Project / Dissertation**						
Total		14	2	14				**21**
Total Teaching Hours		**30**						

POORNIMA UNIVERSITY, JAIPUR
Faculty of Computer Science and Engineering
Faculty of Engineering and Technology

Name of Program: B.Tech. CSE **Duration: 4 Years** **Total Credits: 172**

Teaching Scheme for Batch 2024-28

Semester-II

Course Code	Name of Course	Teaching Scheme			Marks Distribution			Credits
		Lecture (L)	Tutorial (T)	Practical (P)	IE	ESE	Total	
A.		**Major (Core Courses)**						
A.1	**Theory**							
BTXCSA2101 / BTXCSA2102	Engineering Physics / Engineering Chemistry	3	-	-	40	60	100	3
BTXCCE2103	Programming in Python	3	-	-	40	60	100	3
BTXCCV2104/ BTXCEE2105	Basic of Civil Engineering / Basics of Electrical and Electronics Engineering	3	-	-	40	60	100	3
BTXCSA2106/ BTXCME2107	Engineering Mathematics / Basic of Mechanical Engineering	3	-	-	40	60	100	3
A.2	**Practical**							
BTXCSA2201/ BTXCSA2202	Engineering Physics lab / Engineering Chemistry lab	-	-	2	60	40	100	1
BTXCCE2203	Programming in Python Lab	-	-	2	60	40	100	1
BTXCCV2204/ BTXCEE2205	Computer Aided Design (CADD)/ Basics of Electrical and Electronics Engineering Lab	-	-	2	60	40	100	1
BTXCME2206/ BTXCME2207	Workshop Practice/Engineering Graphics	-	-	2	60	40	100	1
BTXCCE2208	Exploratory Project-2	-	-	2	60	40	100	1
B.		**Minor Stream Courses/ Department Electives**						
B.1	**Theory**							
	Professional Elective - I	3	-	-	40	60	100	3
B.2	**Practical**							
	Nil	-	-	-	-	-	-	
C		**Multidisciplinary Courses**						
	MOOC Course-I	1	-	-				2
D		**Ability Enhancement Courses (AEC)**						
BUACHM2206	Language Lab	0	0	2	60	40	100	1
E		**Skill Enhancement Courses (SEC)**						
BULCSE2201	Skill Enhancement Generic Course-II	-	-	2	60	40	100	1
F		**Value Added Courses (VAC)**						
	NIL							
G		**Summer Internship / Research Project / Dissertation**						
		-	-	-				
Total		16	0	14				**24**
Total Teaching Hours		**30**						

POORNIMA UNIVERSITY, JAIPUR
Faculty of Computer Science and Engineering

Name of Program: B.Tech. CSE **Duration: 4 Years** **Total Credits: 172**

Teaching Scheme for Batch 2024-28

Semester-III

Course Code	Name of Course	Teaching Scheme			Marks Distribution			Credits
		Lecture (L)	Tutorial (T)	Practical (P)	IE	ESE	Total	
A.		**Major (Core Courses)**						
A.1	**Theory**							
BCECCE3101	Introduction to Web Technology	3	-	-	40	60	100	3
BCECCE3102	Data Structures and Algorithms	3	-	-	40	60	100	3
BCECCE3103	Operating System	3	-	-	40	60	100	3
BCECCE3104	Relational Database Management System	3	-	-	40	60	100	3
A.2	**Practical**							
BCECCE3201	Introduction to Web Technology Lab	-	-	2	60	40	100	1
BCECCE3202	Data Structures and Algorithms Lab	-	-	2	60	40	100	1
BCECCE3203	Operating System Lab	-	-	2	60	40	100	1
BCECCE3204	RDBMS Lab	-	-	2	60	40	100	1
B.		**Minor Stream Courses/ Department Electives**						
B.1	**Theory**							
	Professional Elective - II	3	-	-	40	60	100	3
B.2	**Practical**							
	-							
C		**Multidisciplinary Courses**						
	MOOC Course-II	1	-	-	40	60	100	2
D		**Ability Enhancement Courses (AEC)**						
BUACHM3209	Professional Skills-I	-	-	2	40	60	100	1
E		**Skill Enhancement Courses (SEC)**						
BULCSE3201	Skill Enhancement Generic Course-III	-	-	2	60	40	100	1
F		**Value Added Courses (VAC)**						
BUVCVD3101	Digital Marketing	2	-	-	40	60	100	2
G		**Summer Internship / Research Project / Dissertation**						
	-	-	-				-	
Total		18		12				**25**
Total Teaching Hours		**30**						

POORNIMA UNIVERSITY, JAIPUR
Faculty of Computer Science and Engineering

Name of Program: B.Tech. CSE **Duration: 4 Years** **Total Credits: 172**

Teaching Scheme for Batch 2024-28

Semester-IV

Course Code	Name of Course	Teaching Scheme			Marks Distribution			Credits
		Lecture (L)	Tutorial (T)	Practical (P)	IE	ESE	Total	
A.		**Major (Core Courses)**						
A.1	**Theory**							
BCECCE4101	Computer Networks	3	-	-	40	60	100	3
BCECCE4102	OOPS With Java	3	-	-	40	60	100	3
BCECCE4103	Software Engineering	3	-	-	40	60	100	3
BCECCE4104	Advance Data Structure	3	-	-	40	60	100	3
A.2	**Practical**							
BCECCE4201	Computer Network Lab	-	-	2	60	40	100	1
BCECCE4202	OOPS With Java Lab	-	-	2	60	40	100	1
BCECCE4203	Frontend Development Lab	-	-	2	60	40	100	1
B.		**Minor Stream Courses/ Department Electives**						
B.1	**Theory**							
	Professional Elective - III	3	-	-	40	60	100	3
B.2	**Practical**							
	Professional Elective Lab - I	-	-	2	60	40	100	1
C		**Multidisciplinary Courses**						
	MOOC Course-III	1	-	-	40	60	100	2
D		**Ability Enhancement Courses (AEC)**						
BUACHM4213	Human Values & Professional Ethics	-	-	2	60	40	100	1
E		**Skill Enhancement Courses (SEC)**						
BULCSE4201	Skill Enhancement Generic Course-IV	-	-	2	60	40	100	1
F		**Value Added Courses (VAC)**						
BUVCVD4102	Business Intelligence	2	-	-	40	60	100	2
G		**Summer Internship / Research Project / Dissertation**						
	-	-	-				-	
Total		18	-	12				**25**
Total Teaching Hours		**30**						

POORNIMA UNIVERSITY, JAIPUR
Faculty of Computer Science and Engineering

Name of Program: B.Tech. CSE **Duration: 4 Years** **Total Credits: 172**

Teaching Scheme for Batch 2024-28

Semester-V

Course Code	Name of Course	Teaching Scheme			Marks Distribution			Credits
		Lecture (L)	Tutorial (T)	Practical (P)	IE	ESE	Total	
A.		**Major (Core Courses)**						
A.1	**Theory**							
BCECCE5101	Design & Analysis of Algorithms	3	-	-	40	60	100	3
BCECCE5102	Advance Java	3	-	-	40	60	100	3
BCECCE5103	Computer Architecture	3	-	-	40	60	100	3
A.2	**Practical**							
BCECCE5201	Design & Analysis of Algorithms Lab	-	-	2	60	40	100	1
BCECCE5202	Advance Java Lab	-	-	2	60	40	100	1
BCECCE5203	Power BI Lab	-	-	2	60	40	100	1
B.		**Minor Stream Courses/ Department Electives**						
B.1	**Theory**							
	Professional Elective - IV	3	-	-	40	60	100	3
	Professional Elective - V	3	-	-	40	60	100	3
B.2	**Practical**							
	Professional Elective Lab - II	-	-	2	60	40	100	1
	Professional Elective Lab - III	-	-	2	60	40	100	1
C		**Multidisciplinary Courses**						
	MOOC Course-IV	1	-	-	40	60	100	2
D		**Ability Enhancement Courses (AEC)**						
BUACHM5214	Professional Skills-II	-	-	2	60	40	100	1
E		**Skill Enhancement Courses (SEC)**						
BULCSE5201	Skill Enhancement Generic Course-V	-	-	2	60	40	100	1
F		**Value Added Courses (VAC)**						
	-	-	-	-				
G		**Summer Internship / Research Project / Dissertation**						
	-	-	-				-	
Total		16	-	14				**24**
Total Teaching Hours		**30**						

POORNIMA UNIVERSITY, JAIPUR
Faculty of Computer Science and Engineering

Name of Program: B.Tech. CSE **Duration: 4 Years** **Total Credits: 172**

Teaching Scheme for Batch 2024-28

Semester-VI

Course Code	Name of Course	Teaching Scheme			Marks Distribution			Credits
		Lecture (L)	Tutorial (T)	Practical (P)	IE	ESE	Total	
A.		**Major (Core Courses)**						
A.1	**Theory**							
BCECCE6101	Information System Security	3	-	-	40	60	100	3
BCECCE6102	Data Mining Techniques and Application	3	-	-	40	60	100	3
BCECCE6103	Salesforce	3	-	-	40	60	100	3
A.2	**Practical**							
BCECCE6201	Data Mining Lab	-	-	2	60	40	100	1
BCECCE6202	Salesforce Lab	-	-	2	60	40	100	1
B.		**Minor Stream Courses/ Department Electives**						
B.1	**Theory**							
	Professional Elective - VI	3	-	-	40	60	100	3
	Professional Elective - VII	3	-	-	40	60	100	3
B.2	**Practical**							
	Professional Elective Lab – IV	-	-	2	60	40	100	1
	Professional Elective Lab - V	-	-	2	60	40	100	1
C		**Multidisciplinary Courses**						
	MOOC Course-V	1	-	-	40	60	100	2
D		**Ability Enhancement Courses (AEC)**						
BUACHM6219	Leadership Communication Lab	-	-	2	60	40	100	1
E		**Skill Enhancement Courses (SEC)**						
BULCSE6201	Skill Enhancement General Courses-VI	-	-	2	60	40	100	1
F		**Value Added Courses (VAC)**						
	-	-	-	-				
G		**Summer Internship / Research Project / Dissertation**						
BCECCE6401	Industrial Training Seminar-I	-	-	2	60	40	100	1
Total		16	-	14				**24**
Total Teaching Hours		**30**						

POORNIMA UNIVERSITY, JAIPUR
Faculty of Computer Science and Engineering

Name of Program: B.Tech. CSE **Duration: 4 Years** **Total Credits: 172**

Teaching Scheme for Batch 2024-28

Semester-VII

Course Code	Name of Course	Teaching Scheme			Marks Distribution			Credits
		Lecture (L)	Tutorial (T)	Practical (P)	IE	ESE	Total	
A.		**Major (Core Courses)**						
A.1	**Theory**							
BCECCE7101	Big Data Analytics	3	-	-	40	60	100	3
A.2	**Practical**							
BCECCE7201	Big Data Analytics Lab	-	-	2	60	40	100	1
BCECCE7202	IoT Lab	-	-	2	60	40	100	1
B.		**Minor Stream Courses/ Department Electives**						
B.1	**Theory**							
	Professional Elective – VIII	3	-	-	40	60	100	3
	Professional Elective - IX	3	-	-	40	60	100	3
B.2	**Practical**							
	Professional Elective Lab – VI	-	-	2	60	40	100	1
	Professional Elective Lab - VII	-	-	2	60	40	100	1
C		**Multidisciplinary Courses**						
	NIL							
D		**Ability Enhancement Courses (AEC)**						
BUACHM7116	Interpersonal Communication & Grooming	-	-	2	60	40	100	1
E		**Skill Enhancement Courses (SEC)**						
BULCSE7201	Skill Enhancement Technical Courses-I	2	-	-	40	60	100	2
F		**Value Added Courses (VAC)**						
	-	-	-	-				
G		**Summer Internship / Research Project / Dissertation**						
BCECCE7301	Minor Project	-	-	4	60	40	100	2
BCECCE7401	Industrial Training Seminar-II			2	60	40	100	1
Total		11	-	16				**19**
Total Teaching Hours		**30**						

POORNIMA UNIVERSITY, JAIPUR
Faculty of Computer Science and Engineering

Name of Program: B.Tech. CSE **Duration: 4 Years** **Total Credits: 172**

Teaching Scheme for Batch 2024-28

Semester-VII

Course Code	Name of Course	Teaching Scheme			Marks Distribution			Credits
		Lecture (L)	Tutorial (T)	Practical (P)	IE	ESE	Total	
A.		**Major (Core Courses)**						
A.1	**Theory**							
	NIL	-	-	-	-	-	-	-
A.2	**Practical**							
	NIL	-	-	-	-	-	-	-
B.		**Minor Stream Courses/ Department Electives**						
B.1	**Theory**							
	NIL	-	-	-	-	-	-	-
B.2	**Practical**							
	NIL	-	-	-	-	-	-	-
C		**Multidisciplinary Courses**						
	NIL	-	-	-	-	-	-	-
D		**Ability Enhancement Courses (AEC)**						
	NIL	-	-	-				
E		**Skill Enhancement Courses (SEC)**						
	NIL	-	-	-	-	-	-	-
F		**Value Added Courses (VAC)**						
	NIL	-	-	-	-	-	-	-
G		**Summer Internship / Research Project / Dissertation**						
BCECCE8301	Major Project			20	60	40	100	10
Total			-	20				**10**
Total Teaching Hours		**20**						

PROFESSIONAL ELECTIVE COURSES
(Session 2024-28)

	For General		For Artificial Intelligence & Data Science		For Cyber Security		For Cloud & Full Stack Development		For Artificial Intelligence & Machine Learning (SAS)	
	Code	**Name**	**Code**	**Name**	**Code**	**Name**	**Code**	**Name**	**Code**	**Name**
Professional Elective-I	BTXECE2111	Introduction to Emerging Technology	BADECE2111	Introduction to AI	BCSECE2111	Introduction to Cyber Security	BCGECE2111	Introduction to Cloud Computing	BAMECE2111	Base SAS Programming
Professional Elective-II	BCEECE3111	Theory Of Computation	BADECE3111	Mathematics for Machine Learning	BCSECE3111	Cyber Criminal Law & IPR	BCGECE3111	Principle of Virtualization	BAMECE3111	SAS Programing in Viya
Professional Elective-III	BCEECE4111	Web Frameworks	BADECE4111	Fundamentals of Machine Learning	BCSECE4111	Ethical Hacking	BCGECE4111	Installation and Configuration of Server	BAMECE4111	SAS® Enterprise Guide®: ANOVA, Regression, and Logistic Regression
Professional Elective-IV	BCEECE5111	Artificial Intelligence	BADECE5111	RPA Tool	BCSECE5111	Network Defence for Cyber Security	BCGECE5111	Cloud Web Services	BAMECE5111	SAS® Visual Text Analytics in SAS®
Professional Elective-V	BCEECE5112	Mobile Application Development	BADECE5112	R Programming	BCSECE5112	Security Audit and risk Management	BCGECE5112	Java Script Framework	BAMECE5112	Data Analytics using R Programming
Professional Elective-VI	BCEECE6111	Advance Web Technology	BADECE6111	Deep Learning	BCSECE6111	Vulnerability Analysis and Penetration Testing	BCGECE6111	Linux Server Administration	BAMECE6111	Deep Learning Using SAS® Software
Professional Elective-VII	BCEECE6112	Software Testing	BADECE6112	NLP & Computer Vision	BCSECE6112	Cyber Forensics	BCGECE6112	No SQL Database	BAMECE6112	Reinforcement Learning
Professional Elective-VIII	BCEECE7111	ASP.Net	BADECE7111	Python for Time Series Data Analysis	BCSECE7111	Malware Analysis & Wireless Security	BCGECE7111	Docker Containers & Kubernetes Fundamentals	BAMECE7111	Optimization Concepts for Data Science and Artificial Intelligence
Professional Elective-IX	BCEECE7112	Cloud Computing	BADECE7112	Introduction to Gen AI & Prompt Engineering	BCSECE7112	Cyber Threat intelligence & Bug Bounty	BCGECE7112	Web Framework using Python	BAMECE7112	Speech & Video Processing

PROFESSIONAL ELECTIVE LABs
(Session 2024-28)

	For General		For Artificial Intelligence & Data Science		For Cyber Security		For Cloud & Full Stack Development		For Artificial Intelligence & Data Science (SAS)	
	Code	**Name**	**Code**	**Name**	**Code**	**Name**	**Code**	**Name**	**Code**	**Name**
Professional Elective Lab -I	BCEECE4211	Web Frameworks Lab	BADECE4211	Fundamentals of Machine Learning Lab	BCSECE4211	Ethical Hacking Lab	BCGECE4211	Installation and Configuration of Server Lab	BAMECE4211	Application of Machine Learning Using SAS(R) Viya Lab
Professional Elective Lab -II	BCEECE5211	Artificial Intelligence Lab	BADECE5211	RPA Tool Lab	BCSECE5211	Network Defence for Cyber-Security Android Lab	BCGECE5211	Cloud Web Services Lab	BAMECE5211	Neural network: Essentials (SAS) Lab
Professional Elective Lab- III	BCEECE5212	Mobile Application Development Lab	BADECE5212	R Programming Lab	BCSECE5212	Database Security Lab	BCGECE5212	Java Script Framework Lab	BAMECE5212	Data Analytics using R Programming Lab
Professional Elective Lab -IV	BCEECE6211	Advance Web Technology Lab	BADECE6211	Deep Learning	BCSECE6211	Vulnerability Analysis and Penetration Testing Lab	BCGECE6211	Linux Server Lab	BAMECE6211	Forecasting Using Model Studio in SAS® Viya®
Professional Elective Lab -V	BCEECE6212	Software Testing Lab	BADECE6212	NLP & Computer Vision Lab	BCSECE6212	Cyber Forensics Lab	BCGECE6212	Mongo DB Lab	BAMECE6212	Reinforcement Learning Lab
Professional Elective Lab- VI	BCEECE7211	ASP.Net Lab	BADECE7211	Python for Time Series Data Analysis Lab	BCSECE7211	Malware Analysis & Wireless Security Lab	BCGECE7211	Docker Containers & Kubernetes Fundamentals	BAMECE7211	Data Exploration & Visualization Lab
Professional Elective Lab- VII	BCEECE7212	Cloud Computing Lab	BADECE7212	Data Handling & Visualization lab	BCSECE7212	Cyber Threat intelligence & Bug Bounty Lab	BCGECE7212	Web Framework using Python Lab	BAMECE7212	Time Series Data Analysis Lab

Comparison according to NEP Guidelines
***TEP 8 Credit**

	Broad Category of Course	Minimum Credit Requirement	
		NEP	PU-CSE
1	Major (Core)	80	93
2	Minor Stream	32	34
3	Multidisciplinary	09	10
4	**Ability Enhancement Courses (AEC)**	08	8
5	Skill Enhancement Courses (SEC)	09	8
6	Value Added Courses common for all UG	06 – 08	8
7	Summer Internship	02 – 04	2
8	Research Project / Dissertation	12	12

Summary Year wise credit distribution: -

		Major	Minor	MD	AEC	SEC	VAC	SI	Project	TEP	
I	Credit	17			1	1	2				21
II	Credit	17	3	2	1	1					24
III	Credit	16	3	2	1	1	2				25
IV	Credit	15	4	2	1	1	2				25
V	Credit	12	8	2	1	1					24
VI	Credit	11	8	2	1	1		1			24
VII	Credit	5	8		1	2		1	2		19
VIII	Credit								10		10
		93	34	10	7	8	6	2	12	0	172

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POORNIMA UNIVERSITY

Member of Association of Indian Universities & Approved by UGC (Govt. of India) under 2(f) & 12(B)

B. TECH - CSE

FIRSY YEAR

SEMESTER: I

COMMON SYLLABUS

For

Cyber Security
Cloud & Full Stack Development
Artificial Intelligence & Data Science
Artificial Intelligence & Machine Learning

BATCH: 2024-28

Code: BTXCSA1101/2101 **Engineering Physics** **3 Credits [LTP: 3-0-0]**

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

CO	At the end of this course, learners will be able to:	Bloom Level	PO Mapping
CO1	Differentiate between the Diamagnetism, Para magnetism and ferromagnetism	L1	PO1, PO2, PO3, PO4, PO6, PO7
CO2	Understand the principle behind Newton's rings interference pattern, its construction using a plano-convex lens and a flat glass plate	L1	PO1, PO2, PO3, PO4, PO6, PO7
CO3	Explain the concept of matter waves and describe their wave-particle duality using the de Broglie wavelength equation.	L1	PO1, PO2, PO3, PO4, PO6, PO7
CO4	Provide a solid foundation for understanding the principles behind many electronic devices and applications like Hall effect, PN Junction	L3	PO1, PO2, PO3,PO4,PO5 PO6, PO7
CO5	Comprehensive foundation in the principles and applications of light amplification, He-Ne laser, and holography	L3	PO1, PO2, PO3, PO4, PO6, PO7

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	-	-	-	-	-	-	-	-	-	3	-	-
CO2	3	3	2	-	-	-	-	-	-	-	-	-	3	-	-
CO3	2	2	2	-	-	-	-	-	-	-	-	-	3	-	-
CO4	2	2	3	-	-	-	-	-	-	-	-	-	2	-	-
CO5	3	2	2	-	-	-	-	-	-	-	-	-	1	-	-
WT. AVG	2.6	2.4	2.2	-	-	-	-	-	-	-	-	-	2.4	-	-

C. OUTLINE OF THE COURSE

Unit No.	Title of the Unit	Time required for the Unit (Hours)
1.	**Material Science**	**09**
2.	**Wave optics**	**09**
3.	**Quantum Mechanics**	**09**
4.	**Semiconductor Physics**	**09**
5.	**Laser and Holography**	**09**

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Material Science**
	• Introduction of Unit • Fundamentals of Crystal structure, • Para-magnetism, Diamagnetism, Ferromagnetism, Ferrimagnetism, Antiferromagnetic, Magnetic Permeability, Magnetization, Paramagnetic Susceptibility of Solid Substances, and Explanation of Hysteresis. • Conclusion of Unit
2.	**Wave optics**
	• Introduction of Unit • Interference of light, Types of Interference, coherent source, methods to produce coherent sources with examples. • Newton's rings: Principle, Construction, working and Applications, Anti Reflection Coating, Fundamental idea about optical fiber, types of fiber, acceptance angle, numerical aperture. • Conclusion of Unit
3.	**Quantum Mechanics**
	• Introduction of Unit • Matter waves and properties. Group and Particle velocities & their relationship. • Compton scattering, wave function and its properties., Energy and momentum operators. • Time dependent and time independent Schrodinger wave equation. • Conclusion of Unit
4.	**Semiconductor Physics**
	• Introduction of Unit • Energy bands in semiconductors, Types of semiconductors, Charge carriers, • Intrinsic and extrinsic materials, Carrier Concentration, Conductivity and Mobility, Hall effect, PN Junction diode, Zener diode, Solar Cells • Conclusion of Unit
5.	**Laser and Holography**
	• Introduction of Unit • Introduction of LASER, Conditions for the Light Amplification, • Population Inversion and metastable state, pumping. • Types of lasers: He-Ne laser and diode laser and their working principle and Holography • Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL

Sr. No	Reference Book	Author	Edition	Publication
1	Materials Science and Engineering	by Raghavan V	Latest	PHI
2	Fundamentals Of Engineering Physics	SS Rawat	Latest	CBH Publications
3	Solid State Physics	By S. O. Pillai	Latest	NEW AGE
4	Concepts of modern physics	Beiser, Arthur	Latest	Mc Graw Hill
5	Introduction to Quantum Mechanics	David J. Griffiths and Darrell F. Schroeter	Latest	Cambridge University Press
Important Web Links				
1	https://jiet.claybits.com/			
2	https://jiet.claybits.com/physics/1-5			
3	https://jiet.claybits.com/physics/3-1			

Code: BTXCSA1102/2102 ENGINEERING CHEMISTRY 3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Enumerate the importance, synthesis and applications of polymers.	L2	PO1, PO2, PO3, PO4, PO6, PO7
CO2	Illustrate the fundamental principles of water chemistry, applications of volumetric and analytical instrumentation	L1	PO1, PO2, PO3, PO4, PO6, PO7
CO3	Explain the fundamental concepts of corrosion, its control and surface modification methods namely electroplating and electroless plating	L2	PO1, PO2, PO3, PO4, PO6, PO7
CO4	Impart the basic knowledge of chemistry and its principles involved in electrochemistry, energy storage devices and its commercial applications	L3	PO1, PO2, PO3,PO4,PO5 PO6, PO7
CO5	Learn about the manufacturing of cement and the chemistry involved in setting and hardening of it and also learn about the suitable use of lubricants	L2	PO1, PO2, PO3, PO4, PO6, PO7

B. Mapping matrix of CO & PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	1	1	-	1	1	-	-	-	-	-	-	-	-
CO2	2	1	2	1	-	2	2	-	-	-	-	-	-	-	-
CO3	3	2	2	1	-	2	2	-	-	-	-	-	-	-	-
CO4	3	1	2	1	1	2	2	-	-	-	-	-	-	-	-
CO5	2	1	1	1	-	2	2	-	-	-	-	-	-	-	-
Wt. AVG	2.6	1.2	1.6	1	1	1.8	1.8						-	-	-

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	olymer Basics and Applications	7
2.	Vater Chemistry and methods of analysis	7
3.	Corrosion and Metal finishing	7
4.	lectrochemistry and energy storage systems	8
5.	ngineering Materials	7

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Polymer Basics and Applications**
	• Introduction of Unit • Introduction to Polymer chemistry:, • Classification of Polymers and Types of polymerization • Plastics: Thermosets and Thermoplastics, Preparation, Properties and Uses of Polyethylene, Bakelite, Teflon, and Nylon, Single use plastic: chemical composition, usage, disposal, and impacts • Nanoplastics and Human Health,Flexible materials: Natural rubber, Vulcanization, Synthetic rubber- Preparation, Properties and Applications of SBR, Buna-N and flexible materials.

	• Conclusion of Unit
2.	**Water Chemistry and Methods of Analysis**
	• Introduction of Unit Water • Sources of water, Impurities in water and effect of impurities • Municipal water supply: Requisites of drinking water, Alkaline Water: Benefits, Side Effects And Dangers • Steps involved in purification of water, Sedimentation, Coagulation, Filtration and Disinfection, Break Point Chlorination • Water Analysis • Hardness of water; Type of hardness , Degree of hardness, Units of hardness ,Disadvantages of hard water, Determination of hardness by Complexometric (EDTA) method , Numericals based on hardness by EDTA method • Treatment of hard water: Lime-soda method, Permutit (zeolite) method, RO water treatment and Deionization or Demineralization method • Conclusion of Unit
3.	**Corrosion and Metal Finishing**
	• Introduction of the Unit • Definition of corrosion and its Significance • Mechanisms of Corrosion: Chemical (Dry) corrosion and Electrochemical (Wet) corrosion • Types of corrosion: Galvanic corrosion, Concentration cell corrosion, Stress corrosion, Pitting corrosion, Methods of protection against corrosion: Galvanization, Sacrificial coatings etc. • Metal finishing: Technological importance, Electroplating and Electroless plating • Conclusion of Unit
4.	**Electrochemistry and Energy Storage Systems**
	• Introduction of the Unit • EMF of cell, Free Energy, Single electrode potential, Nernst equation, Numerical problems based on Nernst Equation. • Electrodes: Calomel electrode and Glass electrode, Energy storage Systems: Introduction, Classification of batteries. Construction, working and applications of Li-ion batteries. • An electrochemical energy system for electric vehicles. Recycling of Lithium-ion batteries by direct cycling Method. • Conclusion & Real life applications.
5.	**Engineering Materials**
	• Introduction of the Unit • Portland Cement; Definition, Manufacturing by Rotary kiln. Chemistry of setting and hardening of cement. Role of Gypsum. • Lubricants: Classification, Mechanism, Properties; Viscosity and viscosity index, flash and fire point, cloud and pour point. Emulsification and steam emulsion number. • Car and motorcycle engine oil: 0W-20 engine oil, 4T engine oil, • Conclusion & Real life applications

RECOMMENDED STUDY MATERIAL:

S. No	Reference Book	Author	Publication
1	ngineering chemistry: A Text book	.K. Jain & K.D. Gupta	aipur Publishing House
2	ngineering chemistry: A Text book	.S. Dara	. Chand & Co
3	ngineering chemistry: A Text book	.C. Jain	hanpat Rai & Sons.

Code: BTXCCE1103 Fundamental of Computer 3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Learn data types, loops, functions, array, pointers, string, structures and files	L2	PO1, PO2, PO3, PO4, PO6, PO7
CO2	Develop conditional and iterative statements to write C programs	L1	PO1, PO2, PO3, PO4, PO6, PO7
CO3	Implement concept of string using array.	L2	PO1, PO2, PO3, PO4, PO6, PO7
CO4	Allocate memory dynamically using pointers	L3	PO1, PO2, PO3,PO4,PO5 PO6, PO7
CO5	Application of file handling and dynamic memory allocation	L2	PO1, PO2, PO3, PO4, PO6, PO7

B. Mapping matrix of CO & PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	-	-	1	-	-	-	-	-	3	-	-
CO2	3	2	3	3	-	-	2	-	-	-	-	-	2	-	-
CO3	2	2	1	1	-	-	2	-	-	-	-	-	2	-	-
CO4	3	1	2	1	-	-	2	-	-	-	-	-	3	-	-
CO5	2	2	1	1	-	-	2	-	-	-	-	-	2	-	-
Wt. AVG	2.6	2	1.8	1.8			1.8						2.4		

D. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to C Programming**	**7**
2.	**Decision Making & Looping**	**7**
3.	**Array and string**	**7**
4.	**Advance programming in C**	**8**
5.	**File handling & Additional features**	**7**

E. DETAILED SYLLABUS

Unit	Unit Details
1	**Introduction to C Programming**
	• Introduction of Unit • Introduction to computer-based problem solving, Program design and implementation issues- Flowcharts & Algorithms. • Types of Languages – Machine language, assembly language, high level languages, Assemblers, Compilers, Interpreters. • Overview of C, Data Types, Constants & Variables, Literals, Operators & Expressions Conclusion & Real-Life Application
2	**Decision Making & Looping**
	• Introduction of Unit • Decision making in C- if statement, if-else statement, Nested if statement, if else if Ladder, Switch case, Loop control in C – for loop, while loop

	• Control flow in C- break, continue and goto statement. Conclusion & Real-Life Application
3	**Array and string**
	• Introduction of Unit • Array- 1D array, 2D array, Scope rules- Local & global variables. • Functions-parameter passing call by value and call by reference, calling functions with arrays, command line argument., String – String in-build functions. Conclusion of the Unit
4	**Advance programming in C**
	• Introduction of Unit • Pointers- The & and * operator, pointer expression, assignments, arithmetic, comparison, arrays of pointers, pointers to pointers, initializing pointers, pointers to functions. • Structures- Basics, declaring, referencing structure elements, array of structures, passing structures to functions, structure pointers. Conclusion of the Unit
5	**File handling & Additional features**
	• Introduction of Unit • File Handling – The file pointer, file accessing functions-fopen, fclose, putc, getc, fprintf, reading and writing into a file • Advance features- storage classes and dynamic memory allocation • C Preprocessor- #define, #include, #undef. • C standard library and header files: Header files, string functions, mathematical functions, Date and Time functions. Conclusion of the Unit

RECOMMENDED STUDY MATERIAL:

S. No	Reference Book	Author	Publication
1.	Letus C, 6th Edition	Yashwant Kanitkar	PBP Publication
2.	The C programming Language	Richie and Kenninghan	BPBPublication,2004
3.	Programming in ANSI C3rd Edition, 2005	E.Balagurusamy	Tata McGraw Hill

Reference Book	
1	The C programming Language Richie and Kenninghan PBP Publication,2004
2	Programming in ANSI C 3rd Edition, 2005 Balaguruswmy Tata McGraw Hill
Online resources	
1	https://www.programiz.com/c-programming/examples
2	https://www.w3resource.com/c-programming-exercises

Code: BTXCCV1104/2104 Basics of Civil Engineering 3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (CO):	At the end of this course, learners will be able to:	Bloom Level
CO1	Understand the various aspect of civil engineering, its novel areas and Career Prospects in Civil Engineering.	L1
CO2	Demonstrate the characteristics of different Construction Materials and constrtuction techniques in Civil Engineering.	L1
CO3	Identify the various building components, and analyze the construction methods and basic principles.	L4
CO4	Understand different types of surveying works required in construction and compare diffenent surveying instruments.	L1
CO5	Understand about the recent advancements in Civil Engineering.	L1

A. MAPPING MATRIX OF CO,PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	-	-	-	-	-	-	-	-	1	-	3
CO2	2	3	1	2	-	-	-	-	-	-	-	-	2	-	3
CO3	3	2	2	2	-	-	-	-	-	-	-	-	1	-	3
CO4	3	3	1	2	-	-	-	-	-	-	-	-	1	-	3
CO5	3	3	2	2	-	-	-	-	-	-	-	-	2	-	3
WT. AVG	2.8	2.6	1.6	1.8									1.6		3

B. OUTLINE OF THE COURSE

Unit No.	**Title of the Unit**	**Time required for the Unit (Hours)**
1.	Introduction to Civil Engineering	**08**
2.	Construction Materials and techniques	**08**
3.	Building Construction	**08**
4.	Basic Surveying	**08**
5.	Advancements in Civil Engineering	**08**

C. DETAILED SYLLABUS

Unit	**Unit Details**
1	**Introduction to Civil Engineering**
	• Introduction • Different disciplines of Civil Engineering, Scope and prospects • Role of a Civil Engineer • Units of measurement, Unit conversion (Length, Area, Volume) • Infrastructure Engineering, Sustainability • Automation and Robotics in Construction • Novel areas in Construction industry • Career Prospects in Civil Engineering
2	**Construction Materials and techniques**
	• Introduction • Cement – Types, properties, grades, Concrete– PCC and RCC, Cement mortar • Stone, Requirements of good building stone, General types of stone used in Construction. • Bricks, Modular and Standard bricks, Characteristics of good brick, Field tests on Bricks, Special bricks –fly ash bricks.

	• Timber, Structure of timber, General properties and uses of good timber, Use of bamboo in construction • Asphalt, bitumen and tar used in construction, their properties and uses • Types of Stone Masonry (Rubble masonry, Ashlar Masonry) and Brick Masonry (English bond and Flemish bond).
3	**Building Construction**
	• Introduction • Classification of Buildings as per National Building Code Group A to I • Types of Constructions- Load Bearing Structure, Framed Structure, Composite Structure • Building Components - Functions of Building Components, Substructure-Foundation, Plinth & Superstructure. • Selection of site for different types of Buildings • Basic principles of building planning. • Pre-engineered steel building, Pre-fabricated buildings
4	**Basic Surveying**
	• Introduction • Survey – Principles, purpose and use • Types & Classification of surveying • Instruments used in chain survey: Chains, Tapes, Arrow, Ranging rod, Line ranger, Offset rod, Open cross staff, Optical square • Ranging: Direct and Indirect Ranging • Methods of chaining, obstacles in chaining. • Errors in chain and tape, Numerical based on errors in length due to incorrect length of chain & tape. • Modern surveying instruments– EDM, Total station, GPS
5	**Advancements in Civil Engineering**
	• Introduction • Smart city and it's features, Plastic Roads • Mass Transportation systems-BRTS, Metro • Rain water harvesting systems, Watershed Management • Green building, Energy efficient building • Development of River fronts, Heritage structures & its conservations • Affordable housing & their features • Building Information Modeling (BIM), 3D Printing Technology for Construction, Advanced Civil Engineering Materials

D. RECOMMENDED STUDY MATERIAL

Sr.No	Reference Book	Author	Edition	Publication
1	Basics of Civil Engineering	S.S. Bhavikatti	Latest	New Age International Publishers
2	Basic Civil Engineering	B C Punmia, Ashok K Jain, Arun K Jain	Latest	Laxmi Publications
3	Basic Civil Engineering	G K Hiraska	Latest	Dhanpat Rai Publication
4	Basic Civil Engineering	Jhonson Victor D and Esther Malini	Latest	Allied Publishers Limited, Madras
5	Basic Civil Engineering	Arunachalam N	Latest	Pratheeba Publishers, Coimbatore
Important Web Links				
1	https://archive.nptel.ac.in/courses/105/106/105106201/			
2	https://onlinecourses.nptel.ac.in/noc22_ce42/preview			

Code: BTXCEE1105/2105 Basics of Electrical and Electronics 3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (CO):	At the end of this course, learners will be able to:	Bloom Level
CO1	Apply basic electrical concepts, including various circuit analysis techniques and fundamentals of theorem to solve numerical problem related to basic electrical circuit.	L4
CO2	Demonstrate the fundamentals of AC circuits and Electric Installation by calculating voltage, current, impedance, and power factors.	L1
CO3	Compare various electrical machines and the process of energy conversion specifying the type and characteristics of the power source supplying these machines, such as AC or DC, voltage levels, frequency, etc.	L3
CO4	Understand the fundamental principles of semiconductor devices such as diodes and transistors, knowing their roles in switching and amplification.	L1
CO5	Understand the working principles of communication systems, Transducers and fundamentals of IoT.	L1

B. MAPPING MATRIX OF CO,PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	3	-	-
CO3	2	2	-	2	-	-	1	-	-	-	-	-	2	-	-
CO4	2	3	-	-	-	-	-	-	-	-	-	-	1	-	-
CO5	2	3	-	-	-	-	-	-	-	-	-	1	2	-	-
WT. AVG	2.2	2.6	1	2			1					1	2.2		

C. OUTLINE OF THE COURSE

Unit No.	Title of the Unit	Time required for the Unit (Hours)
1.	Basic Concepts of Electrical Engineering	08
2.	Alternating Quantities and Electrical Installations	08
3.	Electrical Machines	07
4.	Basic Electronics	08
5.	Communication Systems and IoT	08

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Basic Concepts of Electrical Engineering**
	• **Introduction of Unit** • **Basic circuit elements and sources; Ohms law; Kirchhoff's laws; Series and Parallel connection of circuit elements; Star-delta transformation;**

	• Mesh current analysis; Node voltage analysis; • Theorems: Thevenin's, Maximum power transfer and Superposition theorem • Conclusion of Unit
2.	**Alternating Quantities and Electrical Installations**
	• Introduction of Unit • Alternating voltages and currents: RMS, average, maximum values, Single Phase RL, RC, RLC series circuits, Power in AC circuits, Power Factor, • Three phase balanced systems, Star and delta Connections • Electrical Safety, Fuses and Earthing • Conclusion of Unit
3.	**Electrical Machines**
	• Introduction of Unit • Construction, working principle and applications of DC Machines, Transformers, Three phase Induction motors, Synchronous generators, Single phase induction motors. • Special machines -Stepper motor, Universal motor and BLDC motor • Conclusion of Unit
4.	**Basic Electronics**
	• Introduction of Unit • Characteristics: PN junction diode, Zener diode, BJT, MOSFET; Applications: Rectifier, Voltage regulator, Operational amplifier • Digital Electronics: Boolean algebra, Binary System, Logic Gates and Their Truth Tables. • Conclusion of Unit
5.	**Communication Systems and IoT**
	• Introduction of Unit • Basics of Communication: Types of Communication, Amplitude and Frequency Modulation. • Introduction to Transducers, Thermocouple, RTD, Strain Gauges, Load Cell and Bimetallic Strip. • An overview of Internet of Things-Building blocks of IoT, IoT enabling technologies, Characteristics of IoT systems • Conclusion of Unit

RECOMMENDED STUDY MATERIAL

S.No	Reference Book	Author	Edition	Publication
1	Electrical and Electronic Technology	Edward Hughes et al,	Latest	Pearson Publication
2	Basic Electrical & Electronics Engineering	V. Jagathesan, K. Vinod Kumar & R. Saravan Kumar	Latest	Wiley India
3	Basic Electrical & Electronics Engineering	Van Valkenburge	Latest	Cengage learning
4	Basic Electrical and Electronics Engineering by,	Muthusubrmaniam	Latest	TMH
5	Basic Electrical & Electronics Engineering	Ravish Singh	Latest	TMH
Important Web Links				
1	https://nptel.ac.in/courses/108108076/			
2	https://nptel.ac.in/courses/117103063/			
3	https://nptel.ac.in/courses/108/101/108101091/			

Code: BTXCSA1106 /2106 ENGINEERING MATHEMATICS 3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

(COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Calculating eigen values and eigen vector	L4	PO1, PO5, PO6,
CO2	Determining ordinary differential equation and its types.	L3	PO1. PO2, PO3
CO3	Sketching standard curves using curve tracing.	L3	PO1, PO2, PO3, PO4
CO4	Calculating gamma function and double integration.	L4	PO1, PO2, PO3, PO4,
CO5	Associating Gradient, Divergence and Curl, Directional derivatives	L2	PO1, PO2, PO4

B. Mapping matrix of CO & PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	--	1	1	-	-	-	-	-	-	3	-	-
CO2	3	3	3	-	-	-	-	-	-	-	-	-	3	-	-
CO3	3	2	3	1	-	-	-	-	-	-	-	-	3	-	-
CO4	3	2	1	1	-	-	-	-	-	-	-	-	3	-	-
CO5	2	1		1	-	-	-	-	-	-	-	-	3	-	-
Wt. AVG	2.8	2	2.3	1	1	1	-	-	-	-	-	-	3	-	-

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Matrices	7
2.	Ordinary Differential Equations	8
3.	Applications of Differential Calculus	8
4.	Integral Calculus	8
5.	Introduction Vector Calculus	8

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Matrices**
	1. Introduction of Unit 2. Rank of a Matrix, Normal form of a Matrix, 3. Consistency of systems of linear equations 4. Eigen Values and Eigen Vectors 5. Cayley-Hamilton Theorem (without proof) 6. Conclusion of Unit
2.	**Ordinary Differential Equations**
	• Introduction of Unit • First order and first-degree differential equations-Separable Variables, • Homogenous and reducible to homogenous equation • Linear Equation and reducible to linear form, Exact Equation • Linear differential equations with constant coefficients • Conclusion of Unit
3.	**Applications of Differential Calculus**

	• Introduction of Unit • Asymptotes, Multiple points, Curve tracing for standard Curves (Cartesian Curves only) • Conclusion & Real life applications
4.	**Integral Calculus**
	• Introduction of Unit • Gamma functions and their properties, beta function (only definition) • Double integrals, Double integral by changing into polar form, Areas by Double Integration • Change of order of integration • Conclusion of Unit
5.	**Vector Calculus**
	• Introduction of Unit • Scalar and Vector field, Differentiation and Integration of Vector functions • Gradient, Divergence and Curl, Directional derivatives • Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

Sr.No	Reference Book	Author	Edition	Publication
1.	Higher Engineering Mathematics	B S Grewal	Latest	Khanna Publications, Delhi,
2.	Higher Engineering Mathematics	Ramana, B.V	Latest	Tata McGraw-Hill.
3	Engineering Mathematics: A Tutorial Approach	Ravish R Singh and M Bhatt	Latest	Tata McGraw-Hill
4	Calculus and Analytical Geometry	Thomas and Finney,	Latest	Narosa Publishing, New Delhi
5	Advanced Engineering Mathematics	Erwin Kreyszig	Latest	John Wiley and Sons
Important Web Links:				
1	https://nptel.ac.in/courses/111105134/			
2	https://nptel.ac.in/courses/122/101/122101001/			
3	https://www.classcentral.com/course/swayam-engineering-mathematics-i-13000			

Code: BTXCME1107/ 2107 Basic of Mechanical Engineering 3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Understand the primary manufacturing processes including metal casting, forming, and joining.	L2	PO1, PO2
CO2	Describe the operation and components of internal combustion (IC) engines and evaluate their performance parameters.	L2	PO1, PO2, PO5
CO3	Explain the principles of refrigeration, air conditioning systems, and basics of aerodynamics.	L2	PO1, PO2, PO5
CO4	Understand the architecture, components, and performance factors of electric vehicles.	L2	PO1, PO2, PO5
CO5	Apply ergonomic principles to the design of automotive seats and commercial vehicle cabins.	L3	PO1,PO2, PO3, PO5

B. Mapping matrix of CO & PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	-	-	3
CO2	3	3	-	-	2	-	-	-	-	-	-	-	-	-	3
CO3	3	3	-	-	2	-	-	-	-	-	-	-	-	-	3
CO4	3	3	-	-	2	-	-	-	-	-	-	-	-	-	3
CO5	3	3	2	-	2	-	-	-	-	-	-	-	2	-	3
Wt. AVG	3	2.8	2		2								2		3

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1	Primary Manufacturing Processes	10
2	IC Engines	7
3	Refrigeration and Air Conditioning and Aerodynamics	9
4	Electric Vehicles	7
5	Ergonomics	7

D. DETAILED SYLLABUS

Unit	Unit Details
1	**Primary Manufacturing Processes**
	Metal Casting Process: Introduction to Casting Process, Patterns, Molding. Metal Forming Processes: Introduction to Forging, Rolling, Extrusion, Drawing. Metal Joining Processes: Introduction to Welding Processes, Gas Welding, Arc Welding, Brazing, and Soldering.
2	**IC Engines**
	Introduction, Heat Engine, IC engine components and nomenclature; Classification of ICEngines, SI and CI engine operation - two stroke and four stroke engines - construction - working principle; Engine performance parameters, Valve and port timing diagram.
3	**Refrigeration and Air Conditioning and Aerodynamics**
	Introduction, Clausius statement of second law of thermodynamics, Refrigerator, Heat Pump,COP, Vapour compression refrigeration system (VCRS), Vapour absorption refrigeration system (VARS), Window air-conditioning, Properties and nomenclature of refrigerants. Basicsof aerodynamics, Jet propulsion, Turbojet engine.
4	**Electric Vehicles**
	Architecture of an electric vehicle (EV), Components of an electric car, advantage and limitations EV, Performance Factors of Electric Vehicles, working of traction motor, tractiveeffort, Traction motor characteristics, transmission requirements, Energy storage – batteries.
5	**Ergonomics**
	Introduction to Ergonomics, Ergonomics design of automotive driver seat, Commercialvehicle cabin ergonomics, Goods vehicle layout; Design of Work Areas and Stations - Traffic Areas; Human factors in industrial safety; Ergonomic injuries.

RECOMMENDED STUDY MATERIAL:

Sr.No	Reference Book	Author	Edition	Publication
1	Basics of Mechanical Engineering	Punia	Latest	Dhanpat Rai
2	Basics of MechanicalEngineering	R.K. Rajput	Latest	Laxmi
3	Basics of MechanicalEngineering	DS Kumar	Latest	Kataria
4	IC Engines	V. Ganesan	Latest	Tata McGraw
5	Introduction to Ergonomics	R.S. Bridger	Latest	CRC Press

Code: BTXCCE2103 PROGRAMMING IN PYTHON 3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Understand the basic Python syntax, including data types, Keywords, Variables, and Identifiers.	L1	PO1,PO2, PO3
CO2	Demonstrate basic operators and understand the concept of Decision making, Conditional Statements and loops for problem solving	L2	PO1,PO2, PO3
CO3	Execute and use the concept of Lists, Tuple, Dictionaries, functions, and different python packages.	L3	PO1,PO2, PO5
CO4	Evaluate various object-oriented programming design principles to create structured python class and objects.	L4	PO1,PO2
CO5	Illustrate and Handle files in python which are raised during the execution of Python scripts	L4	PO1,PO3, PO12

B. Mapping matrix of CO & PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	1	-	-	-	-	-	-	-	-	-	2	-	-
CO3	2	3	-	-	1	-	-	-	-	-	-	-	1	-	3
CO4	3	2	-	-	-	-		-	-	-	-	-	2	-	-
CO5	3	-	2	-	-	-	-	-	-	-	-	1	2	1	-
Wt. AVG	2.6	2.5	1.3		1							1	1.8	1	3

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Python Programming	7
2.	Python Operators and Control Flow statements	7
3.	Data Structures, Python Functions and Packages	7
4.	Object Oriented Programming	8
5.	File I/O Handling and Exception Handling	7

D. DETAILED SYLLABUS

Unit	Unit Details
1	**Introduction to Python Programming**
	• Introduction to Unit • What is Python, • Uses of Python Programming Language / Python Applications

	• Features of Python Programming Language • Python-2 and Python-3 differences • Python environment setup — Installation and working of IDE • Running Simple Python scripts to display 'welcome' message. • Python Data Types: Numbers, String, Tuples, Lists, Dictionary. Python building blocks — Identifiers, Keywords, Indention, Variables, Comments • Conclusion of unit
2	**Python Operators and Control Flow statements**
	• Introduction to Unit • Basic Operators: Arithmetic, Comparison/ Relational, Assignment, Logical, Bitwise, Membership, Identity operators, Python Operator Precedence • Control Flow: • Conditional Statements (if, if ... else, nested if) • Looping in python (while loop, for loop, nested loops) • Conclusion of Unit
3	**Data Structures, Python Functions and Packages**
	• Introduction to Unit • Lists, Tuple, Sets • String and Slicing • Use of Python built • User defined functions and its types • Using standard packages (math, scipy, Numpy, pandas) • Conclusion of Unit
4	**Object Oriented Programming**
	• Introduction of Unit • Creating Classes and Objects • Inheritance • Method Overloading and Overriding • Data Hiding • Types of Methods : Instance Methods , Static Methods , Class Methods • Conclusion of Unit
5	**File I/O Handling and Exception Handling**
	• Introduction of Unit • Types of File • File Objects, File Built-in Function, File Built-in Methods • File Built-in Attributes • Read/write operations Reading Text • Errors in Python : Compile-Time Errors ,Runtime Errors , Logical Errors • Regular expressions • Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Reference Book	Author	Publication
	Core Python Programming	Chun, JWesley	Pearson,2007
	Head First Python	Barry,Paul	Orielly,2010
	Learning Python	Lutz, Mark	O Rielly, 2009

Code: BTXCSA1201 /2101 ENGINEERING PHYSICS LAB 1 Credits [LTP: 0-0-2]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course outcomes	On completion of this course, the students will be able:	Bloom Level
CO-1	To perform experiment and determine the wavelength using newton's ring setup & dispersive power of prism for violet, red & yellow color using spectrometer	L4
CO-2	To perform experiment and determine wavelength of prominent lines of mercury by plane diffraction grating using spectrometer	L3
CO-3	To measure the numerical aperture of an optical fiber and the coherent length and coherent time by using He-Ne Laser.	L3
CO-4	Determine the Band Gap of the semiconductor in the form of reverse biased P-N junction diode and determine forward and reverse bias resistance for semiconductor diode.	L5
CO-5	To determine the height of a given line drawn on wall by sextant, determine time constant (both current and voltage graphs are to be plotted)	L5

B. MAPPING OF COURSE OUTCOMES WITH PROGRAMME OUTCOMES

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	2	-	-	-	-	-	-	2	2	-	—	3
CO2	3	3	2	-	-	-	-	-	-	-	2	3	-	—	3
CO3	2	3	2	-	-	-	-	-	-	-	2	3	-	—	3
CO4	3	3	1	2	-	-	-	-	-	-	1	3	-	—	3
CO5	3	3	1	-	-	-	-	-	-	-	2	3	-	—	3
Average	3	2.8	1.6	2	-	-	-	-	-	-	1.8	2.8	-	-	3

C. List of Experiment –

1.	To determine the wavelength of Sodium light by Newton's Ring.
2.	To determine the dispersive power of material of a prism for violet, red & yellow colour of Mercury light with the help of spectrometer.
3.	To determine the wavelength of prominent lines of mercury by plane diffraction grating with the help of spectrometer.
4.	To verify the expression for the Resolving Power of Telescope.
5.	To measure the numerical aperture of an optical fiber by He-Ne Laser.
6.	To determine the coherent length and coherent time by using He-Ne Laser.
7.	To study the variation of a semiconductor resistance with temperature and hence determine the Band Gap of the semiconductor in the form of reverse biased P-N junction diode.
8.	To study the characteristics of semiconductor diode and determine forward and reverse bias resistance.
9.	To determine the height of a given line drawn on wall by sextant.
10.	To study Charging and Discharging of a condenser and determine time constant (both current and voltage graphs are to be plotted)

11.	To determine the high resistance by method of leakage, using a Ballistic Galvanometer.
12.	To specify the specific resistance of a material of a given wire by Carey Foster's bridge.

Virtual Lab Practical's and related link	
1	To find the resolving power of the prism. • https://vlab.amrita.edu/index.php?sub=1&brch=281&sim=1524&cnt=1
2	1. To revise the concept of interference of light waves in general and thin-film interference in particular. 2. To set up and observe Newton's rings. 3. Find the refractive index of given liquid, To determine the wavelength of the given source. https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=1520&cnt=1
3	To calculate the beam divergence and spot size of the given laser beam. **https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=342&cnt=1**
4	**To find the numerical aperture of a given optic fibre and hence to find its acceptance angle.** **https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1**
5	To draw the static current-voltage (I-V) characteristics of a junction diode. **https://vlab.amrita.edu/index.php?sub=1&brch=282&sim=1522&cnt=1**
6	**To construct a zener diode voltage regulator and measure its line and load regulation.** **https://vlab.amrita.edu/index.php?sub=1&brch=282&sim=1207&cnt=1**

Code: BTXCSA1202/ 2102 ENGINEERING CHEMISTRY LAB 1 Credits [LTP: 0-0-2]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Analyze the strength of NaOH and Na_2CO_3 solutions	L2	PO1, PO2, PO3, PO4, PO6, PO7
CO2	Measure hardness and chloride content of water	L1	PO1, PO2, PO3, PO4, PO6, PO7
CO3	Analyze hardness strength of Ferrous Ammonium sulphate solution and CuSO4 solution	L2	PO1, PO2, PO3, PO4, PO6, PO7
CO4	Determine the different properties of lubricating oil	L3	PO1, PO2, PO3,PO4,PO5 PO6, PO7
CO5	Use different instruments & analytical techniques.	L2	PO1, PO2, PO3, PO4, PO6, PO7

B. Mapping matrix of CO, PO & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	1	1	-	1	1	-	-	-	-	-	-	-	-
CO2	2	1	2	1	-	2	2	-	-	-	-	-	-	-	-
CO3	3	2	2	1	-	2	2	-	-	-	-	-	-	-	-
CO4	3	1	2	1	1	2	2	-	-	-	-	-	-	-	-
CO5	2	1	1	1	-	2	2	-	-	-	-	-	-	-	-
Wt. AVG	2.6	1.2	1.6	1	1	1.8	1.8						-	-	-

C. LIST OF EXPERIMENTS

1	To determine the strength of NaOH and Na_2CO_3 in a given alkali mixture.
2	To determine the strength of Ferrous Ammonium sulphate solution with the help of $K_2Cr_2O_7$ solution using diphenyl amine as internal indicator.
3	To determine the hardness of water by EDTA method.
4	To determine the amount of chloride content in drinking water
5	To determine the strength of $CuSO_4$ solution with the help of hypo solution.
6	To determine the acid value of a given oil.
7	To determine the viscosity of a given lubricating oil by Redwood viscometer.
8	To determine the flash and fire point of a given lubricating oil.
9	To determine the cloud and pour point of a given oil.

10	Synthesis of Bakelite.
11	Determination of pH by pH-metric titration.
12	Estimation of an acid (weak/strong) by Conductometric titration.

RECOMMENDED STUDY MATERIAL:

S.No	**Reference Book**	**Author**	**Publication**
1	Engineering Chemistry	Dr. Mahima Tulsian	Jhunjhunuwala Publication
2	Engineering Chemistry Lab Manual	Dr. N. Rama Jyothi	Shine Book publication
3	Chemistry	Dr. Sanjay Sharma	College Book House

Virtual Lab Link: https://www.vlab.co.in/

Code: BTX CCE1203	Programming in C lab	1 Credits [LTP: 0-0-2]

COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Gain concept of functional hierarchical code organization.	L2	PO1, PO2, PO3, PO4, PO6, PO7
CO2	Work with textual information, characters and strings	L1	PO1, PO2, PO3, PO4, PO6, PO7
CO3	Implement file handling concepts	L2	PO1, PO2, PO3, PO4, PO6, PO7
CO4	Implement real time applications using the power of C language features.	L3	PO1, PO2, PO3,PO4,PO5 PO6, PO7
CO5	Overcome and solve possible errors during program execution	L2	PO1, PO2, PO3, PO4, PO6, PO7

. Mapping matrix of CO, PO & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	**3**	**2**	**1**	**2**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**3**	**-**	**-**
CO2	**2**	**3**	**3**	**1**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**
CO3	**3**	**2**	**2**	**3**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**3**	**-**	**-**
CO4	**2**	**1**	**1**	**2**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**-**
CO5	**3**	**1**	**2**	**1**	**-**	**-**	**2**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**
Wt. AVG	**2.6**	**1.8**	**1.8**	**2**			**2**						**2.2**		

IST OF EXPERIMENTS

1	a) Given the values of the variables x, y and z, write a program to rotate their values such that x has the value of y, y has the value of z, and z has the value of x b) Write a program that reads a floating point number and then displays the right-most digit of the integral part of the number.
2	Program to find largest and smallest number from four given number.
3	Program to find whether a year is leap or not
4	Write a C program in which enter any number by the user and perform the operation of Sum of digits of entered number.
5	Write a C Program to convert Decimal number to Binary number
6	Find the sum of this series upto n terms 1+2+3+4+5+6+……..
7	Program to print Armstrong's numbers from 1 to 100.
8	Write a program to convert years into Minute, Hours, Days, Months, Seconds using switch () statements
9	Write a C menu driven program
10	Write a program to generate the various pattern of numbers

1 **1**	a) Write a C Program to print the reverse of an integer number b) rite a C program to perform to perform Matrix addition and multiplication operations.
1 **2**	a) Write a C program to perform the factorial of given number. b) Write a program to determine the length of the string and find its equivalent ASCII codes.
	Virtual Lab
1	1. Click on the square to define a function for calculating the area of a square. 2. Similarly define functions for the other geometrical figures 3. The defined functions are shown in the middle window 4. Now do appropriate function calls in the main program to compute the area of the figure displayed 5. Press execute to execute the code and see the output
2	1. To understand the concept of memory address associated with every variable. 2. To understand pointer referencing and dereferencing. 3. To understand how pointers are useful in dynamic allocation of memory.
3	1. To understand that some problems can be broken down into smaller similar problems. 2. To solve such problems using recursive procedures.

A. RECOMMENDED STUDY MATERIAL

S. No	**Text Books:**	**Author**	**Edition**	**Publication**
	Let us C	Yashwant Kanetkar	6th Edition	PBP Publication
	The C programming Language	Richie and Kenninghan	2nd Edition 2004	PBP Publication,2004
	Programming in ANSI C	E Balaguruswamy	3rd Edition, 2005	Tata McGraw Hill
Reference Book				
	The C programming Language by Richie and Kenninghan, PBP Publication,2004			
	Programming in ANSI C 3rd Edition, 2005 by E.Balagurusamy, Tata McGraw Hill			
Online Resources				
	https://www.programiz.com/c-programming/examples			
	https://cse02-iiith.vlabs.ac.in/List%20of%20experiments.html			
	https://www.w3resource.com/c-programming-exercises			

Code: BTXCCV1204 /2204 Computer Aided Design (CADD) 1 Credits [LTP: 0-0-2]

COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Understand basic functions of commands, their techniques and fundamentals used in drafting and drawing.	L2	PO1,PO2
CO2	Demonstrate the fundamentals of commands through line diagram.	L2	PO1,PO2, PO3
CO3	Compare various shortcut keys and their different use case in drafting.	L4	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO9
CO4	Understand the fundamental principles of Blocks, Hatching, and texts of drawings.	L2	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10,PO11,PO 12
CO5	Creating the layout of 2D plans in the workspace.	L6	PO1,PO2,PO3,PO4, PO5,PO6,PO7,PO8, PO9,PO10,PO11,PO 12

B. Mapping matrix of CO, PO & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	**3**	**1**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**
CO2	**3**	**2**	**2**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**-**	**1**	**-**	**-**
CO3	**3**	**1**	**1**	**1**	**3**	**1**	**2**	**-**	**1**	**-**	**-**	**-**	**1**	**-**	**-**
CO4	**3**	**3**	**2**	**3**	**3**	**2**	**2**	**2**	**2**	**1**	**1**	**1**	**1**	**-**	**-**
CO5	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**3**	**2**	**2**	**2**	**2**	**1**	**-**	**-**
Wt. AVG	**3**	**2**	**2**	**2.3**	**3**	**2**	**2.3**	**2.5**	**1.6**	**1.5**	**1.5**	**1.5**	**1**	**-**	**-**

C.

LIST OF EXPERIMENTS

1	Introduction to AutoCAD and Drawing Tools Draw Different Shapes using Line, Polyline Circle, and Polygon.
2	Draw Different Shapes using Rectangle Use of Dimensions in Circle, rectangles, Line and other shapes.
3	Modify Drawings in AutoCAD using Modification Tools. Offset and Mirror Different Shapes and Lines.
4	Use Trim, Extend & Align, Scale and Stretch Command
5	Use of Text, Line, Block and Conversion Tools.
6	Introduction to Layers, How to add, Modify layers in layer manager.
7	Introduction of Hatch Command in AutoCAD
8	Opening and Modifying properties in AutoCAD
9	2D Plan of 1 BHK Residential Structure
10	2D Pine Flat Dam

11	2D Sewerage Treatment Plant of Poornima University
12	Line Diagram of T shaped Building
	Virtual Lab
1	https://www.autodesk.in/campaigns/autocad-tutorials
2	https://web.autocad.com/login

Code: BTXCEE1205/ 2205 Electrical and Electronics Engineering Lab 1 Credits [LTP: 0-0-2]

COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	***At the end of this course, learners will able to:***	**Bloom level**	**PO Mapping**
CO1	*Assemble and understand the operation of house wiring systems*	L1	PO1, PO2,
CO2	*Gain the ability to prepare and test connections for single-phase and three-phase induction motors using auto transformers*	L2	PO1, PO2, PO3,
CO3	*Develop the skills to connect and measure the electrical parameters (voltage, current, and power) of various lighting systems, such as fluorescent lamps, sodium vapor lamps, and halogen lamps.*	L3	PO1, PO2, PO3
CO4	*Proficient in identifying, testing, and applying various electronic components, such as resistors, conductors, capacitors, diodes , LEDs, LCDs, BJTs,*	L2	PO1, PO2, PO3
CO5	*Verify the truth tables of basic logic gates (AND, OR, NOT, NOR, NAND) and understand their applications.*	L2	PO1, PO2, PO4

. Mapping matrix of CO, PO & PSO

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	**3**	**3**	**2**	**-**									**-**	**-**	**-**
CO2	**3**	**2**	**3**	**-**									**-**	**-**	**-**
CO3	**3**	**2**	**3**	**-**									**-**	**-**	**-**
CO4	**3**	**3**	**2**	**-**									**-**	**-**	**-**
CO5	**3**	**2**	**2**	**2**									**-**	**-**	**-**
Wt. AVG	**3**	**2.4**	**2.4**	**2**									**-**	**-**	**-**

IST OF EXPERIMENTS

1	Assemble house wiring including earthing for 1-phase energy meter, MCB, ceiling fan, tube light, three pin socket and a lamp operated from two different positions. Basic functional study of components used in house wiring.
2	Prepare the connection of ceiling fan along with the regulator and vary the speed.
3	Prepare the connection of single phase induction motor through 1-Phase Auto-transformer and vary the speed.
4	Prepare the connection of three phase squirrel cage induction motor through 3-Phase Autotransformer and vary the speed.
5	Prepare the connection of Fluorescent Lamp, Sodium Vapour and Halogen Lamp and measure voltage, current and power in the circuit.
6	Identification, testing and application of Resistors, Inductors, Capacitors, PN-Diode. Zenger Diode, LED, LCD, BJT, Photo Diode, Photo Transistor, Analog/Digital Multi- Metres and Function/Signal Generator.
7	Measure the frequency, voltage, current with the help of CRO.
8	Assemble the single phase half wave and full wave bridge rectifier & the analyse effect of L, C and L-C filters in rectifiers.
9	Verify the truth table of AND, OR, NOT, NOR and NAND gates

10	Introduction to IOT. Printing of a Hello World on LCD Screen using Arduino
11	Controlling the Light Emitting Diode (LED) with a push button and Interfacing of the Relay with Arduino.
12	Introduction to Raspberry Pi and Programming of available GPIO pins of the corresponding device using native programming language. Interfacing of I/O devices like LED/Switch etc., and testing the functionality.
	Virtual Lab
1	To perform Ohm's Law http://vlabs.iitkgp.ernet.in/be/exp4/index.html
2	Half Wave Rectification http://vlabs.iitkgp.ernet.in/be/exp6/index.html
3	VI Characteristics of Diode http://vlabs.iitkgp.ernet.in/be/exp5/index.html

Code: BTXCME1206 /2206 WORKSHOP PRACTICE 1 Credit [LTP: 0-1-2]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

COURSE OUTCOMES (Cos)	At the end of this course, learner will be able to :	Bloom Level	PO Mapping
CO – 01	Prepare a model of T Lap and T- Briddle Joint through carpentry shop	L3	PO1, PO2, PO3
CO – 02	Understand the making of prototype model through foundry shop	L2	PO1,PO2,PO3
CO – 03	Understand the difference between gas welding and arc welding and their applications.	L2	PO1, PO2, PO3
CO – 04	prepare a model on fitting shop through filling, drilling and tapping operation	L3	PO1, PO2,PO3
CO – 05	Understand the difference between forging, moulding and casting	L2	PO1, PO2, PO3

B. MAPPING MATRIX OF CO, PO, & PSO

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	3	2	1	-	—	—	—	—	—	—	—	—
CO 2	3	1	2	-	—	—	—	—	—	—	—	—
CO 3	3	2	1	-	-	—	—	—	—	—	—	—
CO 4	3	2	1	-	-	—	—	—	—	—	—	—
CO 5	3	2	1	-	-	—	—	—	—	—	—	—
WT. AVG	3	1.8	1.2									

C. DETAILED SYLLABUS

1	**Carpentry Shop** • Timber, definition, engineering applications, seasoning and preservation • Plywood and ply boards
2	**Foundry Shop** • Moulding Sands, constituents and characteristics • Pattern, definition, materials types, core prints • Role of gate, runner, riser, core and chaplets • Causes and remedies of some common casting defects like blow holes, cavities, inclusions
3	**Welding Shop** • Definition of welding, brazing and soldering processes and their applications • Oxyacetylene gas welding process, equipment and techniques, types of flames and their applications • Manual metal arc welding technique and equipment, AC and DC welding • Electrodes: Constituents and functions of electrode coating, welding positions • Types of welded joints, common welding defects such as cracks, undercutting, slag inclusion and boring
4	**Fitting Shop** • Files, materials and classification.
5	**Smithy Shop** • Forging, forging principle, materials • Operations like drawing, upsetting, bending and forge welding • Use of forged parts

Code: BTXCME1207/ 2207	Engineering Graphics Lab	1 Credits [LTP: 0-0-2]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

COURSE OUTCOMES (Cos)	At the end of this course, learner will be able to :	Bloom Level	PO Mapping
CO – 01	Understand basic technical drawing concepts, including lines, lettering, dimensions, and scales, through the creation of sketches and drawings.	L2	PO2, PO4,PO5
CO – 02	Construct conic sections (ellipse, parabola, and hyperbola) on drawing sheets, showcasing practical application skills.	L3	PO4,PO5,
CO – 03	Apply knowledge of projection techniques to represent complex objects, such as regular polygons in different positions.	L3	PO1,PO3,PO4,PO5
CO – 04	Differentiate between types of projections, such as orthographic projections with first and third angle conventions. Proficiently project points, straight lines, and planes in different positions	L3	PO1, PO4
CO – 05	Analyze knowledge of sectional views and machine drawing techniques to communicate design and manufacturing details effectively	L3	PO1, PO4

B. MAPPING MATRIX OF CO, PO, & PSO

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	-	2	-	2	2	—	—	—	—	—	—	—
CO 2	-	-	-	3	2	—	—	—	—	—	—	—
CO 3	3	-	2	2	2	—	—	—	—	—	—	—
CO 4	3	-	-	2	-	—	—	—	—	—	—	—
CO 5	3	-	-	3	-	—	—	—	—	—	—	—
WT. AVG	3	2	2	2.5	2							

C. List of Experiments

1.	• Lines, Lettering and Dimension (Sketch Book) • Scales: Representative Fraction, plain scales, diagonal scales, (In drawing sheet 1)
2.	• Conic Sections: Construction of ellipse, parabola and hyperbola by different methods(in drawing sheet 2)
3.	• Type of Projection, Orthographic projection: first angle and third angle projection (in drawing sheet) • Projection of Points • Projection of Straight lines • Projection of planes: Different positions of plane lamina like: regular polygon (four problems in drawing sheet 3)
4.	• Orthographic Projections (3 Problems in drawing sheet 4)
5.	• Sectional Views (2 Problems) and Riveted joints, lap joints, butt joints, (drawing sheet 5)

RECOMMENDED STUDY MATERIAL

Sr.No	Reference Book	Author	Edition	Publication
1	Engineering Drawing	N. D. Bhatt	Latest	Charotar Publishing House PVT. LTD.
Virtual Lab				
1	https://kdm-iitkgp.vlabs.ac.in/exp/simple-drawing-board/theory.html			
2	https://cse18-iiith.vlabs.ac.in/			

Code: BTXCCE2203 PROGRAMMING IN PYTHON LAB 1 Credits [LTP: 0-0-2]

COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Understand why Python is a useful scripting language for developers.	L2	PO1, PO2, PO3
CO2	Develop problem solving and critical thinking skills in fundamental enable techniques like conditionals and loops.	L3	PO1, PO2, PO3
CO3	Execute python program with concept of functions, List, Dictionary, structure, class & object.	L3	PO1, PO2, PO4
CO4	Integrate the key issues in Python code, develop and experiment with python programming.	L4	PO1, PO2, PO5
CO5	Linking read and write data from/to files in Python and Develop Python programs step-wise by defining functions with turtle.	L4	PO1, PO3, PO5

B. Mapping matrix of CO, PO & PSO

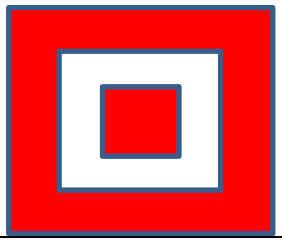
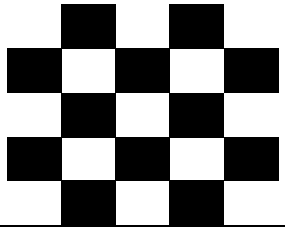
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	-	-	-	-	-	-	-	-	-
CO2	2	3	1	-	-	-	-	-	-	-	-	-
CO3	2	3	-	1	-	-	-	-	-	-	-	-
CO4	3	2	-	-	1	-		-	-	-	-	-
CO5	3	-	2	-	1	-	-	-	-	-	-	-
Wt. AVG	2.6	2.5	1	1	1							

C. Mapping matrix of CO & PSO

	PSO1	PSO2	PSO3	PSO4	PSO5
CO1	3	-	-	-	-
CO2	2	-	-	-	-
CO3	2	-	3	-	-
CO4	2	-	-	-	-
CO5	1	-	-	-	3
Wt. AVG	2		3		3

LIST OF EXPERIMENTS

1	a) Write a python program to check if the number is prime or not. b) Write a python program to find greatest of four numbers entered by user.
2	Write and run a Python program that asks the user for a temperature in Celsius and converts and output the temperature in Fahrenheit.
3	Write python program to perform following operations on Lists: a) Create list b) Access list c) Update list (Add item, Remove item) d) Delete list
4	Here is an algorithm to print out n! (n factorial) from 0! to 19!: 1. Set f = 1

	2. Set n = 0 3. Repeat the following 20 times: a. Output n, "! = ", f b. Add 1 to n c. Multiply f by n Using a for loop, write and run a Python program for this algorithm.
5	Modify the program above using a while loop so it prints out all of the factorial values that are less than 1 billion.
6	Write a python program to find the minimum in the array and find its index in the array.
7	Write a python program to implement bubble sort using function.
8	Write a python program to read 3 subject marks and display pass or failed using class and object.
9	Draw the Target symbol (a set of concentric Squares, alternating red and white) in a graphics window that is 200 pixels wide by 200 pixels high. Hint: Draw the largest circle first in red, then draw the next smaller circle in white, then draw the next smaller circle in red. Graphical objects drawn later appear "on top of" graphical objects drawn earlier. 
10	Create a 5 X 5 rectangle whose top left corner is at (row*5, col*5). (Where is the bottom right corner?) If the sum of the row and col numbers is even, set the fill color of the rectangle to white, otherwise set it to black. Then draw the rectangle. 
	Virtual Lab
1	Simulate python program to perform all Arithmetic Operations.
2	Simulate python program to analyze the concept of Constructor and Inheritance.
3	Simulate python program using Math, datetime, random and operator Module.

Code: BTXCEE1208/ 2208	Exploratory Project	1 Credits [LTP: 0-0-1]

LAB OUTCOMES: After Successful completion of the lab students will be able to-

COURSE OUTCOMES (Cos)	At the end of this course, learner will be able to :	Bloom Level	PO Mapping
CO – 01	Predict a problem of current relevance to society using research and analytical skills.	L2	PO1, PO2, PO3, PO6, PO7, PO9, PO10, PO12
CO – 02	Formulate the problem and identify suitable modeling paradigms through collaborative discussions and research.	L3	PO1,PO2,PO3, PO4, PO6, PO7, PO9, PO10, PO11, PO12
CO – 03	Categorize the problem and identify the solution methodology using systematic problem-solving techniques.	L3	PO1, PO2, PO3, PO4, PO6, PO7, PO9, PO10, PO11, PO12
CO – 04	Simulate and design systems using various modern tools and software applications.	L4	PO1, PO2,PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12
CO – 05	Validate the results and prepare a comprehensive project report.	L4	PO1, PO2,PO3, PO4, PO5, PO6, PO7, PO9, PO10, PO11, PO12

MAPPING MATRIX OF CO, PO, & PSO

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	3	3	1	-	—	2	1	—	1	1	—	2
CO 2	3	3	3	2	—	2	2	—	2	2	1	2
CO 3	3	3	2	2	-	2	2	—	2	2	1	2
CO 4	3	3	3	2	3	2	2	—	3	3	2	3
CO 5	3	3	3	3	3	2	2	—	3	3	2	3
WT. AVG	3	3	2.4	2.25	3	2	1.8	-	2.2	2.2	1.5	2.4

GUIDELINES:

1. The Project group must complete project in all respect (assembly, testing, fabrication, tabulation, test result etc.)
2. The group should maintain a log book of activities. It should have entries related to the worked one, problems faced, solution evolved etc., duly signed by guide.
3. The guides should regularly monitor the progress of the project work.
4. The project work along with project report should be submitted as part of term work in first term on or before the last day of the second term.
5. Project report must be submitted in the prescribed format only. No variation in the format will be accepted.
6. Assessment of the project forward of marks shall be done by the guide and a departmental committee.
7. The guide should be internal examiner for oral examination.
8. The external examiner should be from the related area of the concerned project. He should have experience at degree level / industry.

The evaluation at final oral examination should be done jointly by the internal and external examiner.

Phases:

Project work is divided into the following phases:

Phase I

- Allocation of groups(Max. 4 Members & Min. 2 Members) & guide
- Black board presentation on topics as per the choice & feasibility
- Submission of abstract & synopsis of the project

Phase II

- Procurement of the components
- 2D/3D figure or model
- Paper work like any circuit diagram and tentative cost

Phase III

- Working Model of the project
- Mounting the components
- Final hardware evaluation/presentation
- Submission of the final hardware to the coordinator.

Phase V

- Final report submission (after project exhibition)
- Paper presentation on the selected project in seminars /conferences/journals
- Viva voce

Deadlines of Phases:

The Project will be covered in 13 weeks from starting of semester. The time allocated to each phase is as follow:

Phase -1: Maximum 2 weeks
Phase -2: Maximum 3 weeks
Phase -3: Maximum 6 weeks
Phase- 4: Maximum 2 weeks

Distribution of Marks:-

Total Marks 100

Break up of marks (100)

Performance of Phase 1	:15
Performance of Phase 2	:20
Performance of Phase 3	:20
Performance of Phase 4	:45
Total	:**100**

Note: 1. Performance marks of Phase 1/2/3/4 will be given by Coordinators, Guide and external (if any) on completion of the respective phase.

2. Presentation and demonstration will be taken by Project Coordinator, Guide.
3. Guide feedback will be collected by Project Coordinator.

Code: BUVCVE1102 ENVIRONMENTAL STUDIES 2 Credits [LTP: 2-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Introduction to scope of environmental studies & concept of ecology, ecosystemand biodiversity.	L2	PO7, PO8, PO12,
CO2	Implement innovative ideas of controlling different categories of Environmental Pollution.	L3	PO7, PO8, PO11, PO12,
CO3	Environmental issues & various Environmental Acts, regulations andInternational Agreements.	L2	PO7, PO8, PO11, PO12,
CO4	Social issues related to population, resettlement and rehabilitation of project affected persons. Demonstrate disaster management w.r.t floods, earthquakes, cyclones and landslides.	L2	PO7, PO8, PO11, PO12,
CO5	Determination of local environmental assets with simple ecosystems and identify local flora and fauna	L4	PO7, PO8, PO11, PO12,

B. Mapping matrix of CO & PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	–	–	–	–	–	–	2	2	–	–	–	1	–	–	1
CO2	–	–	–	–	–	–	3	2	–	–	1	1	–	1	1
CO3	–	–	–	–	–	–	2	3	–	–	1	1	–	1	1
CO4	–	–	–	–	–	–	3	2	–	–	1	1	–	1	1
CO5	–	–	–	–	–	–	2	3	–	–	1	1	–	1	1
Wt. AVG							2.4	2.4			1	1		1	1

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Environmental Studies	5
2.	Environmental Pollution and its Control	5
3.	Environmental Policies & Practices	5
4.	Human Communities and the Environment	5
5.	Field Work	4

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction to Environmental studies**
	• Introduction of Unit • Multidisciplinary nature of environmental studies Concept of sustainability and sustainable development.

	• Ecosystem: Structure and function of ecosystem • Energy flow in an ecosystem: food chains, food webs and ecological succession.Case studies\ • Case studies of the following ecosystems: Forest ecosystem, Grassland ecosystem,Desert ecosystem, Aquatic ecosystems, Biodiversity and Conservation • Conclusion of the Unit
2.	**Environmental Pollution and its Control**
	• Introduction of Unit • Environmental pollution: types, causes, effects and controls; Air, water, soil and noise pollution • Nuclear hazards and human health risks • Solid waste management: Control measures of urban and industrial waste. • Pollution case studies • Conclusion of the Unit
3.	**Environmental Policies & Practices**
	• Introduction of Unit • Climate change, global warming, ozone layer depletion, acid rain and impacts on human communities and agriculture • Energy resources: Renewable and non-renewable energy sources, use of alternate energy • sources, growing energy needs, case studies. • Environment Laws: Environment Protection Act; Air (Prevention & Control of Pollution) Act; Water (Prevention and control of Pollution) Act; Wildlife Protection Act; Forest Conservation Act. • International agreements: Montreal and Kyoto protocols and Convention on Biological Diversity (CBD) • Conclusion of the Unit
4.	**Human Communities and the Environment**
	• Introduction of Unit • Human population growth: Impacts on environment, human health and welfare. • Resettlement and rehabilitation of project affected persons; case studies. • Disaster management: floods, earthquake, cyclones and landslides. • Conclusion of the Unit
5.	**Field Work**
	• Introduction of Unit • Visit to an area to document environmental assets: river/ forest/ flora/fauna, etc. • Visit to a local polluted site-Urban/Rural/Industrial/Agricultural. • Study of common plants, insects, birds and basic principles of identification. • Study of simple ecosystems-pond, river, Delhi Ridge, etc. • Conclusion of the Unit

RECOMMENDED STUDY MATERIAL:

S. No	Reference Book	Author	Edition	Publication
1	Environmental Studies	ErachBarucha	Latest	UGC
2	Environmental Studies	Benny Joseph	Latest	Tata McgrawHill
3	Environmental Studies	R. Rajagopalan	Latest	Oxford University Press
4	Principles of Environmental Science and Engineering	P. Venugoplan Rao	Latest	Prentice Hall of India.
5	Environmental Science andEngineering	Meenakshi	Latest	Prentice Hall India.

Your Dreams Our Goal

POORNIMA UNIVERSITY

Member of Association of Indian Universities & Approved by UGC (Govt. of India) under 2(f) & 12(B)

B. TECH - CSE

FIRST YEAR
SEMESTER: II

COMMON SYLLABUS

For

Cyber Security
Cloud & Full Stack Development
Artificial Intelligence & Data Science
Artificial Intelligence & Machine Learning

BATCH: 2024-28

Code: BUACHM1202 FOUNDATION ENGLISH Credits 1 [LTP: 0-0-2

COURSE OUTCOMES:

Course Outcome	At the end of this course, learners will be able to:	Bloom Level
CO1	Practice the fundamental concepts of grammar, including parts of speech, sentence structures, and tenses.	Apply
CO2	Compute common grammatical errors using the rules and principles learned in the course.	Apply
CO3	Construct a robust vocabulary through various exercises for effective sentence formation.	Apply
CO4	Use the principles of effective writing in well-structured paragraphs, articles, and formal and informal letters	Apply
CO5	Examine texts at both beginner and intermediate levels, and effectively answer questions based on the readings, preparing them for competitive exams.	Analyse

A OUTLINE OF THE COURSE

Unit No.	Title of the Unit	Time Required for the Unit (Hours)
1.	Basics of Grammar	**8**
2.	Spotting the Grammatical Errors and Rectification	**2**
3.	Vocabulary Building	**4**
4.	Basics of Writing Skills	**2**
5.	Reading Comprehension	**8**

B DETAILED SYLLABUS

LIST OF LABS	
1	Parts of Speech: Theory & Practice through various Exercises
2	Sentence Structures: Theory & Practice through various Exercises
3	Tenses: Theory & Practice through various Exercises
4	Spotting the Errors: Applying the rules and Practice Questions
5	Vocabulary Building-I: Practice by sentence formation
6	Vocabulary Building-II: Practice by sentence formation
7	Paragraph Writing
8	Article Writing
9	Précis Writing
10	Formal & Informal Letter Writing
11	Reading Comprehension- I: Beginner's level reading and Answering the Questions (Competitive Exams)
12	Reading Comprehension- II: Intermediate's level reading and Answering the Questions (Competitive Exams)

C RECOMMENDED STUDY MATERIAL

Sr.No	ReferenceBook	Author	Edition	Publication
	English Literature	Benjamin W. Griffith	1882	Stuart Langton
	English Literature: Its History and Significance	William J.Long	2012	Maple Press2
	English Grammar and Composition	Wren & Martin	2017	S.Chand Publishing

Code: BUACHM2206 **LANGUAGE LAB** **Credits 1 [LTP: 0-0-2]**

COURSE OUTCOMES:

Course Outcome	At the end of this course, learners will be able to:	Bloom Level
CO1	Identify common errors in spoken and written communication.	Understand
CO2	Employ day-to-day English vocabulary and language proficiency skills.	Apply
CO3	Use sensible official writing and documenting skills.	Apply
CO4	Measure their personal and Technical Communication Skills	Evaluate
CO5	Produce an efficient way of expression in professional and personal spaces.	Apply

A. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Everyday Conversations**	**8**
2.	**Asking for**	**7**
3.	**Reporting/ Describing**	**7**
4.	**Meeting People**	**7**
5.	**Expressing & Talking about**	**7**

B. DETAILED SYLLABUS

Unit	Contents
1.	**Everyday Conversations**
	● Introduction to the Unit ● Introducing self / others ● Weather ● Classroom ● Asking about facilities around ● Describing a person / thing ● Points to cover: Vocabulary, grammar, Construction of sentences, listening ● Methodology: Role plays, Videos, Classroom conversation, worksheets ● Conclusion & Real Life Application
2.	**Asking for**
	● Introduction to the Unit ● Help/ Suggestion/ ideas ● Clarification/ Directions ● Time/ food ● Advice ● Uses ● Points to cover: Vocabulary, grammar, Construction of sentences, listening ● Methodology: Role plays, Videos, Classroom conversation, worksheets ● Conclusion & Real-Life Application
3.	**Reporting/ Describing**
	● Introduction to the Unit ● Incidences ● Personalities ● Experiences

	• Wants/Needs • Intentions • Points to cover: Vocabulary, grammar, Construction of sentences, listening • Methodology: Role plays, Videos, Classroom conversation, worksheets • Conclusion & Real-Life Application
4.	**Meeting People**
	• Introduction to the Unit • Greetings • Starting the Conversation • Small talks • Closing the conversation • Points to cover: Vocabulary, Grammar, Construction of sentences, listening • Methodology: Role plays, Videos, Classroom conversation, worksheet • Conclusion & Real-Life Application
5.	**Expressing & Talking about**
	• Introduction to the Unit • Happiness/Displeasure • Preferences • Doubts • Views • Unawareness • Points to cover: Vocabulary, grammar, Construction of sentences, listening • Methodology: Role plays, Videos, Classroom conversation, worksheets Interests • Different Cultures, Clothes, cars, institutes, situations • Schedules, prices • Points to cover: Vocabulary, grammar, Construction of sentences, listening Methodology: Role plays, Videos, Classroom conversation, worksheets • Conclusion & Real-Life Application

Code: BTXECE2111 Introduction to Artificial Intelligence 2 Credits [LTP: 2-0-0]

COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

(COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Remember how to identify systems with Artificial Intelligence. Demonstrate fundamental understanding of the history of artificial intelligence (AI) and its foundations.	L1	PO2, PO3, PO4, PO12
CO2	Apply appropriate search strategies to solve AI problems. Evaluate the efficiency and effectiveness of different search algorithms.	L3, L5	PO2, PO3, PO4, PO5
CO3	Understand the fundamentals of Constraint Satisfaction Problems (CSP). Explain the importance and applications of CSPs in AI.	L2	PO2, PO3, PO4, PO5
CO4	Analyze adversarial search in game playing and find out optimal solution using different algorithms. Analyzing Different types of games.	L4	PO2, PO3, PO4, PO5
CO5	Demonstrate awareness and a fundamental understanding of various applications of AI techniques in intelligent agents, expert systems, Speech recognition and other machine learning models.	L2	PO2, PO3, PO4, PO12

Mapping matrix of CO & PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	1	2	1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	2	3	2	3	-	-	-	-	-	-	-	-	-	-
CO3	-	2	3	2	3	-	-	-	-	-	-	-	-	-	-
CO4	-	1	3	2	3	-	-	-	-	-	-	-	-	-	-
CO5	-	2	2	1	-	-	-	-	-	-	-	3	-	-	-
Wt. AVG		1.6	2.6	1.6	1.8							1			

. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Artificial Intelligence**	**7**
2.	**Problem solving by Search**	**8**
3.	**Constraint Satisfaction Problems**	**7**
4.	**Adversarial Search and Game Playing**	**7**
5.	**AI Applications**	**7**

3. DETAILED SYLLABUS

Unit	Unit Details
1	**Introduction to Artificial Intelligence**
	● Introduction to Artificial Intelligence ● Definition of Artificial Intelligence, A brief history of Artificial Intelligence ● Why do we study AI?, What is AI? ● Views of AI: Acting Humanly, Thinking Humanly, Thinking Rationally and Acting Rationally, Areas of AI, Agents and environments ● PEAS (Performance measure, Environment, Actuators, Sensors) ● Environment types, Agent types: Simple reflex agents, Model-based reflex agents, Goal-based agents and Utility-based agents, Examples of Agent ● Conclusion of the Unit
2	**Problem solving by Search**
	● Introduction of Unit ● Problem-solving agents, Problem formulation ● Example problems: 8-Puzzle problem and 8-queens problem ● Basic search algorithms, Un-informed search strategies: Breadth-first search, Depth-first search, Depth-limited search, Uniform-cost search and Iterative deepening search ● Informed Search Algorithms: Best-first search, A* search, Hill-climbing search, and Genetic algorithms ● Conclusion of the Unit
3	**Constraint Satisfaction Problems**
	● Introduction to Constraint Satisfaction Problems (CSP) ● Why do we need to consider CSPs? ● Constraint Propagation, CSP Vs Search problems ● Real-world CSPs, Finite vs. Infinite CSP ● CSP as a Search Problem: Backtracking search for CSPs, Forward checking for CSPs and Local search for CSPs ● Conclusion of the Unit
4	**Adversarial Search and Game Playing**
	● Introduction to Adversarial Search and Game Playing ● Games: Definition, Search vs. Games and Game Tree ● Optimal decisions in Games: Mini max algorithm and α-β pruning with example ● Imperfect, real-time decisions, Partially Observable Games ● State-of-the-Art Game Programs: Chess on Deep Blue, Chess on standard PCs, Checkers on Chinook and Backgammon: TD-Gammon ● Conclusion of the Unit
5	**AI Applications**
	● Introduction of Unit ● Language Models, Information Retrieval, Extraction, Natural Language Processing ● Machine Translation, Speech Recognition ● Expert system: Introduction, phases, architecture, Expert system Vs Traditional system, Robot, Hardware, Planning, Moving ● Conclusion of the Unit

RECOMMENDED STUDY MATERIAL:

S. No	Reference Book	Author	Publication
1.	Artificial Intelligence: A Modern Approach	S. Russell and P. Norvig	Prentice Hall
2.	Artificial Intelligence: A Systems Approach (Computer Science),	M. Tim Jones	Jones and Bartlett Publishers
3.	Prolog: Programming for Artificial Intelligence	I. Bratko	Addison-Wesley Educational Publishers Inc

Code: BTXECE2112 INTRODUCTION TO CYBER SECURITY 2 Credits [LTP: 2-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

Course Outcomes (COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Understand the fundamental concepts of cyber security and its significance in the modern digital world.	L2	PO1, PO2, PO8
CO2	Identify and analyze various types of cyber threats and vulnerabilities.	L1	PO1, PO2, PO3, PO4, PO5, PO9
CO3	Implement cryptographic techniques to secure data and communications.	L2	PO1, PO2, PO3, PO4, PO5, PO6
CO4	Develop strategies to secure networks and applications against cyber-attacks.	L3	PO1, PO2, PO3, PO4, PO5 PO6, PO7
CO5	Conduct ethical hacking and penetration testing to assess system security.	L2	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8

B. Mapping matrix of CO & PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	1	-	-	-	1	-	-	-	-	3	-	-
CO2	3	3	2	2	1	-	-	-	1	-	-	-	3	3	2
CO3	3	3	2	2	1	1	-	-	-	-	-	-	3	2	2
CO4	3	3	3	3	2	1	1	-	-	-	-	-	2	2	3
CO5	3	3	3	2	2	1	1	1	-	-	-	-	1	1	-
Wt. AVG	3	2.8	2	2	1.2	0.6	0.4	0.4	0.2	-	-	-	2.4	1.6	1.4

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Cyber Security	7
2.	Cryptography in Cyber Security	7
3.	Network Security Fundamentals	7
4.	Application Security	8
5.	Security Policies	7

5. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction to Cyber Security**
	• Introduction of Unit • Overview of Cyber Security • Importance and Challenges • History of Cyber Security • Cyber Threats and Attack Vectors • Types of Cyber Threats • Attack Vectors • Conclusion of Unit
2.	**Cryptography in Cyber Security**
	• Introduction to Cryptography • Symmetric and Asymmetric Encryption • Hash Functions • Advanced Cryptographic Techniques • Public Key Infrastructure (PKI) • Digital Signatures • Cryptographic Protocols • Conclusion of Unit
3.	**Network Security Fundamentals**
	• Introduction of the Unit • Network Security Principles • Firewalls and Intrusion Detection Systems (IDS) • Virtual Private Networks (VPNs) • Securing Network Infrastructures • Network Segmentation • Secure Network Design • Wireless Network Security • Conclusion of Unit
4.	**Application Security**
	• Introduction of the Unit • Secure Software Development • Web Application Security • Common Vulnerabilities and Exposures (CVEs) • Ethical Hacking and Penetration Testing • Ethical Hacking Principles • Penetration Testing Methodologies • Tools and Techniques • Malware Analysis • Types of Malwares • Analysis Techniques • Conclusion of Unit
5.	**Security Policies**
	• Introduction of the Unit • Security Policies and Governance • Cyber Security Policies • Compliance and Regulations • Risk Management • Emerging Trends in Cyber Security • Internet of Things (IoT) Security • Cloud Security • Artificial Intelligence in Cyber Security • Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Reference Book	Author	Publication
1	INTRODUCTIO N TO CYBER SECURITY	Cma Maithili Malpure, Madhu, Dr Pratap Patil, Niraj Kumar Rai, Dr Kritika Gosa	Redshine Publication
2	Cybersecurity Fundamentals	Rajesh Kumar Goutam	BPB Publisher
3	Mastering Your Introduction to Cyber Security	Dr. Michael C. Redmond PHD	BookLocker Inc

Code: BTXECE2114 Digital Electronics 2 Credits [LTP: 2-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

(Cos)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	Have a thorough understanding of the fundamental concepts and techniques used in digital electronics and To understand and examine the structure of various number systems and its application in digital design.	L2	PO1, PO2, PO3, PO4, PO6, PO7
CO2	Learn the minimization techniques to simply the hardware requirements of digital circuits, implement it, design and apply for real time digital systems.	L4	PO1, PO2, PO3, PO4, PO6, PO7
CO3	Understand the working mechanism and design guidelines of different combinational, sequential circuits and their role in the digital system design.	L3	PO1, PO2, PO3, PO4, PO5, PO6, PO11,PO12
CO4	Ability to identify basic requirements for a design application and propose a cost effective solution.	L4	PO1, PO2, PO3,PO4,PO 5,PO6, PO11,PO12
CO5	Became able to know various types of components- ADC and DAC, memory elements and the timing circuits to generate different waveforms.	L2	PO1, PO2, PO3, PO4, PO5, PO6, PO11,PO12

B. Mapping matrix of CO & PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	2									2		
CO2	3	2	1	2	2	2					1	1	2	2	
CO3	3	2	1	2	2	2					1	1	2	2	
CO4	3	2	1	2	2	2					2	2	2	2	2
CO5	2	-	1	2	2	2					2	2	2	2	2
Wt. AVG	3	1.6	1	2	1.6	1.6					1.2	1.2	2	1.6	0.8

A. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Number System and Logic Gates	7
2.	Decoders, Multiplexers & De-Multiplexers	6
3.	Flip-Flops	7
4.	Registers And Counters	8
5.	Memories And Converters	8

5. DETAILED SYLLABUS

Unit	Unit Details
1.	**Number System and Logic Gates**
	• **Introduction to number systems** – Binary to decimal conversion – Decimal to binary conversion –Octal numbers – Hexadecimal numbers • **Logic gates** – NOT, OR, AND – Universal NAND and NOR gates – EX-OR and EX-NOR gates – De-Morgan's Theorems — 1's complement – 2's complement – Adders (half & full) – Subtractor (half & full). • Conclusion of the Unit
2.	**Decoders, Multiplexers & De-Multiplexers**
	• Introduction of Unit • Basic functions and block diagram of Encoders and decoders. • Basic functions and block diagram of Multiplexers and De-Multiplexers, Different types and ICs. • 4-bit decoder circuits for 7 segment display and other applications. • Conclusion of the Unit.
3.	**Flip-Flops**
	• Introduction of Unit • J-K Flip-Flop • R-S Flip-Flop • D-Type Flip-Flop • T-Type Flip-Flop • Applications of Flip-Flops • Conclusion of the Unit
4.	**Registers And Counters**
	• Introduction to Shift Register • Introduction and basic concepts including shift left and shift right. • Serial in parallel out, serial in serial out, parallel in serial out, parallel in parallel out. • Introduction to Counters (Asynchronous and Synchronous counters) • Binary up/down counters • Ring counter with timing diagram • Conclusion of the Unit
5.	**Memories And Converters**
	• Introduction of Unit • ***Memories*** – ROM, RAM, EPROM, EEPROM – Volatile and non-volatile – Static and dynamic RAM. • **Analog to digital converters** – Parallel Comparator A/D converter – Dual slope converter –Successive approximation method – Counter type converter. • **Digital to analog converters** – Binary weighted D/A converter – R/2R ladder network converter • Conclusion of the Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Digital Principles and Applications	Donald P Leach, Malvino	-	McGraw Hill
2.	Modern Digital Electronics	RP Jain	-	Tata McGraw Hill
3.	Digital Fundamentals	Floyd and Jain	-	Pearsons Education
Reference Book				
1.	Digital Electronics by Rajaraman V., Prentice Hall of India, New Delhi			
2.	Digital Electronics and Applications by Malvino Leach, Tata McGraw Hill Education Pvt Ltd, New Delhi			
Online Resources				
1.	https://archive.nptel.ac.in/courses/108/105/108105132/			
2.	https://onlinecourses.nptel.ac.in/noc22_ee55/preview			

Code: BTXECE2113 INTRODUCTION TO CLOUD 2 Credits [LTP: 2-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

(COs)	At the end of this course, learners will able to:	Bloom level	PO Mapping
CO1	**Understand** core concepts of cloud computing and different types of cloud architectures (public, private, hybrid). Explore principles and architectures of cloud technology and Evaluate cloud solutions from different providers	**Understand**	PO1, PO2, PO3, PO4, PO5
CO2	**Understand** the basic concepts and importance of migrating to the cloud. Identify and apply effective steps for migrating to the cloud.Learn methodologies to mitigate risks and address company concerns during cloud migration. Review and analyze real-world case studies of cloud migration to understand best practices and lessons learned.	**Understand**	PO1, PO2, PO3, PO4, PO7
CO3	**Analyze** skills in selecting scalable applications that align with organizational needs and goals. Understand best practices for selecting cloud services and providers Explore issues for selecting cloud services and providers Analyze commercial and business considerations relevant to cloud cost management and provider selection.	**Analyze**	PO1, PO2, PO3, PO4, PO5
CO4	**Demonstrate** governance principles in cloud computing. Learn about industry standards organizations and groups associated with cloud computing. Understand the importance of IT governance in cloud computing and its implications. Explore cloud governance Examine legal issues Analyze legal considerations in commercial and business contexts.	**Demonstrate**	PO1, PO2, PO3, PO4, , PO7
CO5	**Implemen**t robust service management practices to ensure the smooth operation of cloud services. Initiate cloud adoption with a pilot project to test capabilities and assess feasibility before full-scale deployment.	**Apply**	PO1, PO2, PO3, PO4, PO7

B. Mapping matrix of CO & PO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	2	2	-							3		
CO2	2	3	3	1	-	-	2						1		
CO3	3	2	2	3	2	-							3		
CO4	2	1	1	2	-	-	2						2		
CO5	3	1	2	1	-	-	2						2		
Wt. AVG	2	1	1	2	2		2						2		

A. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction**	**7**
2.	**Cloud Computing Companies and Migrating to Cloud**	**7**
3.	**Cloud Cost Management and Selection of Cloud Provider**	**7**
4.	**Governance in the Cloud**	**8**
5.	**Ten cloud do's and do not's**	**7**

B. DETAILED SYLLABUS

Unit	Unit Details
1	**Introduction**
	● Introduction to Unit ● Introduction to Cloud Computing, ● History and Evolution of Cloud Computing, ● Types of clouds, Private and Public clouds, ● Cloud Computing architecture, ● Cloud computing infrastructure, ● Merits of Cloud computing, ● Practical applications of cloud computing, ● Cloud computing delivery models and services (IaaS, PaaS, SaaS) ● Obstacles for cloud technology, Cloud vulnerabilities, Cloud challenges, ● Conclusion of the Unit
2	**Cloud Computing Companies and Migrating to Cloud**
	● Introduction to Unit, ● Broad Approaches to Migrating into the Cloud, The Seven-Step Model of Migration into a Cloud, ● Efficient Steps for migrating to cloud ● Risks: Measuring and assessment of risks, ● Company concerns Risk Mitigation methodology for Cloud computing, ● Case Studies ● Conclusion of the Unit
3	**Cloud Cost Management and Selection of Cloud Provider**
	● Introduction to Unit ● Assessing the Cloud: software Evaluation, System Testing, Seasonal or peak loading, Cost cutting and cost- benefit analysis, selecting the right scalable application. ● Understanding Best Practices used in selection of Cloud service and providers, ● Clouding the Standards and Best Practices Issue: Interoperability, Portability, Integration, Security, ● Commercial and Business Consideration ● Conclusion of the Unit
4	**Governance in the Cloud**
	● Introduction to Unit ● Industry Standards Organizations and Groups associated with Cloud Computing, ● Need for IT governance in cloud computing ● Cloud Governance Solution: Access Controls, Financial Controls, Key Management and Encryption, Logging and Auditing, API integration ● Legal Issues: Data Privacy and Security Issues, Cloud Contracting models, ● Jurisdictional Issues Raised by Virtualization and Data Location, ● Legal issues in Commercial and Business Considerations ● Conclusion of the Unit
5	**Ten cloud do's and do not's**
	● Introduction to Unit ● Don't be reactive ● do consider the cloud a financial issue ● don't go alone ● do think about your architecture ● don't neglect governance ● don't forget about business purpose

	● do make security the centerpiece of your strategy ● don't apply the cloud to everything don't forget about Service Management ● do start with a pilot project ● Conclusion of the Unit

RECOMMENDED STUDY MATERIAL:

S. No	Reference Book	Author	Publication
1	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg, Andrzej M. Goscinski	John Wiley and Sons Publications, 2011
2	Cloud Computing Theory and Practice	Dan C Marinescu, Elsevier	2013
3	Cloud Computing for Dummies	Judith Hurwitz, Robin Bloor, Marcia Kaufman & Fern Halper	Wiley Publishing, 2010

Code: BTXEME2115 ENGINEERING MECHANICS 2 Credits [LTP: 2-0-0]

COURSE OUTCOME

COURSE OUTCOMES (Cos)	At the end of this course, learner will be able to :	Bloom Level	PO Mapping
CO – 01	**Understand** fundamental mechanics laws and principles to analyse systems of forces and moments accurately.	**L2**	PO1, PO2
CO – 02	**Apply** concepts of machine design and moment of inertia to optimize lifting machines and understand their mechanical behaviour.	L3	PO1, PO2, PO3, PO4
CO – 03	**Analyse** frictional effects in mechanical systems and optimize power transmission efficiency using belt drive principles.	L4	PO1, PO2, PO3, PO4
CO – 04	**Evaluate** particle and rigid body dynamics using kinematic and kinetic principles to predict system behaviour accurately.	L5	PO1, PO2, PO4
CO – 05	**Evaluate** work, power, and impact in mechanical systems, analysing collision impacts effectively using principles of energy conservation.	L5	PO1, PO2, PO4

MAPPING MATRIX OF CO, PO, & PSO

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	2	3	-	-	-	-	-	-	-	-	-	-
CO 2	3	3	3	2	-	-	-	-	-	-	-	-
CO 3	2	3	3	3	-	-	-	-	-	-	-	-
CO 4	2	3	-	3	-	-	-	-	-	-	-	-
CO 5	2	3	-	3	-	-	-	-	-	-	-	-
WT. AVG	2.2	3	3	2.75								

A. OUTLINE OF THE COURSE

Unit No.	Title of the Unit	Time required for the Unit (Hours)
1.	**Fundamentals of Mechanics**	**8**
2.	**Machine & Moment of Inertia**	**8**
3.	**Friction & Belt Drive**	**7**
4.	**Dynamics of Particles**	**8**
5.	**Work, Power & Impact**	**8**

B. DETAILED SYLLABUS

Unit	Unit Details
1.	**Fundamentals of Mechanics**
	• Introduction of Unit • Fundamental laws of mechanics, Principle of transmissibility. • System of forces, Resultant force, Resolution of force. • Moment and Couples, Varignon's Theorem, • Equilibrium, Conditions for equilibrium, Lami's theorem. • Conclusion of Unit
2.	**Machine & Moment of Inertia**
	• Introduction of Unit • Lifting Machines: Mechanical advantage, Velocity Ratio, Efficiency of machine, Ideal machine, Ideal effort and ideal load, Reversibility of machine, Law of machine. • Centroid & Moment of Inertia: Location of centroid and center of gravity, Moment of inertia, Parallel axis and perpendicular axis theorem, Radius of gyration, M.I of composite section- I and L.

	• Conclusion of Unit
3.	**Friction & Belt Drive**
	• Introduction of Unit • Friction: Types of Friction, Laws of friction, Angle of friction, Angle of repose, Numericals on Ladder. • Belt Drive: Types of belts, Types of belt drives, Velocity ratio, Effect of slip on Velocity ratio, Length of belt, Ratio of tensions and power transmission by flat belt drives. • Conclusion of Unit
4.	**Dynamics of Particles**
	• Introduction of Unit • Kinematics of Particles and Rigid Bodies: Velocity, Acceleration, Types of Motion, Equations of Motion, Rectangular components of velocity and acceleration, Angular velocity and Angular acceleration. • Kinetics of Particles and Rigid Bodies: Newton's laws, Linear Momentum, Equation of motion in rectangular coordinate, Equation of motion in plane for a rigid body, D' Alembert principle. • Conclusion of Unit
5.	**Work, Power & Impact**
	• Introduction of Unit • Work, Energy and Power: Work of a force, weight, Power, Efficiency, Energy, Kinetic energy of rigid body, Principle of work and energy. • Impact: Collision of elastic bodies, types of impact, conservation of momentum, Newton's law of collision. • Conclusion of Unit

C. RECOMMENDED STUDY MATERIAL

Sr.No	**Reference Book**	**Author**	**Edition**	**Publication**
1.	Vector Mechanics for Engineers	Beer and Johnston	Latest	Tata McGraw Hill
2.	Engineering Mechanics	D S Kumar	Latest	S K Kataria& Sons
3.	Engineering Mechanics Statics	Meriam, J. L. & Kraige, L. G	Latest	John Wiley & Son
4.	Engineering Mechanics	S. Ramamruthan	Latest	Dhanpat Rai Pub.
5.	Engineering Mechanics	Shames	Latest	Pearson Education
Important Web Links				
1.	https://nptel.ac.in/courses/112103109/			
2.	https://nptel.ac.in/courses/112106286/			
3.	https://freevideolectures.com/course/2264/engineering-mechanics			

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B. TECH - CSE

SECOND YEAR
SEMESTER: III

COMMON SYLLABUS

For

Cyber Security
Cloud & Full Stack Development
Artificial Intelligence & Data Science
Artificial Intelligence & Machine Learning

BATCH: 2024-28

A: Major (Core Courses)
A.1: Theory

Code :BCECCE3101 Introduction to Web Technology 3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Relate internet with WWW and various internet protocols such as FTP, TCP/IP, SMTP, and Telnet.
CO – 02	Understand	Implement single page applications in Angular.
CO – 03	Apply	Implement HTML, CSS, and XML in a web page.
CO – 04	Analyze	Install a PHP web server and perform basic procedures in php, such as working with loops, data type, session, and cookies.
CO – 05	Evaluate	Develop a basic web application and perform basic administrative procedures on the web server, such as tuning communication parameters, denying access to certain domains, and interpreting an access log.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	3	1	-	-	-	-	-	-	-	2	-	-
CO2	-	2	3	1	2	-	-	-	-	-	-	-	-	2	-
CO3	-	1	1	2	3	-	-	-	-	-	--	-	-	-	3
CO4	-	1	2	2	3	-	-	-	-	-	-	-	-	1	-
CO5	-	-	3	1	2	-	-	-	-	-	-	-	-	-	1
WT. AVG		2		2	2									1.5	2

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to HTML And Internet**	8
2.	**HTML& CSS**	8
3.	**HTML5, CSS3**	8
4.	**XML**	6
5.	**Practical Website Development**	6

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to HTML And Internet**
	● Introduction, History of internet, Internet Design Principles, Internet Protocols - FTP,TCP/IP,SMTP, Telnet, etc., Client Server Communication, Web System architecture ● Evolution of the Web, Web architectures, Web clients and servers, Static and Dynamic Web Applications, Front end and back end web development. ● HTML, CSS, JS, XML; HTTP, secure HTTP, etc; URL, Web Services – SOAP, REST ● Conclusion of the Unit
2.	**HTML & CSS**
	● Introduction to Html, Html Document structure, Html Editors, Html element/tag & attributes,Designing simple page - Html tag, Head tag, Body tag; ● More HTML Tags - Anchor tag, Image tag, Table tag, List tag, Frame tag, Div tag ; Html forms -Input type, Text area, Select , Button, Images. ● Introduction to CSS, Syntax, Selectors ,Embedding CSS to Html, Formatting fonts, Text & background colour, Inline styles, External and Internal Style Sheets, Borders & boxing ● Conclusion of the Unit

3.	**HTML5, CSS3**
	● Introduction to HTML5. ● Introduction to CSS3, New features, Local storage, Web Sockets, Server events, Canvas, Audio & Video, Geo location, Micro data, Drag and Drop. Browser life cycle and browser renderingstages. Service workers ● Conclusion of the Unit
4.	**XML**
.	● Introduction to XML ● Difference b/w Html & XML, XML editors. ● XML Elements & Attributes XML DTD. ● XML Schema, XML Parser. ● Document Object Model (DOM), XML DOM. ● Conclusion of the Unit
5	**. Practical web site development**
.	● Commonly used Web Servers and browsers, Setting up a server and domain name, website typesand structures, ● Web authoring tools, Web hosting, website maintenance, generating traffic to your website. ● Conclusion of the Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Practical Web Design for Absolute Beginners	AdrianW. West	2016	Apress 2016
2.	Introducing Web Development	Jorg Krause	2017	Apress 2017
3.	HTML &CSS: The Complete Reference	Thomas Powell	2010	McGrawHill, FifthEdition.
Reference Book				
1	HTML and CSS: Design and Build Websites – by Jon Duckett			
2.	Head First HTML and CSS: A Learner's Guide to Creating Standards-Based Web Pages – by Elisabeth Robson & Eric Freeman Publisher- ORELLY			
Online Resources				
1	https://www.w3schools.com/html/html_links.asp			
2	https://www.tutorialrepublic.com/html-tutorial/html-links.php			

Code :BCECCE3102	Data Structures and Algorithms	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Define fundamental data structure concepts, including classification, time and space complexity, dynamic memory allocation, and recursion.
CO – 02	Understand	Contrast basic search techniques, such as sequential and binary search, along with sorting algorithms (Bubble sort, Selection sort, Insertion sort, Merge sort, Quicksort).
CO – 03	Apply	Examine stack and queue concepts, including operations, representation, and types.
CO – 04	Analyze	Explain linked lists, their components, types, and operations. Estimate advantages, disadvantages, and real-life applications of linked lists.
CO – 05	Evaluate	Validate tree and graph concepts, including binary trees, search trees, heaps, and graph applications. Covering tree terminology, array representation, tree traversal, and graph algorithms like Depth First Search and Breadth First Search.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	2	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2.2	2.6	1.4	1.2									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Data structures	8
2.	Searching and Sorting	8
3.	Stack and Queue	8
4.	Linked List	9
5.	Tree Graphs and their Applications	7

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	Introduction to Data structures
	• Introduction to Unit • Definition, • Classification of data structures: primitive and non-primitive • Elementary data organization • Time and space complexity of an algorithm (Examples), String processing. • Definition of dynamic memory allocation • Accessing the address of a variable, Declaring and initializing pointers - • Accessing a variable through its pointer, Meaning of static and dynamic memory allocation, Memory allocation functions: malloc(), calloc(), free() and realloc(). • Recursion – Definition, advantages, Writing Recursive programs – Binomial coefficient, Fibonacci,GCD. • Conclusion and Real Life Applications of unit

2.	**Searching and Sorting**
	• Introduction to Unit • Basic Search Techniques - Sequential search, Iterative and Recursive methods, Binary search:Iterative and Recursive methods, Comparison between sequential and binary search. • Sorting: General back ground and definition- Bubble sort, Selection sort, Insertion sort, Merges ort, Quick sort • Conclusion and Real Life Applications of unit
3.	**Stack and Queue**
	• Introduction to Unit • Stack – Definition • Array representation of stack • Operations on stack: Infix, prefix and postfix notations • Conversion of an arithmetic expression from Infix to postfix • Applications of stacks., Definition of queue, Array representation of queue • Types of queue: Simple queue, Circular queue, Double ended queue (deque), Priority queue, • Operations on all types of Queues • Conclusion and Real Life Applications of Unit
4.	**Linked List**
.	• Introduction of Unit • Definition of linked list, Components of linked list • Representation of linked list, Advantages and Disadvantages of linked list • Types of linked list: Singly linked list, doubly linked list, Circular linked list • Operations on singly linked list: creation, insertion, deletion, search and display • Conclusion and Real Life Applications of Unit
5	**Tree Graphs and their Applications**
.	● Introduction to Unit, Definition : Tree ● Binary tree, Complete binary tree, Binary search tree, Red-Black Tree, AVL Tree ● Heap, Tree terminology: Root, Node, Degree of a node and tree, Terminal nodes, Non-terminal nodes,Siblings, Level, Edge, Path, depth, Parent node, ancestors of a node ● Binary tree: Array representation of tree, Creation of binary tree. ● Traversal of Binary Tree: Preorder, Inorder and postorder. ● Comparison between Red-Black Tree & AVL Tree ● Graphs, Application of Graphs ● Depth First search, Breadth First search. ● Conclusion and Real Life Applications of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Schaum's outline series Data structures	Lips chutz	Latest	TMH.
2	Data Structures and program designing using'C'	Robert Kruse	Latest	Pearson Education
Reference Book				
1	Introduction to Data Structures in C by- Kamthane Pearson Education2005			
2.	Data Structures Using C by- Bandyo Padhyay Pearson Education			
Online Resources				
1	https://www.gatevidyalay.com/data-structures/			
2	https://www.youtube.com/watch?v=QBrDsG3MTkw			
3	https://www.tutorialspoint.com/data_structures_algorithms/index.htm			

Code :BCECCE3103	Operating System	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Memorizing the fundamental concepts, structures, evolution of operating systems and effectively manage system resources and design efficient system programs.
CO – 02	Understand	Demonstrate process and thread management, inter-process communication, mutual exclusion, and process scheduling to design efficient and reliable system solutions.
CO – 03	Apply	Determining deadlock characterization, prevention, avoidance, detection, and recovery strategies to ensure system reliability and stability. Top of Form
CO – 04	Analyze	Organizing memory management principles including segmentation, paging, virtual memory, and page replacement algorithms, for optimized system performance.
CO – 05	Evaluate	Reviewing mastering file system fundamentals and implementation techniques for efficient data storage, retrieval.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	3	1	1	–	–	–	–	–	–	–	–	1	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	1	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	1	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	-	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	1.8	2	1.4	1.2									1.6		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Operating System Overview	08
2.	Process Management	08
3.	Process Deadlocks	08
4.	Memory Management	09
5.	File Management	07

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	Operating System Overview
	• Introduction of Unit • Definition, Two views of operating system, Evolution of operating system, Types of OS. • System Call, Handling System Calls, System Programs, Operating System Structures, • The Shell, Open Source Operating Systems • Conclusion of Unit
2.	Process Management
	• Introduction of Unit • Process v/s Program, Multi-programming, Process Model, Process States, Process ControlBlock. • Threads, Thread v/s Process, User and Kernel Space Threads.

	• Inter Process Communication, Race Condition, Critical Section • Implementing Mutual Exclusion: Mutual Exclusion with Busy Waiting • Interrupts, Lock Variables, Strict Alteration, Peterson's Solution, Test and Set Lock. • Sleep and Wake-up, Semaphore, Monitors, Message Passing. • Classical IPC problems: Producer Consumer, Sleeping Barber, Dining Philosopher • Problem • Process Scheduling: Goals, Batch System Scheduling (First-Come First-Served, Shortest Job First, Shortest Remaining Time Next), Interactive System Scheduling (Round-Robin Scheduling,Priority Scheduling, Multiple Queues), Overview of Real Time System Scheduling • Conclusion of Unit
3.	**Process Deadlocks**
	• Introduction of Unit • Introduction, Deadlock Characterization, Preempt able and Non-preempt able Resources • Resource – Allocation Graph, Conditions for Deadlock. • Handling Deadlocks: Ostrich Algorithm, Deadlock prevention, Deadlock Avoidance. • Deadlock Detection (For Single and Multiple Resource Instances), Recovery From • Deadlock (Through Preemption and Rollback) • Conclusion of Unit
4.	**Memory Management**
.	• Introduction of Unit • Introduction, Mono programming vs. Multi-programming, Modeling Multiprogramming,Multiprogramming with fixed and variable partitions, Relocation and Protection. • Memory management (Bitmaps & Linked-list), Memory Allocation Strategies • Virtual memory: Paging, Page Table, Page Table Structure, Handling Page Faults, TLB's • Page Replacement Algorithms: FIFO, Second Chance, LRU, Optimal, LFU, Clock, WS- Clock,Concept of Locality of Reference, Belady's Anomaly • Segmentation: Need of Segmentation, its Drawbacks, Segmentation with Paging(MULTICS) • Conclusion of Unit
5	**File Management**
.	• Introduction of Unit • File Overview: File Naming, File Structure, File Types, File Access, File Attributes, File Operations,Single Level, two Level and Hierarchical Directory Systems, File System Layout. • Implementing Files: Contiguous allocation, Linked List Allocation, Linked List • Allocation using Table in Memory, Inodes. • Directory Operations, Path Names, Directory Implementation, Shared Files • Free Space Management: Bitmaps, Linked List • Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Operating system concepts	Silberschatz, Galvin, Gagne	8th edition	John Wiley and Sons
2	Modern Operating System	A.S.Tanenbaum	2nd Edition	Pearson
Reference Book				
1	"Head First Java" by Kathy Sierra			
Online Resources				
1	https://www.coursera.org/courses?query=operating%20system			
2	https://onecompiler.com/java			

Code :BCECCE3104	Relational Database Management System	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall the basic principles of database management systems. Define key database concepts such as data, information, database, DBMS, RDBMS, and SQL.
CO – 02	Understand	Understand architecture, features and functionalities of RDBMS.
CO – 03	Apply	Describe transaction processing and concurrency control concepts.
CO – 04	Analyze	Implement constraints for maintaining data integrity, Execute SQL commands and Apply database concepts and techniques to solve real-world problems.
CO – 05	Evaluate	Explain the advantages of using PL/SQL in the context of database programming, Assess the usage of variables and constants in PL/SQL, understanding their role in program logic and data storage.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	2	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2.2	2.6	1.4	1.2									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Database Management System**	7
2.	**RDBMS**	7
3.	**SQL**	7
4.	**PL/SQL**	8
5.	**Oracle, Trigger and wrapping**	7

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Database Management System**
	• Introduction to Database Management System • Characteristics of database approach • Advantages of DBMS • Schemas: Three schema architecture - The external level, the conceptual level and the internal level. • Data Independence, Database languages and Interfaces • Roles of Database Administrator • ACID Property of Transaction • Introduction to Data Models (Hierarchical, Network and Relation) • Entity type, Entity sets, Attributes and keys. • The ER Model: ER Diagram & Database design with the ER Model • Conclusion of the Unit
2.	**RDBMS**
	• Introduction to Distributed Database • Classification of DBMS • Introduction to RDBMS, Relational Model –Concepts

	• Relational operations (Insert, delete, update, select, project, rename, union, intersection, minus, Join,division) • Transactions and ER mapping Examples • 12 Golden rules of dr. E.F.Codd • Normalization of RDBMS (1NF, 2NF, 3NF and 4NF) and inference rules. • Conclusion of the Unit
3.	**SQL**
	• Introduction to Unit • DBMS v/s RDBMS • Introduction to SQL: Data types, Constraints • Commands in SQL: Create table, Drop command, Alter Queries in SQL • Statements in SQL (Insert, delete and update) • Features of SQL • Manipulation of data, Tables in SQL • Conclusion of the Unit
4.	**PL/SQL**
.	• Introduction to PL/SQL • Approaches to database programming: with function calls, Embedded SQL using CURSORs,Dynamic SQL, SQL commands in Java, Retrieving multiple triples using Iterators • Advantages of PL/SQL • Features of PL/SQL :Blocks structure, Error handling, Input and output designing, variables andconstant, data abstraction, control structures and subprogram • Fundamentals of PL/SQL : character sets, lexical, delimeters, identifiers, declarations, scope andvisibility, Static and dynamic and static SQL, Implicit and explicit locking • Conclusion of the Unit
5	**Oracle, Trigger and wrapping**
.	• Introduction to Oracle, Trigger and wrapping • Functions/responsibilities of DBA • Oracle product details • Oracle files, System and User process • Oracle Memory • Protecting data: Oracle backup & recovery • Triggers - types, uses, data access for triggers • PL/SQL Packages and Wrapping • Conclusion of the Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	System ConceptsSQL, PL/SQL	S. Sudarshan, Henry F. Korth, Avi Silberschatz	6th Edition	McGraw HillBpb
2	Oracle Complete	Ivan Bayross		Bpb
3	System ConceptsSQL, PL/SQL	Kevin Loney		McGraw HillBpb
Reference Book				
1	PL/SQL, best practices, Bpb Publications, Steven Feuerstein			
2.	The Oracle Cook Book, Bpb Publications, Liebschuty			
3.	Oracle A Beginners Guide, TMH Publication, Michael Abbey, Michael J.Corey			
Online Resources				
1	https://www.tutorialspoint.com/sql/sql-rdbms-concepts.htm			
2	https://nptel.ac.in/courses/106106093			
3	https://www.coursera.org/learn/introduction-to-relational-databases			

A.2: Practical

Code :BCECCE3201 Introduction to Web Technology Lab 2 Credits [LTP: 0-0-1]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	use the fundamental concepts of web development including HTML, CSS, and JavaScript. List different web development frameworks and libraries used for building interactive web applications.
CO – 02	Understand	Explain the client-server architecture and how it relates to web application development. Summarize the importance of user experience (UX) and user interface (UI) design principles in web development.
CO – 03	Apply	Develop interactive web applications using HTML, CSS, and JavaScript. Apply responsive design techniques to ensure compatibility with various devices and screen sizes.
CO – 04	Analyze	Analyze and debug code to identify and resolve issues in interactive web applications.
CO – 05	Evaluate	Evaluate the performance and efficiency of different web development techniques and tools. Critique the usability and accessibility of web applications based on established standards and guidelines.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	2	2	1	2	-	1	-	2	1	1	1	3	3	2
CO2	-	2	2	1	2	-	1	-	2	1	1	1	3	3	2
CO3	-	2	2	1	2	-	1	-	2	1	1	1	3	3	2
CO4	-	2	2	1	2	-	1	-	2	1	1	1	3	3	2
CO5	-	2	2	1	2	-	1	-	2	1	1	1	3	3	2
WT. AVG		2	2	1	2		1		2	1	1	1	3	3	2

C. LIST OF EXPERIMENTS

1	Hello World Web Page (a) Create a web page using basic HTML features like tags, attributes, elements and page title(b)How to install and configure a web server
2	Create a My Profile Page (a) Using text boxes, check boxes, radio buttons and submit buttons. (b) Design a web page using CSS include the following: 1. Control the repetition of image with back ground-repeat property. 2. Define style for links asa: link, b:active,c:hover,d:visited. 3. Add customized cursors for links.
3	Create a My Profile Page (a) A more functional web page by making use of headings, paragraphs, lists, images and links. (b) Design a web page using CSS include the following: 1. Use different font styles. 2. Set back ground image for both the page and single elements on the page.
4	Create XML Http Request and retrieve data from a text file and an XML file.
5	Create the following webpage: (a) Show the class time table in a tabular format. (b) Create a web page using HTML to show your geolocation.
6	Create a webpage using HTML for audio and video player.
7	Create a log in registration form using PHP.
8	Develop a PHP web page to manipulating files such as creating ,writing, reading and uploading.
9	Create a dynamic web page by using PHP conditional operators, loops and strings to create an dynamictime table page.
1	Develop a PHPweb application track the user as how many times visited and last visited time

0	
11	Develop a static website–I.
12	Develop a static website–II.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Practical Web Design for Absolute Beginners	AdrianW. West	2016	Apress 2016
2	Introducing Web Development	Jorg Krause	2017	Apress2017
3	HTML & CSS: The Complete Reference	Thomas Powell	2010, Fifth Edition	McGrawHill,
Reference Book				
1	HTML and CSS: Design and Build Websites – by Jon Duckett			
2.	Head First HTML and CSS: A Learner's Guide to Creating Standards-Based Web Pages – by Elisabeth Robson & Eric Freeman Publisher- ORELLY			
Online Resources				
1	https://www.w3schools.com/html/html_links.asp			
2	https://www.tutorialrepublic.com/html-tutorial/html-links.php			

Code :BCECCE3202	Data Structures and Algorithms Lab	2 Credits [LTP: 0-0-1]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Introduction to Data Structures, Recall fundamental data structures such as arrays, linked lists, stacks, and queues.
CO – 02	Understand	Understand various algorithm design techniques, including greedy algorithms, dynamic programming, and divide and conquer.
CO – 03	Apply	Apply data structure implementation techniques in programming assignments to solve problems efficiently.
CO – 04	Analyze	Analyze the time and space complexity of algorithms using Big O notation to evaluate their efficiency.
CO – 05	Evaluate	Evaluate algorithm performance through empirical analysis and optimize algorithms for better efficiency.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	-	2	1	-	-	-	-	-	-	-	2	-	-
CO2	1	2	3	2	-	-	-	-	-	-	-	-	1	-	-
CO3	1	3	3	1	-	-	-	-	-	-	-	-	2	-	-
CO4	2	-	1	1	-	-	-	-	-	-	-	-	2	-	-
CO5	1	1	2	-	-	-	-	-	-	-	-	-	1	-	-
WT. AVG	1.4	1.75	2.25	1.5	1								1.6		

C. LIST OF EXPERIMENTS

1	Use a recursive function to find a) GCD of two numbers. b) Use a recursive function to find the Fibonacci series. c) Factorial d) Binomial Coefficient
2	Perform the following: a) Insert an integer into a given position in an array. b) Deleting an integer from an array.
3	Perform the following: a) Write a program for linear search b) Write a program for Binary search c) Write a program to sort N numbers using bubble sort.
4	Perform the following: a) Write a program to sort N numbers using insertion sort. b) Write a program to sort N numbers using selection sort. c) Write a program to sort N numbers using bubble sort.
5	Write a program to sort N numbers using quick sort.
6	Write a program to sort N numbers using merge sort.
7	Write a C program to create Stack using array.
8	Write a C program to create queue using array.
9	Write a program to create a linked list and to display it.
10	Inserting a node into a singly linked list on various position beginning, after given location and end.
11	Deleting a node into a singly linked list on various position beginning, after given location and end.
12	Write a C program to create stack and queue using linked list.
13	Creating a binary search tree and traversing it using inorder, preorder and postorder.
14	Write a C program to implement graph.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Data Structures and Algorithm Analysis in C	Weiss	2001	Pearson Education
2	Schaum's outline series Data structures	Lipschutz		Tata McGraw-Hill
3	Data Structures and program designing using 'C'	Robert Kruse		Pearson
	Data Structures Using C	Bandyopadhyay	1999	Pearson Education
Reference Book				
1	Data Structures Using C, Pearson Education, Tenenbaum.			
2.	Introduction to Data Structures in C, Pearson Education 2005, Kamthane			
3	Data Structures using C and C++, Pearson Education, Langsam, Ausenstein Maoshe & M. TanenbaumAaron.			
Online Resources				
1	https://www.programiz.com/dsa			
2	https://www.geeksforgeeks.org/data-structures/			
3	https://www.codechef.com/certification/data-structures-and-algorithms/prepare			

Code :BCECCE3201	Operating System Lab	2 Credits [LTP: 0-0-1]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Remember the operating system concepts
CO – 02	Understand	Understand the fundamental concepts of operating systems.
CO – 03	Apply	Gain practical experience in operating system concepts.
CO – 04	Analyze	Develop problem-solving skills related to OS design.
CO – 05	Evaluate	Learn to analyze and evaluate different OS architectures.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	1	1	-	-	-	-	-	1	1	-	-	1	1	-
CO2	-	1	1	-	-	-	-	-	1	1	-	-	2	2	-
CO3	2	2	2	2	-	-	-	-	1	2	2	-	2	2	3
CO4	3	3	3	3	-	-	-	-	1	3	3	3	3	3	3
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
WT. AVG	2.5	1.75	1.75	2.5					1	1.75	2.5	3	2	2	3

C. LIST OF EXPERIMENTS

1.	A.	Write a shell script that accept a file name starting and ending line numbers as arguments and display all the lines between given line no.
	B.	Write a shell script that delete all lines containing a specified word.
2.	A.	Write a shell script that displays a list of all the files in the current directory.
	B.	Write a shell script that accept a list of file names as arguments count and report the occurrence of each word.
3.	A.	Write a shell script to find the factorial of given integer.
	B.	Write a shell script that list the all files in a directory.
4.		Implement in C the following Linux commands using system calls a) cat b) mv
5.		Write a C program to emulate the Linux ls-l command?
6.		Write a C program to simulate any of the following non-preemptive CPU scheduling algorithms to find turnaround time and waiting time. a) FCFS b) SJF c) Round Robin d) Priority
7.		Write a C program to simulate producer-consumer problem using semaphores.
8.		Write a C program to simulate the concept of Dining-Philosophers problem.
9.		Write a C program to simulate any of the following contiguous memory allocation techniques a) Worst-fit b) Best-fit c)First-fit
10.		Write a C program to simulate any of the following page replacement algorithm a) FIFO b) LRU c) LFU
11.		Write a C program to simulate any of the following file allocation strategies a) Sequential b) Indexed c) Linked
12.		Write a C program to simulate any of the following disk scheduling algorithms a) FCFS b) SCAN c) C-SCAN

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Operating system concepts	Silberschatz, Galvin, Gagne	8th edition	John Wiley and Sons
2	Modern Operating System	A.S.Tanenbaum	Second Edition	Pearson
Reference Book				
1	Operating Systems-S Halder, Alex A Aravind Pearson Education Second Edition 2016			
Online Resources				
1	*https://www.coursera.org › courses › query=operating s...*			
2	*https://www.javatpoint.com › best-courses-for-the-oper...*			
3	*https://hackr.io › tutorials › learn-operating-systems*			

Code :BCECCE3204	RDBMS Lab	2 Credits [LTP: 0-0-1]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall fundamental concepts of relational databases including tables, rows, columns, and SQL queries.
CO – 02	Understand	Understand database design principles including normalization, indexing, and entity-relationship modeling.
CO – 03	Apply	Apply SQL queries for data manipulation tasks such as inserting, updating, deleting, and querying data.
CO – 04	Analyze	Analyze SQL query performance using query execution plans and optimize queries for better performance.
CO – 05	Evaluate	Evaluate database security measures including user authentication, authorization, and encryption techniques.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	-	2	1	-	-	-	-	-	-	-	2	-	-
CO2	1	2	3	2	-	-	-	-	-	-	-	-	1	-	-
CO3	1	3	3	1	-	-	-	-	-	-	-	-	2	-	-
CO4	2	-	1	1	-	-	-	-	-	-	-	-	2	-	-
CO5	1	1	2	-	-	-	-	-	-	-	-	-	1	-	-
WT. AVG	1.4	1.75	2.25	1.5	1								1.6		

C. LIST OF EXPERIMENTS

1	To setup and removal phases of a Student database using the basic Data Definition Language (DDL)commands: 1. CREATE 2. ALTER 3. DROP 4. RENAME 5. TRUNCATE
2	The routine operation of the Employee database like retrieve, insert and modify by basic DataManipulation Language (DML) commands: 1. INSERT 2. UPDATE 3. DELETE
3	To Retrieve data from one or more tables using DATA RETRIEVAL LANGUAGE (DRL) commandsSELECT FROM SELECT - FROM –WHERE SELECT - FROM -GROUP BYSELECT - FROM -ORDER BY JOIN using SELECT - FROM - ORDER BYJOIN using SELECT - FROM - GROUP BYUNION INTERSET MINUS
4	DATA CONTROL LANGUAGE (DCL) and TRANSATIONAL CONTROL LANGUAGE (TCL) commands. Creating objects: tables, views, users, sequences, Collections etc.Privilege management through the Grant and Revoke commands Transaction processing using Commit and Rollback Save points.
5	Queries for following functions Conversion functions (to_char, to_number and to_date) string functions (Concatenation, lpad, rpad, ltrim, rtrim, lower, upper, initcap, length, substr and instr), date functions (Sysdate, next_day, add_months, last_day, months_between, least, greatest, trunc,round, to_char, to_date)
6	Simple queries: selection, projection, sorting on a simple table for employee databaseSmall-large number of attributes Distinct output values Renaming attributes Computed attributes

	Simple-complex conditions (AND, OR, NOT)Partial Matching operators (LIKE, %, _, *, ?) ASC-DESC ordering combinations Checking for Nulls
7	To manipulate data items and returning the results using Group functions or Aggregate functions andSingle Row or scalar functions: Group functions or Aggregate functions: Sum(), Avg(), Min(), Max() and Count() Single Row or scalar function: Abs(), Power(), Sqrt(), Round(), Exp(), Greastest(), Least(), Mod(), Floor(), Sign() and Log().
8	Multi-table queries (JOIN OPERATIONS)Simple joins (no INNER JOIN) Aliasing tables – Full/Partial name qualificationInner-joins (two and more (different) tables) Inner-recursive-joins (joining to itself) Outer-joins (restrictions as part of the WHERE and ON clauses) Using where & having clauses
9	Write Nested queries to retrieve the name of each employee who has a dependent with the same firstname and same sex as the employee using following Nested queries. In, Not In Exists, Not Exists Dynamic relations (as part of SELECT, FROM, and WHERE clauses)
10	Write a query to make a list of all project numbers for projects that involve an employee whose last nameis 'Smith', either as a worker or as a manager of the department that controls the project using the following Set Oriented Operations Union Difference IntersectionDivision
11	PL/SQL Programming using the following Programs using named and unnamed blocks Programs using Cursors, Cursor loops and records
12	PL/SQL Programming using Creating stored procedures, functions and packagesError handling and Exception Triggers and auditing triggers

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Database System Concepts	S. Sudarshan, Henry F. Korth, Avi Silberschatz	6th Edition	McGraw Hill
2	SQL, PL/SQL	Ivan Bayross		Bpb
3	Oracle Complete Reference	Kevin Loney		Bpb
Reference Book				
1	PL/SQL, best practices, Bpb Publications, Steven Feuerstein			
2.	The Oracle Cook Book, Bpb Publications, Liebschuty			
3	Oracle A Beginners Guide, TMH Publication, Michael Abbey, Michael J.Corey			
Online Resources				
1	https://www.tutorialspoint.com/sql/sql-rdbms-concepts.htm			
2	https://nptel.ac.in/courses/106106093			
3	https://www.coursera.org/learn/introduction-to-relational-databases			

B: Minor Stream Courses/ Department Electives
B1: Theory - Professional Elective - I

Code :BCEECE3111	Theory Of Computation	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall and memorize basic concepts of discrete mathematics, principle of mathematical induction etc.
CO – 02	Understand	Demonstrate an understanding of finite state machines and different machines & Regular Expression.
CO – 03	Apply	Apply Grammar concept for context free Grammar along with BNF & CNF.
CO – 04	Analyze	Examine automata concept on push down machine with CNF Forms.
CO – 05	Evaluate	Appraise the Model of Computation in the Turing machine with Chomsky hierarchy.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	2	–	–	–	–	–	–	–	–	1	–	–
CO4	2	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	1	2	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2.2	2.6	1.4	1.6									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Review of Mathematical Theory	07
2.	Regular Languages and Finite Automata	08
3.	Context free grammar (CFG)	08
4.	Pushdown Automata, CFL And NCFL	07
5.	Turing Machine (TM)	07

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Review of Mathematical Theory**
	• Introduction of Unit • Sets, Functions, Logical statements, Proofs, Relations, • Languages, Principal of Mathematical Induction, • Strong Principle, Recursive Definitions, • Structural Induction. • Conclusion of Unit
2.	**Regular Languages and Finite Automata**
	• Introduction of Unit • Regular Expressions, Regular Languages, Application of Finite Automata, • Automata with output - Moore machine & Mealy machine, • Finite Automata, Memory requirement in a recognizer, • Definitions, union- intersection and complement of regular languages, Non Deterministic FiniteAutomata, • Conversion from NFA to FA, Non Deterministic Finite Automata, Conversion of NFA to NFA

	• Kleene's Theorem, Minimization of Finite automata, Regular And Non Regular Languages –pumping lemma. • Conclusion of Unit
3.	**Context free grammar (CFG)**
	• Introduction of Unit • Definitions and Examples, Unions Concatenations And Kleene's of Context free language, • Regular Grammar for Regular Language, Derivations and Ambiguity , • Unambiguous CFG and Algebraic Expressions, BacosNaur Form (BNF), Normal Form – CNF. • Conclusion of Unit
4.	**Pushdown Automata, CFL And NCFL**
.	• Introduction of Unit • Definitions, Deterministic PDA, Equivalence of CFG and PDA & Conversion, • Pumping lemma for CFL, Intersections and Complements of CFL, Non-CFL. • Conclusion of Unit
5	**Turing Machine (TM)**
.	• Introduction of Unit • TM Definition, Model Of Computation, • Turing Machine as Language Acceptor, • TM that Compute Partial Function, Church Turning Thesis, • Combining TM, Variations Of TM, Non Deterministic TM, Universal TM, • Recursively and Enumerable Languages, Context sensitive languages and Chomsky hierarchy. • Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Theory of Computer Science: Automata, Languages andComputation	Mishra & Chandrashekhar	3rd	PHI
2	An Introduction to Formal Languages and Automata	Peter Linz	6th	Mass Market Paperback
Reference Book				
1	Introduction to Languages and the Theory of Computation, 4th by John Martin, Tata Mc Graw Hill			
2.	Introduction to computer theory By Deniel I. Cohen , Joh Wiley & Sons, Inc			
3	Compiler Design By Alfred V Aho, Addison Weslley			
Online Resources				
1	http://en.wikipedia.org/wiki/Theory_of_computation			
2	http://meru.cecs.missouri.edu/courses/cecs341/tc.html			
3	https://www.geeksforgeeks.org/introduction-of-theory-of-computation/			

Code :BADECE3111	Mathematics for Machine Learning	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Understand and Analyze	To understand the basic concepts of Probability and Statistics such as Descriptive Statistics, Probability Distributions and use them to analyze the real-world data.
CO – 02	Understand and Evaluate	To understand the basic and advanced concepts in Linear Algebra such as Vectors, Singular and Non-Singular Matrices, Linearly Dependent, Independent, Orthogonal and Eigen Vectors as well as Eigen Values, Matrix Factorizations.
CO – 03	Understand and Apply	To understand the basic concepts in Differential Calculus such as Derivatives, Chain Rule and Constrained as well as Unconstrained Optimization Algorithms and apply them to solve real world optimization problems.
CO – 04	Apply	To apply advanced concepts of Probability and Statistics as well as Differential Calculus, to perform parameter estimation of prevalent Probability Distributions used in Data Science, Machine Learning and AI, using sample data.
CO – 05	Apply	To apply advanced concepts of Linear Algebra to perform Transformation of different types of data such as Tabular Data, Text and Images.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	3	-	-	1	-	2	-	-	3	-	1
CO2	3	3	-	-	3	-	-	-	-	-	-	-	3	-	1
CO3	3	3	-	1	3	-	-	-	-	-	-	-	3	-	1
CO4	3	3	3	3	3	2	2	1	1	1	-	1	3	-	2
CO5	3	3	3	3	3	-	2	2	1	1	-	1	3	-	3
WT. AVG	3	3	3	2.33333	3	2	2	1.5	1	1.5		1	3		1

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Fundamental Concepts of Probability and Statistics**	6
2.	**Applied Concepts in Linear Algebra**	6
3.	**Applied Concepts in Differential Calculus**	8
4.	**Introduction to Statistical Learning**	8
5.	**Introduction to Feature Engineering**	8

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Fundamental Concepts of Probability and Statistics**
	• Introduction of Unit. • Applied Probability and its fundamental concepts. • Introduction to Continuous and Discrete Probability Distributions. • Introduction to Binomial, Poisson, Normal, Standard Normal and Multivariate Normal Distributions. • Conclusion of the unit.
2.	**Applied Concepts in Linear Algebra**
	• Introduction of Unit. • Orthogonal, Linearly Dependent, Independent Vectors and Vector Spaces.

	• Gaussian Elimination and Singular Matrices. • LU and QR Factorization. • Conclusion of Unit.
3.	**Applied Concepts in Differential Calculus**
	• Introduction of Unit. • Linear and Quadratic Approximation. • Unconstrained Optimization Algorithms: Gradient Descent, Momentum Update, Adagrad, RMS Prop and Adam. • Constrained Optimization: Method of Lagrange Multipliers • Conclusion of the Unit
4.	**Introduction to Statistical Learning**
.	• Introduction of Unit. • Maximum Likelihood Estimation. • Parameter Estimation of parameter of Binomial Distribution. • Parameter Estimation of parameter of Poisson Distribution. • Parameter Estimation of parameters of Normal Distribution. • Parameter Estimation of parameters of Multivariate Normal Distribution. • Conclusion of the Unit
5	**Introduction to Feature Engineering**
.	• Introduction of Unit. • Eigen and Singular Value Decomposition. • Principal Components Analysis applied to Tabular and Visual Data. • Latent Semantic Analysis applied on Textual Data. • Conclusion of the Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Applied Probability and Statistics for Engineers	George C. Unger and Douglas C. Montgomery	7th Editi on	Wiley
2	Linear Algebra and its Applications	Gilbert Strang	4th Editi on	Cengage Learning
3	Linear and Non-Linear Optimization	Stephen G. Nash and Ariela Sofer Igor Griva	2nd Editi on	SIAM
Reference Book				
1	Introduction to Probability and Statistics by William Mendenhall, Robert J. Beaver and Barbara M. Beaver, 16th Edition, 2016, Cengage Learning			
2.	Introduction to Applied Linear Algebra: Vectors, Matrices and Least Squares by Stephen Boyd, Lieven Vanderberghe, 1st Edition, Cambridge University Press			
Online Resources				
1	https://docs.scipy.org/doc/scipy/reference/stats.html			
2	https://www.numpy.org			

Code :BCSECE3111	Cyber Criminal Law & IPR	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	To remember theoretical and cross-disciplinary approaches (criminological, political, legal and information security/management)
CO – 02	Understand	Understand the structure, mechanics and evolution of the Internet in the context of emerging crime threats and technological and other trends in cyberspace.
CO – 03	Apply	To apply structure, mechanics and evolution of the Internet in the context of emerging crime threats and technological and other trends in cyberspace.
CO – 04	Analyze	Analyse and assess the impact of cybercrime on government, businesses, individuals and society.
CO – 05	Evaluate	Evaluate the effectiveness of cyber-security, cyber-laws (e.g. the Budapest Convention) and other countermeasures against cybercrime and cyber warfare.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	—	—	—	—	—	—	—	—	2	—	—
CO2	3	2	1	2	—	—	—	—	—	—	—	—	2	—	—
CO3	2	3	2	1	—	—	—	—	—	—	—	—	1	—	—
CO4	2	2	2	1	—	—	—	—	—	—	—	—	2	—	—
CO5	2	3	1	1	—	—	—	—	—	—	—	—	2	—	—
WT. AVG	2.2	2.6	1.4	1.2									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Cyber Crime Introduction	07
2.	Introduction to Cyber Laws	08
3.	Indian Cyber Laws: Related Issues and Provisions	08
4.	Computer Forensics	07
5.	IPR in Cyberspace	07

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Cyber Crime**
	• Introduction of Unit • Introduction to Conventional Crime and Cyber Crime • Classification of Conventional and Cyber Crime, Difference between Conventional and Unconventional Cyber Crime • Cyber Criminal Mode and Manner of Committing Cyber Crime • Reasons for Cyber Crime, Cyber law and IPR, Need for cyber law, Evolution of key terms and concepts • Cyber Crimes and Jurisprudence of Cyber Law • Introduction of IT Act, Computer crime prevention measures • Conclusion of Unit
2.	**Introduction to Cyber Laws**
	• Introduction of Unit • Provisions in Indian Laws in dealing with Cyber Crimes and its critical analysis, Information Technology Act, 2000, Penalties and Offences under IT Act • Offences related with Digital Signature and Electronic Signature under IT Act, Statutory

	Provisions • Establishment of Authorities under IT Act and their functions, powers such as Controller, Certifying Authorities ,Cyber Regulation Appellate Tribunal, Adjudicating officer • Conclusion of Unit
3.	**Indian Cyber Laws: Related Issues and Provisions**
	• Introduction of Unit • International Organizations and their roles such as ICANN,URDP,WTO and TRIPS • Evolution of IT Act, Digital/ Electronic Signature- Analysis in the background of Indian Laws • E-Commerce: Issues and provisions in Indian Law • E-Governance: concept and practicality in India, E-Taxation issues in Cyberspace • Conclusion of Unit
4.	**Computer Forensics**
.	• Introduction of Unit • Introduction to Computer Forensics, Subdivisions, Steps of Computer Forensics, need for computer forensics • Analyzing the suspected Computer, Incidence Response, Cyber Forensics and Digital Evidence • Digital Forensic Science , Digital Forensics Life Cycle, Chain of custody concept and Challenges in Computer Forensics, Forensic Imaging • Conclusion of Unit
5	**IPR in Cyberspace**
.	• Introduction of Unit • Concept of Copyright and Patent in Cyberspace, Copyright in the Digital Medium • Copyright in Computer Programmes, Copyright and WIPO Treaties • Concept of Patent Right, Relevant provisions of Patent Act 1970 • Data Storage on a Hard Drive, Data Storage on a Hard Drive, Hard Disk Drive addressing • Hard Disk Drive addressing, File corruption and Recovery, Fundamentals of drive imaging • Cloning and Issues in Imaging • Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Cyber security - understanding cyber crimes,computer forensics and legal perspectives	Nina godbole and sunit belapure	1st	WILEY
2	Computer Forensics: Principles And Practices	Linda Volonino, Reynaldo Anzaldua And Jana Godwin	1st	PEARSON
Reference Book				
1	System Forensics Ankit Fadia,Boonlia, Prince Komal, 1st Vikas Publication			
2.	Cyber law in India, Farooq Ahmad 1st Pioneer Publishers, New Delhi			
	Information technology law and practice, Sharma Vakul, Universal Law Publishing Co Ltd			
Online Resources				
1	http://www.vjolt.net/vol12/issue3/v12i3_a1-Azam.pdf			
2	https://www.wipo.int/export/sites/www/amc/en/docs/wipointaudrp.pdf			
3	http://www.iibf.org.in/documents/Cyber-Laws-chapter-in-Legal-Aspects-Book.pdf			

Code :BCGECE3111	Principle of Virtualization	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Remember and comprehend the fundamentals of Python programming, virtualization concepts including types
CO – 02	Understand	Understanding of the components of vSphere 6.0, including its architecture, vCenter Server features, certificate management, alarms, monitoring features, template management, linked mode deployment, storage features.
CO – 03	Apply	Able to apply the features of vSphere and NSX to effectively manage and optimize virtualized environments, implement resource management strategies such as vMotion
CO – 04	Analyze	Analyze data center challenges, including availability, scalability, management, optimization, application upgrade, and cloud-related issues
CO – 05	Evaluate	Evaluate resource optimization and management strategies within the vSphere environment.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	2	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2.2	2.6	1.4	1.2									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction**	7
2.	**Components of vSphere 6.0**	7
3.	**Features of vSphere and NSX**	9
4.	**VSphere Solutions to Data Center Challenges and vSphere Security**	8
5.	**Resource optimization and resource management**	8

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Python Programming**
	• Introduction of Unit • Introduction to Virtualization - Types of virtualization - Difference between cloud and virtualization -Physical infrastructure and virtual infrastructure - Virtualization approaches • Partitioning - Hosting - Isolation - Hardware independence - Virtual machine – Hypervisor - Typesof hypervisor - Virtual machine manager - Types of hypervisor - Introduction to datacenter virtualization Esxi - Difference between Esxi and Esx - Versions of Esxi • Conclusion of Unit
2.	**Components of vSphere 6.0**
	• Introduction of Unit • Components of VMware vSphere - vSphere 6.0: Overview and Architecture - Topology of vSphere

	6.0 Data Center - vSphere 6.0 Configuration MaximumsvCenter Server - vCenter • Server Features - Certificate Management - Alarms and Alerts - Monitoring Features- Template Management - Linked Mode Deployment - Storage Features in vSphere – Shared Storage - Storage Protocols - Datastores - Virtual SAN - Virtual Volumes – Networking, Features in vSphere -Virtual Networking - Virtual Switches and its types • Conclusion of Unit
3.	**Features of vSphere and NSX**
	• Introduction of Unit • vSphere Resource Management Features - vMotion - Distributed Resource Scheduler (DRS) Distributed Power Management (DPM) - Storage vMotion - Storage DRS - Storage I/O Control - Network I/O Control - vSphere Availability Features - vSphere Data Protection -High Availability -Fault Tolerance - vSphere Replication - Introduction to NSX. • Conclusion of Unit
4.	**VSphere Solutions to Data Center Challenges and vSphere Security**
.	• Introduction of Unit • Challenges - Availability Challenges - Scalability Challenges - Management Challenges - Optimization Challenges - Application Upgrade Challenges - Cloud Challenges - Security -Describe the features and benefits of VMware Platform Services Controller • host access and authorization - Secure ESXi - vCenter Server - and virtual machines- Upgrade ESXiand vCenter Server instances • Conclusion of Unit
5	**Resource optimization and resource management**
.	• Introduction of Unit • Network Optimization - Configure and manage vSphere distributed switches - Migrate virtual machines from standard switches to distributed switches - Explain distributed switch features suchas port mirroring - LACP - QoS tagging - and NetFlow - CPU Optimization – Explain the CPU scheduler operation - NUMA support - and other features that affect CPU performance • Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Virtualization Essentials Paperback	Matthew Portnoy	Latest	Wiley Publications
2	VMware Cookbook Paperback	Troy- Shroff	Latest	O'Reilly
Reference Book				
1	Nelson Ruest, Danielle Ruest, Virtualization, A Beginner's Guide, McGraw Hill, 2009			
2.	Cloud Computing: Concepts Technology & Architecture By Thomes Erl			
Online Resources				
1	https://www.javatpoint.com/virtualization-in-cloud-computing			
2	https://www.tutorialspoint.com/virtualization2.0/virtualization2.0_tutorial.pdf			

Code :BAMECE3111	SAS Programing in Viya	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Remember and apprehend the SAS platform, and running SAS code on SAS computer server.
CO – 02	Understand	Understand the accessing of DBMS date, modifying Data step code to run in SAS cloud Analytic services (CAS)
CO – 03	Apply	Try to apply the various procedure in SAS Viya and running CAS-embedded SAS procedure
CO – 04	Analyze	Analyze how SQL code can modify to run CAS and column alteration.
CO – 05	Evaluate	Evaluate the CAS action and understand CASL.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	1	3	2	2	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	2	2	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2	2.8	1.6	16									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Connect to Cloud Analytic Services (CAS).Access and use caslibs**	07
2.	**Load SAS data sets, CSV files, and Microsoft Excel files into CAS.**	08
3.	**Save SASHDAT files. Modify Base SAS programs to execute in CAS.**	07
4.	**Modify SQL procedure code to execute in CAS using FedSQL.**	08
5.	**Use CAS-enabled functions and procedures.**	07

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Connect to Cloud Analytic Services (CAS).Access and use caslibs**
	Introducing the SAS Viya platform. SAS Viya programming interfaces. SAS Viya servers and processing environments Overview of running SAS code on the SAS Compute Server. SAS Viya Compute Server overview. Running SAS 9 Code on the Compute Server in SAS Viya. CAS fundamentals. Understanding caslibs.
2.	**Load SAS data sets, CSV files, and Microsoft Excel files into CAS.**
	• Introduction of Unit • Loading data to in-memory tables. • Accessing DBMS data.

	• Saving and dropping in-memory tables • Modifying DATA step code to run in SAS Cloud Analytic Services (CAS) • .Conclusion of Unit
3.	**Save SASHDAT files. Modify Base SAS programs to execute in CAS.**
	• Introduction of Unit • Introduction to SAS procedures in SAS Viya. • Running CAS-enabled SAS procedures. • Conclusion of Unit
4.	**Modify SQL procedure code to execute in CAS using FedSQL.**
.	• Introduction of Unit • Modifying SQL code to run in CAS. • Column data types in CAS. • Conclusion of Unit
5	**Use CAS-enabled functions and procedures.**
.	• Introduction to CASL. • Using CAS actions.

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	The Little SAS Enterprise Guide Book	Susan Slaughter & LoraDelwiche	Late st	SAS
2	SAS Certification Prep Guide		3rd	SAS Institute
Reference Book				
1	Learn SAS By example A Programmer Guide By Ron Cody			
Online Resources				
1	https://www.sas.com/de_de/training/courses/learning-formats/e-learning.html			

D: Ability Enhancement Courses (AEC)

Code : BUACHM3209	PROFESSIONAL SKILLS-I	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Apply	Produce a professional attitude and approach toward assigned tasks.
CO – 02	Understand	Comprehend the formal elements of specific genres of organizational written communication: reports, proposals, memorandums, web pages, wikis, blogs, and business letters.
CO – 03	Apply	Present a clear and fluent demonstrative, informative, and persuasive presentation.
CO – 04	Apply	Illustrate preparedness for any type of interview from classic one-on-one interviews to panel interviews, Phone/Skype interviews, and Behavioral/Situational.
CO – 05	Apply	Use the bargaining techniques and strategies of inventing options for mutual gain and negotiating from bargaining to closing.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	1	3	2	2	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	2	2	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2	2.8	1.6	16									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Professional Attitude & Approach**	6
2.	**Professional Writing-I**	8
3.	**Presentation Skills: Structure Study**	2
4.	**Interview Skills & Group Discussion**	4
5.	**Negotiation Skills & Time Management**	4

D. List of Labs

LIST OF LABS	
1.	Professional & Ethical Approaches: Degree of adherence, Business world & meeting deadlines
2.	Job Hunting and Networking: Skill Branding & Usage of Online Platforms
3.	Trust Building & Cultural Etiquettes
4.	Professional Writing-I: Direct-Indirect approaches to Business Writing-Five main stages of writing Business Messages.
5.	Professional Email Writing
6.	Resume Building-I: Difference between C.V. & Resume, formats, points to cover, practice sessions
7.	E-Learning & E-Content Development-I
8.	Presentation Skills: format & structure of presentations, using tools & techniques
9.	Job Interviews I: Preparation and Presentation
10.	Advanced Group Discussion – I
11.	Negotiation Skills & and Conflict Resolution-I
12.	Professional Code of Ethics & Effective Time Management

E: Skill Enhancement Courses (SEC)

Code : BULCSE3201	Skill Enhancement Generic Course-III	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

- Understand basic problems based on arithmetic and soft skills area which are asked in aptitude test taken by companies
- Effectively solve these problems by applying the knowledge earned.
- Actively participate in group discussion / meetings / interviews and prepare & deliver presentations.
- Become more effective individual through goal/target setting, self-motivation and practicing creative thinking.
- Function effectively in multi-disciplinary and heterogeneous teams through the knowledge of team work, Inter-personal relationships, conflict management and leadership quality

B. RECOMMENDED

LIST OF LABS	
1.	Objective Building, Parts of speech, Nouns, Numbers & Genders, Importance of soft skills
2.	Logarithms, Number Theory
3.	Tenses
4.	Number system- Fractions & Decimals
5.	Stress Management Techniques, Critical Thinking
6.	Modal Verbs & Conditional Tense, Working under pressure
7.	Boosting brain power for fast learning & unlearning
8.	Pronouns, Adverbs & Adjectives
9.	Emotional Intelligence, 5 levels of listening
10.	Remainder Theoram
11.	Points, lines & angles
12.	Article Writing

F: Value Added Courses (VAC)

Code : BUVCVD3101	Digital Marketing	2 Credits [LTP: 2-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Understand Digital Marketing, its scope, objectives, opportunities and it challenges.
CO – 02	Understand	Understand Digital Strategy building & learn its effectiveness.
CO – 03	Apply	Applying alternatives for Dynamic organization to ensure their success in highly competitive sale environment and to analyze the concept of Internet marketing and its applications
CO – 04	Analyze	Analyze the digital tools effectively for Social Media Marketing.
CO – 05	Evaluate	Develop an understanding toward E-mail marketing and its various application

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	1	3	2	2	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	2	2	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2	2.8	1.6	16									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	An Overview of Digital Marketing	6
2.	Digital Marketing Planning and Structure	6
3.	Internet Marketing	6
4.	Social Media Marketing	6
5.	E-mail marketing and Applications	6

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**An Overview of Digital Marketing**
	• Introduction of Unit: Introduction to Digital Marketing • Different Ways to Market Your Business Online, Evolution of Digital Marketing, Status of Digital Marketing in India • How Digital Marketing Works, Traditional vs. Digital Marketing, New Trends for Online Marketers • Digital Marketing Strategies, 6 Cs of Digital Marketing • Impact of Digital Marketing on Business, Benefits of Digital Marketing, Drawbacks of Digital Marketing • Internet Marketing in India – Challenges • Conclusion of Unit
2.	**Digital Marketing Planning and Structure**
	• Introduction of Unit • Creating initial digital marketing plan • Target group analysis, In bound vs Outbound Marketing,

	• Content Marketing, Understanding Traffic, Understanding Leads Strategic Flow for Marketing Activities. • WWW, Domains, Buying a Domain, Website Language & Technology, Core • Objective of Website and Flow • One Page Website, Strategic Design of Home Page, Optimization of Web sites, • Application of Word Press in Digital Marketing, Application of CSS, HTML & Java Script for web page design • Conclusion of Unit
3.	**Internet Marketing**
	• Introduction of Unit • Marketing and Internet, Market place to Marketspace • Online buyer behavior, suppliers, Intermediaries Websites • Types of Websites, Web portals like: B2B, B2C,C2B,C2C, B2E(Business to Employee) • Social Networking, The promise and challenges of online marketing • The Indian Internet Marketing Mix, Significance of Internet marketing. • Traditional vs. Online Marketing • Conclusion of Unit
4.	**Social media Marketing**
.	• Introduction of Unit • Introduction of Social Media Marketing, How Social media marketing works • Different components or Tools for Social Media Marketing • Facebook Marketing, Google Ad Words, YouTube Marketing, Content Marketing • Meme marketing, Affiliate Marketing, LinkedIn, Twitter, Instagram • Keywords with SEO marketing- On page Search Engine Optimisation, Off page SEO, why search • Engine marketing., SEM and its application, Benefits of SEM • Blogging as a marketing strategy, Types of Blogs, What is Blogging? Benefits of Blogging. Pitfalls of Blogging. • Conclusion of Unit
5	**E-mail Marketing and Applications**
.	• Introduction of Unit • Introduction of E-mail marketing, E-mail Marketing - What is it? Why do it and How? • Types of E-mail Marketing • Comparison to Traditional Mail, Opt-in E-mail Advertising • How to deal with Spam Filter, Choosing your metrics , Tracking Landing Pages • Topl0 Benefits of E-mail Marketing • E-mail-Marketing Strategy Checklist, Effective E-mail Marketing Techniques • Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Digital Marketing	Dave Chaffey	7th	Pearson
2	Social Media Marketing All-in-one Dummies	Jan Zimmerman, Deborah Ng	4th	John Wiley & Sons Inc

Your Dreams Our Goal

POORNIMA UNIVERSITY

Member of Association of Indian Universities & Approved by UGC (Govt. of India) under 2(f) & 12(B)

B. TECH - CSE

SECOND YEAR
SEMESTER: IV

COMMON SYLLABUS

For

Cyber Security
Cloud & Full Stack Development
Artificial Intelligence & Data Science
Artificial Intelligence & Machine Learning

BATCH: 2024-28

A: Major (Core Courses)
A.1: Theory

Code : BCECCE4101	Computer Networks	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall fundamental principles of networking, including network types, devices, topologies, and transmission media, facilitating basic network analysis and troubleshooting.
CO – 02	Understand	Interpret and comprehend key protocols at the Presentation and Application Layers, and fostering a solid understanding of their functions and support within networked environments.
CO – 03	Apply	Implement Transport Layer protocols (TCP, UDP), Network Layer protocols (IP), as well as basic concepts of routing and switching.
CO – 04	Analyze	Break-down and deduce the concepts of Data Link Layer principles along with understanding error detection and correction protocols and wireless security protocols for effective network analysis and implementation.
CO – 05	Evaluate	Assess and dissect the WAN concepts, switching techniques, and various methods of internet and LAN connectivity, enabling them to design and deploy efficient network infrastructures.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	-	-	-	-	-	-	-	-	-	-	1	-	-
CO2	1	3	-	1	-	-	-	-	-	-	-	-	-	-	-
CO3	2	1	3	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	3	-	3	-	-	-	-	-	-	-	-	-	2	-
CO5	-	2	3	2	-	-	-	-	-	-	-	-	-	-	-
WT. AVG	2	2	3	2									1	2	

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Networking Fundamentals & Internet**	09
2.	**Basics Presentation & Application Layer**	07
3.	**Basics of Transport layer & Network, Layer**	08
4.	**Basics of Data Link Layer**	07
5.	**Basics of WAN Technology**	07

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Networking Fundamentals & Internet**
	• Basics of Network & Networking, Types of Networks: LAN, MAN, WAN, Peer-to-Peer & Client/Server, Workgroup Vs. Domain, Network Topologies. The Internet, Network Devices- NIC, Hub, Switch, Bridge, Router, Gateways, Firewall, Repeater, CSU/DSU, and modem, Introduction of OSI model, and TCP/IP Model, Comparison between OSI model & TCP/IP model. Physical Layer: Types of Transmission Media, Communication Modes, Wiring Standards and Cabling- straight through cable, crossover cable, rollover cable, Media connectors (Fibre optic, Coaxial, and TP etc.) Switching Methods (Circuit/Packet Switching) Uni-cast, Multicast, Broadcast • Conclusion & Real Life Application
2	**Basics Presentation & Application Layer**

.	
	• Presentation Layer protocols:-TLS, SSL, MIME • Application Layer: Functions and support, Application Layer Protocols: DHCP, DNS, HTTP/HTTPS, FTP, TFTP, SFTP, Telnet, Email: SMTP, POP3/IMAP, NTP. • Conclusion & Real Life Application
3.	**Basics of Transport layer & Network, Layer**
	• Transport Layer: Transmission Control Protocol(TCP), User Datagram Protocol (UDP), Overview of Ports & Sockets • Network Layer: Internet Protocol (IP), IP standards, versions, functions, The IPv4 Datagram Format, IPv4 addressing, IPv4 address Classes, IPv4 address types, Default Gateway, Public & Private IP Address, methods of assigning IP address, Subnet Mask and subneting, IPv6 address, types, assignment, Data encapsulation, Introduction to Routing and Switching concepts. • Conclusion &Real Life Application
4.	**Basics of Data Link Layer**
.	• Application of Data Link Layer: Framing and Error detection and correction. Stop and Wait protocol, Sliding Window protocols Go-Back-N Protocol, Channel allocation problem, Multiple access protocols: ALOHA, Carrier sense multiple access protocols. Wireless Networking, Types of Wireless Networks: Ad-hoc mode, Infrastructure mode, wireless LAN standards: IEEE 802.11a, IEEE 802.11b, IEEE 802.11g, wireless security Protocols: WEP,WPA, 802.1X. • Conclusion & Real Life Application
5	**Basics of WAN Technology**
.	• What Is a WAN?, WAN Switching, WAN Switching techniques Circuit Switching, Packet Switching etc., Connecting to the Internet : PSTN, ISDN, DSL, CATV, Satellite-Based Services, Last Mile Fibre, Cellular Technologies • Connecting LANs: Leased Lines, SONET/SDH, Packet Switching, Remote Access: Dial-up Remote Access, Virtual LAN, Virtual Private Networking • Conclusion &Real Life Application

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Computer Network	AndrewS. Tanenbaum	2013	Pearson
2.	Computer Networking: Top Down Approach	Kurose. Ross	2017	Pearson
Reference Book				
1	Networking All in One – Doug Lowe 7th edition Publisher- Wiley			
Online Resources				
1	https://www.edx.org/learn/computer-networking			
2	https://www.youtube.com/watch?v=VwN91x5i25g			

Code: BCECCE4102	OOPs with JAVA	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall Java's history, and deploy object-oriented programming principles, and basic concepts and fostering competency in Java development.
CO – 02	Understand	Interpret and proficiently utilize control statements, methods, recursion, access control, nested classes, command line arguments, and inheritance, fostering advanced Java programming skills.
CO – 03	Apply	Implement access protection, interfaces, and exception handling, enabling proficient application development with robust error handling.
CO – 04	Analyze	Break-down and deduce the concepts of Muti-threading, alongside understanding fundamental AWT package concepts and event handling, enhancing Java application development skills.
CO – 05	Evaluate	Assess and grasp the concepts of JDBC for database connectivity and implement Java 8 features enhancing their skills in database-driven Java applications.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	-	1	-	-	-	-	-	-	-	-	-	1	-	-
CO2	1	2	2	-	-	-	-`	-	-	-	-	-	1	-	-
CO3	2	2	3	-	-	-	-	-	-	-	-	-	1	-	-
CO4	1	2	3	-	-	-	-	-	-	-	-	-	1	-	-
CO5	1	2	2	-	-	-	-	-	-	-	-	-	1	-	-
WT. AVG	1.4	2	2.2										1		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction**	08
2.	**Object Oriented Programming**	09
3.	**Exception Handling and Multithreading**	09
4.	**Java Collections Framework and I/O**	07
5.	**JAVA Database Connectivity (JDBC)**	07

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction**
	• History and Features of Java • Java Development Kit (JDK) and Java Runtime Environment (JRE) • Java Virtual Machine (JVM) Architecture • Structure of a Java Program, Data Types, Variables, and Operators, Control Flow Statements (if, else, switch, loops), Arrays and Strings • Concepts of OOP: Classes, Objects, and Methods, Constructors and Destructors, this keyword, Static Members • Conclusion of unit
2.	**Object Oriented Programming**
	Inheritance: • Types of Inheritance in Java, super keyword, Method Overriding • Final Classes and Methods

	• Polymorphism, Compile-time vs Runtime Polymorphism • Method Overloading, Dynamic Method Dispatch • Encapsulation and Abstraction, Access Modifiers, Packages and Interfaces • Abstract Classes and Methods, Interface Implementation • Conclusion of Unit
3.	**Exception Handling and Multithreading**
	• Introduction of Unit • Exception Handling, Types of Exceptions • Try, Catch, Finally Blocks, Throw and Throws Clauses • Custom Exceptions, Multithreading: • Introduction to Threads, Life Cycle of a Thread • Creating Threads: Extending Thread Class vs Implementing Runnable Interface • Synchronization in Java, Inter-thread Communication and methods • Conclusion of the Unit
4.	**Java Collections Framework and I/O**
.	• Introduction of Unit • Collections Framework: • Overview of Collections, List, Set, and Map Interfaces • ArrayList, LinkedList, HashSet, TreeSet, HashMap, and TreeMap • Iterators and ListIterators, File I/O:, File Class and Methods • Byte Streams and Character Streams, Reading and Writing Files • Serialization and Deserialization • Conclusion of the Unit
5	**JAVA Database Connectivity(JDBC)**
.	• Introduction of Unit • Database connectivity – JDBC architecture and Drivers. • JDBC API - loading a driver, connecting to a database, creating and executing JDBC statements, handling SQL exceptions. • Accessing result sets: types and methods. An example - JDBC application to query a database. • Conclusion of the Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	The complete reference Java –2	Herbert Schildt	V Edition,	TMH.
2.	SAMS teach yourself Java – 2	Rogers Cedenhead and Leura Lemay	3rd Edition,	Pearson Education
Reference Book				
1	Object Oriented Programming with Java PUBLISHER PHI by M.T. Somashekara (Author), D.S. Guru (Author), K.S. Manjunatha (Author)			
2	**“**Head First Java” by Kathy Sierra			
Online Resources				
1	https://www.programiz.com/java-programming/online-compiler/			
2	https://www.tutorialspoint.com/compile_java_online.php			
3.	https://onecompiler.com/java			

Code: BCECCE4103	Software Engineering	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall basic concepts of software life cycle models to develop realtime projects.
CO – 02	Understand	Demonstrate an understanding of the procedural design methods to architect software systems.
CO – 03	Apply	Apply and articulate Software Testing Principles.
CO – 04	Analyze	Analyze existingSoftware Quality Management Standards.
CO – 05	Evaluate	Evaluate cost estimation and risk analysis in project management.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	2	–	–	–	–	–	–	–	–	1	–	–
CO4	2	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2.2	2.6	1.4	1.4									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Software Engineering Fundamentals**	07
2.	**Software Project Planning**	08
3.	**Software Design and UML**	08
4.	**Software Testing**	07
5.	**AGILE Project Management**	07

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Software Engineering Fundamentals**
	● Introduction of Unit ● Software Engineering - A layered Technology, The importance of software, software myths, software engineering paradigms ● Software Process Models: Linear Sequential Model, Prototyping Model, RAD Model ● Evolutionary Software Process Models: Incremental Model, Spiral Model Component Assembly Model, Formal Methods, Fourth-Generation Techniques. ● Conclusion of Unit
2.	**Software Project Planning**
	● Introduction of Unit ● Software Project Planning, Size Estimation, Cost Estimation, Models, Static, single variable models, Static, Multivariable Models, COCOMO, The Putnam Resource Allocation Model, ● Risk Identification and Projection: RMMM, Project scheduling and Tracking. ● Software Design Process, Design Principles, and Design Concepts: Effective Modular Design, Design Heuristics, Design Documentation, ● Design Methods: Data Design, Architectural Design, Interface Design, Human Computer Interface Design, Procedural Design. Case Study for Design of any Application Project. ● Conclusion of Unit
3.	**Software Design and UML**
	● Introduction of Unit ● Unified Modeling Language, Basic structures and modeling classes, common modeling

	techniques, relationships, common mechanism, class diagrams. ● Advanced structured modeling, advanced classes and relationships, interfaces, types and roles, instances and object diagram. ● Basic behavioral Modeling: Use cases, use case diagrams, Interaction diagram, Activity diagrams, state chart diagrams, component diagrams, deployment diagrams, patterns and frame works. ● Conclusion of Unit
4.	**Software Testing**
.	● Introduction of Unit ● S/W Testing Fundamentals, Unit, integration, system testing, black box and white box testing Incremental testing, formal proof of correctness, software matrix ● Automated Testing: Introduction to Automated testing, Software testing with automated tools ● Conclusion of Unit
5	**AGILE Project Management**
.	● Introduction of Unit ● Agile Programming- Introduction, Flavors of Agile Development, Agile Manifesto, Refactoring Techniques, Limitations of the Agile Process. ● Agile Modeling: Introduction, Agile Modeling – Principles, Comparing Waterfall and Agile Modeling ● Scrum Methodology- The roles of Scrum, Project Artifacts, Meetings, Advantages of Scrum. ● Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Software Engineering: A Practitioner‟s Approach	Roger S Pressman, Bruce R Maxim	8th Edition	TMH.
2.	Software engineering	Ian Sommerville	9th Edition	Addison Wesley Longman
Reference Book				
1	Grady Booch, James Rumbaugh, IvarJacobson.,” The Unified Modeling Language User Guide”, 2nd Edition, 2017			
2.	James Rumbaugh. MichealBlaha “Object oriented Modeling and Design with UML”, 2011			
3.	Ali Behforooz, Hudson, “Software Engineering Fundamentals”, Oxford, 2009			
4.	Charles Ritcher, “Designing Flexible Object Oriented systems with UML”, TechMedia , 2008			
Online Resources				
1	https://nptel.ac.in/courses/106105182			
2	https://www.w3schools.in/sdlc/software-development-life-cycle-sdlc			

Code :BCECCE4104	Advance Data Structure	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Define fundamental data structure concepts, including classification, time and space complexity, dynamic memory allocation, and recursion.
CO – 02	Understand	Contrast basic search techniques, such as sequential and binary search, along with sorting algorithms (Bubble sort, Selection sort, Insertion sort, Merge sort, Quicksort).
CO – 03	Apply	Examine stack and queue concepts, including operations, representation, and types.
CO – 04	Analyze	Explain linked lists, their components, types, and operations. Estimate advantages, disadvantages, and real-life applications of linked lists.
CO – 05	Evaluate	Validate tree and graph concepts, including binary trees, search trees, heaps, and graph applications. Covering tree terminology, array representation, tree traversal, and graph algorithms like Depth First Search and Breadth First Search.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	2	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2.2	2.6	1.4	1.2									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Hashing**	07
2.	**Priority Queues (Heaps)**	08
3.	**Trees**	07
4.	**Graphs Algorithms**	07
5.	**Disjoint Sets and String Matching**	07

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Hashing**
	● Introduction to Hashing ● Hash Function ● Separate Chaining ● Hash Tables without linked lists: Linear Probing, Quadratic Probing, Double Hashing, Rehashing, Hash Tables in the Standard Library ● Universal Hashing ● Extendible Hashing. ● Conclusion of Unit
2.	**Priority Queues (Heaps)**
	● Introduction to Priority Queues (Heaps) ● Model ● Simple implementations ● Binary Heap: Structure Property, Heap Order Property, ● Basic Heap Operations: insert, delete, Percolate down

	• Other Heap Operations • Introduction toBinomial Queues • Binomial Queue Structure • Binomial Queue Operations • Implementation of Binomial Queue • Priority Queues in the Standard Library. • Conclusion of Unit
3.	**Trees**
	• Introduction to Trees • AVL: Single Rotation, Double Rotation • B-Trees • Multi-way Search Trees – 2-3 Trees • Searching for an Element in a 2-3 Tree • Inserting a New Element in a 2-3 Tree • Deleting an Element from a 2-3 Tree • Red-Black Trees • Properties of red-black trees: Rotations, Insertion, Deletion. • Conclusion of Unit
4.	**Graphs Algorithms**
.	• Introduction to Graphs Algorithms • Elementary Graph Algorithms: Topological sort • Single Source Shortest Path Algorithms: Dijkstra's, Bellman-Ford, All-Pairs Shortest Paths: Floyd- Warshall's Algorithm • Conclusion of Unit
5	**Disjoint Sets and String Matching**
.	• Introduction to Disjoint Sets • Equivalence relation • Basic Data Structure • Simple Union and Find algorithms • Smart Union and Path compression algorithm. • Introduction to String Matching • The naive string-matching algorithm • The Rabin-Karp algorithm • The Knuth-Morris-Pratt algorithm. • Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Data Structures and Algorithm Analysis in C++	Mark Allen Weiss	4 th Edition	Pearson
2.	Introduction to Algorithms	Thomas H Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein	3 rd Edition	The MIT Press.
Reference Book				
1	Fundamentals of Computer Algorithms, 2nd Edition, 2009, University Press Pvt. Ltd, Ellis Horowitz, Satraj Sahani and Raja sekharam.			
2.	Advanced Data Structures, Oxford University Press, 2018, ReemaThareja, S. Rama Sree.			
Online Resources				
1	https://www.coursera.org/learn/advanced-data-structures			
2	https://ocw.mit.edu/courses/6-851-advanced-data-structures-spring-2012/			
3.	https://nptel.ac.in/courses/106106133			
4.	https://www.mooc-list.com/search/node?keys=Advanced+Data+Structures			
5.	https://freevideolectures.com/course/2279/data-structures-and-algorithms			

A.2: Practical

Code: BCECCE4201	Software Engineering Lab	2 Credits [LTP: 0-0-1]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall key software engineering concepts, such as software development life cycle models (e.g., Waterfall, Agile), software requirements engineering, software design principles, and software testing techniques.
CO – 02	Understand	Understand the principles and methodologies of software engineering, including requirements analysis, design patterns, version control systems (e.g., Git), and software documentation practices.
CO – 03	Apply	Apply software engineering techniques by participating in team-based software development projects, where students work collaboratively to develop software solutions, adhering to project management methodologies (e.g., Scrum, Kanban) and utilizing relevant development tools and frameworks.
CO – 04	Analyze	Analyze software requirements documents, architectural designs, and codebases to identify potential flaws, inefficiencies, or areas for improvement, and propose solutions or enhancements to address them.
CO – 05	Evaluate	Evaluate the quality and effectiveness of software solutions developed during the lab, considering factors such as functionality, reliability, scalability, maintainability, and user satisfaction, and provide constructive feedback for further refinement.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	-	-	-	-	-	-	-	-	-	1	1	-	-
CO2	2	1	2	-	-	-	-	-	-	-	-	1	2	-	-
CO3	1	1	-	-	-	-	-	-	-	-	-	1	1	-	-
CO4	1	1	-	-	-	-	-	-	-	-	1	1	1	-	-
CO5	1	1	-	-	-	-	-	-	-	-	-	1	11	-	-
WT. AVG	1.2	1	2								1	1	3.2		

C. LIST OF EXPERIMENTS

1	Prepare a SRS document in line with the IEEE recommended standards
2	Draw the Entity relationship diagram of a project.
3	Develop DFD Model (Level 0, Level 1 DFD and data dictionary) of the sample problem (Useof a CASE tool required). (1 class)
4	Develop Structured design for the DFD model developed. (1 class)
5	Develop UML Use case diagram for a problem (Use of a CASE tool any of Rational rose, ArgoUML, or Visual Paradigm etc.
6	To draw the structural view diagram for the system: Class diagram, object diagram.
7	To draw the behavioral view diagram: State-chart diagram, Activity diagram.
8	To perform the behavioral view diagram for the suggested system : Sequence diagram, Collaboration diagram
9	To perform various testing using the testing tool Junit for unit testing for a sample code of the suggested system
10	To Prepare time line chart/Gantt Chart/PERT Chart for selected software project.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Software Engineering	K.K.Aggarwal & Yogesh Singh	2005	New Age International
2.	An Integrated Approach to Software Engineering	Pankaj Jalote,	Second Edition	Springer
Reference Book				
1.	Software engineering, Roger S Pressman			
Online Resources				
1.	https://www.javatpoint.com/software-engineering-tutorial			
2.	https://www.geeksforgeeks.org/software-engineering/			
3.	https://www.tutorialandexample.com/software-engineering-tutorial			

Code :BCECCE4202 **OOPs with JAVA Lab** **1 Credits [LTP: 0-0-2]**

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Remember the basics of object-oriented programming using JAVA
CO – 02	Understand	Understanding of fundamental object-oriented programming (OOP) concepts such as classes, objects, inheritance, polymorphism, encapsulation, and abstraction.
CO – 03	Apply	Apply the concept of classes, Java, JDK Components and develop Simple Java Programs
CO – 04	Analyze	Analyze object-oriented programming principles in Java
CO – 05	Evaluate	Develop Simple Java Programs using inheritance and Exception handling.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	1	-	-	-	-	-	1	2	-	-	1	2	1
CO2	1	1	1	-	-	-	-	-	1	2	-	-	1	2	1
CO3	2	2	2	2	-	-	-	-	1	2	-	-	2	2	2
CO4	2	2	3	3	-	-	-	-	1	3	2	-	2	2	2
CO5	3	3	3	3	3	-	-	-	1	3	3	3	3	3	3
WT. AVG	1.8	1.8	2	2.6666667	3				1	2.4	2.5	3	1.8	2.2	1.8

C. LIST OF EXPERIMENTS

1	A. Write a program to print “Hello World” in Java. B. Write a program to add two numbers C. Write a program to demonstrate the different access specifiers D. Write a program which uses different packages
2	A. Write a program to demonstrate inheritance, abstraction, encapsulation and Polymorphism. B. Write a program to find the factorial of n numbers C. Write a program to calculate Fibonacci series D. Write a program to add n numbers and series
3	A. Write a program to create an array and store elements into the array. B. Write a program to find the sum of elements in an array C. Write a program to demonstrate switch case, if, if-else and for loop
4	A. Write a program to demonstrate the working of methods. B. Write a program which has four methods – add(), subtract(), multiply() and divide() and demonstrate a simple console calculator. C. Write a program to accept command line arguments and display them to the user
5	A. Write a program to create a package. B. Write a program to handle different exceptions
6	A. Write a program to demonstrate try-catch, throw and throws. B. Write a program for user defined exception
7	A. Write a program to read a file Write a program to write into a file
8	Write a program to demonstrate client server communication (socket programming)
9	Write a program to create threads and manipulate them
10	Write a program to create a user interface to check user authentication.
11	Write a program to create a registration form and save the details into a file
12	Write a program to save and fetch the details from database

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	The complete reference Java – 2	Herbert Schildt	5th Edition,	TMH.
2	SAMS teach yourself Java – 2	Rogers Cedenhead and Leura Lemay	3rd Edition,	Pearson Education
Reference Book				
1	The complete reference Java –2			
2.	SAMS teach yourself Java – 2			
Online Resources				
1	https://www.programiz.com/java-programming/online-compiler/			
2	https://www.tutorialspoint.com/compile_java_online.php			

Code: BCECCE4203	Frontend Development Lab	1 Credits [LTP: 0-0-2]

A. **COURSE OUTCOMES**

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	A free frontend framework for faster and easier web development
CO – 02	Understand	Reduce the amount of code and you write to build rich user interface applications
CO – 03	Apply	Increase the reliability and maintainability of UI by using data binding
CO – 04	Analyze	We can analyze website component and their traifc
CO – 05	Evaluate	Use the Features of shopify to animated and attractive web site

. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	1	2	1								1		3
CO2	2		3										2		2
CO3		2		1		2								2	
CO4	3		2				1						3		1
CO5		3		3										1	
WT. AVG	2	2	2	2	1	2	1						2	1.5	2

. LIST OF EXPERIMENTS

1	Develop a web page with suitable HTML elements to demonstrate Bootstrap framework classes for headings.
2	Develop a Web page with suitable HTML elements to demonstrate Bootstrap framework classes for text and fonts.
3	Develop web pages with suitable HTML elements to demonstrate Bootstrap framework classes for Lists, colors, and bootstrap buttons
4	Develop a Web page with suitable HTML elements to demonstrate Bootstrap Framework classes for Borders, Margins, and Paddings
5	Develop web pages with suitable HTML elements to demonstrate Bootstrap framework classes for Containers
6	Develop web pages with suitable HTML elements to demonstrate Bootstrap framework classes for Grid System
7	Develop Web pages with suitable HTML elements to demonstrate Bootstrap framework classes for Images, Cards and Gutters
8	Develop a Web Page with suitable HTML elements to demonstrate Bootstrap framework classes for Nav and Footers
9	Create a responsive web application using Bootstrap and Test it on Postman.
10	Develop a web application using Vercel.
11	Develop and deploy a web application using Shopify.
12	Deploy the Web application created in experiment 9 on a remote web server

. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Practical Bootstrap	Panos Matsinopoulos	2020	Apress
2.	Shopify: Beginner to Pro Guide – The Comprehensive Guide	Timothy Short	2016	Kindle
3.	Angular 6 for Enterprise Ready-Web Applications			

Reference Book	
4.	Node.js 8 the Right Way: Practical, Server-Side JavaScript That Scales
5.	**Pro Angular**
Online Resources	
6.	https://academy.shopify.com/
7.	https://getbootstrap.com/
8.	https://vercel.com/

B: Minor Stream Courses/ Department Electives
B.1: Theory- Professional Elective - III

Code: BCEECE4111 **WEB FRAMEWORKS** **3 Credits [LTP: 3-0-0]**

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Understand	To Understand the fundamentals of Web Frameworks
CO – 02	Understand	To Understand the basics of Flask Framework
CO – 03	Understand and apply	To understand and apply advanced Flask Features for web development
CO – 04	Understand	To Understand the basics of Django Framework
CO – 05	Understand and apply	To understand and apply advanced Django and RESTful APIs for web development.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	-	-	1	–	–	–	–	-	-	1	-	1	-
CO2	2	2	1	-	2	–	–	–	–	-	2	2	2	2	-
CO3	3	3	2	-	3	–	–	–	–	-	3	2	3	2	2
CO4	3	3	3	-	3	–	–	–	–	-	3	3	3	3	2
CO5	3	3	3	3	2	–	–	–	–	-	-	1	3	3	3
WT. AVG	2.4	2.4	2.25	3	2.2						2.66	1.8	2.75	2.2	2.33

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Web Frameworks	6
2.	Introduction to Flask	8
3.	Advanced Flask Features	8
4.	Introduction to Django	6
5.	Advanced Django and RESTful APIs	8

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction to Web Frameworks**
	• Introduction of Frameworks • Web Application Frameworks MVC Architectures • Overview of Python frameworks and their importance • Comparison of popular Python frameworks (e.g., Django, Flask, Pyramid) • Setting up a Python environment for framework development • Conclusion of the Unit
2.	**Introduction to Flask**
	• Overview of Flask and its features • Setting up the Flask development environment • Creating and structuring a basic Flask application • Understanding URL routing and dynamic URLs • Implementing variable rules in Flask

	● Handling HTTP methods like GET, POST, etc.
3.	**Advanced Flask Features**
	● Working with Jinja2 templates for dynamic HTML ● Serving and managing static files ● Understanding and using the request object ● Processing and validating form data ● Implementing cookies and managing sessions ● Handling redirects, custom error pages, and message flashing ● Introduction to Flask extensions, focusing on SQLAlchemy for database integration
4.	**Introduction to Django**
.	● Introduction to Django and its role in back-end web development. ● Understanding the HTTP protocol and its importance in web development. ● Learning the MVC model and its application in Django (MTV model). ● Setting up and using virtual environments for Django projects. ● Exploring Django project structure and the built-in admin interface. ● Working with generic views for rapid development. ● Creating and managing HTML templates. ● Configuring URL dispatcher and routing in Django
5	**Advanced Django and RESTful APIs**
.	● Building custom views and handling GET/POST methods. ● Implementing a URL shortener and managing the user model. ● Adding logic in templates and querying models. ● Serving static files and deploying Django applications. ● Introduction to Django REST framework for building APIs. ● Creating and customizing generic views and serializers in Django REST. ● Handling JSON data and building RESTful APIs. ● Filtering models, working with images, and API authentication with tokens. ● Using Postman for API testing and managing related models. ● Deploying web APIs and using API endpoints. ● Deploying a Django REST project.

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	PHP: The Complete Reference	Steven Holzner	1 July 2017	TMH
Reference Book				
1	Learning PHP, MySQL & JavaScript with j Query, CSS & HTML5 – 1 January 2015			
Online Resources				
1	https://www.w3schools.com/php/			
2	https://www.tutorialspoint.com/php/index.htm			

Code: BADECE4111	Fundamentals of Machine Learning	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Understand and Analyze	Demonstrate and distinguish between types of machine learning techniques
CO – 02	Understand and Evaluate	Examine the correlation coefficient for a given data set
CO – 03	Understand and Apply	Implement applications with classification and clustering techniques
CO – 04	Apply	Examine the uses of appropriate performance metrics of machine learning
CO – 05	Apply & Understand	Modify existing machine learning algorithms to improve classification efficiency

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	1	2	-	-	-	-	-	-	-	-	-	-	-
CO3	2	3	2	1	-	-	-	-	-	-	-	-	-	-	-
CO4	2	2	2	1	-	-	-	-	-	-	-	-	-	-	-
CO5	2	3	1	1	-	-	-	-	-	-	-	-	-	-	-
WT. AVG	2.2	2.6	1.4	1.2	-	-	-	-	-	-	-	-	-	-	-

C. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction to Machine Learning**
	● Introduction to Machine Learning ● Definition of Machine Learning ● Working principles of Machine Learning ● Classification of Machine Learning : Supervised Learning, Unsupervised Learning, Reinforcement Learning ● Supervised Learning: Classification and Regression ● Unsupervised Learning: Clustering and Association ● Reinforcement Learning ● Types of Reinforcement learning : Positive Reinforcement and Negative Reinforcement ● Working of Reinforcement learning ● Markov Decision Process ● Reinforcement Learning Algorithms: Q-Learning and State Action Reward State action (SARSA) ● Application of Reinforcement Learning ● Conclusion of Unit
2.	**Regression**
	● Introduction to Regression ● Types of Regression: Linear regression, Logistics regression, Ridge Regression, Lasso Regression, Bayesian Linear Regression and Polynomial Regression ● Regression and Correlation

	• Crosstabs and Scatterplots • Pearson's r • Regression – Finding The line • Regression – Describing the line • Contingency Tables • Conclusion of Unit
3.	**Classification**
	• Introduction of Unit • Classification model building • Types of Classification Algorithm: Binary Classification and Multi Class Classification • Logistic Regression • k-Nearest Neighbors • Decision Trees • Random Forest • Support Vector Machine • Naïve bayes • Handling imbalanced datasets • Feature engineering and selection techniques for classification problems • Conclusion of Unit
4.	**Clustering**
.	• Introduction to unit • K-Means Clustering: • Hierarchical Clustering: • Density-Based Clustering (DBSCAN) • Gaussian Mixture Models (GMM) • Spectral Clustering: • Conclusion of Unit
5	**Performance Metrics**
.	• Introduction of Performance metrics • Performance metrics for Regression: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-Squared, Adjusted R-squared • Performance metrics for classification: Accuracy, Confusion Matrix, Precision, Recall, F1 score, ROC AUC, Kappa, MCC (Matthews Correlation Coefficient) and Log-loss. • Performance metrics for clustering: Silhouette Score, Rand Index, Adjusted Rand Index, Mutual Information, Calinski-Harabasz Index and Davies-Bouldin Index • Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Machine Learning–An Algorithmic Perspective, Chapmanand Hall	Stephen Marsland	Second Edition	CRCPress
2.	Machine Learning–An Algorithmic Perspective, Chapmanand Hall	Stephen Marsland	Second Edition	CRCPress
Reference Book				
4.	IntroductiontoMachineLearning3e(AdaptiveComputationandMachineLearningSeries) ,ThirdEdition, MITPress, Ethem Alpaydin			
5.	Machine Learning: The Art and Science of Algorithmsthat Make Sense of Data,1stEdition,Cambridge UniversityPress, PeterFlach			
Online Resources				
8.	https://github.com/adeshpande3/Machine-Learning-Links-And-Lessons-Learned			

Code: BCSECE4111	Ethical Hacking	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Describe basic concepts and importance of information security
CO – 02	Understand	Evaluate threats to information security, analyze their impact and propose suitable countermeasures
CO – 03	Apply	Apply various aspects of securing network infrastructure and importance of classifying information
CO – 04	Analyze	Interpret student to understand the concepts of IT security, Threats, Vulnerabilities, Impact and control measures.
CO – 05	Evaluate	Create asset management along with the objective to create awareness in Digital Rights Management.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	2	1	3	--	--	-	-	-	-	-	-	3	1	1
CO2	1	2	1	3	--	--	-	-	-	-	-	-	3	2	1
CO3	1	1	2	3	-	--	-	-	-	-	-	-	3	3	1
CO4	1	2	1	3	-	--	-	-	-	-	-	-	3	2	1
CO5	2	2	1	3	--	--	-	-	-	-	-	-	3	1	2
WT. AVG	1.2	1.8	1.2	3									3	1.8	1.2

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Ethical Hacking**	7
2.	**Hacking methodologies**	7
3.	**Web Application hacking**	9
4.	**Database and Network hacking**	8
5.	**Report writing & Mitigation**	8

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction to Ethical Hacking**
	• Introduction of Unit • Hacking vs. Ethical Hacking • Hacking Methodologies • Foot printing and Scanning • Trojans and Viruses • Black Box vs. White Box Techniques • Conclusion of Unit
2.	**Hacking methodologies**
	• Introduction to Unit • Denial of Service and DDOS • Sniffers, Session Hijacking and Hacking Web Servers • Attacks on CIA triad • Conclusion of the Unit
3	**Web Application hacking**

.	
	• Introduction of Unit • Web Application Vulnerabilities • Web Techniques Based Password Cracking • Web Jacking and its tools • Phishing and its tools • Conclusion of Unit
4.	**Database and Network hacking**
.	• Introduction of Unit • SQL Injection • Hacking Wireless Networking • Worms and Physical Security • Evading IDS and Firewalls • Conclusion of Unit
5	**Report writing & Mitigation**
.	• Introduction of Unit • Introduction to Report Writing & Mitigation • Requirements for low level reporting & high level reporting of Penetration testing results • Demonstration of vulnerabilities and Mitigation of issues identified including tracking • Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Basic of Hacking and Penetration	Patrick Engerbrestson	First Edition	2010
2	Certified Ethical Hacker All-in-One	Matt Walker	First Edition	2011
Reference Book				
1	Hacking Exposed - Stuart McClure, Joel Scambray, George Kurtz			
2.	Gray Hat Hacking: The Ethical Hacker's Handbook, Fifth Edition (NETWORKING & COMM - OMG)			
Online Resources				
1	https://www.coursera.org/courses?query=ethical%20hacking			
2	https://www.tutorialspoint.com/ethical_hacking/ethical_hacking_tutorial.pdf			
3	https://repo.zenk-security.com/Magazine%20E-book/EN-Ethical%20Hacking.pdf			

Code: BCGECE4111	Installation and Configuration of Server	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Elaborating the basic concepts of selection, configuration and installation of Server.
CO – 02	Understand	Articulate and planning of server, its configuration along with various types of data base and server.
CO – 03	Apply	Explore knowledge about server role in the real environment.
CO – 04	Analyze	Discover and configure Print and Document Services and Servers for Remote Management using Server
CO – 05	Evaluate	Articulate and planning of Local storage, its configuration along with various types of file systems and Formats.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	2	-
CO3	-	-	2	2	-	-	-	-	-	-	-	-	-	-	2
CO4	-	2	-	-	-	-	-	-	-	-	-	-	-	-	1
CO5	-	2	-	-	-	-	-	-	-	-	-	-	-	-	2
WT. AVG	3	2	2	2									3	2	1.666667

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction of Server**	07
2.	**Installing and Configuring Window's Servers 2022**	08
3.	**Installing and Configuring RHEL 9.1 Servers**	08
4.	**Installing and Configuring SQL Server 2022**	07
5.	**Installing and Configuring Oracle Database 21c server**	07

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction of Server**
	● Introduction of Unit ● Introduction-characteristics –types of servers-c/s ● Building blocks-c/s OS: base services-extended services- GUI vs. ● OOUI. Base Middleware: NOS-RPC-Peer-to-Peer ● Communication-sockets-SQL database servers-SQL ● Database server architecture-trigger, Stored procedures-rules. ● Conclusion of Unit
2.	**Installing and Configuring Window's Servers 2022**
	● Introduction of Unit ● Introduction of window's server ,Edition, Supporting Server Role, ● Installing Windows Server 2012: ● System Requirement, Types of Window's server Core Capabilities, Upgrade paths ● Installing Windows Server and Migration Tools ● Configuring Servers: Post-Installation Tasks, Configuring NIC Teaming, Using Roles ● Manager, ● Configuring Servers: Active Directory ,Active Directory Federation, Network Policy

	Access Services (NPAS),Dynamic Host Configuration Protocol ● (DHCP) Server, ● Domain Name System (DNS) Server, Web & Application Servers, Printer, File Services Server, FTP ● Conclusion of Unit
3.	**Installing and Configuring RHEL 9.1 Servers**
	● Introduction of Unit ● Introduction of RHEL Server, System Requirement, Type of Linux Server ,Capabilities of Linux Server ● Installing RHEL Server, Repository installation. ● Configuring Servers : FTP, Domain Name System (DNS) Server, Web & Application Servers. ● Conclusion of Unit
4.	**Installing and Configuring SQL Server 2022**
.	● Introduction of Unit ● Introduction of SQL server, Hardware Requirements ● SQL Server installation, Completing the SSMS Install ● Connecting to your Database, Creating a Database for your Assignments ● Dual Booting Versus Using a Virtual Machine ● Conclusion of Unit
5	**Installing and Configuring Oracle Database 21c server**
.	● Introduction of Unit ● Introduction of Oracle Database 21c server, Hardware Requirements ● Oracle Database 21c server installation in Linux or Windows ● Create User , Connect User, Create Data base, Data base Connect to User ● Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Fedora 9 and Red Hat Enterprise Linux Bible	Christopher Negus		
2	Windows Server 2022 Beginners Guide	Howard J. Wall	2022	
Reference Book				
1	Windows Server 2022 & Powershell, Sara Perrott, Dummies			
2.	Red Hat Enterprise Linux 8 Administration, Miguel Perez Colino, Pablo Iranzo, Packt Publishing			
3.	Oracle 19c Database Administration, Tanveer A			
Online Resources				
1	https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/9			
2	https://github.com/PacktPublishing/Red-Hat-Enterprise-Linux-RHEL-9-Administration			
3.	https://docs.oracle.com/en/cloud/saas/supply-chain-management/22b/index.html			

Code: BAMECE4111	SAS® Enterprise Guide®: ANOVA, Regression, and Logistic Regression	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	
CO – 02	Understand	
CO – 03	Apply	
CO – 04	Analyze	
CO – 05	Evaluate	

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	2	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2.2	2.6	1.4	1.2									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction**	7
2.	**Components of vSphere 6.0**	7
3.	**Features of vSphere and NSX**	9
4.	**VSphere Solutions to Data Center Challenges and vSphere Security**	8
5.	**Resource optimization and resource management**	8

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction to Python Programming**
	• Introduction of Unit • Introduction to Virtualization - Types of virtualization - Difference between cloud and virtualization - Physical infrastructure and virtual infrastructure - Virtualization approaches • Partitioning - Hosting - Isolation - Hardware independence - Virtual machine – Hypervisor - Types of hypervisor - Virtual machine manager - Types of hypervisor - Introduction to datacenter virtualization Esxi - Difference between Esxi and Esx - Versions of Esxi • Conclusion of Unit
2.	**Components of vSphere 6.0**
	• Introduction of Unit • Components of VMware vSphere - vSphere 6.0: Overview and Architecture - Topology of vSphere 6.0 Data Center - vSphere 6.0 Configuration MaximumsvCenter Server - vCenter • Server Features - Certificate Management - Alarms and Alerts - Monitoring Features- Template Management - Linked Mode Deployment - Storage Features in vSphere – Shared Storage - Storage Protocols - Datastores - Virtual SAN - Virtual Volumes – Networking, Features in vSphere - Virtual Networking - Virtual Switches and its types • Conclusion of Unit

3.	**Features of vSphere and NSX**
	● Introduction of Unit ● vSphere Resource Management Features - vMotion - Distributed Resource Scheduler (DRS) Distributed Power Management (DPM) - Storage vMotion - Storage DRS - Storage I/O Control - Network I/O Control - vSphere Availability Features - vSphere Data Protection -High Availability - Fault Tolerance - vSphere Replication - Introduction to NSX. ● Conclusion of Unit
4.	**VSphere Solutions to Data Center Challenges and vSphere Security**
.	● Introduction of Unit ● Challenges - Availability Challenges - Scalability Challenges - Management Challenges - Optimization Challenges - Application Upgrade Challenges - Cloud Challenges - Security - Describe the features and benefits of VMware Platform Services Controller ● host access and authorization - Secure ESXi - vCenter Server - and virtual machines- Upgrade ESXi and vCenter Server instances ● Conclusion of Unit
5	**Resource optimization and resource management**
.	● Introduction of Unit ● Network Optimization - Configure and manage vSphere distributed switches - Migrate virtual machines from standard switches to distributed switches - Explain distributed switch features such as port mirroring - LACP - QoS tagging - and NetFlow - CPU Optimization – Explain the CPU scheduler operation - NUMA support - and other features that affect CPU performance ● Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Virtualization Essentials Paperback	Matthew Portnoy	Latest	Wiley Publications
2	VMware Cookbook Paperback	Troy- Shroff	Latest	O'Reilly
Reference Book				
1	Nelson Ruest, Danielle Ruest, Virtualization, A Beginner's Guide, McGraw Hill, 2009			
2	Cloud Computing: Concepts Technology & Architecture By Thomes Erl			
Online Resources				
1	https://www.javatpoint.com/virtualization-in-cloud-computing			
2	https://www.tutorialspoint.com/virtualization2.0/virtualization2.0_tutorial.pdf			

B.2: Practical- Professional Elective Lab - I

Code: CEECE4211	Web Frameworks Lab	3 Credits [LTP: 3-0-0]

E. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Understand and Analyze	Understand web framework for Web Development
CO – 02	Understand and Evaluate	Reduce the amount of code you write to build rich user interface applications
CO – 03	Understand and Apply	Increase the reliability and maintainability of UI by using Flask and Django
CO – 04	Apply	Modularize your code with the custom services and directives.
CO – 05	Apply & Understand	To understand and apply advanced Django and RESTful APIs for web development

F. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1	2	1	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	2	3	-	-	-	-	-	-	-	-	1	1	-
CO3	3	3	3	3	-	2	-	-	-	-	-	-	1	2	-
CO4	3	1	2	1	-	-	-	-	-	-	-	-	2	3	-
CO5	2	3	3	3	-	2	-	-	-	-	-	-	3	3	-
WT. AVG	2.8	2.2	2.4	2.2		2							1.75	2.25	

G. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction to Machine Learning**
	● Introduction to Machine Learning, Definition of Machine Learning ● Working principles of Machine Learning ● **Basic terminology and concepts (e.g., features, labels, overfitting, underfitting)** ● Classification of Machine Learning : Supervised Learning, Unsupervised Learning, Reinforcement Learning ● Supervised Learning: Classification and Regression ● Unsupervised Learning: Clustering and Association ● Reinforcement Learning, Types of Reinforcement learning : Positive Reinforcement and Negative Reinforcement ● Working of Reinforcement learning ● Markov Decision Process ● Reinforcement Learning Algorithms: Q-Learning and State Action Reward State action (SARSA), Application of Reinforcement Learning ● Conclusion of Unit
2.	**Regression**
	● Introduction to Regression ● Types of Regression: Linear regression, Logistics regression, Ridge Regression, Lasso Regression, Bayesian Linear Regression and Polynomial Regression ● Regression and Correlation, Crosstabs and Scatterplots ● Pearson's r, Regression – Finding The line ● Regression – Describing the line, Contingency Tables ● Conclusion of Unit
3.	**Classification**

	• Introduction of Unit • Classification model building • Types of Classification Algorithm: Binary Classification and Multi Class Classification • Logistic Regression, k-Nearest Neighbors, Decision Trees • Random Forest, Support Vector Machine, Naïve bayes • Handling imbalanced datasets • Feature engineering and selection techniques for classification problems • Conclusion of Unit
4.	**Clustering**
	• Introduction to unit • K-Means Clustering, Hierarchical Clustering: • Density-Based Clustering (DBSCAN), Gaussian Mixture Models (GMM) • Spectral Clustering:, Conclusion of Unit
5	**Performance Metrics**
	• Introduction of Performance metrics • Performance metrics for Regression : Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R-Squared, Adjusted R-squared • Performance metrics for classification: Accuracy, Confusion Matrix, Precision, Recall, F1 score, ROC AUC, Kappa, MCC (Matthews Correlation Coefficient) and Log-loss. • Performance metrics for clustering : Silhouette Score, Rand Index, Adjusted Rand Index, Mutual Information, Calinski-Harabasz Index and Davies-Bouldin Index • Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Mastering Django: A Beginner's Guide (Mastering Computer Science)	Sufyan bin Uzayr	2022	
2.	Flask Web Development 2e: Developing Web Applications with Python	Miquel Grinberg		
Reference Book				
1	Mastering Django : George, Nigel			
2	Two Scoops of Django: Audrey Roy Greenfeld			
Online Resources				
3	https://www.interviewbit.com/blog/flask-vs-django/			

Code :BADECE4211	Fundamentals of Machine Learning Lab	2 Credits [LTP: 0-0-1]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Apply statistical methods for data exploration and visualization.
CO – 02	Understand	Utilize R for data preprocessing, cleaning, and transformation.
CO – 03	Apply	Implement R packages for machine learning and predictive modeling.
CO – 04	Analyze	Apply and analyze regression, classification, and clustering techniques.
CO – 05	Evaluate	Evaluate and communicate results from data analysis.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	3	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	2	2	-
CO3	-	-	3	-	-	-	-	-	-	-	-	-	-	-	2
CO4	-	-	-	3	2	-	-	-	-	-	-	-	-	-	2
CO5	-	-	-	-	2	-	-	-	-	-	-	-	-	-	
WT. AVG	2.5	3	3	3	2								2.5	2	2

C. LIST OF EXPERIMENTS

1	Read a dataset into a Pandas DataFrame and display the first few rows using the "head" function in Python.
2	Work with Pandas and XlsxWriter.
3	Work with CSV files in Python and apply preprocessing techniques such as Scaling, Normalization, Binarization, Standardization, and Data Labeling, as well as divide the data into train and test split.
4	Implement Simple Linear Regression to predict house prices using datasets from any data source.
5	Implement Logistic Regression to predict car prices in Python.
6	Use Logistic Regression to recognize hand-written digits (0 to 9) by loading the dataset from any data source in Python.
7	Build a classifier on a set of email data and determine if they are spam or not using SVM in Python.
8	Implement Decision Tree classifier on the Pima Indian Diabetes dataset in Python.
9	Implement Random Forest classification in Python on the Iris dataset from its web link.
10	Use Naïve Bayes with training examples of individuals categorized into high, medium, and low credit-worthiness in Python.
11	Implement K-means clustering on a simple digits dataset. K-means will try to identify similar digits without using the original label information in Python.
12	Implement the Mean-Shift algorithm on a 2D dataset containing 4 different blobs in Python.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Machine Learning - An Algorithmic Perspective	Stephen Marsland	2nd Edition	
2	Data Mining: Concepts and Techniques	Jiawei Han and Micheline Kamber	2nd Edition	Elsevier
3	Introduction to Machine Learning	Nils J. Nilsson		
Reference Book				
1	Introduction to Machine Learning, 3rd Edition (Adaptive Computation and Machine Learning Series)	Ethem Alpaydin	Third Edition	MIT Press
2	Machine Learning: The Art and Science of Algorithms that Make Sense of Data	Peter Flach	1st Edition	Cambridge University Press
Online Resources				
1	https://nptel.ac.in/courses/106106182			

Code :BCSECE4211	Ethical Hacking Lab	2 Credits [LTP: 0-0-1]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Ability to translate end-user requirements into system and software requirements
CO – 02	Understand	Ability to generate a high-level design of the system from the software requirements
CO – 03	Apply	Will have experience and/or awareness of testing problems and will be able to develop a simple testing report
CO – 04	Analyze	Learn Trojans Creating and Analyzing
CO – 05	Evaluate	Use different Data recovery and password protecting tool's for network security and analysis

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	–	3	3	3	–	3	–	–	3	–	–	–
CO2	3	3	2	–	3	3	3	–	3	–	–	3	–	–	–
CO3	3	3	–	–	3	3	3	–	3	–	–	3	–	–	–
CO4	3	3	–	–	3	3	3	–	3	–	–	3	–	–	–
CO5	3	3	–	–	3	3	3	–	3	–	–	3	–	–	–
WT. AVG	3	3	2		3	3	3		3			3			

C. LIST OF EXPERIMENTS

1	Passive Reconnaissance using —Who is‖ and Online tools
2	Active Reconnaissance using —Sampad‖ and web site details
3	Full Scan, Half Open Scan and Stealth scan using —nmap‖
4	UDP and Ping Scanning using —Advance Lan Scanner‖ and —Superscan‖
5	Packet crafting using —Packet creator‖ tools
6	Exploiting NetBIOS vulnerability
7	Password Revelation from browsers and social networking application
8	Creating and Analyzing spoofed emails
9	Creating and Analyzing Trojans I
10	Creating and Analyzing Trojans II
11	OS password cracking
12	Data recovery and password protecting techniques for Pen drive and CD

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Basic of Hacking and Penetration	Patrick Engerbrestson	First Edition	2010
2	Certified Ethical Hacker All-in-One	Matt Walker	First Edition	2011
Reference Book				
1	Hacking Exposed - Stuart McClure, Joel Scambray, George Kurtz			
2.	Gray Hat Hacking: The Ethical Hacker's Handbook, Fifth Edition (NETWORKING & COMM - OMG)			
Online Resources				
1	https://www.coursera.org/courses?query=ethical%20hacking			
2	https://www.tutorialspoint.com/ethical_hacking/ethical_hacking_tutorial.pdf			
3	https://repo.zenk-security.com/Magazine%20E-book/EN-Ethical%20Hacking.pdf			

Code: BCGECE4211	Installation and Configuration of Server Lab	2 Credits [LTP: 0-0-1]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Ability to use the basic concepts of server hardware, software, and networking protocols. List the components required for setting up a server lab environment. Recognize common server operating systems and their features.
CO – 02	Understand	Explain the importance of proper server configuration in a network environment. Differentiate between various server roles and services and their functionalities.
CO – 03	Apply	Demonstrate proficiency in setting up a server lab environment using virtualization technologies. Apply troubleshooting techniques to resolve common server installation and configuration issues.
CO – 04	Analyze	Analyze network requirements and select appropriate server hardware and software configurations. Evaluate the security implications of different server configurations and implement best practices.
CO – 05	Evaluate	Critique different approaches to server setup and configuration based on performance and scalability factors. Assess the effectiveness of server lab configurations in meeting specified business or organizational needs.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	3	3	-	3	3	2	3	3	3	2	2
CO2	-	-	-	-	3	1	-	1	2	2	3	3	2	3	3
CO3	-	-	-	-	3	1	-	1	3	2	2	3	3	3	2
CO4	-	-	-	-	3	1	-	1	2	3	2	3	3	3	2
CO5	-	-	-	-	3	1	-	1	2	3	3	3	3	3	3
WT. AVG					**3**	**1.4**		**1.4**	**2.4**	**2.4**	**2.6**	**3**	**2.8**	**2.8**	**2.4**

C. LIST OF EXPERIMENTS

1. Installation windows Server 2012.
2. Configuration for Windows Server.
3. Configuration Local Storage for Windows Server.
4. Configuration File and Share Access for Windows Server.
5. Configuration Print and Document Services for Windows Server.
6. Configuration windows server for Remote Management.
7. Creating Virtual Machine in Windows Server.
8. Configuration and Setting Virtual Machine.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Fedora 9 and Red Hat Enterprise Linux Bible	Christopher Negus		
2.	Windows Server 2022 Beginners Guide	Howard J. Wall	2022	
Reference Book				
1	Windows Server 2022 & Powershell, Sara Perrott, Dummies			
2	Red Hat Enterprise Linux 8 Administration, Miguel Perez Colino, Pablo Iranzo, Packt Publishing			
3	Oracle 19c Database Administration, Tanveer A			
Online Resources				
1	https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/9			
2	https://github.com/PacktPublishing/Red-Hat-Enterprise-Linux-RHEL-9-Administration			
3	https://docs.oracle.com/en/cloud/saas/supply-chain-management/22b/index.html			

Code: BAMECE4211	Application of Machine Learning Using SAS(R) Viya Lab	2 Credits [LTP: 0-0-1]

Introduction : Machine learning in business decision making, Essentials of supervised prediction. Introduction to SAS .
Data Preparation Data exploration, Feature extraction., Input transformations., Feature selection.,Variable clustering .,Best practices.,Selecting your algorithm.
Decision Trees and Ensembles of Trees Introduction.,Tree-structure models.,Recursive partitioning.,Pruning.,Ensembles of trees.
Neural Networks Introduction.,Network architecture.,Network learning and optimization
Support Vector Machines and Additional Topics Large-margin linear classifier.,Methods of solution.,Nonlinear classifier: Kernel Trick.,Additional tools.
Model Assessment and Deployment Model assessment and comparison.,Model deployment.

LIST OF EXPERIMENTS

1	Discuss the problem of machine learning in business decision making.
2	Essentials of supervised prediction.
3	Data exploration, Feature extraction
4	Input transformations
5	Feature selection
6	Variable clustering
7	Decision tree introduction
8	Tree structure model
9	Recursive partitioning, Pruning
10	Ensembles of trees
11	Neural Networks Introduction, Network architecture
12	Network Learning and Optimization
13	Support vector machine: Large-margin linear classifier
14	Methods of solution
15	Non linear classifier: Kernel Trick
16	Additional tools
17	Model assessment and Deployment: Model assessment and comparison
18	Model deployment

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Virtualization Essentials Paperback	Matthew Portnoy	Latest	Wiley Publications
2	VMware Cookbook Paperback	Troy- Shroff	Latest	O'Reilly
Reference Book				
1	Nelson Ruest, Danielle Ruest, Virtualization, A Beginner's Guide, McGraw Hill, 2009			
2.	Cloud Computing: Concepts Technology & Architecture By Thomes Erl			
Online Resources				
1	https://www.javatpoint.com/virtualization-in-cloud-computing			
2	https://www.tutorialspoint.com/virtualization2.0/virtualization2.0_tutorial.pdf			

D: Ability Enhancement Courses (AEC)

Code : BUACHM4213	HUMAN VALUES & PROFESSIONAL ETHICS	Credits [LTP: 0-0-1]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Understand	Explain the fundamental human values, including love, compassion, truth, non-violence, righteousness, peace, service, and renunciation.
CO – 02	Apply	Construct a positive self-esteem, assertive personality, and the ability to manage ambitions and desires in a balanced manner.
CO – 03	Analyse	Compare the differences between personal and professional ethics, and ethical principles in various professional scenarios.
CO – 04	Apply	Practice emotional intelligence skills, including self-awareness, self-regulation, internal motivation, empathy, and social skills.
CO – 05	Apply	Show essential life skills and value-based education to achieve personal and professional success.

B. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Human Values**	**6**
2.	**Study of Self**	**6**
3.	**Introduction to Professional Ethics**	**8**
4.	**Emotional Intelligence**	**2**
5.	**Life Skills & Value Education**	**2**

C. List of Labs

LIST OF LABS	
1.	Human Values: Love & Compassion
2.	Truth, Non-Violence, Righteousness
3.	Peace, Service, Renunciation (Sacrifice)
4.	Self-Esteem: Do's and Don’ts to develop positive self-esteem
5.	Self-Assertiveness: Development of Assertive Personality
6.	Ambition & Desire: Self & Body (concepts & differences)
7.	Professional Ethics: Personal & Professional Ethics
8.	Emotional Intelligence: Skill Building for Strengthening the Elements of Self-awareness, Self-regulation, Internal motivation, Empathy, Social skills
9.	Governing Ethics & Ethics Dilemma
10.	Profession, Professionalism & Professional Risks
11.	Professional Accountabilities & Professional Success
12.	Life Skills & Value Education

E: Skill Enhancement Courses (SEC)

Code : BULCSE4201	Skill Enhancement Generic Course-IV	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

- Understand basic problems based on arithmetic and soft skills area which are asked in aptitude test taken by companies
- Effectively solve these problems by applying the knowledge earned.
- Actively participate in group discussion / meetings / interviews and prepare & deliver presentations.
- Become more effective individual through goal/target setting, self-motivation and practicing creative thinking.
- Function effectively in multi-disciplinary and heterogeneous teams through the knowledge of team work, Inter-personal relationships, conflict management and leadership quality

B.

LIST OF LABS	
1.	Averages, Mean, Median and Mode
2.	Cognitive learning theory, Body Language basics
3.	Heights & Distances
4.	Sitting Arrangements
5.	Fill Ups(Grammar based)
6.	Error Detection, Confusing words
7.	Alphanumeric Series
8.	Verbal Analogy, One word substitution
9.	Dices
10.	Sentence Correction, Subject-Verb agreement
11.	Statement & Assumptions, Setting SMART goals,
12.	Persuasion Skills, Interview Preparation

F: Value Added Courses (VAC)

Code : BUVCVD4102	Business Intelligence	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Gain knowledge of Business Intelligence
CO – 02	Understand	Elements of Business Intelligence Solutions
CO – 03	Apply	Build business projects
CO – 04	Analyze	Generate and manage BI reports
CO – 05	Evaluate	BI Deployment, Administration & Security.

B. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Business Intelligence**	6
2.	**Elements of Business Intelligence Solutions**	6
3.	**Building the BI Project**	6
4.	**Reporting Authoring**	6
5.	**BI Deployment, Administration & Security**	6

C. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction to Business Intelligence**
	• Introduction of the Unit • Understanding the scope of today's BI solutions and how they fit into existing infrastructure. Describe BI, its components & architecture. • The future of BI, better experience for all business users. • The Functional Area of BI Tools, Query Tools and Reporting. • OLAP and Advanced Analytic • Conclusion of the Unit
2.	**Elements of Business Intelligence Solutions**
	• Introduction of the Unit • Reports & ad hoc queries. • Dashboards & Scorecards development. • Metadata, Real time monitoring capabilities. • BI portals, web applications, Desktop applications. • Conclusion & Real life applications • Conclusion of the Unit
3.	**Building the BI Project**
	• Introduction of the Unit • Planning the BI project, Project Resources, • Collecting User Requirements, • Validating BI Requirements, BI Design and Development • Conclusion of the Unit
4.	**Reporting Authoring**

.	• Introduction of the Unit • Building reports with relational vs Multidimensional data models. • Types of Reports – List, crosstabs, Statistics, Chart, map, financial etc. • Data Grouping & Sorting, Filtering Reports. • Conditional formatting, Adding Summary Lines to Report • Conclusion of the Unit
5	**BI Deployment, Administration & Security**
.	• Introduction of the Unit • BI Architecture • Expanding BI Authentication Authorization, Access Permissions, Groups and Roles. • Manage Status & Monitoring. • Back Up and Restore • Conclusion of the Unit

I. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Business Intelligence	Mark Whitehorn , Mary Whitehorn	Ist	(IBM ICE Publication).
2	Data Strategy: How To Profit From A World Of Big Data, Analytics And The Internet Of Things	Bernard Marr	2nd	Kogan Page
3	The Data Detective: Ten Easy Rules to Make Sense of Statistics	Tim Harford	Latest	Riverhead Books
Reference Book				
1	From Big Data to Big Profits: Success with Data and Analytics	Russell Walker	Latest	Oxford University Press

Your Dreams Our Goal

POORNIMA UNIVERSITY

Member of Association of Indian Universities & Approved by UGC (Govt. of India) under 2(f) & 12(B)

B. TECH - CSE

THIRD YEAR
SEMESTER: V

COMMON SYLLABUS

For

Cyber Security
Cloud & Full Stack Development
Artificial Intelligence & Data Science
Artificial Intelligence & Machine Learning

BATCH: 2024-28

A: Major (Core Courses)

Code : BCECCE5101	Design & Analysis of Algorithms	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Apply Divide and conquer. Greedy algorithm design techniques
CO – 02	Understand	To handle the dynamic programming concept with solving real word problem
CO – 03	Apply	To manage of different Pattern matching algorithms.
CO – 04	Analyze	Apply randomize algorithms

B. MAPPING MATRIX OF CO, PO, & PSO

	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	-	-	-	-	-	-	-	-	-	-	-
CO2	2	2	1	2	-	-	-	-	-	-	-	-	-	-	-
CO3	2	3	2	1	-	-	-	-	-	-	-	-	-	-	-
CO4	2	2	2	1	-	-	-	-	-	-	-	-	-	-	-
CO5	2	3	1	1	-	-	-	-	-	-	-	-	-	-	-
WT. AVG	2	2.6	1.4	1.2	-	-	-	-	-	-	-	-	-	-	-

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction	06
2.	Dynamic Programming, Branch and Bound	06
3.	Pattern Matching and Assignment Problem	08
4.	Randomized Algorithm	08
5.	NP-Hard and NP-Complete Problem	08

D. DETAILED SYLLABUS

Unit	Unit Details
1.	Introduction
	□ Introduction to Unit □ Algorithm Specification, Algorithm Complexity and Order Notations. □ Divide and Conquer Method : General Method, Binary Search, Merge Sort, Quick sort and strassen's matrix multiplication algorithm. □ Greedy Method: General method, Knapsack Problem, Job Sequencing, Optimal Merge Patterns and Minimal Spanning Tree: Prim's, Kruskal's Algorithm □ Conclusion of Unit
2.	Dynamic Programming, Branch and Bound
	● Introduction to Unit ● Dynamic Programming: Matrix Chain Multiplication, Longest Common ● Subsequence and 0/1Knapsack Problem, All pairs shortest path, Flow shop scheduling ● Branch And Bound: Traveling Salesman Problem, Bounding, FIFO Branch and Boun d, ● Backtracking:The8-queensproblem,Hamiltoniancycles Comparison between Dynamic, Backtracking and Branch Bound ● Conclusion of Unit
3.	Pattern Matching and Assignment Problem
	● Introduction to Unit ● Pattern Matching Algorithms: Naïve and Rabin Karp string matching algorithms, KMP Matcher and Boyer Moore Algorithms.

	● Assignment Problems: Formulation of Assignment and Quadratic assignment Problem. ● Conclusion of Unit.
4.	Randomized Algorithm
.	● Introduction of Unit. ● Probabilistic Analysis & Randomized Algorithms: Las Vegas algorithm, Monte Carlo algorithms for Min-Cut, randomized algorithm for 2- SAT. ● Problem definition of Multi commodity flow, Flow shop scheduling and Network capacity ● Assignment problems. ● Conclusion of Unit
5	NP-Hard and NP-Complete Problem
.	● Introduction of Unit. ● Definitions of P, NP-Hard and NP-Complete Problems. Decision Problems. Cook's Theorem. Proving NP- Complete Problems - Satisfiability problem and Vertex Cover Problem. ● Approximation Algorithms for Vertex Cover and Set Cover Problem ● Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Fundamentals of Computer Algorithms	E.Horowitz & S.Sahani	Latest	Galgotia Publications
2.	Introduction to Algorithms	Corman, Leiserson & Rivest	Latest	MIT Press
Reference Book				
1	Algorithm Analysis & Design, Goodrich, Tamassia, Wiley			
2.	Computer Algorithms, Introduction to Design and Analysis, SaraBasse, A. V. Geider			
Online Resources				
1	https://www.javatpoint.com/daa-tutorial			
2	https://www.guru99.com/design-analysis-algorithms-tutorial.html			
3	https://www.geeksforgeeks.org/system-design-tutorial/			

Code : BCECCE5102	Advance Java	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Understand and apply fundamental concepts of Java Servlets and their role in web development.
CO – 02	Understand	Develop dynamic web applications using JavaServer Pages (JSP) and integrate them with servlets.
CO – 03	Apply	Implement JavaServer Faces (JSF) for building rich user interfaces and manage user interactions.
CO – 04	Analyze	Develop database application using hibernate
CO – 05	Evaluate	Develop enterprise-level applications with a focus on dependency injection and aspect-oriented programming using Spring Framework

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	-	3	-	-	-	-	-	-	-	-	-	3	1	-
CO2	-	-	-	3	-	-	-	-	-	-	-	-	-	1	-
CO3	-	-	3	1	-	-	-	-	-	-	-	-	-	2	1
CO4	-	2	-	-	-	-	-	-	-	-	-	-	-	2	-
CO5	-	1	-	2	-	-	-	-	-	-	-	-	2	-	-
WT. AVG															

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Java Servlets	08
2.	Java Server Pages (JSP)	08
3.	Java Server Faces	07
4.	Hibernate	08
5.	Springs	08

D. DETAILED SYLLABUS

1.Unit	Unit Details
1.	**Java Servlets**
	● Introduction of Unit ● Servlets and HTTP Servlets, Filters, Security, Servlet Life Cycle, Servlets for the World Wide Web, Requests, Responses, and Headers, GET and POST, HTTP, Deploying a Servlet, Web Application Deployment Descriptor Structure, Servlet Configuration, Http Servlet Request/Response, Servlet Context, Session Management, ● Case Study ● Conclusion of Unit
2.	**Java Server Pages (JSP)**
	● Introduction of Unit: JavaBeans, Custom Tags and JSP Fragments, JSP Life Cycle, The Difference Between Servlets and JSP, JSP Syntax and Semantics, Elements and Template Data, JSP Configuration, Standard JSP Actions, Attributes, Comments, Quoting and Escape Characters, Exception Handling, JavaBeans and the JSP Expression Language, JSP Standard Tag Library, Custom Tag Libraries, Database Connectivity, building a Complete Web Application, Using AJAX with JSP for Dynamic Content. ● Case Study ● Conclusion of Unit
3.	**Java Server Faces**
	● Introduction of Unit: features, life cycle, manage Beans, UI Components- input Text, output Text, form, command Button, input Text Area, input Hidden, input File, Bean, Validation, facelets, JSF JDBC, JSF with controllers, architectural overview of

	application developed with JSF and JSP, validator tag, data tables, Advanced UI Components and AJAX Integration in JSF. ● Case Study ● Conclusion of Unit
4.	**Hibernate**
.	● Introduction of Unit: advantages, features, Architecture, Environment, Life Cycle, ORM Tool, Sessions, Session factory, Persistent Class, MVC, Hibernate Query language, Criteria Query, Mapping Types, Annotations, Query Language, Native SQL, Caching Strategies in Hibernate, Using Hibernate with Spring. ● Case Study ● Conclusion of Unit
5	**Springs**
.	● Introduction of Unit: Architecture, Environment Setup, IoC Containers, Bean Definition and Lifecycle, Dependency Injection, Setter Injection, Injecting Inner Beans and Collections, Event Handling in Spring, Spring MVC Framework, Aspect-Oriented Programming (AOP) in Spring, Spring Boot for Rapid Application Development, ● Case Study ● Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Servlets and Java Server Pages	Jayson Falkner		Kevin Jones-2003
2.	Beginning Hibernate	Joseph B. Ottinger, Jeff Linwood, Dave Minter		Apress-2014
Reference Book				
1	Professional Java Development with the Spring Framework, Rod Johnson, 8th edition – Wiley			
2.	Core Java Server Faces, David M. Geary, 2004 – 3rd Edition-Prentice Hall			
Online Resources				
1	https://www.simplilearn.com/resources-to-learn-java-programming-article			
2	https://www.docdroid.net/mY1yTPu/advancedjavaprogrammingbyuttamkumarroy-pdf			
3	https://www.edureka.co/blog/advanced-java-tutorial			

Code : BCECCE5103	Computer Architecture	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Understand the fundamental concepts and importance of digital logic in computing and electronics.
CO – 02	Understand	Understand the organization of processors including general and stack registers and will learn about different types of instructions.
CO – 03	Apply	Apply the knowledge of register transfer and computer arithmetic to real-life computing problems and solutions.
CO – 04	Analyze	Describe the theoretical concepts of micro-operations and control units to practical applications in computer systems and microprocessors.
CO – 05	Evaluate	Apply the knowledge of memory and input-output organization to real-world computing scenarios and understand the importance of efficient memory and input-output management.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	-	3	-	-	-	-	-	-	-	-	-	3	1	-
CO2	-	-	-	3	-	-	-	-	-	-	-	-	-	1	-
CO3	-	-	3	1	-	-	-	-	-	-	-	-	-	2	1
CO4	-	2	-	-	-	-	-	-	-	-	-	-	-	2	-
CO5	-	1	-	2	-	-	-	-	-	-	-	-	2	-	-
WT. AVG															

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Basics Of Digital Logics	8
2.	Register Transfer and Computer Arithmetic	8
3.	Basic Computer Organization	7
4.	Micro-operation and Micro Programmed Control Unit	8
5.	Memory & Input-Output	8

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Basics Of Digital Logics**
	• Introduction of Unit • Number systems : Binary number system, Octal & Hexa-decimal number system, Conversion of Number System, r's & (r-1)'s, , **Signed number representation**, Binary arithmetic Operations (Addition & Subtraction) • Logic Gates: AND, OR, NOT GATES and their Truth tables, NOR, NAND & XOR gates. • Boolean algebra: AND, OR, Inversion, Basic Boolean Law's, Demorgan's theorem, Minimization techniques: K -Map, Sum of Product & Product of Sum, • Conclusion & Real Life Application
2.	**Register Transfer and Computer Arithmetic**
	• Introduction of Unit

	● Register Transfer Language, Register Transfer, **Buses, Bus architecture, Types of buses and Bus arbitration**, Bus and Memory Transfer: Three state bus buffers, Memory Transfer. ● **Half Adder, Full Adder, Parallel Adder, look Ahead carry adder** ● Multiplication Algorithms (Booth algorithm), Division Algorithms, **IEEE Standard for Floating Point Numbers** ● Conclusion & Real Life Application
3.	**Basic Computer Organization**
	● Introduction of Unit ● Processor organization, general registers organization, stack organization. ● Instruction types, formats, instruction cycles and sub cycles (fetch and decode etc), **pipelining, Addressing Modes** ● Conclusion & Real Life Applications
4.	**Micro-operation and Micro Programmed Control Unit**
.	● Introduction of Unit ● Logic Micro-operations: List of Logic micro operations, Shift Micro-operations (excluding H/W implementation), Arithmetic Logic Shift Unit. ● Control Memory, Address Sequencing, Conditional branching, Mapping of instruction, Subroutines. ● Hardwired and micro programmed control unit ● Conclusion & Real Life Application
5	**Memory & Input-Output**
.	● Introduction of Unit ● Memory Organization: Memory Hierarchy, Main Memory, Auxiliary Memory, ● Associative Memory, Cache Memory, Virtual Memory, **Mapping and replacement** ● Input – Output Organization: Peripheral devices, Input – Output interface, Modes of Data Transfer: Priority Interrupt, Direct Memory Access Introduction of Multiprocessors: Characteristics of multi-processors. ● Conclusion &Real Life Application

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Computer System Architecture	Morris Mano		PHI
2.	Computer Organization and Architecture	William Stallings		PHI
Reference Book				
1	Digital Computer Electronics: An Introduction to Microcomputers, Malvino, TMH			
2.	PC Hardware in a Nutshell Barbara Fritchman Thompson, Robert Bruce, Thompson, O'Reilly, 2nd Edition , 2010			
3.	Fundamentals of Computer Organization and Architecture, Mostafa AB-EL-BARR and Hesham EL-REWNI by John Wiley and Sons			
Online Resources				
1	https://www.javatpoint.com/computer-organization-and-architecture-tutorial			

Code :BCGCCE5201	Design & Analysis of Algorithms Lab	1 Credits [LTP: 0-0-2]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall basic concepts of Algorithms, complexity and methods like divide and conquer and greedy Algorithms.
CO – 02	Understand	Demonstrate an understanding of the Dynamic Programming to resolve min and max related Daily life problems.
CO – 03	Apply	Apply Pattern matching algorithms to find out patterns along with the Assignment Problem.
CO – 04	Analyze	Examine randomized algorithms to find desired solutions.
CO – 05	Evaluate	Assess the concept of NP-Hard and NP-Complete Problem for high computational problems.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	-	2	1	-	-	-	-	-	-	-	2	-	-
CO2	1	2	3	2	-	-	-	-	-	-	-	-	1	-	-
CO3	1	3	3	1	-	-	-	-	-	-	-	-	2	-	-
CO4	2	-	1	1	-	-	-	-	-	-	-	-	2	-	-
CO5	1	1	2	-	-	-	-	-	-	-	-	-	1	-	-
WT. AVG	1.4	1.75	2.25	1.5	1								1.6		

C. LIST OF EXPERIMENTS

1	Write a C program to implement the Stack using arrays. Write Push(), Pop(),and Display() methods to demonstrate its working. Write a C program to implement the Queue using arrays. Write Enqueue(), Dqueue(), Display() & Peak() methods to demonstrate its working.
2	Write a C program to implement a Merge sort algorithm to a list of elements for different values of n and determine the time required to sort the elements. Write a C program to sort a list of elements using the quick sort algorithm. The elements can be read from a file.
3	Write a C program to find the product of two matrix of size (2X2) using Strassen's Matrix multiplication algorithm. Implement 0/1 Knapsack problem using Dynamic Programming in C.
4	WAP to find the Minimum Cost Spanning Tree (MCST) of a given connected undirected graph using Kruskal's Algorithm. WAP in C to find the Minimum Cost Spanning Tree (MCST) of a given connected undirected graph using Prim's algorithms.
5	Write a C program to find the shortest paths between nodes in a undirected graph using Dijkstra's algorithm. Write a C program to find the shortest paths between nodes in a directed graph using BelmanFord algorithm.
6	Write a C program to implement N Queen's problem using Back Tracking.
7	WAP in C to find a subset of a given set S = {sl , s2,.....,sn} of n positive integers whose sum is equal to a given positive integer d.
8	Write a C program to Print all the nodes reachable from a starting node in a digraph using BFS method.
9	Check whether a graph is connected or not using DFS method.
10	Write a C Program to Search a given pattern in a input String.
11	Write a C program to implement all pairs shortest paths problem using Floyd's algorithm.
12	Write a C program to find out the probability of generating any random no using Monte Carlo algorithm.

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Mastering Algorithms with C	Kyle Loudon	Latest	O'Reilly
2.	Algorithms Illuminated (Part 3): Greedy Algorithms and Dynamic Programming	Tim Roughgarden	2014	Kindle
Reference Book				
1.	Data Structures and Algorithms, Made Easy by NarasimhaKarumanchi, Kindle Edition			
Online Resources				
1.	https://www.sanfoundry.com/c-program			
2.	https://www.thecrazyprogrammer.com/2015/03/c-program-for-n-queens-problem-using-backtracking.html			

Code : BCECCE5202	Advance Java Lab	1 Credits [LTP: 0-0-2]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall Java syntax, language features, and basic programming concepts.
CO – 02	Understand	Understand advanced Java concepts such as multithreading, networking, and GUI programming.
CO – 03	Apply	Apply Java programming skills to develop complex applications, implement algorithms, and solve problems.
CO – 04	Analyze	Analyze and debug Java code, identify and resolve errors, and optimize program performance
CO – 05	Evaluate	Evaluate Java solutions in terms of efficiency, maintainability, and adherence to coding standards.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	-	1	-	-	-	-	-	-	-	-	-	1	-	-
CO2	1	2	2	-	-	-	-`	-	-	-	-	-	1	-	-
CO3	2	2	3	-	-	-	-	-	-	-	-	-	1	-	-
CO4	1	2	3	-	-	-	-	-	-	-	-	-	1	-	-
CO5	1	2	2	-	-	-	-	-	-	-	-	-	1	-	-
WT. AVG	1.6	1.6	1.6	2	1								1	1.6	

C. LIST OF EXPERIMENTS

1	Develop dynamic web application to display current system date and time using servlets
2	Create a servlet that generates a login page with proper HTML UI elements (e.g., username, password fields, and a submit button).
3	Develop dynamic web application to manage product (prod Id, name, category, price) details using servlets. This app must have following pages . Home page . Product adding page . Product editing page . Product displaying page
4	Develop a JSP page that uses JavaBeans to display user information.
5	Write JSP program to implement custom tag with name <product>, which display product (prodId, name, category, price) details
6	Enhance previous JSP program to fetch data from database
7	Develop Rich Internet Applications to manage product and user details using struts and database
8	Develop Hibernate application to manage product details like insert, update, delete and display from database using HQL
9	Develop Spring based dynamic web application to manage courses, students in a college environment using Web MVC framework and JDBC
10	Transfer a file from one system to another system by the network
11	Develop Chat Server using Java.
12	Develop a comprehensive project that incorporates servlets, JSP, JavaBeans, custom tags, database connectivity, Struts, Hibernate, and Spring MVC to create a complete web application.

I. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Advanced Java Programming	B.Prasanalakshmi	1st	CH Publishers
2.	Advanced Java Programming	Uttam K Roy	1st	Oxford University Press
3.	Advanced Java Technology -A Conceptual Approach	A.A.Puntambekar	1st	Technical Publications
Reference Book				
1.	Advanced Java Coding Problems: Best Advanced Coding Problems with Explanation and Solutions, by Pratap Divyansh			
2.	Advanced Java Optimization Techniques, by Jason Arnold			
Online Resources				
1.	https://www.simplilearn.com/resources-to-learn-java-programming-article			
2.	https://www.docdroid.net/mY1yTPu/advancedjavaprogrammingbyuttamkumarroy-pdf			

Code : BCECCE5203	Power BI LAb	1 Credits [LTP: 0-0-2]

A. **COURSE OUTCOMES**

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Able to understand power BI, develop medium-difficulty applications of power BI
CO – 02	Understand	Implement power BI with conditionals and Dax Query
CO – 03	Apply	Apply the concept of operators and dashboard.
CO – 04	Analyze	Implement Read and write data from/to files in power BI
CO – 05	Evaluate	Develop power BI programs step-wise by defining functions

. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	2	3	-	-	-	-	-	-	-	-			
CO2	3	2	3	2	-	-	-	-	-	-	-	-			
CO3	2	3	1	1	-	-	-	-	-	-	-	-			
CO4	2	2	3	3	-	-	-	-	-	-	-	-			
CO5	2	3	2	1	-	-	-	-	-	-	-	-			
WT. AVG	2.2	2.6	2.2	2											

. LIST OF EXPERIMENTS

1	Getting started with power BI Software using data file formats, connecting with power BI.
2	Creating basic charts (line, bar charts, Tree maps), using the BPB Publication me panel.
3	Power BI calculations, SUM, AVG, and aggregate features, creating custom calculations and fields.
4	Applying new data calculations, formatting, menus visualizations specific parts of the view in power BI.
5	Editing and formatting axes, manipulating data in power BI data, pivoting data.
6	Structuring our data, sorting and filtering power BI data.
7	Advanced visualization tools filters, using the detail size panels.
8	Customizing filters, using and customizing tooltips, and formatting data with colors.
9	Creating dashboards storytelling, creating first dashboard. Design for different displays, adding interactivity to dashboard.
10	Power BI file types, publishing to power BI online, and sharing visualizations, printing, and exporting.
11	Creating custom charts, cyclical data and circular area charts, dual axis charts.
12	Foundations for building custom dashboard for visualizations with multiple tiles.

. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Power BI: A Complete Step-by-Step Guide for Beginners in Understanding Power BI Kindle Edition	Power BI Kindle Edition	2007	BPB Publication
2	*Mastering Power BI*	Power BI Kindle Edition	2010	BPB Publication
Reference Book				
1	*Mastering Power BI*			
Online Resources				
1	https://learn.microsoft.com/en-us/power-bi/connect-data/desktop-data-sources			
2	https://learn.microsoft.com/en-us/power-bi/connect-data/service-get-data			

B. Minor Stream Courses/ Department Electives
B.1 Theory: Professional Elective - IV

Code :BCEECE5111	Artificial Intelligence	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Understand Foundational Principles and Concepts of AI.
CO – 02	Understand	Develop problem-solving techniques using various search algorithms
CO – 03	Apply	Apply CSP techniques and adversarial search for game playing
CO – 04	Analyze	Analyze Knowledge Representation and Reasoning in AI models
CO – 05	Evaluate	Explore AI applications in various domains

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	-	-	2	2	-	1	-	1	-	2	2	-	2
CO2	1	2	1	1	2	-	-	-	2	-	-	2	1	2	-
CO3	2	3	2	1	3	-	-	1	2	-	-	2	3	1	2
CO4	2	3	-	2	3	-	-	2	2	-	-	2	2	2	-
CO5	3	3	3	3	3	2	-	2	-	1	-	2	2	1	2
WT. AVG	1.8		1.5	1.8	2.3	2.0		1.3	2.0	1.0		2.0	2.0	1.3	2.0

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to AI	7
2.	Problem Solving	8
3.	Constraint Satisfaction Problems	7
4.	Knowledge Representation and Reasoning	7
5.	AI Applications	7

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to AI**
	• Introduction of Unit • What is AI? • The Foundations of Artificial Intelligence • The History of Artificial Intelligence and State of the art of AI • Approaches of AI -Thinking humanly, Acting Humanly, Thinking rationally, Acting rationally • Agents, Environment, Agent types and Environment types, Structure of

	Agent ● AI vs. Machine Learning vs. Deep Learning ● Conclusion & Real-Life Application
2.	**Problem Solving**
	● Introduction of Unit ● Problem-solving agent ● Problem formulation with 8 puzzle problem example ● Un-informed Search Algorithms: Breadth-first search, Depth-first search, Depth-limited search, Uniform-cost search and Iterative deepening search ● Informed (Heuristic) Search Algorithms: Best-first search, A* search ● Local Search Algorithm: Genetic algorithms ● Conclusion & Real-Life Application
3.	**Constraint Satisfaction Problems**
	● Introduction of Unit ● Constraint Satisfaction Problems (CSP), Real-world CSPs ● Backtracking search for CSPs, Forward checking for CSPs ● Adversarial Search and Game Playing, Games: Definition, Game Tree and Types ● Optimal decisions in Games: Mini max algorithm and α-β pruning with example ● State-of-the-Art Game Programs: Chess on Deep Blue, Chess on standard PCs, Checkers on Chinook and TD-Gammon ● Conclusion of the Unit
4.	**Knowledge Representation and Reasoning**
.	● Introduction of Unit ● Knowledge Based Agent, Representation, Types ● Logic and Propositional Logic - Syntax and semantics of propositional logic ● First-Order Logic - Syntax and semantics of first-order logic ● Inference in First-Order Logic ● Forward Chaining and backward chaining in AI ● Probabilistic representation using Bayesian Networks and Hidden Markov Model ● Conclusion of the Unit
5	**AI Applications**
.	● Introduction of Unit ● AI in Healthcare, Education, Agriculture ● Machine Translation, Speech Recognition ● Expert system: Introduction, phases, architecture ● Robotics and AI, Future of AI ● Language used in AI ● Conclusion of the Unit

I. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Artificial Intelligence: A Modern Approach	S. Russell and P. Norvig	Forth Edition	Pearson
2.	"Artificial Intelligence: Foundations of Computational Agents"	David L. Poole and Alan K. Mackworth	Second Edition	Cambridge University Press
Reference Book				
3.	Artificial Intelligence: A Systems Approach (Computer Science), Jones and Bartlett Publishers, Inc.;First Edition, M. Tim Jones.			
4.	"Artificial Intelligence: Structures and Strategies for Complex Problem Solving" by George F. Luger			
Online Resources				
5.	https://www.javatpoint.com/artificial-intelligence-ai			
6.	https://onlinecourses.nptel.ac.in/noc20_cs42/preview			

Code :BADECE5111	RPA Tools	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall RPA concepts and terminology
CO – 02	Understand	Explain the principles and techniques of RPA
CO – 03	Apply	Demonstrate the use of UI Path Studio for automation tasks
CO – 04	Analyze	Analyze workflows and data structures in RPA projects
CO – 05	Evaluate	Evaluate the efficiency and effectiveness of RPA solutions

B. MAPPING MATRIX OF CO, PO, & PSO

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2				-	-	-	-	-	-	-	2	1	
CO2	3	2	3	2	1	-	-	-	-	-	-	-		2	3
CO3	3	1	3	3	2	-	-	-	-	-	-	-			1
CO4	3		2	1	3	-	-	-	-	-	-	-			1
CO5	3	2	3	2	3	-	-	-	-	-	-	-			
WT. AVG	3	1.75	2.75	2	2.25								2	1.5	1.6666667

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Robotic Process Automation concepts**	**8**
2.	**Sequence and Data Manipulation**	**8**
3.	**Overview of UiPath**	**7**
4.	**Control Flow Activities and Selectors**	**7**
5.	**Automation**	**6**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Robotic Process Automation concepts**
	☐ Introduction to Unit ☐ Introduction to RPA: Scopes and techniques of RPA, About UiPath, The future of automation ☐ Record and Play: Record and Play, UiPath stack, Installing and Learning UiPath studio, Task recorder ☐ Conclusion of unit
2.	**Sequence and Data Manipulation**

	Introduction to Unit Sequence: Sequence, Flowchart and Control Flow, sequencing the workflow, control flow, various types of loops and decision making, step by step example using sequence, Flowchart and Control Flow. Data Manipulation: Variables and scope, Collections, Arguments, Collections, Clipboard management, File operation, CSV/Excel to data table and vice versapass, break, else. Conclusion of Unit
3.	**Overview of UiPath**
	Introduction to Unit User Interface Components: Ribbon, Toolbars Access, Library panel, project panel, Outline panel, locals panel, Debugging, Recording, Workflow execution, context menu, properties panel, Designer panel, Universal search bar. Workflow Design and UiPath Studio: Layout diagrams, Type of Decisions, switch activity, Flow Decision, Flow switch, Naming conventions, managing variables in studio, types of variables, Managing arguments, Argument panel, Types of recording, Automatic recording, Basic, web and desktop automatic recording, Manual recording, Data scraping, Screen scrapping and its methods. Conclusion of Unit
4.	**Control Flow Activities and Selectors**
.	Introduction of Unit Control Flow Activities and Selectors: Assign activity, Delay activity, While activity, Do while activity, If activity, Switch activity, For each activity and Break activity. Selectors and input/output methods: Simple selectors, Generation of Dynamic selectors, passing the variables in selectors, Input methods, Output methods, Full text, Native, OCR. Built-In Class Attributes Conclusion of Unit
5	**Automation**
.	Introduction of Unit Excel and PDF Automation: Reading and working with rows of excel, Looping with excel, Working with PDF and excel files, retrieving data from web. Email Automation: Outlook Email activity, Get IMAP mail activity, Get POP3 mail message, get exchange mail activity, sending and receiving mail messages. Orchestrator: Dashboard, Robots, Processes, Jobs, Queues, Schedules, Transaction clause regular expressions Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Learning Robotic Process Automation with UiPath	Alok Mani Tripathi	Latest	Packt
2	Intelligent Control: A stochastic optimization approach	Amitava Chatterjee, Anjan Rakshit, and Kaushik Das Sharma	Latest	Springer edition
3	Learning Robotic Process Automation with UiPath	Alok Mani Tripathi	Latest	Packt
Online Resources				
1	https://www.uipath.com/			
2	https://www.udemy.com/course/robotic-process-automation/			

Code :BCSECE5111	Network Defence for Cyber- Security	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall basic concepts and importance of different network topologies.
CO – 02	Understand	Interpreting threats to information security, logical and conceptual model that defines network communication.
CO – 03	Apply	Apply and articulateabout the different network security techniques.
CO – 04	Analyze	Correlatingknowledge of different network defense tools
CO – 05	Evaluate	Experimenting with different network monitoring tools and provide security to network.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	2	-
CO3	-	-	3	2	-	-	-	-	-	-	-	-	-	2	-
CO4	-	-	-	1	-	-	-	-	-	-	-	-	-	3	2
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
WT. AVG	2	3	3	1.5									3	2.3	2

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Networks**	**07**
2.	**Computer Networks**	**08**
3.	**Network Security**	**08**
4.	**Tools for Network defense**	**07**
5.	**Network Monitoring**	**07**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Networks**
	• Introduction of Unit • Importance of Networks • LAN,WAN,MAN • Types of cables in computer networks • Network Topologies • Conclusion of Unit
2.	**Computer Networks**
	• Introduction to Unit • OSI and TCP/IP model • Protocols and Port, IPv4, IPv6, MAC address • Attacks on CIA triad • Conclusion of the Unit
3.	**Network Security**

	• Introduction of Unit • Access Rights Management • End Point Security • Firewalls • Intrusion Prevention/Detection • Network Access Control • Security Monitoring • Wireless Security • Conclusion of Unit
4.	**Tools for Network defense**
.	• Introduction of Unit • Open source tools for Network defense • Wireshark, Sort, Nessus • Air-crack, Wi-Fi pumpkin, Cain & Abel • TCP dump, Splunk • Conclusion of Unit
5	**Network Monitoring**
.	• Introduction of Unit • Network Breach • Firewalls and its types • DMZ and its importance • Malware analysis, SSL/ TLS, UTM/SIEM • Software Defined Networking (SDN) vulnerabilities • Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Network Security And Cyber Defense (CNSS)	Gurpej Singh	2021	Notion Press
2	Basic of Hacking and Penetration	Patrick Engerbrestson	First Edition	2010
1	Hacking: A Beginners' Guide to Computer Hacking, Basic Security, And Penetration Testing, John Slavio			
2	Hacking Exposed - Stuart McClure, Joel Scambray, George Kurtz			
1	https://www.javatpoint.com/ethical-hacking-tutorial			
2	https://www.guru99.com/ethical-hacking-tutorials.html			

Code :BCGECE5111	Cloud Web Services	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Understand Cloud Computing and Amazon Web Services.
CO – 02	Understand	Understand AWS Technologies like EC2(Elastic Compute Cloud) and Create Servers and AMI (Amazon Machine Image)
CO – 03	Apply	Deploy Web Applications with the help of AWS Elastic Beanstalk. And Understand Cloud Watch for create alarms and alerts
CO – 04	Analyze	Understand AWS Cloud Security with the help of IAM (identity Access management)
CO – 05	Evaluate	Understand and create AWS Storage (S3,Buckets,) and

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	2	2	3
CO2	3	-	2	-	-	-	-`	-	-	-	-	-	1	2	3
CO3	2	-	2	-	-	-	-	-	-	-	-	-	1	2	3
CO4	2	1	-	2	-	-	-	-	-	-	-	-	1	2	3
CO5	3	-	2	-	-	-	-	-	-	-	-	-	1	2	3
WT. AVG	2.6	1.5	2	2									1.2	2	3

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Cloud Computing and Amazon Web Services**	8
2.	**Introduction to EC2**	8
3.	**Web Applications**	7
4.	**AWS Security**	7
5.	**AWS Storage**	6

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Cloud Computing and Amazon Web Services**
	• Introduction of Unit • Introduction to Cloud Computing, Cloud Service Delivery Models (IAAS, PAAS, SAAS), • Cloud Deployment Models (Private, Public, Hybrid and Community), • Cloud Computing Security, • Introduction to Amazon Web Services, AWS Storage Options, • AWS Compute Options, AWS Database Options, AWS Workflow Automation • AWSOrchestration Options, AWS Systems Management and Monitoring Options, • AWS Virtual, Private Cloud Introduction

	• Conclusion of Unit
2.	**Introduction to EC2**
	• Introduction of Unit • Introduction To EC2, Instance Types And Uses, Auto scaling Instances, • Amazon Machine Images (AMIS), Modifying Existing Images, • Creating New Images of Running Instances, • Converting An Instance Store AMI To An EBS AMI, • Instances Backed By Storage Types, • Conclusion of Unit
3.	**Web Applications**
	• Introduction of Unit • Introduction to Elastic Beanstalk, Deploying Scalable Application On AWS, • Selecting And Launching An Application Environment, • Provisioning Application Resources with Cloud formation, • Introduction to Cloud Watch, Describe Amazon Cloud Watch metrics and alarms, • AWS Messaging Services(SNS,SQS,SES). • Conclusion of Unit
4.	**AWS Security**
.	• Introduction of Unit • Introduction to AWS Security, Describe Amazon Identity and Access Management (IAM), • AWS Directory Service, AWS Key Management Service • Securing Data at Rest and In Motion • Conclusion of Unit
5	**AWS Storage**
.	• Introduction of Unit • Amazon Storage, S3 Storage Basics, Buckets and Objects, • Creating A Web Server Using S3 Endpoints, Managing Voluminous Information with EBS • Glacier Storage Service , Describe Amazon Dynamo, • Understand key aspects of Amazon RDS, Launch an Amazon RDS instance, • Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg, Andrzej M. Goscinski		John Wiley and Sons Publications
2	Cloud Computing For Dummies	Fern Halper,Robin Bloor		
Reference Book				
1	Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online - Michael Miller - Que 2008			
2.	Cloud Computing: Web-Based Applications That Change the Way You Work by micheal Miller			
Online Resources				
1	https://www.javatpoint.com/aws-tutorial			
2	https://www.w3schools.com/aws/index.php			

Code :BAMECE5111	SAS® Visual Text Analytics in SAS®	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Use the point-and-click interface of Model Studio and SAS Visual Text Analytics.
CO – 02	Understand	Explore collections of text documents to discover key topics. Interpret term maps.
CO – 03	Apply	Identify key textual topics automatically in your large document collections.
CO – 04	Analyze	Create robust models for categorizing the content according to your organization's specific needs.
CO – 05	Evaluate	Create, modify, and enable (or disable) custom concepts and test linguistic rule definitions with validationchecks within the same interactive GUI.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO10	PO11	PO12	PSO 1	PSO 2	PSO 3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	2	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	**2.2**	**2.6**	**1.4**	**1.2**									**1.8**		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to SAS Visual Text Analytics	
2.	SAS Visual Text Analytics Demonstrations	
3.	**SAS Visual Text Analytics Nodes**	
4.	**Concept and Category Rule Definitions**	
5.	**Case Studies**	

DETAILED SYLLABUS

1. Unit	Unit Details
1.	Introduction to SAS Visual Text Analytics
	▪ SAS Visual Text Analytics. ▪ Language challenges (self-study).
2.	SAS Visual Text Analytics Demonstrations
	● Converting documents for analysis using SAS Data Explorer. ● Creating a project with no predefined concepts. ● Creating a project with custom concepts.
3.	**SAS Visual Text Analytics Nodes**
	● Projects. ● Concepts and terms. ● Machine-generated topics. ● Categories. ● Document scoring.
4.	**Concept and Category Rule Definitions**
.	● SAS Visual Text Analytics rules. ● SAS Visual Text Analytics concept rules. ● SAS Visual Text Analytics demo category rules.
5	**Case Studies**
.	● Retrieving information about anxiety and depression from drug reports. ● Automatic categorization of ASRS incident reports. ● Retrieving mortgage complaints from the CFPB customer complaints data.

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Cloud Computing: Principles andParadigms	Rajkumar Buyya, James Broberg, Andrzej M. Goscinski		John Wiley and Sons Publications
2.	Cloud Computing For Dummies	Fern Halper,Robin Bloor		
Reference Book				
3.	Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online -Michael Miller - Que 2008			
4.	Cloud Computing: Web-Based Applications That Change the Way You Work by micheal Miller			

B.1 Theory: Professional Elective - V

Code : BCEECE5112	Mobile Application Development	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Remember data management techniques such as Shared Preferences, SQLite Database, and Firebase.
CO – 02	Understand	Grasp the concepts of user experience, including application context, intents, layouts, and user input controls.
CO – 03	Apply	Apply cross-platform development tools like Ionic, PhoneGap, and Xamarin Studio for designing user interfaces.
CO – 04	Analyze	Analyze frameworks and features of cross-platform development tools like Ionic, PhoneGap, and Xamarin.
CO – 05	Evaluate	Assess the effectiveness of layouts, user input controls, and Android Manifest settings in user experience design.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO 1	PO 2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3
CO1	2	2	2	-	-	-	-`	-	-	-	-	-	1	-	-
CO2	2	2	2	-	-	-	-	-	-	-	-	-	1	-	-
CO3	1	2	3	1	-	-	-	-	-	-	-	-	3	-	-
CO4	1	2	3	3	-	-	-	-	-	-	-	-	3	-	-
CO5	2	2	2	-	-	-	-`	-	-	-	-	-	1	-	-
WT. AVG	1.6	2.0	2.4	2.0									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Android	08
2.	User Experience	09
3.	Background Processing	10
4.	Data Management	07
5.	Introduction to cross platform application development	06

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Android**
	• Introduction of Unit • Introduction to mobile application development,Android platform,Android Architecture Android SDK , Android Development Tools (ADT) • Android Virtual Devices (AVDs),Emulators,Dalvik Virtual Machine,Difference between JVM and DVM • Steps to install and configure Android Studio and SDK • Understanding project structure • Installing and running applications on Android Studio

	• Conclusion of Unitt
2.	**User Experience**
	• Introduction of Unit • Application Context: Activities, Services, Intents • Receiving and Broadcasting Intents • Android Manifest File and its common settings,Intent Filter,Permissions. • Layouts: Linear and Relative Layouts • Android User Input Controls : Button, Text Field, Seek bar, Checkbox, Radio Button, Toggle Button • Conclusion of Unit
3.	**Background Processing**
	• Introduction of Unit • Creating background tasks: AsyncTask, AsyncTaskLoader; • Network Connections. • Programming paradigms • Application Components Part 2: Services – bound/unbound services, Starting and stopping • services, Broadcast receivers, Content providers. • Triggering, scheduling and optimizing background tasks: Notifications, Alarms, • Transferringdata between Activities • Google API • Conclusion of Unit.
4.	**Data Management**
.	• Introduction of Unit • Data Access and Storage: Shared Preferences • App settings, Files & the Android File system, • SQLite Database, Loaders • Firebase. Programming paradigms • Content Providers and Content Resolvers • Conclusion of Unit
5	
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RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Android Programming: The Big Nerd Ranch Guide	Bill Phillips, Chris StewartKristinMarsicano, Brian Gardner	4th Edition	Big Nerd Ranch Guides
Reference Book				
1	Android Programming: The Big Nerd Ranch Guide			
	Pragmatic Flutter: Building Cross-Platform Mobile Apps for Android, iOS, Web & Desktop			
Online Resources				
1	https://www.youtube.com/watch?v=fis26HvvDII			
2	https://www.mygreatlearning.com/mobile-app-development/free-courses			
3	https://www.udacity.com/course/new-android-fundamentals--ud851			

Code : BADECE5112	R Programming	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall various data structures learned earlier and learn various data structures functionality in R.
CO – 02	Understand	Demonstrate an understanding of functions associated with various types of data structures used in R.
CO – 03	Apply	Apply to read and test different file format into R
CO – 04	Analyze	Analyse data by statistical approach
CO – 05	Evaluate	Evaluate data analysis by graphical approach and non-parametric testing of hypothesis for real world problems in R

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	1	-	-	-	-	-	-	-	-	-	2	1	-
CO2	2	2	2	-	-	-	-`	-	-	-	-	-	1	-	-
CO3	2	2	2	-	-	-	-	-	-	-	-	-	1	-	-
CO4	1	2	3	1	-	-	-	-	-	-	-	-	3	-	-
CO5	1	2	3	3	-	-	-	-	-	-	-	-	3	-	-
WT. AVG	1.8	2.0	2.2	2.0									2.0		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to R Environment	6
2.	Data Structures and Control Statements	8
3.	I/O operations and String Manipulations	7
4.	R for Summary Statistics and Parametric Tests	8
5.	R for Graphs, Nonparametric Tests and ANOVA	7

D. DETAILED SYLLABUS

1 Unit	Unit Details
1.	**Introduction to R Environment**
	History and development of R Statistical computing programming language. Installing R and R studio. Getting started with R. Creating new working directory. Changing existing working directory. Installing the available packages. Calling the installed packages, Variable definition in R. Simple functions, vector definition and logical expressions Matrix calculation and manipulation using matrix data types.
2.	**Data Structures and Control Statements**
	Introduction to different data types, vectors, atomic vectors, types and tests, coercion, lists, list indexing, Function applying on the lists, adding and deleting the elements of lists, attributes, name and factors, matrices and arrays, Matrix indexing, filtering on matrix, generating a covariance matrix. lapply( ) and sapply( ) on data frames, Control statements.
3.	**I/O operations and String Manipulations**
	Introduction to I/O functions in R Using of scan( ), readline ( ) function,

	Comparison and usage of scan and readline function, Reading different format files into R: text file, CSV file Statistical package files, xls and xlsx files. Converting from one format to another using in built function, Writing different file format in to the local machine directory, Basics of string manipulations – grep ( ), nchar ( ), paste( ), sprintf( ), substr( ), regexpr( ), strsplit( ), Testing of file name with given suffix.
4.	**R for Summary Statistics and Parametric Tests**
.	Descriptive statistics – summary statistics for vectors, making contingency tables, creating contingency tables from vectors. Testing tables and flat table objects, cross tables, testing cross tabulation, recreating original data from contingency tables, switching class, mean (arithmetic, geometric and harmonic), Median, mode for raw and grouped data, measure of dispersion – range, standard deviation, variance, coefficient of variation, testing of hypothesis – small sample test, large sample test – for comparing mean, proportion, variance (dependent and independent samples).
5	**Testing of Hypothesis: Non - Parametric Test**
.	Introduction to graphs, Box-Whisker Plot, Scatter plots, pairs plots, line chart, Pie Chart, Bar Charts Non-parametric test: The Wilcoxon U-Test (Mann-Whitney): One and Two-Sample U-Test, Tests for association: Chi Square Tests, Yates Correction for 2X2 Tables, single category goodness of fit tests, Analysis of Variance for one-way variation and two variation

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Beginning R: The statistical Programming Language	Dr. Mark Gardener	John Wiley & Sons, Inc.	1.
2.	The art of R programming	Norman Matloff	no starch Press, San Francisco	2.
3.	Introduction to Probability and Statistics for Engineers and Scientists	Owen Jones, Robert Maillardet and Andrew Robinson	CRC Press	3.
4.	The R Book	Hadley Wickham	CRC Press	4.
5.	Advanced R	Ken Black	Sixth Edition, John Wiley & Sons, Inc.	5.

Code : BCSECE5112	Security Audit and risk Management	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Learn and recall the fundamentals of processes and models used in the entertainment software industry of Game Technology
CO – 02	Understand	Understand the history of gaming hardware, including the evolution of console architecture over the decades.
CO – 03	Apply	Apply their knowledge of types and variations of input devices, including touch devices, controllers, keyboards, and mice
CO – 04	Analyze	Analyze the functions of a GPU in games, understanding its role in rendering graphics and enhancing gaming experiences
CO – 05	Evaluate	Evaluate the need for multi-threading in games, considering its importance in enhancing performance and responsiveness.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	1	-	-	-	-	-	-	-	-	3	2	3
CO2	3	3	2	2	-	-	-	-	-	-	-	-	3	2	2
CO3	2	2	3	1	-	-	-	-	-	-	-	-	2	1	1
CO4	2	2	1	-	-	-	-	-	-	-	-	-	2	1	-
CO5	2	1	1	-	-	-	-	-	-	-	-	-	1	-	-
WT. AVG	2.4	2.2	1.8	1.3									2.2	1.5	2.0

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Foundation for IT Audit, Assurance and Process**	**07**
2.	**Computer Assisted Audit Tools and Techniques**	**08**
3.	**Managing Information technology Audit System**	**08**
4.	**Security consideration for ERP Applications**	**07**
5.	**Risk Management**	**07**

D. DETAILED SYLLABUS

1 Unit	Unit Details
1.	**Foundation for IT Audit, Assurance and Process**
	Assurance Services - Need for Assurance - Characteristics of Assurance Services-Types of Assurance Services ECommerce and Electronic Funds Transfer - Future of electronic payment system. Audit Standards - Types of Auditors and their functions - Internal Audit Function and External Auditor. Audit Plan - Developing an Audit Schedule - Audit Budget - Preliminary Review - Audit Findings - Analysis Re-examination - Verification - Recommendations - Communication Strategy Conclusion of Unit
2.	**Computer Assisted Audit Tools and Techniques**
	Auditor Productivity Tools Data and Resource Management Flowcharting Techniques - Flowcharting as an analysis tool Developing Audit Data Flow Diagrams Appropriateness of flowcharting techniques Computer assisted tools for operational reviews Web Analysis tools

	Conclusion of Unit
3.	**Managing Information technology Audit System**
	Evaluating IT Audit Quality Criteria for assessing the audit Criteria for assessing the auditor Best Practices in IT Audit Planning IT Governance: Performance Measurement Metrics and Management - Metric Reporting and Independent Assurance Conclusion of Unit
4.	**Security consideration for ERP Applications**
.	Information Security Policy Security Standards - ISO 27002 and National Institute of Standards and Technology Information Security Controls Information Owner Responsibilities - Third- Party Responsibilities Intranet/Extranet Security Identity Theft E-Commerce Application Security as a strategic and structural problem Planning and Control Approach to E-Commerce Security Management Internet Security and Mobile Computing Security ERP Data Warehouse-Data Warehouse integrity checklist ERP-Security features of the basic component. Conclusion of Unit
5	**Risk Management**
.	Introduction to risk Source and evaluation of risks Risk management Evaluation of Risk Management Strategies Risk model Credit risk measurement and management Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Information Technology Control and Audit	Sandra Senft, Frederick Gallegos, Aleksandra Davis	4th	CRC Press, 2012.
2	Derivatives & Risk Management	R.P. Rustagi	Latest	Taxmann
Reference Book				
1	Information System Audit and Assurance, D P Dube, V P Gulati, Tata Mc-Graw Hill, 2008			
2	Micheal E.Whitman, Herbert J.Mattor, "Principles of Information Security", Course Technology, Delmar Cengage Learning, Fourth Edition, 2012.			
3	Jennifer L.Bayuk, Jason Healey, Paul Rohmeyer and Marcus Sachs, "Cyber Security Policy Guidebook", John Wiley Sons, Kindle Edition, 2012			
Online Resources				
1	http://www.isaca.org/			
2	https://www.youtube.com/watch?v=w0I4M82c1dc			
3	https://www.youtube.com/watch?v=yqgGyvw_diA			

Code: BCGECE5112	Java Script Framework: Node Js and Angular Js	3 Credits [LTP: 3-0-0]

COURSE OUTCOME

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	To understand the fundamental concepts of Node JS
CO – 02	Understand	To configure and Node js Module in applications
CO – 03	Apply	To understand Angular Js and its use cases
CO – 04	Analyze	To create and analyze the controller and filter
CO – 05	Evaluate	To implement module and service

B. CO-PO-PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	-	-	1	-	-	-	-	-	-	1	-	1	-	-
CO2	3	1	1	-	2	-	-	-	-	-	3	2	2	2	-	-
CO3	1	2	2	-	3	-	-	-	-	-	2	3	1	3	2	-
CO4	2	3	3	-	3	-	-	-	-	-	3	3	3	2	2	-
CO5	3	2	3	3	2	-	-	-	-	1	1	2	2	3	3	-

C. OUTLINE OF THE COURSE

Unit No.	Title of The Unit	Time required for the Unit (Hours)
1.	**Introduction : Node Js**	07
2.	**Node JS Modules**	08
3.	**Introduction: Angular JS**	08
4.	**Controllers Filters**	07
5.	**Modules and Services**	07

A. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction : Node Js**
	Introduction of Unit Introduction to Node JS, Features of Node.js, Who Uses Node.js? Advantages of Node JS, Traditional Web Server Model, Node.js Process Model Conclusion of Unit
2.	**Node JS Modules**
	Introduction of Unit Functions, Buffer- Creating Buffers, Writing to Buffers, Reading from Buffer, Convert Buffer to JSON, Concatenate Buffers, Compare Buffers, Copy Buffer, Slice Buffer, Buffer Length Module, Module Types -Core Modules, Local Modules, Module. Exports Open a File, Get File Information – Reading a File, Writing a File, Closing a File, Truncate a File, Delete a File, Create a Directory, Read a Directory, Remove a Directory Conclusion of Unit
3.	**Introduction: Angular JS**
	Introduction of Unit Introduction to AngularJS ,MVC Architecture, Conceptual Overview, Setting up the Environment, First Application, Understanding ng attributes Expressions and Data Biding: Number and String Expressions,Object Binding and Expressions,Working with Arrays,Forgiving Behavior,Understanding Data binding Working with Directives: Conditional Directives,Styles Directives,Mouse and Keyboard Events Directives Conclusion of Unit
4.	**Controllers and : Filters**
	Introduction of Unit

	Understanding Controllers:Programming Controllers & $scope object,Adding Behavior to a Scope Object,Passing Parameters to the Methods,Having Array as members in Controller Scope. Nested Controllers and Scope Inheritance.Multiple Controllers and their scopes Filters:Built-In Filters, Uppercase and Lowercase Filters ,Currency and Number Formatting Filters,OrderBy Filter, Filter Filter,Creating Custom Filter Conclusion of Unit
5.	**Modules and Services**
	Introduction of Unit Forms:Using Simple Form, Working with Select and Options, Input Validations, Using CSS classes, Form Events, Custom Model update triggers, Custom Validations Modules:, Module Loading and Dependencies, Recommended Setup of Application, Creation vs Retrieval. Services: Understanding Services, Developing Creating Services,Using a Service, Injecting Dependencies in a Service Routing:Introduction to SPA,Creating HTML Templates,Configuring Route Provider. Conclusion of Unit

c. RECOMMENDED STUDY MATERIAL

S. No	Text Books:	Author	Edition	Publication
1	Node.js in Action	Mike Cantelon, Marc Harter	Latest	Manning
2	Mastering Node.js: Build robust and scalable real-time server-side web applications efficiently	Sandro Pasquali (Author), Kevin Faaborg	Latest	Packt
3	**ang-book: The Complete Guide to Angular**	Carlos Taborda	2018	
4.	**Angular: From Theory To Practice**	Asim Hussain		
Reference Book				
3.	**Angular 6 for Enterprise Ready-Web Applications**			
4.	Node.js 8 the Right Way: Practical, Server-Side JavaScript That Scales			
5.	**Pro Angular**			
Online Resources				
6.	https://www.tutorialspoint.com/angularjs/index.htm			
7.	https://www.simplilearn.com/tutorials/nodejs-tutorial/nodejs-backend			
8.	https://www.mygreatlearning.com/academy/learn-for-free/courses/introduction-to-angularjs1			

Code : BAMECE5112	Data Analytics using R Programming	3 Credits [LTP: 3-0-0]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Defining the Big Data technology and its popular platform Hadoop.
CO – 02	Understand	Explain the Distributed Storage Systems and its implementation in Hadoop HDFS framework
CO – 03	Apply	Apply Mapping and Reducing algorithms to solvethe analytical problems on Big Data using MapReduce
CO – 04	Analyze	Analyze the Big Data using Hive HQL and PIG scripting programming languages.
CO – 05	Evaluate	Evaluate the Big Data technology with Machine Learning using R programming language.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	2	1	-
CO2	3	2	3	2	-	-	-	-	-	-	--	-	-	2	-
CO3	3	1	3	3	-	-	-	-	-	-	-	-	1	1	-
CO4	3	-	2	1	-	-	-	-	-	-	-	-	-	-	3
CO5	3	2	3	2	-	-	-	-	-	-	-	-	--	3	-
WT. AVG	3.0	1.8	2.8	2.0									1.5	1.8	3.0

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Big Data And Hadoop**	07
2.	**HDFS(Hadoop Distributed File System)**	08
3.	**Map Reduce**	08
4.	**Hadoop Eco System**	07
5.	**Data Analytics with R**	07

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Big Data And Hadoop**
	• Introduction of Unit • The Design of HDFS, HDFS Concepts, Command Line Interface, • Hadoop file system interfaces, Data flow, • Data Ingest with Flume and Scoop and Hadoop archives, • Hadoop I/O: Compression, Serialization, Avro and File-Based Data structures • Conclusion of Unit
2.	**HDFS (Hadoop Distributed File System)**
	• Introduction of Unit • The Design of HDFS, HDFS Concepts, Command Line Interface, • Hadoop file system interfaces, Data flow, • Data Ingest with Flume and Scoop and Hadoop archives, • Hadoop I/O: Compression, Serialization, Avro and File-Based Data structures • Conclusion of Unit
3.	**Map Reduce**
	• Introduction of Unit • Anatomy of a Map Reduce Job Run, Failures, • Job Scheduling, Shuffle and Sort, • Task Execution, Map Reduce Types and Formats, Map Reduce Features. • Conclusion of Unit
4.	**Hadoop Eco System**
.	• Introduction of Unit

	• Pig : Introduction to PIG, Execution Modes of Pig, • Comparison of Pig with Databases, Grunt, Pig Latin, • User Defined Functions, Data Processing operators. Hive : Hive Shell, • Hive Services, Hive Metastore, Comparison with Traditional Databases, • HiveQL, Tables, Querying Data and User Defined Functions. Hbase : HBasics, Concepts,Clients, Example, Hbase Versus RDBMS. • Big SQL : Introduction • Conclusion of Unit
5	**Data Analytics with R**
.	• Introduction of Unit • Machine Learning: Introduction, Supervised Learning, • Unsupervised Learning, Collaborative Filtering, • Big Data Analytics with BigR, • Conclusion with R

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Hadoop: The Definitive Guide	Tom White	Third Editon	O'reily
2.	Big Data Analytics	Seema Acharya, Subhasini Chellappan	2015	Wiley
Reference Book				
1.	Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007.			
2.	Jay Liebowitz, "Big Data and Business Analytics" Auerbach Publications, CRC press (2013)			
3.	Tom Plunkett, Mark Hornick, "Using R to Unlock the Value of Big Data: Big Data Analytics with Oracle R			
Online Resources				
1.	http://www.bdbanalytics.ir/media/1121/big-data-analytics_turning-big-data-into-big-money.pdf			
2.	https://www.techtarget.com/searchbusinessanalytics/definition/big-data-analytics			
3.	https://www.tutorialspoint.com/hadoop/hadoop_big_data_overview.htm			

B.2: Practical - Professional Elective Lab - II

Code : BCEECE5211	Artificial Intelligence Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall key concepts and algorithms in artificial intelligence, including machine learning, neural networks, and search algorithms.
CO – 02	Understand	Understand the theoretical foundations and principles behind various artificial intelligence techniques.
CO – 03	Apply	Apply artificial intelligence algorithms and tools to solve real-world problems and tasks.
CO – 04	Analyze	Analyze and evaluate the performance of artificial intelligence models and algorithms through experimentation and testing.
CO – 05	Evaluate	Evaluate the effectiveness and limitations of artificial intelligence solutions in different application domains.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	-	-	2	-	-	-	-	-	-	-	1	-	3
CO2	2	1	-	1	1	-	-	-	-	-	-	-	1	-	3
CO3	-	2	2	1	3	-	-	-	-	-	-	-	2	-	3
CO4	1	2	3	3	2	-	-	-	-	-	-	-	1	-	3
CO5	2	1	1	2	2	-	-	-	-	-	-	-	2	-	3
WT. AVG	1.7	1.4	2	1.7	2								1.4		3

C. LIST OF EXPERIMENTS

1	Introduction to PROLOG. Installation and working on PROLOG.
2	Write a program in PROLOG to implement simple facts and rules and perform various types of queries on them.
3	Write a program in PROLOG to design a pedigree chart like a Family-tree or a Food Query Chart.
4	Write a program in PROLOG to implement various types of objects and line segments.
5	Write a program in PROLOG to implement LIST as a Data Structure.
6	Write a program in PROLOG for computation of Recur.sive Functions, like, Factorial, Fibonacci numbers, etc.
7	Write a program in PROLOG to solve Monkey-Banana Problem.
8	Write a program which behaves like a Knowledge Based Expert System for Medical Diagnosis.
9	Write a program in PROLOG to solve n-queens Problem.
10	Write a program in PROLOG to solve Water-Jug Problem.
11	Introduction to PROLOG. Installation and working on PROLOG.
12	Write a program in PROLOG to implement simple facts and rules and perform various types of queries on them.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	"Prolog Programming for Artificial Intelligence"	Ivan Bratko	4nd Edition	Addison-Wesley
2	“Programming in Prolog”	W.P. Clocksin, C.S. Mellish	4th Edition	Springer
3	"Prolog Programming for Artificial Intelligence"	Ivan Bratko	4nd Edition	Addison-Wesley
Reference Book				
1	"The Art of Prolog: Advanced Programming Techniques" by Leon Sterling and Ehud Shapir			
2.	"Clause and Effect: Prolog Programming for the Working Programmer" by William F. Clocksin			
Online Resources				
1	Logic programming - Wikipedia			
2	Prolog Tutorial - javatpoint			
3	Prolog Tutorial (tutorialspoint.com)			

Code : BADECE5211	RPA Tool Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall RPA concepts and terminology
CO – 02	Understand	Explain the principles and techniques of RPA
CO – 03	Apply	Demonstrate the use of UI Path Studio for automation tasks
CO – 04	Analyze	Analyze workflows and data structures in RPA projects
CO – 05	Evaluate	Evaluate the efficiency and effectiveness of RPA solutions

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2				-	-	-	-	-	-	-	2	1	
CO2	3	2	3	2	1	-	-	-	-	-	-	-		2	3
CO3	3	1	3	3	2	-	-	-	-	-	-	-			1
CO4	3		2	1	3	-	-	-	-	-	-	-			1
CO5	3	2	3	2	3	-	-	-	-	-	-	-			
WT. AVG	3	1.7	2.7	2	2.2								2	1.5	1.6

C. LIST OF EXPERIMENTS

1	Install UiPath Studio and record a task.
2	Implement sequence in UiPath Studio.
3	Implement flowchart in UiPath Studio.
4	Implement sequence using the Assign activity, Do While activity, Delay activity, If activity.
5	Implement sequence using the Switch activity, While activity, For Each activity, Break activity.
6	Generate dynamic selector and pass the variables in selectors.
7	Implement and run example of screen scrapping through UiPath Studio.
8	Manipulate a set of data using UiPath Studio / Excel.
9	Set up and automate Excel and PDF through UiPath Studio.
10	Set up and automate Email using UiPath Studio.
11	Enable and implement text based automation using UiPath Studio.
12	Organize a process through orchestration using UiPath Studio.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Robotic Process Automation-Guide to building robot	Richard Murdoch	1st Edition	
2	Robotic Process Automation and Risk Mitigation: The Definitive Guide	Mary C. Lacity and Dr. Leslie P. Willcocks		SB Publishing, 2017
3	Introduction to robotic process automation	Frank Casale		Institute for Robotic Process Automation,
Reference Book				

1	“Learning Robotic Process Automation with UiPath” by Alok Mani Tripathi, Packt
2.	“The Robotic Process Automation Handbook: A Guide to Implementing RPA Systems” by Tom Taulli, Apress, 2020
Online Resources	
1	https://www.uipath.com/
2	https://www.udemy.com/course/robotic-process-automation/

Code : BCSECE5211	Network Defense for Cyber- Security Android Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Gain the knowledge of basic concepts and importance of different network topologies.
CO – 02	Understand	Evaluate threats to information security, logical and conceptual model that defines network communication.
CO – 03	Apply	Learn about the different network security techniques.
CO – 04	Analyze	Gain the knowledge of different network defense tools.
CO – 05	Evaluate	Acquire knowledge of different network monitoring tools and provide security to network.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	-	-	-	-	-	-	-	-	-	-	3	-	-
CO2	-	3	-	-	-	-	-	-	-	-	-	-	-	2	-
CO3	-	-	3	2	-	-	-	-	-	-	-	-	-	2	-
CO4	-	-	-	1	-	-	-	-	-	-	-	-	-	3	2
CO5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
WT. AVG	2.0	3.0	3.0	1.5									3.0	2.3	

C. LIST OF EXPERIMENTS

1	Network Scanning and Enumeration
2	Vulnerability Scanning
3	Simulate an attack to identify and exploit vulnerabilities.
4	Test the strength of password policies.
5	Identify and exploit common web application vulnerabilities.
6	Analyze malware behavior and understand its impact.
7	Detect and respond to network intrusions.
8	Test the effectiveness of email security measures and user awareness.
9	Practice responding to cybersecurity incidents.
10	Investigate network traffic to identify malicious activities.
11	Configure firewall rules and test their effectiveness.
12	Design and implement a secure network architecture using cisco packet tracer.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Network Security And Cyber Defense (CNSS)	Gurpej Singh	2021	Notion Press
2	Basic of Hacking and Penetration	Patrick Engerbrestson	First Edition	2010
Reference Book				
1	Hacking: A Beginners' Guide to Computer Hacking, Basic Security, And Penetration Testing, John Slavio			
2.	Hacking Exposed - Stuart McClure, Joel Scambray, George Kurtz			
Online Resources				
1	https://www.javatpoint.com/ethical-hacking-tutorial			
2	https://www.guru99.com/ethical-hacking-tutorials.html			
3	https://www.youtube.com/watch?v=dz7Ntp7KQGA			

Code : BCGECE5211	Cloud Web Services Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Articulate the main concepts, key technologies, strengths and limitations of cloud computing.
CO – 02	Understand	Explain the key and enabling technologies that help in the development of cloud.
CO – 03	Apply	Make use of NIST cloud computing architecture to solve architecture design challenges
CO – 04	Analyze	Explain the core issues of cloud computing such as resource management and security.
CO – 05	Evaluate	Install and use current cloud technologies. Ilustrate and choose the appropriate technologies, algorithms and approaches for implementation and use of cloud.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	2	1	1	-	-	-	-	1	2	2	-
CO2	-	-	-	-	2	1	1	-	-	-	-	1	2	2	-
CO3	-	-	-	-	3	2	2	1	-	-	-	1	2	3	-
CO4	-	-	-	-	2	1	1	-	-	-	-	1	2	2	-
CO5	-	-	-	-	3	2	2	1	-	-	-	1	2	3	-
WT. AVG					2.4	1.4	1.4	1				1	2	2.4	

C. LIST OF EXPERIMENTS

1	Create account on AWS, **Setting Up an EC2 Instance,** launching a virtual server using Amazon Elastic Compute Cloud (EC2). Configuring security groups, key pairs, and Elastic IP addresses.
2	Creating S3 storage bucket
3	Hosting a static website on Amazon Simple Storage Service (S3), Configuring bucket policies and static website hosting.
4	Deploying a sample web application using Elastic Beanstalk, Managing application versions and environment configurations.
5	Creating and configuring an Amazon Relational Database Service (RDS) instance. Connecting to the database using a client.
6	Creating a Lambda function to process data. Triggering Lambda functions with AWS services like S3 or DynamoDB.
7	Setting up an Auto Scaling group and an Elastic Load Balancer (ELB). Configuring scaling policies and monitoring performance.
8	Creating and managing Identity and Access Management (IAM) users, groups, and roles. Assigning permissions and policies.
9	Setting up CloudWatch alarms and dashboards. Configuring logs for EC2 instances and other AWS services.
10	Designing and implementing a VPC with subnets, route tables, and Internet Gateways. Configuring security groups and Network Access Control Lists (NACLs).
11	Setting up MFA for IAM users. Enforcing MFA policies.
12	Implementing Backup and Recovery with AWS Backup

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Cloud Computing: Principles and Paradigms	RajkumarBuyya, James Broberg, Andrzej M. Goscinski		John Wiley and Sons Publications
2	Cloud Computing For Dummies	Wesley J. Chun, Prentice Hall	1999	McGraw Hill International Edition
Reference Book				
1	Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online – Michael Miller - Que 2008			

Code : BAMECE5211	Neural network: Essentials (SAS) Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Understand the core concept of neural networks
CO – 02	Understand	Essential practices of real-world application
CO – 03	Apply	Discover how to adjust the models essential parameters to solve different types of business challenges
CO – 04	Analyze	Build autoencoders for a predictive model
CO – 05	Evaluate	Perform an intelligent automatic search of the model hyperparameter values

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	1	-	-	-	-	-	-	-	-	-	1	-	-
CO2	-	2	-	-	-	-	-	-	-	-	-	-	2	-	-
CO3	-	-	2	2	-	-	-	-	-	-	-	-	1	-	-
CO4	-	-	3	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	2	-	1	-	-	-	-	-	-	-	-	1	-	-
WT. AVG	3.0	2.0	2.0	1.5									1.4		

C. LIST OF EXPERIMENTS

1	Programmatically build neural networks in SAS 9.4 and SAS Viya.
2	Modify neural networks' parameters for better performance.
3	Conduct automatic search for neural networks' hyperparameters through genetic algorithm.
4	Enhance data with autoencoders and synthetic observations.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Cloud Computing: Principles and Paradigms	RajkumarBuyya, James Broberg,Andrzej M. Goscinski		John Wiley and Sons Publications
2	Cloud Computing For Dummies	Wesley J. Chun, PrenticeHall	1999	McGraw Hill International Edition
Reference Book				
3	Cloud Computing: Web-Based Applications That Change the Way You Work and Collaborate Online –Michael Miller - Que 2008			

B.2: Practical - Professional Elective Lab - III

Code : BAMECE5212 **Mobile Application Development Lab** **1 Credits [LTP: 0-0-2]**

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Students will be able to recall and identify key concepts, tools, and techniques used in mobile application development.
CO – 02	Understand	Students will understand the fundamental principles of mobile application design and development.
CO – 03	Apply	Students will be able to apply programming concepts and techniques to develop functional mobile applications.
CO – 04	Analyze	Students will analyze user requirements and translate them into technical specifications for mobile app development.
CO – 05	Evaluate	Students will evaluate the usability and effectiveness of mobile applications based on user feedback and testing.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	1	1	-	3	2	1	-	1	-	-	1	2	-	-
CO2	-	1	1	-	3	2	1	-	1	-	-	1	2	-	-
CO3	-	1	1	-	3	2	2	1	1	1	-	1	3	-	-
CO4	-	1	1	-	3	2	2	1	1	1	-	1	3	-	-
CO5	-	1	1	-	3	2	2	1	1	2	1	1	3	-	-
WT. AVG		**1**	**1**		**3**	**2**	**1.6**	**1**	**1**	**1.3**	**1**	**1**	**2.6**		

C. LIST OF EXPERIMENTS

1	Installation of Android studio.
2	Develop a program to display Hello World on screen.
3	Develop a program to implement linear layout and absolute layout.
4	Develop a program to implement frame layout, table layout and relative layout
5	Develop a program to implement Text View and Edit Text
6	Develop a program to implement Auto Complete Text View.
7	Develop a program to implement Button, Image Button and Toggle Button.
8	Develop a program to implement login window using above UI controls.
9	Develop a program to implement Checkbox.
10	Develop a program to implement Radio Button and Radio Group.
11	Develop a program to implement Progress Bar.
12	Develop a program to implement List View, Grid View, Image View and Scroll View
13	Develop a program to implement Custom Toast Alert
14	Develop a program to implement Date and Time Picker.
15	Develop a program to create an activity
16	Develop a program to implement new activity using explicit intent and implicitintent.
17	Develop a program to implement content provider
18	Develop a program to implement service
19	Develop a program to implement broadcast receiver.
20	Develop a program to implement sensors
21	Develop a program to build Camera.
22	Develop a program for providing Bluetooth connectivity
23	Develop a program for animation
24	Perform Async task using SQLite.
25	Create sample application with login module. (Check username and password) On successful login, Change TextView —Login Successful‖. And on login fail, alert user using Toast —Login fail‖
26	Create login application where you will have to validate username and password till the username and password is not validated, login button should remain disabled.
27	Develop a program to a) Send SMS b)Receive SMS

28	Develop a program to send and receive e-mail.
29	Deploy map based application.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Hello, Android Introducing Google's Mobile Development Platform	Ed Burnett	4th Edition, 2015	The Pragmatic Bookshelf
2.	Android Application Development in 24 Hours	Sams Teach Yourself	4th Edition, 2016	Sams
Reference Book				
1.	Head First Android Development: A Brain-Friendly Guide			
2.	Android Programming for Beginners by John Horton, 2016.			
Online Resources				
1	https://www.tutorialspoint.com/android			
2	https://www.tutorialspoint.com/android/android_advanced_tutorial.pdf			

Code : BADECE5212	R Programming Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Apply statistical methods for data exploration and visualization.
CO – 02	Understand	Utilize R for data preprocessing, cleaning, and transformation.
CO – 03	Apply	Implement R packages for machine learning and predictive modeling.
CO – 04	Analyze	Apply and analyze regression, classification, and clustering techniques.
CO – 05	Evaluate	Evaluate and communicate results from data analysis.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	3	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	2	2	-
CO3	-	-	3	-	-	-	-	-	-	-	-	-	-	-	2
CO4	-	-	-	3	2	-	-	-	-	-	-	-	-	-	2
CO5	-	-	-	-	2	-	-	-	-	-	-	-	-	-	
WT. AVG	2.5	3	3	3	2								2.5	2	2

C. LIST OF EXPERIMENTS

1	
2	Install and configure R, set working directory. Install Packages and calling installed packages R studio environment and functionalities of R studio Implement basic R operations (data input, missing values, importing data into R using different formats : xlsx, CSV, Text files) Use R as a calculator Explore various functionalities of dataframes. Create data set using data frames, list and tables. Create the contingency table for the given raw data. Create the interactive user input code line in r using readline ( ) function. Create the contingency table for the given vector format data. Convert the contingency table to original format of the given data. Analyse and give interpretation of summary statistics for the given data. Calculate mean, median and mode for the grouped data and compare the results for the given data. Analyse the given data for non-parametric tests and give the interpretations. Use R for test the given data In order to compare the effectiveness of two sources of nitrogen, namely ammonium chloride (NH4Cl) and urea, on grain yield of Coarse cereal, an experiment was conducted. The results on the grain yield of Coarse Cereal (kg/plot) under the two treatments are given below. NH4Cl : 13.4, 10.9, 11.2, 11.8, 14.0, 15.3, 14.2, 12.6, 17.0, 16.2, 16.5, 15.7. Urea : 12.0, 11.7, 10.7, 11.2, 14.8, 14.4, 13.9, 13.7, 16.9, 16.0, 15.6, 16.0. Assess which source of nitrogen is better for Coarse Cereal.

3	
4	Before an increasing in exercise duty on tea, 800 persons out of a sample of 1000 persons were found to be tea drinkers. After an increasing in duty, 800 people were tea drinkers in a sample of 1200 people. Using SE of a proportion, state whether there is a significant decrease in consumption of tea after the increase in the exercise duty. Use R for test the given data A health status survey in a few villages revealed that the normal serum protein value of children in that locality is 7.0 g/100ml. A group of 16 children who received high protein food for a period of six months had serum protein values shown below. Can we consider that the mean serum protein level of those who were fed on high protein diet is different from that of the general population? S.No. (Child No.) 1 2 3 4 5 6 7 8 Protein level (g%) 7.10 7.70 8.20 7.56 7.05 7.08 7.21 7.25 S.No. (Child No.) 9 10 11 12 13 14 15 16 Protein level (g%) 7.36 6.59 6.85 7.90 7.27 6.56 7.93 8.56 Students were selected to training. Their performance was noted by giving a test and the marks recorded out of 50. They were given effective 6 months training and again they were given a test and marks were recorded out of 50. Students 1 2 3 4 5 6 7 8 9 10 Before training 25 20 35 15 42 28 26 44 35 48 After training 26 20 34 13 43 40 29 41 36 46 By applying the t-test can it be concluded that the students have benefited by the training? 100 individuals of a particular race were tested with an intelligence test and classified into two classes. Another group of 120 individuals belong to another race were administered the same intelligence test and classified into the same two classes. The following are the observed frequencies of the two races: Race Intelligence elligent Non-intelligent Total Race I 42 58 100 Race II 55 65 120 Total 97 123 220 Test whether the intelligence is anything to do with the race. Obtain the correlation coefficient between the heights of father(X) and of the son (Y) from the following data X 65 66 67 68 69 70 71 72 Y 67 68 65 68 72 72 69 71 And also test its significance. Using R functions. Consider the inbuilt data set cars. Find Correlation between possible variables and pairwise correlation Find regression line between appropriate variables Display the summary statistics and comment on the results

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Beginning R: The statistical Programming Language	Dr. Mark Gardener		John Wiley & Sons, Inc
2.	The R Book	Michael J. Crawley		Wiley & Sons, Inc
Reference Book				
1	R graphics cookbook: practical recipes for visualizing data. O'Reilly Media, 2018, Chang, Winston.			
Online Resources				
1.	https://www.simplilearn.com/			
2.	https://www.w3schools.com/			

Code : BCSECE5212	Database Security Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	To implement security features in DBMS scenarios.
CO – 02	Understand	Formulate query, using SQL, solutions to a broad range of query contains DML DDL & DCLlikecreate,delete, Drop commands and data update problems.
CO – 03	Apply	Use a desktop database package in PI/SQL to create, populate, maintain, and query a database including Courser & Locking
CO – 04	Analyze	Implement security in database schema
CO – 05	Evaluate	Learn implementation of secure database in real life.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	1	-	-	-	-	-	-	-	-	-	1	1	1
CO2	2	2	1	-	-	-	-	-	-	-	-	-	2	2	1
CO3	3	3	2	1	-	-	-	-	-	-	-	-	3	2	2
CO4	3	3	3	2	-	-	-	-	-	-	-	-	3	3	2
CO5	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
WT. AVG	2.4		2.0	2.0									2.8	2.7	1.5

C. LIST OF EXPERIMENTS

1	Installation of SQL Server on the Microsoft Windows Server 2008 R2 operating system.
2	Demonstration of SQL Injection Vulnerability.
3	Demonstrate Broken Authentication and Session Management vulnerability in database
4	plementing server-side Input Validation over client-side input validation.
5	Create and use application roles.
6	e and implement the schema for security.
7	ate and use symmetric and asymmetric encryption keys.
8	Create and use certificates
9	n and implement to encrypt data with symmetric keys.
10	Learn and implement to encrypt data with asymmetric keys and use certificates.
11	Learn and use DML trigger for auditing data modification.
12	Learn and use DDL trigger for auditing data modification.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Microsoft SQL Server 2012 Security Cookbook	Rudi Bruchez	First published: September 2012	Packt Publishing Ltd.
2	Securing SQL Server, DBAs Defending the Database	Peter A. Carter	2018	Apress
Reference Book				
1	Fernández E. B., Summers R. C., and Wood C.. "Database security and integrity. The Systems Programming Series." Addison-Wesley; First. Edition, (1981).			
2.	Blokdyk Gerardus, " Database security A Complete Guide", 5starcooks publications. 2019 Edition			
Online Resources				
1	https://www.infosecinstitute.com/skills/learning-paths/database-security/			
2	https://www.youtube.com/watch?v=Q5aTUc7c4jg&list=PLOspHqNVtKAAXDobTc9kBWwnfgzNV2k_a			

Code : BCGECE5212	Java Script Framework: Angular and node.js Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Understand working operation of JS
CO – 02	Understand	Reduce the amount of code you write to build rich user interface applications
CO – 03	Apply	Increase the reliability and maintainability of UI by using data binding
CO – 04	Analyze	Modularise your code with the custom services and directives

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	-	-	1	-	-	-	-	-	-	1	-	1	-
CO2	3	1	1	-	2	-	-	-	-	-	3	2	2	2	-
CO3	1	2	2	-	3	-	-	-	-	-	2	3	1	3	2
CO4	2	3	3	-	3	-	-	-	-	-	3	3	3	2	2
WT. AVG	1.8	0.0	2.0	0.0	2.3						2.7	2.3	1.5	2.5	2.0

C. LIST OF EXPERIMENTS

1	Write a Node.js script that creates a simple HTTP server. The server should listen on port 3000 and respond with the text "Hello, World!" when accessed.
2	Create a Node.js module named calculator.js that exports functions for addition and subtraction. Then, create another file app.js to import the module and use its functions to perform addition of two numbers and log the result.
3	Write a Node.js script that reads the content of a file named input.txt and writes that content to a new file named output.txt.
4	Write a Node.js script that performs a simple GET request to fetch data from the JSONPlaceholder API (https://jsonplaceholder.typicode.com/todos/1) and logs the data to the console.
5	Write a Node.js script using the Express framework to create a simple API with two routes:
6	Perform the following task on Student tables. ● Implement functions to perform SQL JOIN operations to retrieve data from multiple related tables. ● Implement sorting functionality to sort data based on different columns. ● Write queries to filter data based on specific conditions. ● Group data based on certain criteria and perform aggregate functions like COUNT, SUM, AVG, etc.
7	Write a program Looping Over Data
8	Write a program Formatting Data for Display
9	Write a program More Reusable Components
10	Write a program Responding to an Event
11	Write a program Create a Form to Edit Your Data
12	Write a program Communicating from Child to Parent Component

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	**ng-book: The Complete Guide to Angular**	Carlos Taborda	2018	
2.	**Angular: From Theory To Practice**	Asim Hussain		
3.	**Angular 6 for Enterprise Ready-Web Applications**			
Reference Book				
4.	Node.js 8 the Right Way: Practical, Server-Side JavaScript That Scales			
5.	Pro Angular			
Online Resources				
6.	https://www.tutorialspoint.com/angularjs/index.htm			
7.	https://www.w3schools.com/angular/			
8.	https://www.tutorialspoint.com/angularjs/index.htm			

Code : BAMECE5212	Data Analytics using R Programming Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Apply statistical methods for data exploration and visualization.
CO – 02	Understand	Utilize R for data preprocessing, cleaning, and transformation.
CO – 03	Apply	Implement R packages for machine learning and predictive modeling.
CO – 04	Analyze	Apply and analyze regression, classification, and clustering techniques.
CO – 05	Evaluate	Evaluate and communicate results from data analysis.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	3	-	-	-	-	-	-	-	-	-	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	2	2	-
CO3	-	-	3	-	-	-	-	-	-	-	-	-	-	-	2
CO4	-	-	-	3	2	-	-	-	-	-	-	-	-	-	2
CO5	-	-	-	-	2	-	-	-	-	-	-	-	-	-	
WT. AVG	2.5	3	3	3	2								2.5	2	2

C. LIST OF EXPERIMENTS

	PART - A
1	Install and configure R, set working directory.
2	Install Packages and calling installed packages.
3	R studio environment and functionalities of R studio
4	Implement basic R operations (data input, missing values, importing data into R using different formats : xlsx, CSV, Text files)
5	Use R as a calculator/
6	Explore various functionalities of dataframes.
7	Create data set using data frames, list and tables.
8	Calculate the remainder after dividing 31079 into 170166719.
9	Calculate the interest earned after 5 years on an investment of $2000,
10	Assuming an interest rate of 3% compounded annually.
11	Use R to calculate the area of a circle with radius 7 cm.
12	Do you think there is a difference between 48:14ˆ2and 48:(14ˆ2)?
13	Usingrep()and seq()as needed, create the vectors? 0000011111222223333344444 and 12345123451234512345123345
14	Create the vector ## [1]00011110001111000111100011110001 1 ## [34] 1 1 and convert it to a factor. Identify the levels of the result, and then change the level labels to obtain the factor: ## [1] Male Male Male Female Female Female Female Male Male ## [10] Male Female Female Female Female Male Male Male Female ## [19] Female Female Female Male Male Male Female Female Female ## [28] Female Male Male Male Female Female Female Female ## Levels: Male FemaleExplore various functionalities of plots
	PART - B
15	Create the contingency table for the given raw data.
16	Create the interactive user input code line in r using readline ( ) function.

17	Create the contingency table for the given vector format data.
18	Convert the contingency table to original format of the given data.
19	Analyse and give interpretation of summary statistics for the given data.
20	Calculate mean, median and mode for the grouped data and compare the results for the given data.
21	Analyse the given data for non-parametric tests and give the interpretations.
22	Use R for test the given data In order to compare the effectiveness of two sources of nitrogen, namely ammonium chloride (NH4Cl) and urea, on grain yield of Coarse cereal, an experiment was conducted. The results on the grain yield of Coarse Cereal (kg/plot) under the two treatments are given below. NH4Cl : 13.4, 10.9, 11.2, 11.8, 14.0, 15.3, 14.2, 12.6, 17.0, 16.2, 16.5, 15.7. Urea : 12.0, 11.7, 10.7, 11.2, 14.8, 14.4, 13.9, 13.7, 16.9, 16.0, 15.6, 16.0. Assess which source of nitrogen is better for Coarse Cereal.
23	Use R to test the given data and interpret the results. In a health survey of school children, it is found that the mean hemoglobin level of 55 boys is10.2 per 100ml with a SD of 2.1. Can we consider this group as taken from a population with a mean of 11.0 g/100ml?
24	In a hearing survey among 246 town school children, 36 were found with conductive hearing loss and among 349 village school children 61 were found with conductive hearing loss. Does this present any evidence that conductive hearing loss is as common among town children as among village children?
25	In an experiment to compare two types of Goat foods A and B, the following results of increase in weight were observed in Goats. Assuming the two samples are independent can we conclude food B is better than food A?
26	Before an increasing in exercise duty on tea, 800 persons out of a sample of 1000 persons were found to be tea drinkers. After an increasing in duty, 800 people were tea drinkers in a sample of 1200 people. Using SE of a proportion, state whether there is a significant decrease in consumption of tea after the increase in the exercise duty.
27	Use R for test the given data A health status survey in a few villages revealed that the normal serum protein value of children in that locality is 7.0 g/100ml. A group of 16 children who received high protein food for a period of six months had serum protein values shown below. Can we consider that the mean serum protein level of those who were fed on high protein diet is different from that of the general population?
28	Students were selected to training. Their performance was noted by giving a test and the marks recorded out of 50. They were given effective 6 months training and again they were given a test and marks were recorded out of 50. By applying the t-test can it be concluded that the students have benefited by the training?
29	100 individuals of a particular race were tested with an intelligence test and classified into two classes. Another group of 120 individuals belong to another race were administered the same intelligence test and classified into the same two classes. The following are the observed frequencies of the two races:

Goat No.	1	2	3	4	5	6	7
Increase in weightduetoA	49	53	51	52	47	50	52
Increase in weightduetoB	52	55	52	53	50	54	54

S.No. (Child No.)	1	2	3	4	5	6	7	8
Proteinlevel(g%)	7.10	7.70	8.20	7.56	7.05	7.08	7.21	7.25

S.No. (Child No.)	9	10	11	12	13	14	15	16
Proteinlevel(g%)	7.36	6.59	6.85	7.90	7.27	6.56	7.93	8.56

Farmers	1	2	3	4	5	6	7	8	9	10
Before training	25	20	35	15	42	28	26	44	35	48
After training	26	20	34	13	43	40	29	41	36	46

	<table><tr><td rowspan="2">Race</td><td colspan="3">Intelligence</td></tr><tr><td>Intelligent</td><td>Non-intelligent</td><td>Total</td></tr><tr><td>Race I</td><td>42</td><td>58</td><td>100</td></tr><tr><td>Race II</td><td>55</td><td>65</td><td>120</td></tr><tr><td>Total</td><td>97</td><td>123</td><td>220</td></tr></table> Test whether the intelligence is anything to do with the race.
28	Obtain the correlation coefficient between the heights of father(X) and of the son (Y) from the following data <table><tr><td>X</td><td>65</td><td>66</td><td>67</td><td>68</td><td>69</td><td>70</td><td>71</td><td>72</td></tr><tr><td>Y</td><td>67</td><td>68</td><td>65</td><td>68</td><td>72</td><td>72</td><td>69</td><td>71</td></tr></table> And also test its significance. Using R functions.
29	Analyze the given data for analysis of variance and interpret the same for all the possible values.
30	Consider the inbuilt data set cars. ● Find Correlation between possible variables and pairwise correlation ● Find regression line between appropriate variables Display the summary statistics and comment on the results

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
3.	Beginning R: The statistical Programming Language	Dr. Mark Gardener		John Wiley & Sons, Inc
4.	The R Book	Michael J. Crawley		Wiley & Sons, Inc
Reference Book				
1	R graphics cookbook: practical recipes for visualizing data. O'Reilly Media, 2018, Chang, Winston.			
Online Resources				
3.	https://www.simplilearn.com/			
4.	https://www.w3schools.com/			

D: Ability Enhancement Courses (AEC)

Code : BUACHM5214	PROFESSIONAL SKILLS-II	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Produce the skills for reporting the experience, education, & expertise sections on social media.
CO – 02	Understand	Use correct grammar, tone, and formatting in formal writing.
CO – 03	Apply	Examine the interview and presentation's weak spots and areas for improvement.
CO – 04	Analyze	Estimate basic factors such as personal skills & abilities and their role in career.
CO – 05	Evaluate	Employ negotiation and team skills dynamics.

B. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Personal Branding**	**2**
2.	**Professional Writing-II**	**8**
3.	**Presentation Skills: Professional Setting**	**2**
4.	**Job Interview & Group Discussion: Preparation by Mock Practice**	**4**
5.	**Negotiation Skills, Team Management & Professional Awareness**	**8**

C. List of Labs

LIST OF LABS	
1.	Personal Branding: Its best practices
2.	Professional Writing II: Abstract Writing, Statement of purpose and other formal documents
3.	Expanding Professional Vocabulary
4.	Resume Building-II: Revising & Updating
5.	E-Learning & E-Content Development-II
6.	Presentation Skills in Professional Setting
7.	Job Interviews II: Preparation and Presentation for Mock Interviews
8.	Advanced Group Discussion-II: Analysis of professional GD Videos and Practices on Topics/Video/Article based topics
9.	Negotiation Skills & and Conflict Resolution II
10.	Change and Transition Management
11.	Team Building Strategies: Project Management
12.	Career Awareness & Productive Mindset

E: Skill Enhancement Courses (SEC)

Code : BULCSE5201	Skill Enhancement Generic Course-V	1 Credits [LTP: 0-0-2]

COURSE OUTCOMES: On completion of the course a student will be able to:

- Understand basic problems based on arithmetic and soft skills area which are asked in aptitude test taken by companies
- Effectively solve these problems by applying the knowledge earned.
- Actively participate in group discussion / meetings / interviews and prepare & deliver presentations.
- Become more effective individual through goal/target setting, self-motivation and practicing creative thinking.
- Function effectively in multi-disciplinary and heterogeneous teams through the knowledge of team work, Inter-personal relationships, conflict management and leadership quality

LIST OF ACTIVITIES	
1	Problems on Age, Cause & Effect
2	Career Development, Stress Management
3	Conflict Management, Data Interpretation
4	Sitting Arrangements
5	Written Communication, Behavioral interview skills
6	Error Detection, Confusing words
7	Number series, Speed, Time & distance
8	Linear Equations, Points, lines & angles
9	Allegations & Mixtures, Data sufficiency
10	Articles & Prepositions, Modal Verbs & Conditional Tense
11	Pronouns, Adverbs & Adjectives, Emotional Intelligence
12	Managing pressure & maintaining balance

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B. TECH - CSE

THIRD YEAR
SEMESTER: VI

COMMON SYLLABUS

For

Cyber Security
Cloud & Full Stack Development
Artificial Intelligence & Data Science
Artificial Intelligence & Machine Learning

BATCH: 2024-28

A: Major (Core Courses)

Code : BCECCE6101	Information System Security	3 Credits [LTP: 3-0-0]

F. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Explain conventional cryptographic techniques and their applications in securing information.
CO – 02	Understand	Apply symmetric and asymmetric cryptographic techniques such as DES, AES, RSA, and IDEA to encrypt and decrypt data
CO – 03	Apply	Identify common program security issues such as buffer overflows, viruses, and man-in-the-middle attacks, and explain their potential impact.
CO – 04	Analyze	Implement various internet security protocols such as SSL, TLS, and SET to protect online transactions and communications.
CO – 05	Evaluate	Design a comprehensive network security strategy incorporating various controls and mechanisms to address network threats and ensure security.

G. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	1	-	-	-	-	-	-	-	-	3	2	1
CO2	3	2	1	-	-	-	-	-	-	-	-	-	3	2	1
CO3	3	3	2	1	-	-	-	-	-	-	-	-	3	2	2
CO4	3	3	3	2	-	-	-	-	-	-	-	-	3	3	2
CO5	3	2	2	2	-	-	-	-	-	-	-	-	3	2	2
WT. AVG	**3**	**2.4**	**1.8**	**1.5**									**3**	**2.2**	**1.6**

H. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Information Security	7
2.	Encryption and Authentication Techniques.	8
3.	Risk Management	8
4.	Internet Security	7
5.	Network Security	7

I. DETAILED SYLLABUS

1. Unit	Unit Details
1.	Introduction to Information Security
	Introduction to Information Security: Attacks, Vulnerability, Security Goals, Security Services and mechanisms, Conventional Cryptographic Techniques: Conventional substitution and transposition ciphers, One-time Pad, Block cipher and Stream Cipher, Steganography . Conclusion of the Unit
2.	Encryption and Authentication Techniques.
	Symmetric and Asymmetric Cryptographic Techniques : DES, AES, RSA algorithms, International Data Encryption Algorithm (IDEA), Digital Certificates, Private Key Management, The PKIX Model Authentication and Digital Signatures Conclusion of the Unit

3.	Risk Management
	Key management – Kerberos, Program Security : Nonmalicious Program errors – Buffer overflow, Incomplete mediation, Time-of-check to Time-of- use Errors, Viruses, Trapdoors, Salami attack, Man-inthe- middle attacks, Covert channels Conclusion of the Unit
4.	Internet Security
.	Internet Security Protocols: Introduction, Basic Concepts, Secure Socket Layer (SSL), Transport Layer Security (TLS), Secure Hyper Text Transfer Protocol (SHTTP), Time Stamping Protocol (TSP), Secure Electronic Transaction (SET), SSL Versus SET, 3D Secure Protocol, Electronic Money, Email Security Wireless Application Protocol (WAP) Security Conclusion of the Unit
5	Network Security
.	Security in Networks : Threats in networks, Network Security Controls – Architecture, Encryption, Content Integrity, Strong Authentication, Access Controls, Wireless Security, Honeypots, Traffic flow security, Firewalls – Design and Types of Firewalls, Personal Firewalls, IDS, Email Security – PGP,S/MIME Conclusion of the Unit

J. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Information Security Risk Analysis	Thomas R.Peltier	Third Edition,Pub: Auerbach,2012	Information Security Risk Analysis
2.	Mark Stamp's Information Security: Principles and Practice (WIND)	DevenN.Shah, Wiley(2009)	Wiley,2009	MarkStamp's Information Security:Principles And Practice (WIND)
3.	Information Systems Security: Security Management, Metrics, Frameworks and Best Practices	NinaGodbole	Wiley,1sted;2008	Information systems Securi ty:Security Mana gement, Metric s, Frameworks and Best Practices
Reference Book				
1	Cryptography and Network security, Atul Khate			
2.	Cryptography And Network Security Principles And Practice, Fourth or Fifth Edition, William Stallings, Pearson			
Online Resources				
1	https://www.programiz.com/c-programming/examples			
2	https://www.w3resource.com/c-programming-exercises			

Code : BCECCE6102 Data Mining Techniques and Application 3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Innovative methods and concepts to understand the types of the data to be mined and present a general classification of tasks.
CO – 02	Understand	Recognize the preprocessing methods for any given raw data.
CO – 03	Apply	Demonstrating interesting patterns, measurement and rule based data from large amounts of data.
CO – 04	Analyze	Integrate and evaluate suitable data mining algorithms to build analytical applications.
CO – 05	Evaluate	Comprehend the organization of data in the form of data warehouse and advanced concepts.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	1	1	-	-	-	-	-	-	-	1	3	2	-
CO2	3	3	1	2	-	-	-	-	-	-	-	1	2	3	-
CO3	3	3	1	2	-	-	-	-	-	-	-	1	3	3	-
CO4	3	3	2	2	-	-	-	-	-	-	-	1	3	3	-
CO5	3	3	2	2	-	-	-	-	-	-	-	1	3	3	-
WT. AVG	3	3	1.4	1.8								1	2.8	2.8	

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Data Mining**	**07**
2.	**Classification**	**08**
3.	**Cluster Analysis**	**07**
4.	**Association Rule Mining and Visualization**	**08**
5.	**Data warehousing**	**07**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Data Mining**
	Introduction to Data Mining Data Mining Tasks, Components of Data Mining Algorithms Data Mining supporting Techniques, Major Issues in Data Mining Measurement and Data, Data Preprocessing, Data sets Conclusion of Unit
2.	**Classification**
	Introduction to Classification Basic Concepts, Decision Tree induction, Bayes Classification Methods Rule Based Classification, Model Evaluation and Selection, Techniques to Improve Classification Accuracy, Classification: Advanced concepts Bayesian Belief Networks, Classification by Back Propagation Support Vector Machine, Classification using frequent patterns. Conclusion of Unit
3.	**Cluster Analysis**
	Introduction to Cluster Analysis Basic concepts and Methods, Partitioning methods

	Hierarchical methods, Density Based Methods Grid Based Methods, Evaluation of Clustering Advanced Cluster Analysis: Probabilistic model based clustering, Clustering High Dimensional Data, Clustering Graph and Network Data, Clustering with Constraints. Conclusion of Unit
4.	**Association Rule Mining and Visualization**
.	Introduction to Association Rule Mining Large Item sets, Basic Algorithms, Parallel and Distributed Algorithms Comparing Approaches, Incremental Rules Advanced Association Rule Techniques Measuring the Quality of Rules, Introduction to Visualization Visualization of Multidimensional Data Diagrams for Multidimensional visualization Visual Data Mining, Data Mining Applications, Case Study: WEKA. Conclusion of Unit
5	**Data warehousing**
.	Introduction to Data warehousing Data warehousing components, Multi dimensional data model Data warehouse architecture, Data warehouse implementation Mapping the data warehouse to multiprocessor architecture OLAP, Need, Categorization of OLAP Tools Introduction to Data Cube, Data Cube Technology: Efficient Methods for Data Cube Computation Exploration and Discovery in Multidimensional Databases Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Data Mining Concepts and Techniques	Jiawei Han and Micheline Kamber	Third Edition	Elsevier
2.	Principles(Adaptive Learning) of Data Computation and MiningMachine	MiningMachine	David J. Hand, Heikki Padhraic Smyth	Mannila
3.	Data Mining: Introductory and Advanced Topics	Margaret H Dunham	Latest	Pearson
Reference Book				
1	Data Mining Concepts and Techniques, Author Jiawei Han and Micheline Kamber, August 2000			
2.	Principles of Data Mining (Adaptive Computation and Machine Learning), David J. Hand, Heikki Mannilaand Padhraic Smyth			
3	Data Mining: Introductory and Advanced Topics, Margaret H Dunham, Pearson			
Online Resources				
1	https://www.educba.com/data-mining-concepts-and-techniques/			
2	https://nptel.ac.in/courses/106105174			
3	https://onlinecourses.swayam2.ac.in/cec20_cs12/preview			

Code : BCECCE6103	Salesforce	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	To understand the fundamental concepts of Customer Relationship Management (CRM) and Salesforce architecture.
CO – 02	Understand	To configure and customize Salesforce applications, including objects, fields, and data management.
CO – 03	Apply	To understand the security architecture of Salesforce and to apply security controls to protect data within Salesforce.
CO – 04	Analyze	To create and analyze reports and dashboards to support business decision-making
CO – 05	Evaluate	To implement process automation using workflows, process builder, and flows.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	-	-	1	-	-	-	-	-	-	1	-	1	-
CO2	2	2	1	-	2	-	-	-	-	-	2	2	2	2	-
CO3	3	3	2	-	3	-	-	-	-	-	3	2	3	2	2
CO4	3	3	3	-	3	-	-	-	-	-	3	3	3	3	2
CO5	3	3	3	3	2	-	-	-	-	1	2	3	3	3	3
WT. AVG	2.4	2.4	2.3	3.0	2.2					1.0	2.5	2.2	2.8	2.2	2.3

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Cloud Computing & Salesforce and CRM basics	6
2.	Data Modelling and UI customization	8
3.	Data Security	8
4.	Reports And Dashboards	6
5.	Automation & Workflows	8

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Cloud Computing & Salesforce and CRM basics**
	Introduction to Cloud computing Deployment models of cloud computing, Cloud service models Introduction to Salesforce, CRM, benefits of CRM History and Evolution of Salesforce, Types of Salesforce Environments
2.	**Data Modelling and UI customization**
	Apps & Tabs in Salesforce Fields in Salesforce, Global picklist and dependency in salesforce Introduction to Salesforce Objects: Standard objects & custom objects Object Relationships (Master-Detail, Lookup, Junction) Formula Fields & Roll-up summary field., Managing Records: Creating and Editing Records Page Layouts, Mini page layouts, Publisher Layouts, Search, List Views, and Reports
3.	**Data Security**
	Introduction to Data Security concepts: encryption, authentication, and authorization.

	Level of Data Access in Salesforce User Management (Control Access to the Organization), Object Level Security in Salesforce (Profile Level) Field Level Security in Salesforce, Record Level Security in Salesforce Managing Password Policies, Restrict Login Access by IP Address Restrict Login Access by Tim, Permission Sets in Salesforce Organization Wide Default in Salesforce (OWD) Role Hierarchy in Salesforce, Sharing Rules in Salesforce, Manual Sharing.
4.	**Reports And Dashboards**
.	Introduction to Reports Types of Reports (Tabular, Summary, Matrix, Joined) Creating and Customizing Reports, Report Filters, Formulas, and Summaries Dashboards in Salesforce, Creating and Managing Dashboards Adding and Customizing Dashboard Components Dynamic Dashboards
5	**Automation and Workflows**
.	Introduction to Process Automation Benefits of Automation in Salesforce Workflow Rules, Creating Workflow Rules Workflow Actions (Email Alerts, Tasks, Field Updates) Time-Dependent Workflows, Approval Processes Creating Approval Processes, Approval Steps and Actions Managing and Monitoring Approval Requests

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Salesforce For Dummies	Tom Wong, Liz Kao, and Matt Kaufman	7th Edition	For Dummies
2.	Learning Salesforce Visual Workflow and Process Builder	Rakesh Gupta and Johann Furmann	1st Edition	Packt Publishing
3.	Salesforce Handbook	Wes Nolte and Jeff Douglas	1st Edition	CreateSpace Independent Publishing Platform
Reference Book				
1	"Practical Salesforce.com Development Without Code" by Philip Weinmeister			
2.	"Advanced Apex Programming for Salesforce.com and Force.com" by Dan Appleman			
Online Resources				
1	**URL**: Trailhead by Salesforce			
2	**URL**: Salesforce Documentation			

Code : BCECCE6201	Data Mining Lab	1 Credits [LTP: 0-0-2]

E. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall fundamental concepts of data mining such as data preprocessing, classification, clustering, association rule mining, and anomaly detection.
CO – 02	Understand	Understand the principles and techniques of data mining, including supervised and unsupervised learning algorithms, pattern discovery, and model evaluation methods.
CO – 03	Apply	Apply data preprocessing techniques such as data cleaning, normalization, and dimensionality reduction to prepare data for mining.
CO – 04	Analyze	Analyze data using classification algorithms such as decision trees, logistic regression, and support vector machines to predict categorical outcomes.
CO – 05	Evaluate	Evaluate the performance of data mining models using appropriate evaluation metrics such as accuracy, precision, recall, F1-score, and ROC curves.

F. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	3	-	-	1	-	2	-	1	-	-	1	3	-	2
CO2	-	3	-	-	1	-	2	-	1	-	-	1	3	-	2
CO3	-	3	-	-	1	-	2	-	1	-	-	1	3	-	2
CO4	-	3	-	-	1	-	2	-	1	-	-	1	3	-	2
CO5	-	3	-	-	1	-	2	-	1	-	-	1	3	-	2
WT. AVG		**3**			**1**		**2**		**1**			**1**	**3**		**2**

G. LIST OF EXPERIMENTS

1	Demonstration of preprocessing on dataset student.arff
2	Demonstration of preprocessing on dataset labor.arff
3	Demonstration of Association rule process on dataset contactlenses. arff using aprioris algorithm
4	Demonstration of Association rule process on dataset test.arff using apriori algorithm
5	Demonstration of classification rule process on dataset student.arff using j48 Algorithm
6	Demonstration of classification rule process on dataset employee.arff using j48 algorithm
7	Demonstration of classification rule process on dataset employee.arff using id3 algorithm
8	Demonstration of classification rule process on dataset employee.arff using naïve bayes algorithm
9	Demonstration of clustering rule process on dataset iris.arff using simple k-means
10	Demonstration of clustering rule process on dataset student.arff using simple kmeans
11	Usage of WEKA for visualization of data set student.arff.
12	Usage of WEKA for visualization of data set employee.arff.

H. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Data Mining Concepts and Techniques	Jiawei Han and MichelineKamber	Third Edition	Elsevier
2	Principles of Data Mining (Adaptive Computationand Machine Learning)	David J. Hand, Heikki Mannilaand Padhraic Smyth	Latest	Springer
3	Data Mining: Introductory and Advanced Topics	Margaret H Dunham	Latest	Pearson Education , 2006
Reference Book				

1	Data Mining Concepts and Techniques, Jiawei Han and Micheline Kamber,Third Edition Elsevier
2.	Principles of Data Mining (Adaptive Computation and Machine Learning), David J. Hand, Heikki Mannila andPadhraic Smyth
3	Data Mining: Introductory and Advanced Topics, Margaret H Dunham Latest,Pearson Education, 2006
Online Resources	
1	https://www.javatpoint.com/data-mining
2	https://nptel.ac.in/courses/106105174

Code : BCECCE6202	Salesforce Lab	2 Credits [LTP: 0-0-1]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	To understand the fundamental concepts of Customer Relationship Management (CRM) and Salesforce architecture.
CO – 02	Understand	To configure and customize Salesforce applications, including objects, fields, and data management.
CO – 03	Apply	To understand the security architecture of Salesforce and to apply security controls to protect data within Salesforce.
CO – 04	Analyze	To create and analyze reports and dashboards to support business decision-making
CO – 05	Evaluate	To implement process automation using workflows, process builder, and flows.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	-	-	1	-	-	-	-	-	-	1	-	1	-
CO2	2	2	1	-	2	-	-	-	-	-	2	2	2	2	-
CO3	3	3	2	-	3	-	-	-	-	-	3	2	3	2	2
CO4	3	3	3	-	3	-	-	-	-	-	3	3	3	3	2
CO5	3	3	3	3	2	-	-	-	-	1	2	3	3	3	3
WT. AVG	2.4	2.4	2.3	3.0	2.2					1.0	2.5	2.2	2.8	2.2	2.3

C. LIST OF EXPERIMENTS

1	Salesforce Platform Overview, Navigating the Salesforce interface, Setting up a Salesforce Developer Account
2	Creating custom objects and fields, Understanding standard objects, Field types and properties
3	Creating relationships (Lookup)
4	Creating Master-Detail Relationship
5	Creating and managing user profiles, Role hierarchy and sharing rules
6	Configuring field-level security, Creating and assigning permission sets
7	Setting up organization-wide defaults, Creating and managing sharing rules
8	Creating and activating workflow rules, Workflow actions (tasks, email alerts, field updates)
9	Building custom reports, Using report types and filters
10	Creating and customizing dashboards, Adding components to dashboards

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Salesforce For Dummies	Tom Wong, Liz Kao, and Matt Kaufman	7th Edition	For Dummies
2	Learning Salesforce Visual Workflow and Process Builder	Rakesh Gupta and Johann Furmann	1st Edition	Packt Publishing
3	Salesforce Handbook	Wes Nolte and Jeff Douglas	1st Edition	CreateSpace Independent Publishing Platform
Reference Book				
1	"Practical Salesforce.com Development Without Code" by Philip Weinmeister			
2.	"Advanced Apex Programming for Salesforce.com and Force.com" by Dan Appleman			
Online Resources				
1	**URL**: Trailhead by Salesforce			
2	**URL**: Salesforce Documentation			

B: Minor Stream Courses/ Department Electives
B1: Theory - Professional Elective - VI

Code : BCEECE6111	Advance Web Technology	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	To understand the fundamental concepts of Node JS
CO – 02	Understand	To configure and Node js Module in applications
CO – 03	Apply	To understand Angular Js and its use cases
CO – 04	Analyze	To create and analyze the controller and filter
CO – 05	Evaluate	To implement module and service

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	-	-	1	-	-	-	-	-	-	1	-	1	-
CO2	3	1	1	-	2	-	-	-	-	-	3	2	2	2	-
CO3	1	2	2	-	3	-	-	-	-	-	2	3	1	3	2
CO4	2	3	3	-	3	-	-	-	-	-	3	3	3	2	2
CO5	3	2	3	3	2	-	-	-	-	1	1	2	2	3	3
WT. AVG	2.0		2.0	3.0	2.2					1.0	2.3	2.2	2.0	2.7	2.0

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction : Node Js**	07
2.	**Node JS Modules**	08
3.	**Introduction: Angular JS**	08
4.	**Controllers Filters**	07
5.	**Modules and Services**	07

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction : Node Js**
	Introduction of Unit Introduction to Node JS, Features of Node.js, Who Uses Node.js? Advantages of Node JS, Traditional Web Server Model, Node.js Process Model Conclusion of Unit
2.	**Node JS Modules**
	Introduction of Unit Functions, Buffer- Creating Buffers, Writing to Buffers, Reading from Buffer, Convert Buffer to JSON, Concatenate Buffers, Compare Buffers, Copy Buffer, Slice Buffer, Buffer Length Module, Module Types -Core Modules, Local Modules, Module. Exports Open a File, Get File Information – Reading a File, Writing a File, Closing a File, Truncate a File, Delete a File, Create a Directory, Read a Directory, Remove a Directory

	Conclusion of Unit
3.	**Introduction: Angular JS**
	Introduction of Unit Introduction to AngularJS ,MVC Architecture, Conceptual Overview, Setting up the Environment, First Application, Understanding ng attributes Expressions and Data Biding: Number and String Expressions,Object Binding and Expressions,Working with Arrays,Forgiving Behavior,Understanding Data binding Working with Directives: Conditional Directives,Styles Directives,Mouse and Keyboard Events Directives Conclusion of Unit
4.	**Controllers and : Filters**
.	Introduction of Unit Understanding Controllers:Programming Controllers & $scope object,Adding Behavior to a Scope Object,Passing Parameters to the Methods,Having Array as members in Controller Scope. Nested Controllers and Scope Inheritance.Multiple Controllers and their scopes Filters:Built-In Filters, Uppercase and Lowercase Filters ,Currency and Number Formatting Filters,OrderBy Filter, Filter Filter,Creating Custom Filter Conclusion of Unit
5	**Modules and Services**
.	Introduction of Unit Forms:Using Simple Form, Working with Select and Options, Input Validations, Using CSS classes, Form Events, Custom Model update triggers, Custom Validations Modules:, Module Loading and Dependencies, Recommended Setup of Application, Creation vs Retrieval. Services: Understanding Services, Developing Creating Services,Using a Service, Injecting Dependencies in a Service Routing:Introduction to SPA,Creating HTML Templates,Configuring Route Provider. Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Node.js in Action	Mike Cantelon, Marc Harter	Latest	Manning
2.	Mastering Node.js: Build robust and scalable real-time server-side web applications efficiently	Sandro Pasquali (Author), Kevin Faaborg	Latest	Packt
3.	**ang-book: The Complete Guide to Angular**	Carlos Taborda	2018	
4	**Angular: From Theory To Practice**	Asim Hussain		
Reference Book				
1	**Angular 6 for Enterprise Ready-Web Applications**			
2.	Node.js 8 the Right Way: Practical, Server-Side JavaScript That Scales			
3	**Pro Angular**			
Online Resources				
1	https://www.tutorialspoint.com/angularjs/index.htm			
2	https://www.simplilearn.com/tutorials/nodejs-tutorial/nodejs-backend			
3	https://www.mygreatlearning.com/academy/learn-for-free/courses/introduction-to-angularjs1			

Code : BADECE6111	Deep Learning	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Understand the foundational concepts of deep learning and its applications in computer vision, enabling students to grasp the underlying principles of convolutional neural networks (CNNs) and their architecture.
CO – 02	Understand	Develop practical skills in implementing and training deep learning models for image classification, enabling students to build accurate classifiers for different visual recognition tasks.
CO – 03	Apply	Gain proficiency in object detection techniques, allowing students to create systems capable of identifyingand localizing multiple objects within images
CO – 04	Analyze	Acquire expertise in semantic segmentation, enabling students to perform pixel-wise image segmentation foradvanced computer vision tasks like medical image analysis.
CO – 05	Evaluate	Master face detection and tracking algorithms, empowering students to design real-time systems capable of detecting and tracking faces in video streams

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	2	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2.2	2.6	1.4	1.2									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Deep Learning basics for computer Vision**	8
2.	**Image Classification**	8
3.	**Object Detection**	7
4.	**Semantic Segmentation**	8
5.	**Face detection and tracking**	8

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Deep Learning basics for computer Vision**
	Introduction to Unit Introduction to deep learning and its applications in computer vision. Fundamentals of convolutional neural networks (CNNs) and their architecture. Activation functions, pooling layers, and regularization techniques for CNNs. Optimizers and loss functions for training deep learning models. Hands-on implementation of a basic image classification model Conclusion of unit
2.	**Image Classification**

	Introduction to Unit Understanding image classification tasks and dataset preparation. Training deep learning models for image classification with various architectures. Fine-tuning pre-trained models for custom classification tasks. Evaluating model performance using metrics like accuracy and confusion matrix. Project: Building an image classification system for a specific domain Conclusion of unit
3.	**Object Detection**
	Introduction to unit Introduction to object detection and its applications. Region-based CNNs (R-CNN), Fast R-CNN, and Faster R-CNN for object detection. Single Shot Multibox Detector (SSD) and You Only Look Once (YOLO) for real-time detection. Handling multi-class object detection and overlapping objects. Project: I Conclusion of unit
4.	**Semantic Segmentation**
.	Understanding semantic segmentation and its importance in computer vision. Fully Convolutional Networks (FCNs) for pixel-wise image segmentation. U-Net architecture for medical image segmentation and other applications. Evaluating segmentation models using metrics like Intersection over Union (IoU). Project: Building a semantic segmentation model for a specific use case
5	**Face detection and tracking**
.	Introduction to face detection and tracking algorithms. Haar Cascades for face detection and Viola-Jones algorithm. Deep learning-based face detection using Single Shot Multibox Detector (SSD) and RetinaNet. Face tracking techniques using Kalman filters or correlation-based tracking. Project: Developing a real-time face detection and tracking system.

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	"Deep Learning forComputer Vision"	Rajalingappaa Shanmugamani	Latest	Packt Publishing
2.	"Computer Vision:Algorithms and Applications"	Richard Szeliski	Latest	Springer
Reference Book				
1	Deep Learning"Author: Ian Goodfellow, Yoshua Bengio, and Aaron CourvillePublisher: MIT Press			
Online Resources				
1	https://www.javatpoint.com/deep-learning			
2	https://www.simplilearn.com/tutorials/deep-learning-tutorial			

Code : BCSECE6111	Vulnerability Analysis and Penetration Testing	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Learn the concepts and foundations of computer security, Network security, cyber security
CO – 02	Understand	Identify Web Application Vulnerabilities and its types
CO – 03	Apply	Identify Vulnerabilities of IT Systems. Acquire the knowledge of basic security tools risk
CO – 04	Analyze	Practice the risk analysis on Web vulnerability scanner -acunetix tool
CO – 05	Evaluate	Acquire the knowledge of basic security tools risk

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	—	—	—	—	—	—	—	—	2	—	—
CO2	3	2	1	2	—	—	—	—	—	—	—	—	2	—	—
CO3	2	3	2	1	—	—	—	—	—	—	—	—	1	—	—
CO4	2	2	2	1	—	—	—	—	—	—	—	—	2	—	—
CO5	2	3	1	1	—	—	—	—	—	—	—	—	2	—	—
WT. AVG	**2.2**	**2.6**	**1.4**	**1.2**									**1.8**	**2.2**	**2.6**

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Fundamentals of Vulnerability Analysis**	**8**
2.	**Vulnerability Identification**	**8**
3.	**Vulnerability Analysis**	**7**
4.	**Risk Assessment**	**8**
5.	**Remediation**	**8**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Fundamentals of Vulnerability Analysis**
	Introduction of Unit Benefits, Methodology, Vulnerability Assessment, Reasons for Vulnerability Existence Steps for Vulnerability Analysis Conclusion of Unit
2.	**Vulnerability Identification**
	Introduction of Unit Web Application Vulnerabilities - Types, SQL-Injection, Blind Injection Detection, Cross-Site Scripting, Broken Authentication & Session Management, Insecure Direct Object References, Failure to restrict URL, Remote Code Execution. Network Protocols and Vulnerabilities - Overview of basic networking infrastructure and network protocols, IP, TCP, Routing protocols, DNS Conclusion of Unit
3.	**Vulnerability Analysis**
	Introduction of Unit Automatic metrics calculation and Tools: Networking security monitoring tools, Encryption tools, Web Vulnerability Scanner

	Web Vulnerability Scanner set-up and Configuration Vulnerability Assessment using Acunetix(Vulnerability assessment tool) Conclusion of Unit
4.	**Risk Assessment**
	Introduction of Unit Information Security Risk Assessment Quantitative and Qualitative Risk Assessment in Cyber Security Case study on Information Security Risk analysis Case study on Web Vulnerability scanners - acunetix Conclusion of Unit
5.	**Remediation**
	Introduction of Unit Network Defenses - Network defense tools, secure protocols, Firewalls, VPNs, Tor, I2P, Intrusion Detection and filters, Host - Based IDS vs Network-Based IDS, dealing with unwanted traffic: denial of service attacks Malicious Software and Software Security - Malicious Web, Internet Security Issues, Types of Internet Security Issues, Computer viruses, Spyware, Key-Loggers, Secure coding, Electronic and Information Warfare. Mobile platform security models- Android, ios Mobile platform security models, Detecting Android malware in Android markets Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Network Security Essentials: Applications and Standards	William Stallings	4th Edition	Prentice Hall,
2.	Introduction to Computer Security	Michael T Goodrich and Roberto Tamassia	2011	Addiaon Wesley
Reference Book				
1	William Stallings, Network Security Essentials: Applications and Standards, Prentice Hall, 4 th edition, 2010			
2.	Alfred J.Menezes, Paul C. van OOrschot and Scott A. Vanstone, Handbook of Applied Cryptography, CRC Press, 2001.			
3	Computer Networks, PHI, Tanenbaum, Andrew S,			
Online Resources				
1	https://www.edx.org/course/cybersecurity-risk-management			
2	https://www.udemy.com/topic/computer-network/			
3	https://www.researchgate.net/publication/344803532_information_security_Risk_Analysis			
4	https://www.acuntetix.com			

Code : BCGECE6111	Linux Server Administration	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Defining of Linux Server.
CO – 02	Understand	Predicating server technology resources.
CO – 03	Apply	Determining Installing, configuring, and managing a Linux server and relevant services and applications.
CO – 04	Analyze	Distinguishing the importance of maintaining a secure Linux server.
CO – 05	Evaluate	ValidatingFile system management

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	-	-	1	-	-	-	-	-	-	1	-	1	-
CO2	3	1	1	-	2	-	-	-	-	-	3	2	2	2	-
CO3	1	2	2	-	3	-	-	-	-	-	2	3	1	3	2
CO4	2	3	3	-	3	-	-	-	-	-	3	3	3	2	2
CO5	3	2	3	3	2	-	-	-	-	1	1	2	2	3	3
WT. AVG	2.0	2.0	2.0	3.0	2.2					1.0	2.3	2.2	2.0	2.7	2.0

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Linux	07
2.	Package, User and group Management & File System Management	08
3.	Configuration Server and file system & Configuring Server for Security	08
4.	Managing system and infrastructure services Configuring Server for Security	07
5.	OpenSSH and Linux Security & Setting up Web Services	07

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Linux**
	Introduction of Unit History of Linux, Open source licensing , Unix Vs Linux - Flavors of Linux - Benefits and characteristics of Linux - Installation of Linux, Linux booting process , Log in and switch users in multiuser run levels , Shell and bash features , Linux kernel - sudo vs su ,Date and time configuration – Linux run levels. **Directories and files:** Directory structure , System directory , Absolute path and relative path ,Creating and removing directory ,Working with directory , Links – hard link and soft link , Input and output redirection , Filters and pipes ,Locate - read - and use system documentation including man page Working as root, working with the Shell, Using Bash to Best Effect, Managing Bash with Key Sequences, Performing Basic File System Management Tasks, Working with Directories, Working with Files, Viewing the Content of Text Files, Finding Files That Contain Specific Text , Creating Empty Files, Piping and Redirection, Piping, Redirection, Finding Files, Working with Vi Editor: Vi Modes, Saving and Quitting, Cut, Copy, and Paste, Deleting Text. Getting Help: Using man to Get Help,Getting Information on Installed Packages Conclusion of Unit

2.	**Package, User and group Management & File System Management**
	Introduction of Unit RPM ,YUM ,Archive , Compress - unpack and uncompressed files using tar , star , gzip - and bzip2 , Create, delete ,and modify local user accounts , Change passwords for local user accounts , Create - delete - and modify local groups and group memberships , Changing owner and modes. Mounting Disks, Using the mount Command, Un-mounting Devices, Automating Mounts with /etc/fstab, Checking File System Integrity, Working with Links: Working with Symbolic Links, Working with Hard Links. Configuring Storage, Comparing File Systems, Creating File Systems, Working with Logical Volumes Conclusion of Unit
3.	**Configuration local storage and file system**
	Introduction of Unit List ,create , delete , and partition type for primary , extended - and logical partitions - Create and remove physical volumes Assign physical volumes to volume groups , Create and delete logical Volumes. , Create ,mount ,unmount ,ext2 , ext3 and ext4 file systems. Mount ,unmount and LUKS-encrypted file systems ,Access control list. Setting Up User Accounts, Commands for User Management, Managing Passwords, Modifying and Deleting User Accounts, Configuration Files, Creating Groups, Commands for Group Management, /etc/group, Using Group Passwords, Managing the User's Shell Environment, Configuring Permissions, Read, Write, and Execute: The Three Basic Linux Permissions, Permissions and the Concept of Ownership, Working with Advanced Linux Permissions, Setting Permissions, Using umask to Set Default Permissions for New Files, Working with Access Control Lists, Preparing the File System for ACLs, ACL Limitations, Applying File Attributes, Apply Quota to Allow a Maximum Amount of Files, Installing the Quota Software, Preparing the File System for Quota, Initializing Quota, Setting Quota for Users and Groups, Configuring Administrator Tasks with sudo. Conclusion of Unit
4.	**Managing system and infrastructure services**
.	Introduction of Unit Managing system services - Shutting down - suspending and hibernating the system - Controlling systemd on remote machine - Creating and modifying systemd unit files DHCP Configuration - HTTP server Configuration - FTP server Configuration , Samba server Configuration Configuring DNS, Methods of Name Resolution, Structure of the DNS Hierarchy, Introducing Forward and Reverse DNS, Configuring DNS, Configuring Reversed Lookup, Testing Your Name Server, Mail server Configuration ,Configuring DHCP, Understanding the DHCP Protocol, Creating the DHCP Server Configuration, The DHCP Process, The /etc/dhcp/dhcpd.conf Configuration File, Advanced DHCP Configuration Options, NTP server Configuration ,NFS server Configuration ,Configuring NTP, How NTP Works, Configuring a Stand-Alone NTP Time Server, Pulling or Pushing the Time, Configuring an NTP Client, Checking NTP Synchronization Status, Customizing Your NTP Server, Starting Services with xinetd, Setting up Xinetd, Setting Up Mail Servers, Clients Conclusion of Unit
5	**OpenSSH and Linux Security & Setting up Web Services**
.	Introduction of Unit OPENSSH - The SSH Protocol, Configuring OpenSSH and Starting an OpenSSH Server KeyBased Authentication in OpenSSH - OpenSSH Clients Using the ssh Utility ,scp Utility and sftp Utility ,Configure firewall settings using system-configfirewall or iptables ,Set enforcing and permissive modes for SELinux - List and identify SELinux file and process context Setting up Apache, Apache Components, Starting, Stopping, and Testing the Apache Web Server, The Structure of the Apache Configuration Files, Checking the Configuration, Working with Virtual Hosts, Configuring Virtual Hosts, Managing Access to the Web Server, Configuring Host-Based Access Restrictions, Configuring User-Based Access Restrictions Enabling HTTPS, Creating a Self-Signed

	Certificate, Configuring Apache to Use the Self-Signed Certificate Using PHP, Setting Up MySQL, Setting the MySQL Root Password, Creating a MySQL Database, Configuring a Squid Proxy Server, Installing a Squid Proxy Cache, Configuring Squid Access Control Policies, Configuring User Authentication Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	RHCSA/RHCE Red Hat Linux Certification Study Guide Exams EX200 & EX300	Orsaria, Jang		McGraw-Hill Education
2.	Red Hat RHCSA/RHCE 7 Cert Guide: Red Hat Enterprise Linux 7 (EX200 and EX300)	Sander Van Vugt		Phi Learning Pvt Ltd
Reference Book				
1	Linux Pocket Guide: Essential Commands			
2.	Linux Administration			
3	Linux: The Complete Reference			
Online Resources				
1	https://www.udemy.com/topic/linux-administration/free/			
2	https://training.linuxfoundation.org/resources/			
3	https://intellipaat.com/linux-administration-training/			

Code : BAMECE6111	Deep Learning Using SAS® Software	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Define and understand deep learning.
CO – 02	Understand	Build models using deep learning techniques.
CO – 03	Apply	Apply models to score (inference) new data.
CO – 04	Analyze	Modify data for better analysis results.
CO – 05	Evaluate	Search the hyperparameter space of a deep learning model.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	-	-	-	-	-	-	-	-	2	-	-
CO2	3	2	1	2	-	-	-	-	-	-	-	-	2	-	-
CO3	2	3	2	1	-	-	-	-	-	-	-	-	1	-	-
CO4	2	2	2	1	-	-	-	-	-	-	-	-	2	-	-
CO5	2	3	1	1	-	-	-	-	-	-	-	-	2	-	-
WT. AVG	2.2	2.6	1.4	1.2									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Deep Learning	8
2.	Convolutional Neural Networks	8
3.	Recurrent Neural Networks	8
4.	Tuning a Neural Network	9
5.	Additional Topics	7

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Deep Learning**
	Introduction to neural networks. Introduction to deep learning., ADAM optimization., Dropout. Batch normalization., Autoencoders Building level-specific autoencoders (self-study)
2.	**Convolutional Neural Networks**
	Applications, Input layers., Convolutional layers. Padding, Pooling layers., Traditional layers Types of skip-layer connections, Image pre-processing and data enrichment. Training convolutional neural networks
3.	**Recurrent Neural Networks**
	Introduction Recurrent neural networks overview, Sub-types of recurrent neural networks. Time series analysis using recurrent neural networks. Sentiment analysis using recurrent neural networks

4.	**Tuning a Neural Network**
.	Selecting hyperparameters. Hyperband.
5	**Additional Topics**
.	Types of transfer learning. Transfer learning basics, Transfer learning strategies. Transfer learning with unsupervised pretraining. Customizations with FCMP.

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Professional Nosql	Shashank Tiwari		Wrox
2.	MongoDB in Action	KYLE BANKER PETER	Second Edition	Manning
Reference Book				
3.	NoSQL for Dummies, Adam Fowler, John Wiley & Sons, Inc			
4.	NoSQL Distilled, Pramod J. Sadalage & Martin Fowler, Pearson Education, Inc.			
5.	Making Sense of NoSQL, Dan McCreary& Ann Kelly, Manning Shelter Island			
Online Resources				
6.	https://www.javatpoint.com/nosql-databases			
7.	https://www.tutorialspoint.com/mongodb/index.htm			

B1: Theory - Professional Elective - VII

Code : BCEECE6112	Software Testing	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall fundamentals of software testing
CO – 02	Understand	Demonstrate clearly and effectively use the software testing terminology
CO – 03	Apply	Applysoftware testing in structured, organized, and dynamic ways
CO – 04	Analyze	Assess, Develop and validate a test plan and know about static and regression testing.
CO – 05	Evaluate	Evaluate test plans and test cases designed &use automatic testing tools.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	2	-	-	-	-	-	1	1	-	3	2	2
CO2	2	2	3	2	-	-	-	-	-	-	-	-	2	3	-
CO3	2	3	2	2	-	1	1	-	-	-	-	-	3	2	-
CO4	2	2	2	2	-	2	-	-	-	1	-	-	3	2	-
CO5	2	2	2	2	-	-	-	-	-	-	-	-	3	1	3
WT. AVG	2.2	2.2	2.0	2.0		1.5	1.0			1.0	1.0		2.8	2.0	2.5

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Testing Methodology**	08
2.	**Software Testing Terminology**	08
3.	**Dynamic Testing Techniques**	08
4.	**Static and Regression Testing**	09
5.	**Managing the Test Process and Test Automation**	07

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Testing Methodology**
	Introduction to Effective Software Testing, Evolution of Software Testing, Software Testing Myths, Goals of Software Testing, Psychology for Software Testing, Software Testing Definitions, Model for Software Testing, Role of tester- testing as a process Overview of Testing maturity model Defects -Hypothesis and tests
2.	**Software Testing Terminology**
	Definitions, Life Cycle of a Bug, Bug Classification based on SDLC, Testing Principles, Software Testing Life Cycle (STLC), Software Testing Methodology, Software Testing Strategy, Test Strategy Matrix,

	Verification and Validation, Verification and Validation Activities, How to verify Requirements and Objectives, Verification of High level Design, Verification of Data Design, Verification of Architectural Design, Verification of Low level Design, Unit Verification.
3.	**Dynamic Testing Techniques**
	Dynamic Testing : Black Box Testing Techniques, Boundary Value Analysis, Boundary value checking, Equivalence Class Testing, Identification of Equivalence classes, State Table based Testing, Decision Table based Testing, Cause Effect Graphing based Testing White Box Testing : Need of White box testing, Logic Coverage Criteria, Basis Path Testing, Control Flow Graph, Path Testing, Terminology, Cyclomatic Complexity, Formulae based on Cyclomatic complexity. Guidelines for Basis Path Testing, Applications of Path Testing, GraphMatrices Graph Matrix, Connection Matrix, Loop Testing, Data Flow Testing, Static Data flow testing, Dynamic Data flow testing, Mutation Testing, Mutation Testing Process.
4.	**Static and Regression Testing**
	Static Testing: Inspections, Inspection Process, Walkthroughs, Technical Reviews, Unit ValidationTesting, Integration Testing Types of Incremental Integration Testing, Pair-wise Integration, Path Based Integration, Function Testing, System Testing, Performance Testing, Usability Testing, performing the system tests, Acceptance Testing. Regression Testing: Progressive vs Regression Testing, Regression testing produces quality software Regression Testability, Objectives of Regression Testing, Regression Testing Types Regression Testing Techniques, Selective Retest Techniques, Strategy for Test Case Selection,Regression Test selection Techniques, Evaluating Regression Test Selection Techniques, Minimization Technique, Regression Test Prioritization, Types of Test case Prioritization, Prioritization Techniques, Prioritization Techniques, Code-based test case prioritization Vs coverage based test case prioritization.
5.	**naging the Test Process and Test Automation**
	Test Management, Test Organization, Test Planning, Test Plan Hierarchy, Master Test Plan, Verification Test Plan, Validation Test Plan, Unit Test Plan, Integration Test Plan, Function Test Plan, System Test Plan, Acceptance Test Plan, Detailed Test Design and Test Specifications, Test Log, Test Reports, Software Metric, Testing Metrics for Monitoring and Controlling the Testing Process, Testing Process Maturity Models. Automation and Testing Tools, Need of Automation, Categorization of Testing Tools Static and Dynamic Testing Tools, Testing Activity Tools, Selection of Testing Tools Costs incurred in Testing s, Guidelines for Automated Testing, Overview of some commercial Testing Tools.

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Software Testing-Principle and Practices	Naresh Chauhan	3rd	Oxford
2.	Fundamentals of Software Engineering,	RajibMall	PHI	2018
Reference Book				
1.	The Art of Software Testing, 3rd Edition by Glenford J. Myers, Corey Sandler, Tom Badgett.			
2.	Software Testing, 2nd Edition by Ron Patton			
Online Resources				
1.	https://www.javatpoint.com/software-testing-tutorial			
2.	https://www.guru99.com/software-testing.html			

Code : BADECE6112	NLP & Computer Vision	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	To understand the complete text preprocessing pipeline used to create vocabulary from the textual data and apply it to create vocabulary.
CO – 02	Understand	To understand the various architectures of RNNs: Sequence to Vector, Sequence to Sequence as well as Encoder-Decoder Architecture with Attention Mechanism and apply them on various NLP applications such as Sentiment Classification, Language Translation, Text Summarization, Drug Discovery, etc.
CO – 03	Apply	To delve deep into the Transformer Architecture, understanding its nuts and bolts. Applying the various concepts learned during this journey to improve other architectures and using these architectures to solve various problems in the domain of NLP.
CO – 04	Analyze	To apply the several concepts learned while delving deep into Transformers, for making transition towards Large Vision Models such as Vision Transformers.
CO – 05	Evaluate	To apply the Large Vision Models to solve various problems in Computer Vision.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	2	2	2	-	-	-	-	-	-	-	-	2	-	-
CO2	2	2	2	2	-	-	-`	-	-	-	-	-	2	-	-
CO3	2	1	2	2	-	-	-	-	-	-	-	-	2	-	-
CO4	2	2	3	3	-	-	-	-	-	-	-	-	2	-	-
CO5	2	2	3	3	-	-	-	-	-	-	-	-	2	-	-
WT. AVG	1.8	1.8	2.4	2.4									2.0		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Text Preprocessing	6
2.	RNNs for NLP	6
3.	Delving deep into Transformers	8
4.	Large Vision Models for Computer Vision	8
5.	Different Large Vision Models to solve problems	8

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Text Preprocessing**
	Introduction of Unit Various stages involved in Text Preprocessing: Tokenization, Normalization, Contractions Removal, Unnecessary tokens removal and stemming, POS Tagging and Lemmatization. State Of the Art techniques to create tokens: Byte Pair Encoding and Word Piece Tokenizer. Conclusion & Real-Life Application.
2.	**RNNs for NLP**
	Introduction of Unit. Different non contextual word embedding schemes: LSA, Word2Vec, GloVE, LDiA. Recurrent Neurons, Their types: LSTM, GRU.

	Various RNN Architectures: Sequence to Vector, Sequence to Sequence, Encoder-Decoder with Attention Mechanism, Embedding Layers. Conclusion & Real-Life Applications.
3.	**Delving deep into Transformers**
	Introduction of Unit. Extending the concept of Attention to an advanced architecture called Transformers. What are Transformers? How are they different from RNNs? Several Transformer Architectures to Large Language Models. Using several dominant Transformer architectures to solve various problems in NLP. Conclusion and Real-Life Applications.
4.	**Large Vision Models for Computer Vision**
.	Introduction of Unit. Introduction to various Large Vision Models: CLIP, Landing Lens and Vision Transformers. Training and Fine-tuning various Large Vision Models. Large Vision Models for various Computer Vision Applications. Application of various Large Language Models in Real-Life scenarios. Conclusion of the Unit.
5	**Different Large Vision Models to solve problems**
.	Introduction of Unit Delving deep into CLIP and understanding the nuts and bolts of it. Delving deep into Landing Lens and understanding the nuts and bolts of it. Delving deep into Vision Transformers and understanding the nuts and bolts of it. Applying various Large Vision Models to solve problems in Computer Vision. Conclusion of the Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Deep Learning for Natural Language Processing	Stephan Raaijmakers	1st Edition	Manning Publications
2.	Getting Started with Natural Language Processing	Ekatrina Kochmar	1st Edition	Manning Publications
3.	Build a Large Language Model from Scratch	Sebastian Raschka	1st Edition	Manning Publications
4.	Deep Learning for Vision Systems	Mohammed Elgendy	1st Edition	Manning Publications
Reference Book				
1	Natural Language Processing in Action by Hobson Lane, Maria Dyshel, 2nd Edition, 2024, Manning Publications.			
2.	Transfer Learning for Natural Language Processing by Paul Azunre, 1st Edition, 2021, Manning Publications			
Online Resources				
1	https://www.tensorflow.org			
2	https://www.pytorch.org			
3	https://www.keras.io			
4	https://huggingface.co			

Code : BCSECE6112	Cyber Forensics	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Apply various aspects of securing network infrastructure and importance of classifying information
CO – 02	Understand	Evaluate
CO – 03	Apply	Create asset management along with the objective to create awareness in Digital RightsManagement.
CO – 04	Analyze	Gain knowledge about basic concepts and importance of information security
CO – 05	Evaluate	Synthesize student to understand the concepts of IT security, Threats, Vulnerabilities, Impact andcontrol measures.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	3	2	1	2	–	–	–	–	–	–	–	–	2	–	–
CO3	2	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	2	2	2	1	–	–	–	–	–	–	–	–	2	–	–
CO5	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	2.2	2.6	1.4	1.2									1.8		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Cyber Forensics**	**08**
2.	**Storage Devices & Data Recover Methods**	**08**
3.	**Forensics Techniques I**	**08**
4.	**Forensics Techniques II**	**09**
5.	**Cyber Law**	**07**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1	**Introduction to Cyber Forensics**
	Introduction of Unit Introduction to Computer Forensics Forms of Cyber Crime First Responder Procedure- Non-technical staff, Technical Staff, Forensics Expert and Computer Investigation procedure Conclusion of Unit
2	**Storage Devices & Data Recovery Methods**
	Introduction of Unit Storage Devices- Magnetic Medium, Non-magnetic medium and Optical Medium. Working of Storage devices-Platter, Head assembly, spindle motor. Data Acquisition Data deletion and data recovery method and techniques Conclusion of Unit
3	**Forensics Techniques I**
	Introduction of Unit Windows forensic Linux Forensics

	Mobile Forensics Steganography, Application Password cracking-Brute force Conclusion of Unit
4	Forensics Techniques II
	Introduction of Unit Dictionary attack Rainbow attack Email Tacking – Header option of SMTP, POP3, IMAP Arsenal – Surveillance Tools Conclusion of Unit
5	**Cyber Law**
	Introduction of Unit Corporate espionage Evidence handling procedure Chain of custody Main features of Indian IT Act 2008 (Amendment) Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Guide to Computer Forensics and Investigations	B. Nelson	FirstEdition	Cengage,2010 BBS
2	Hacking Exposed Computer Forensics	Aaron Philipp, David Cowen, Chris Davis	First Edition	McGraw Hill-2011
Reference Book				
1	Introduction to Computer Forensics and Digital Investigation, Rauf gauney			
2	MariE-Helen Maras, “Computer Forensics: Cybercriminals, Laws, and Evidence”, Jones & Bartlett Learning; 2nd Edition, 2014			
Online Resources				
1	https://www.techtarget.com/searchsecurity/definition/computer-forensics			
2	https://intellipaat.com/blog/what-is-cyber-forensics/			
3	https://www.educba.com/cyber-forensics/			

Code : BCGECE6112	No SQL Database	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Describe, compare and use the four types of NoSQL Databases (Document-oriented, KeyValue Pairs, Column-oriented and Graph).
CO – 02	Understand	Explain detailed architecture, define objects, load data, query data and performance tune Column- oriented
CO – 03	Apply	NoSQL databases.
CO – 04	Analyze	Explain detailed architecture, define objects, load data, query data and performance tune Document- oriented NoSQL databases.
CO – 05	Evaluate	Analyze an understanding of the detailed architecture, define objects, load data, query data and performance tune Key-Value Pair NoSQL databases.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	3	1	1	–	–	–	–	–	–	–	–	2	–	–
CO2	2	2	–	2	–	1	–	–	–	–	–	–	2	1	–
CO3	1	3	2	1	–	–	–	–	–	–	–	–	1	–	–
CO4	2	1	2	1	–	–	–	–	–	–	–	–	2	1	–
CO5	2	3	1	1	–	–	–	–	–	–	–	–	2	–	–
WT. AVG	1.6	2.4	1.5	1.2		1.0							1.8	1.0	

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to NoSQL**	**08**
2.	**MongoDB**	**08**
3.	**Key-Value & Document Based Databases**	**08**
4.	**Column-Oriented & Graph Based Databases**	**09**
5.	**Search Engine**	**07**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to NoSQL**
	Introduction of Unit Understanding NoSQL Databases, History of NoSQL, Features of NoSQL, Scalability, Cost, Flexibility, NoSQL Business Drivers, Classification and Comparison of NoSQL Databases, Consistency – Availability - Partitioning (CAP), Limitations of Relational Databases, Comparing NoSQL with RDBMS Managing Different Data Types, Columnar, Key-Value Stores, Triple and Graph Stores, Document, Search Engines, Hybrid NoSQL Databases, Applying Consistency Methods, ACID, BASE, Polyglot persistence, Need for NoSQL, Advantages Conclusion of unit
2.	**MongoDB**
	Introduction of Unit Introduction to MongoDB , Advantages of Mongo DB, Data Modelling Program using MongoDB, Constructing Queries, Replication Deployment Conclusion of unit
3.	**Key-Value & Document Based Databases**

	Introduction of Unit Introduction to Key-Value Databases, Key Value Store, Essential Features, Consistency, Transactions, Partitioning, Scaling, Replicating Data, Versioning Data, How to construct a Key, Using Keys to Locate Values, Hash Functions, Store data in Values, Use Cases. Introduction to Document Databases, Supporting Unstructured Documents, Document Databases Vs. Key-Value Stores, Basic Operation on Document database, Partition, Sharding, Features, Consistency, Transactions, Availability, Scaling, Use Cases. Conclusion of unit
4.	**Column-Oriented & Graph Based Databases**
	Introduction of Unit Introduction to Column Family Database, Features, Architectures, Differences and Similarities to Key Value and Document Database, Consistency, Transactions, Scaling, Use Cases, Introduction to Graph Databases, Advantages, Features, Consistency, Transactions, Availability, Scaling, Graph & Network Modelling, Properties of Graphs and Noes, Types of Graph, Undirected and directed Graph, Flow Network, Bipartite Graph, Multigraph, Weighted Graph Conclusion of unit
5.	**Search Engine**
	Introduction of Unit Common Feature of Search Engine, Dissecting a Search Engine, Search versus query, Web crawlers, Indexing, Searching, indexing Data Stores, Altering, Using Reverse queries, Use Cases, Types of Search Engine, Elastic Search Conclusion of unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Professional Nosql	Shashank Tiwari		Wrox
2.	MongoDB in Action	KYLE BANKER PETER	Second Edition	Manning
Reference Book				
3.	NoSQL for Dummies, Adam Fowler, John Wiley & Sons, Inc			
4.	NoSQL Distilled, Pramod J. Sadalage & Martin Fowler, Pearson Education, Inc.			
5.	Making Sense of NoSQL, Dan McCreary& Ann Kelly, Manning Shelter Island			
Online Resources				
6.	https://www.javatpoint.com/nosql-databases			
7.	https://www.tutorialspoint.com/mongodb/index.htm			

Code : BAMECE6112	Reinforcement Learning	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall the fundamental concepts and terminologies associated with reinforcement learning and Markov decision processes, including elements of reinforcement learning, Markov decision processes, and policies.
CO – 02	Understand	Understand the principles and frameworks of reinforcement learning, including the history of reinforcement learning, Markov decision processes, value functions, and solution techniques such as dynamic programming and model-free methods.
CO – 03	Apply	Apply efficient solution frameworks in reinforcement learning, including batch reinforcement learning algorithms, kernel-based approximate dynamic programming, and neural fitted Q iteration.
CO – 04	Analyze	Analyze constructive and representational directions in reinforcement learning, such as function approximation, hierarchical approaches, and evolutionary computation for reinforcement learning.
CO – 05	Evaluate	Evaluate the probabilistic models for self and others in reinforcement learning, including Bayesian reinforcement learning, partially observable Markov decision processes, and game theory applications.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	1	-	-	-	-	-	-	-	-	-	3	-	1
CO2	-	2	1	-	-	-	-	-	-	-	-	-	-	1	-
CO3	-	1	2	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	1	-	2	-	-	-	-	-	-	-	-	-	-	-
CO5	-	-	-	1	-	-	-	-	-	-	-	-	-	1	-
WT. AVG	3.0	1.3	1.3	1.5									3.0	1.0	1.0

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Reinforcement Learning and Markov Decision Process**	6
2.	**Efficient Solution Framework**	8
3.	**Constructive- Representational Directions**	8
4.	**Probabilistic Model For Self and Other**	7
5.	**Domain and Background**	7

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Reinforcement Learning and Markov Decision Process**
	oduction- Reinforcement Learning mples OF Reinforcement Learning-Elements of Reinforcement Learning- mple: Tic-Tac-Toe - History of Reinforcement Learning -Learning Sequential decision Making-A Formal Frame Work on Markov Decision Process and Policies-Value Function and Bellman Equations-Solving Markov Decision Process namic Programing Model Based Solution Technique-Reinforcement Learning Model Free Solution Technique
2.	**Efficient Solution Framework**

	oduction- The Batch Reinforcement Learning Problem- Foundations of Batch Reinforcement Learning Algorithms ch Reinforcement Learning Algorithms: Kernel-Based Approximate Dynamic Programming- Fitted Q Iteration- Least-Squares Policy Iteration- Identifying Batch Algorithms. ory of Batch Reinforcement Learning- Neural Fitted Q Iteration (NFQ)- Batch Reinforcement Learning for Learning in Multi-agent Systems p Fitted Q Iteration. Least-Squares Methods for Approximate Policy Evaluation- Least-Squares Policy Iteration- Performance Guarantees.
3.	**Constructive- Representational Directions**
	nforcement learning in continuous state and action space: Function Approximation- Approximate Reinforcement Learning. ving Relational and first-order logical Markov decision: Introduction to sequential decision in relational Reinforcement Learning- model based solution techniques- modelfree solution- rarchical Approaches- Approaches to hierarchical reinforcement learning – Evolutionary computation for Reinforcement Learning: Neuro-evolution - Hybrids- Coevolution.
4.	**Probabilistic Model For Self and Other**
.	esian Reinforcement Learning: Model free Bayesian Reinforcement Learning - Model based Bayesian Reinforcement Learning- Partially observable Markov decision process: Decision making in partially observable environments- model based techniques- dictively defined representation of state: PSRs- Learning a PSR model- Game theory and multi agent Reinforcement Learning – Reinforcement Learning in Repeated games- Sequential games.
5	**Domain and Background**
.	nforcement Learning in games- challenges of applying Reinforcement Learning to games nforcement Learning in Robotics: challenges in robot REINFORCEMENT LEARNING ndations of Robotic Reinforcement Learning- tractability through simulation, representation and prior knowledge.

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Neural Networks	Ruchi sehrawat		Neural Networks
2.	Deep Learning	Ian Goodfellow and Yoshua		Deep Learning
3.	Chris Bishop's Pattern recognition and machine learning	Christopher M. Bishop		Chris Bishop's Pattern recognition and machine learning
Reference Book				
1	Deep Learning Methods andApplications by Deng & Yu's monograph			

B: Minor Stream Courses/ Department Electives Labs
B2: Practical - Professional Elective - IV

Code : BCEECE6211	Advance Web Technology Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	use the fundamental concepts of web development including HTML, CSS, and JavaScript. List different web development frameworks and libraries used for building interactive web applications.
CO – 02	Understand	Explain the client-server architecture and how it relates to web application development. Summarize the importance of user experience (UX) and user interface (UI) design principles in web development.
CO – 03	Apply	Develop interactive web applications using HTML, CSS, and JavaScript. Apply responsive design techniques to ensure compatibility with various devices and screen sizes.
CO – 04	Analyze	Analyze and debug code to identify and resolve issues in interactive web applications.
CO – 05	Evaluate	Evaluate the performance and efficiency of different web development techniques and tools. Critique the usability and accessibility of web applications based on established standards and guidelines.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	2	2	1	2	-	1	-	2	1	1	1	3	3	2
CO2	3	2	2	1	2	-	1	-	2	1	1	1	3	3	2
CO3	2	2	2	1	2	-	1	-	2	1	1	1	3	3	2
CO4	-	2	2	1	2	-	1	-	2	1	1	1	3	3	2
CO5	-	2	2	1	2	-	1	-	2	1	1	1	3	3	2
WT. AVG	2.5	2	2	1	2		1		2	1	1	1	3	3	2

C. LIST OF EXPERIMENTS

1	Write a Node.js script that creates a simple HTTP server. The server should listen on port 3000 and respond with the text "Hello, World!" when accessed.
2	Create a Node.js module named calculator.js that exports functions for addition and subtraction. Then, create another file app.js to import the module and use its functions to perform addition of two numbers and log the result.
3	Write a Node.js script that reads the content of a file named input.txt and writes that content to a new file named output.txt.
4	Write a Node.js script that performs a simple GET request to fetch data from the JSONPlaceholder API (https://jsonplaceholder.typicode.com/todos/1) and logs the data to the console.
5	Write a Node.js script using the Express framework to create a simple API with two routes:
6	Perform the following task on Student tables. Implement functions to perform SQL JOIN operations to retrieve data from multiple related tables. Implement sorting functionality to sort data based on different columns. Write queries to filter data based on specific conditions. Group data based on certain criteria and perform aggregate functions like COUNT, SUM, AVG, etc.
7	Write a program Looping Over Data
8	Write a program Formatting Data for Display
9	Write a program More Reusable Components
10	Write a program Responding to an Event
11	Write a program Create a Form to Edit Your Data
12	Write a program Communicating from Child to Parent Component

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Introduction to Reinforcement Learning, 2nd Edition	Introduction to Reinforcement Learning, 2nd Edition	Introduction to Reinforcement Learning, 2nd Edition	Introduction to Reinforcement Learning, 2nd Edition
2	Neuro Dynamic Programming	Neuro Dynamic Programming	Neuro Dynamic Programming	Neuro Dynamic Programming
Reference Book				
1	V. S. Borkar, Stochastic Approximation: A Dynamical Systems Viewpoint, Hindustan Book Agency, 2009. ISBN-13: 978-0521515924			
2.	Deep Learning. Ian Goodfellow and Yoshua Bengio and Aaron Courville. MIT Press. 2016.ISBN-13: 978-0262035613.			
Online Resources				
1	NPTEL Course in Reinforcement Learning: https://onlinecourses.nptel.ac.in/noc22_cs75/preview			
2	https://www.youtube.com/watch?v=FgzM3zpZ55o			

Code : BADECE6211	Deep Learning Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall fundamental concepts of deep learning including neural networks, activation functions, and backpropagation.
CO – 02	Understand	Understand various deep learning architectures such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and generative adversarial networks (GANs).
CO – 03	Apply	Apply deep learning frameworks such as TensorFlow or PyTorch to develop and train neural network models for various tasks such as image classification and natural language processing.
CO – 04	Analyze	Analyze the performance of deep learning models using evaluation metrics and tuning hyperparameters to optimize model performance.
CO – 05	Evaluate	Evaluate ethical implications of deep learning including biases in data and models, privacy concerns, and fairness in deployment.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	-	2	1	-	-	-	-	-	-	-	2	-	-
CO2	1	2	3	2	-	-	-	-	-	-	-	-	1	-	-
CO3	1	3	3	1	-	-	-	-	-	-	-	-	2	-	-
CO4	2	-	1	1	-	-	-	-	-	-	-	-	3	-	-
CO5	1	1	2	-	2	-	-	-	-	-	-	-	1	-	-
WT. AVG	1.4	1.75	2.25	1.5	1.5								1.8		

C. LIST OF EXPERIMENTS

1	Implement a basic CNN for image classification.
2	Train a CNN model on a custom image dataset.
3	Fine-tune a pre-trained CNN for new categories.
4	Build an object detection model using SSD.
5	Implement real-time object detection using YOLO.
6	Develop a semantic segmentation model using FCNs.
7	Implement U-Net for medical image segmentation.
8	Evaluate model performance using IoU and accuracy.
9	Build a face detection system using Haar Cascades.
10	Implement face detection with a deep learning model
11	Develop a face recognition system using embeddings.
12	Implement face tracking using Kalman filters.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
3.	"Deep Learning for Computer Vision"	Rajalingappaa Shanmugamani	Latest	Packt Publishing
4.	"Computer Vision: Algorithms and Applications"	Richard Szeliski	Latest	Springer
Reference Book				
2.	"Deep Learning"Author: Ian Goodfellow, Yoshua Bengio, and Aaron Courville Publisher: MIT Press			
Online Resources				
3.	https://www.javatpoint.com/deep-learning			
4.	https://www.simplilearn.com/tutorials/deep-learning-tutorial			

Code : BCSECE6211	Vulnerability Analysis and Penetration Testing Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Remember Recall common types of vulnerabilities, attack vectors, and mitigation techniques.
CO – 02	Understand	Understand Understand the principles and methodologies of vulnerability analysis, including vulnerability assessment tools, techniques, and best practices.
CO – 03	Apply	Apply vulnerability assessment tools and methodologies to identify, prioritize, and remediate vulnerabilities in computer systems and networks.
CO – 04	Analyze	Analyze the results of vulnerability assessments to assess the severity and potential impact of vulnerabilities on system security.
CO – 05	Evaluate	Evaluate Evaluate the effectiveness of vulnerability mitigation strategies and recommend appropriate countermeasures to reduce risk and enhance security posture.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	-	-	-	-	2	-	-	3	-	-	-	-	-	-
CO2	-	-	-	-	-	-	3	-	2	2	-	-	-	-	-
CO3	-	2	-	2	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	2	-	3	-	1	-	-	-	-	-
CO5	-	2	-	2	2	-	-	1	-	-	2	-	1	-	-
WT. AVG	2	2		2	2	2	3	2	2.5	1.5	2		1		

C. LIST OF EXPERIMENTS

1	Write a program in python/Java/Java script to implement Private and Public key encryption.
2	Write a program in python/Java/Java script to implement Private and Public key decryption
3	Write a program in python to implement encryption and decryption of RSA – Public Key Cryptography Algorithm.
4	Installation of Intrusion detection System (IDS) tool.
5	Using IDS tool monitor the network and system traffic for any suspicious activity.
6	Installation Acunetix tool to perform web vulnerability analysis.
7	Case study on Threat and Vulnerability analysis of the email system.
8	Case study on Web application Vulnerability analysis using web Vulnerability scanners - acunetix .
9	Write test cases for Penetration Testing to perform Vulnerability Assessment.
10	roject on Vulnerability Analysis and Cyber Risk.(or))r): Project on Vulnerability Analysis on Network Security. Or): Project on Vulnerability Analysis on Computer Security.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Network Security Essentials: Applications and Standards	William Stallings	4 th Edition	Prentice Hall,
2	Introduction to ComputerSecurity	Michael T Goodrichand Roberto Tamassia	2011	Addiaon Wesley
Reference Book				
1	Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software MichaelSikorski, Andrew Honig publisher Williampollock			
Online Resources				
1	https://www.edx.org/course/cybersecurity-risk-management			
2	https://www.udemy.com/topic/computer-network/			
3	https://www.researchgate.net/publication/344803532_information_security_Risk_Analysis			

Code : BCGECE6211	Linux Server Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall basic Linux commands and their functions.
CO – 02	Understand	Explain the purpose and functionality of different Linux commands
CO – 03	Apply	Execute Linux commands to perform specific tasks such as file manipulation, directory navigation, and process management
CO – 04	Analyze	Analyze complex shell scripts to understand their functionality and logic
CO – 05	Evaluate	Assess the efficiency and effectiveness of shell scripts in solving specific problems.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	1	1	-	2	2	1	-	1	-	1	1	2	2	-
CO2	-	1	1	-	2	2	1	-	1	-	1	1	2	2	-
CO3	-	1	1	-	2	2	2	-	1	-	1	1	3	2	-
CO4	-	1	1	-	2	2	2	-	1	-	1	1	3	2	-
CO5	-	1	1	-	2	2	2	-	1	-	1	1	3	2	-
WT. AVG		1	1		2	2	1.6		1		1	1	2.6	2	

C. LIST OF EXPERIMENTS

Configure the following tasks & verify it. (Hint - use grep/cut/tr/sed)
List the lines containing "/sbin/nologin" from the /etc/passwd file.
List only lines of output from ps, which lists running processes thatcontain the string "init".
Display the list of GIDs from /etc/passwd file.
Alter all the letters that starts from range "a-f" to "A-F" in /etc/passwdfile.
Create an alias named eth0:0 using below credentials in RHEL 5 and verify it.
3. (a) IP ADRESS = 172.16.0.1 (b) 255.255.0.0
(c) Default Gateway = 172.16.0.254 (d) DNS 1 = 4.2.2.1
Configure password policy for user john with below arguments in RHEL 5. Afterconfiguration verify the policy applied.
Minimum password age = 4 days
Maximum password age = 15 days
Inactive days = 2 days
Account Expiration date = 6 months from today
Configure the following tasks:
Add user accounts to your system: Joshua, alex, dax, bryan, zak, ed andmanager. Assign each user this password: 123@iMs.
Add the groups to your system: sales with GID: 1000, HR with GID:1100 and web with GID: 1200.
Add Joshua and alex to the sales group, dax and bryan to the HR group,zak and ed to the web group and add manager to all of these groups.
Login with each user & verify using id command that they are in theappropriate groups.
Use ACL to accomplish these tasks:
Create groups named Admin and Web.
Create users named John and Jimmy.
Create a new directory named /depts/tech/. Change the permission sothat root is the owner and Admin is the group owner.
Use ACL to give full permission for /depts/tech/ to the Web group.

Allow John read/execute but not write permission on the /depts/tech/directory.
Allow Jimmy full permission on the /depts/tech/ directory.

	You are tasked with finding all SUID & SGID files under the / directories. Configure your system that boots to run level 3 by default. Configure X serverusing command in run level 3. Devise a ps command that does the following. (Hint: sort/ps/top) List all processes. For each process, prints the information which displays the percentageof CPU usage, the process ID & name of the command that created it. The output is sorted by the %cpu value from highest to lowest Explain the suid, sgid & sticky bit permission with example Customize the Bash prompt as per given tasks (Hint - PS1) Display the current value of primary prompt string. Changes prompt to print a static string "ITIMS -'". Restore the original prompt. Insert the bash history prompt special character "\!" between thehostname and dollar-sign. Configure given tasks for package management: (Hint: use rpm command) Check whether ftp package is installed or not. If it is not installed, install it & verify it. Display the configuration files available through this package. Be sure that ftp service must be enabled at startup. Use rpm queries to answer the following questions. What files are in the "initscripts" package? Which installed packages have "gnome" in their names? Which RPM provides /etc/inittab? Prepare a cron job that take the backup of /home at 5:00pm on every Saturday. Change your system date to 1:00pm March 1990. Copy /etc/fstab file to /tmp directory as newfstab file. The user owner is Jack andgroup owner is admin. Give full permission to user owner and read, write permission to group owner. No permission to others. Configure your system such that SELinux must be in enforcing mode and firewallis enables and ssh service is not allowed through your firewall. Configure ftp server such that anonymous can download and upload the data to ftpserver. Deny users John and Carel to access the ftp server. Note that your ftp server must be accessible in your private network only. It can't be accessible in another network. Create a RAID level 1 on /dev/md0 device by creating two equal partitions of 100MB size and mount it on /data. The RAID device must be mounted at the timeof system startup. Configure LVM in volume group named volgrp by creating 2 partitions of 100MB size and mount it on /exam directory. The initial size of LVM is approximately40MB and after extending the size of LVM is 80MB. Configure the DHCP server such that your DHCP server will able to provide IPconfiguration to 65 systems in your network. Configure the station as NFS server such that /share directory is shared & only accessible in your network. This NFS share should be automatically mounted onremote client using autofs. On remote client system, NFS share should be mounted on /data/share directory.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	RHCSA/RHCE Red Hat Linux Certification Study Guide Exams EX200 & EX300	Orsaria, Jang	McGraw-Hill Education	RHCSA/RHCE Red Hat Linux Certification Study Guide Exams EX200 & EX300

2	Red Hat RHCSA/RHCE 7 Cert Guide: Red Hat Enterprise Linux 7(EX200 and EX300)	Sander Van Vugt	Phi Learning Pvt Ltd	Red Hat RHCSA/RHCE 7 Cert Guide: Red Hat Enterprise Linux 7(EX200 and EX300)

Code : BAMECE6211	Forecasting Using Model Studio in SAS® Viya®	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Automatically create and fit custom forecast models using structured analytic workflows or pipelines
CO – 02	Understand	Visualize modeling data using attribute variables.
CO – 03	Apply	Refine forecast models to improve forecast accuracy.
CO – 04	Analyze	Apply overrides-generated forecasts
CO – 05	Evaluate	Generate forecast data sets for deployment

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	-	-	-	-	-	-	-	-	-	-	1	1	-	-
CO2	1	-	-	-	-	-	-	-	-	-	-	1	1	-	-
CO3	1	1	1	-	-	-	-	-	-	-	-	1	1	-	-
CO4	1	1	1	-	-	-	-	-	-	-	-	1	1	-	-
CO5	1	1	1	-	-	-	-	-	-	-	-	1	1	-	-
WT. AVG	**1**	**1**	**1**									**1**	**1**		

C. LIST OF EXPERIMENTS

1	Introduction and Data Visualization: Creating a forecasting project and loading the data
2	Visualizing the modeling data using attribute variables
3	Pipeline Essentials: Basic modeling with pipelines
4	Pipeline templates and pipeline comparison
5	Accuracy statistics and forecast model selection
6	Families of models supported.
7	Hierarchical Forecasting
8	Time series data creation and forecast reconciliation
9	Combined models
10	Honest assessment
11	Post-forecasting Functionality: ▪ Overrides
12	Exporting generated tables
13	Adjustments to statistical forecasting
14	In-Line Code Access and Overview (Appendix): Code overview
15	Accommodating Event Variables in a Model Studio Project (Appendix)
16	Adding event variables in the TSMODEL procedure

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	The Complete Reference Linux		Seventh Edition	Mc graw hill
2.	Linux Bible	Christopher Negus	Kindle Edition	Wiley
Reference Book				
1.	Windows 10 Portable, Wiley, Paul Mac Fedrics.			
2.	Desktop OS for Expert, Sayan Banerjee and Swati Goel			
Online Resources				
1.	**https://www.youtube.com/watch?v=BGjTboXjH28**			
2.	**https://www.youtube.com/watch?v=g5d0dfq_Ew8**			

B2: Practical - Professional Elective - V

Code : BCEECE6212	Software Testing Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall fundamental concepts of software testing, including types of testing (e.g., unit testing, integration testing, system testing, acceptance testing), testing strategies (e.g., black-box testing, white-box testing), and testing techniques (e.g., equivalence partitioning, boundary value analysis).
CO – 02	Understand	Understand the principles and objectives of software testing, such as ensuring software quality, detecting defects, and validating software functionality against requirements.
CO – 03	Apply	Apply software testing techniques by writing test cases, test plans, and test scripts for various software modules or components, utilizing testing frameworks (e.g., JUnit, Selenium) and tools (e.g., TestNG, JIRA) to automate testing processes and execute test suites.
CO – 04	Analyze	Analyze software requirements documents and system specifications to derive test scenarios and identify testable features, ensuring comprehensive coverage of software functionality and validation of non-functional requirements (e.g., performance, security).
CO – 05	Evaluate	Evaluate the effectiveness of software testing strategies and methodologies employed during the lab, assessing their ability to detect defects, improve software quality, and ensure compliance with user requirements, and provide recommendations for refining testing approaches.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	-	2	-	-	-	-	-	-	-	-	-	-	2
CO3	3	2	2	2	-	-	-	-	-	-	-	-		-	2
CO4	3	2	-	2	-	-	-	-	-	-	-	-	-	-	2
CO5	3	2	2	2	-	-	-	-	-	-	-	-	-	-	2
WT. AVG	**3**	**2**	**2**	**2**											**2**

C. LIST OF EXPERIMENTS

1	Find the test cases and coverage using JaButi.Tool. Write a program that calculates the area and perimeter of the circle. And find the Coverage &Test Cases of that program. Write a program which read the first name and last name from console and matching withexpected result. Write a program that takes three double numbers from the java console representing , respectively, the three coefficients a,b, and c of a quadratic equation.
2	Write a program that reads commercial website URL from a url from file .you should expect that the URLstarts with www and ends with .com. retrieve the name of the site and output it. For instance, if the user inputs www.yahoo.com, you should output yahoo. After that find the test cases and coverage using JaButi.
3	Write a program that reads two words representing passwords from the java console and outputs the number of character in the smaller of the two. For example, if the words are open and sesame, then the output should be 4, the length of the shorter word, open. And test this program using JaButi.
4	Analyze the performance of following website using JMeter. Site Amazon Website Amazon.com Type shopping Flip kart Flipkart.com shopping Railway reservation Irctc.co.in Ticket booking site Train searching Erail.in Train searching
5	Calculate the coverage analysis of given programs using Eclemma Free open source Tool.

	Write a program that takes three double numbers from the java console representing ,respectively, the three coefficients a,b, and c of a quadratic equation. Write a program that calculates the area and perimeter of the circle. And find the Coverage &Test Cases of that program. Write a program which read the first name and last name from console and matching with expected result.
6	Find the Pass or fail test cases using unit testing tool Junit.
7	Find the Pass or fail test cases using unit testing tool TestNG.
8	Calculate the mutation score of programs using jumble Tool Write a program that calculates the area and perimeter of the circle. And find the Coverage &Test Cases of that program. Write a program which read the first name and last name from console and matching withexpected result. Write a program that takes three double numbers from the java console representing , respectively, the three coefficients a,b, and c of a quadratic equation.
9	Generate Test sequences and validate using Selenium tool for given websites below: Site Amazon Website Amazon.com Type shopping Flip kart Flipkart.com shopping Railway reservation Irctc.co.in Ticket booking site Train searching Erail.in Train searching
10	Write a program for a calculator and find the test case and coverage and Def-use-graph.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Software Testing-Principle and Practices	Naresh Chauhan	3rd	Oxford
2.	Fundamentals of Software Engineering,	RajibMall	PHI	2018
Reference Book				
1.	The Art of Software Testing, 3rd Edition by Glenford J. Myers, Corey Sandler, Tom Badgett.			
2.	Software Testing, 2nd Edition by Ron Patton			
Online Resources				
1.	https://www.javatpoint.com/software-testing-tutorial			
2.	https://www.guru99.com/software-testing.html			

Code : BADECE6212	NLP & Computer Vision Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	To understand the complete text preprocessing pipeline used to create vocabulary from the textual data and apply it to create vocabulary.
CO – 02	Understand	To understand the various architectures of RNNs: Sequence to Vector, Sequence to Sequence as well as Encoder-Decoder Architecture with Attention Mechanism and apply them on various NLP applications such as Sentiment Classification, Language Translation, Text Summarization, Drug Discovery, etc.
CO – 03	Apply	To delve deep into the Transformer Architecture, understanding its nuts and bolts. Applying the various concepts learned during this journey to improve other architectures and using these architectures to solve various problems in the domain of NLP.
CO – 04	Analyze	To apply the several concepts learned while delving deep into Transformers, for making transition towards Large Vision Models such as Vision Transformers.
CO – 05	Evaluate	To apply the Large Vision Models to solve various problems in Computer Vision.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	1	-	-	-	-	-	-	-	-	-	3	-	3
CO2	2	2	1	-	-	-	-	-	-	-	-	-	3	-	3
CO3	3	3	2	1	-	-	-	-	-	-	-	-	3	-	3
CO4	3	3	3	2	-	-	-	-	-	-	-	-	3	-	3
CO5	3	3	3	3	-	-	-	-	-	-	-	-	3	-	3
WT. AVG															

C. LIST OF EXPERIMENTS

1	Extracting vocabulary of Sentiment 140 Dataset, with the complete text preprocessing pipeline to convert text into tokens.
2	Extracting vocabulary of Sentiment 140 Dataset. using Word Piece Tokenizer and Byte Pair Encoding as tokenizers.
3	Training a Language Translation Model on English to Indic Language Samanantar Dataset using Seq2Seq Encoder-Decoder LSTM RNN with Attention Mechanism.
4	Training an Abstractive Summarization Model on CNN Daily Mail News Text Summarization dataset using Seq2Seq Encoder-Decoder LSTM RNN with Attention Mechanism.
5	Create a resource efficient chatbot using open-source Meta AI Llama LLM.
6	Adding domain specific knowledge to Meta AI Llama LLM, using Retrieval Augmented Generation (RAG) to void hallucinations.
7	Fine tuning the Meta AI Llama LLM based chatbot using Supervised Fine-Tuning techniques such as LoRA and RLHF.
8	Build a custom chatbot using Langchain and Chat GPT.
9	Training a Vision Transformer model-based Image Classifier on a Medical Imaging Dataset.
10	Training a Vision Transformer based Image Segmentation model such as SegFormer or MaskFormer to perform Image Segmentation on Medical Imaging Dataset.
11	Finetuning a pre-trained DETR model to perform multiclass object detection on any Dataset.
12	

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Speech and language processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition	Jurafsky D. and Martin J. H	2nd	Upper Saddle River, NJ: Prentice- Hall, 2008
2.	Natural Language Processing with Python	Edward Loper, Ewan Klein, and	1st	Pearson Education
Reference Book				
1.	Speech and language processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition			
2.	Computer Vision: Models, Learning, and Inference			
Online Resources				
1.	https://www.nlp.com/nlp-online-course/			
2.	https://www.futurelearn.com/courses/cloudswyft-msft-natural-language-processing-advanced			

Code : BCSECE6212	Cyber Forensics Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Introduction to Digital Forensics ,Recall fundamental concepts and principles of digital forensics, including evidence acquisition and preservation techniques.
CO – 02	Understand	Understand various file system analysis techniques used in digital forensics investigations, such as recovering deleted files and analyzing file metadata.
CO – 03	Apply	Apply data carving techniques to extract and recover deleted or fragmented files from storage media during forensic investigations.
CO – 04	Analyze	Analyze network traffic logs to identify suspicious activities, such as unauthorized access attempts or data exfiltration.
CO – 05	Evaluate	Evaluate chain of custody procedures and documentation to ensure the integrity and admissibility of digital evidence in legal proceedings

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	-	2	1	-	-	-	-	-	-	-	2	-	-
CO2	1	2	3	2	-	-	-	-	-	-	-	-	1	-	-
CO3	1	3	3	1	-	-	-	-	-	-	-	-	2	-	-
CO4	2	-	1	1	-	-	-	-	-	-	-	-	2	-	-
CO5	1	1	2	-	-	-	-	-	-	-	-	-	1	-	-
WT. AVG	1.4	1.75	2.25	1.5	1								1.6		

C. LIST OF EXPERIMENTS

1	Physical Collection of electronic evidence using forensic standards
2	Dismantling and re-building PCs in order to access the storage media safely
3	Boot sequence and Power On Self-Test mode analysis
4	Examination of File systems of Windows
5	Examination of File systems of Linux
6	Examination of File systems of Mac
7	Analysing Word processing and Graphic file format
8	Network data sniffing and analysing
9	Password and encryption techniques
10	Internet forensic and Malware analysis
11	Data recovery techniques for hard drive
12	Data recovery techniques for Pen drive and CD

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	PROFESSIONAL NoSQL	Shashank Tiwari		Wrox
2.	MongoDB in Action	KYLE BANKER PETER	Second Edition	Manning
Reference Book				
1.	NoSQLfor Dummies, AdamFowler, JohnWiley&Sons,Inc			
2.	NoSQLDistilled, PramodJ.Sadalage&Martin Fowler, PearsonEducation, Inc.			
3.	MakingSenseof NoSQL, DanMcCreary& Ann Kelly, ManningShelterIsland			
4.	NoSQLforMereMortals, DanSullivan, PearsonEducation			

Code : BCGECE6212	Mongo DB Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	define objects, load data, query data and performance tune Column-oriented NoSQL databases.
CO – 02	Understand	Create objects, load data, query data and performance tune Document-oriented NoSQL databases by analyzing the detailed architecture
CO – 03	Apply	Apply NoSQL database development tools and progra
CO – 04	Analyze	Analyze, compare and use the four types of NoSQL Databases (Document-oriented, KeyValue Pairs, Column-oriented and Graph).
CO – 05	Evaluate	Perform hands-on NoSql database lab assignments that will allow students to use the four NoSQL database types via products such as Cassandra, Hadoop Hbase

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1	1	-	3	2	1	-	-	-	-	-	2	2	-
CO2	2	1	1	-	3	-	1	-	-	-	-	-	2	2	-
CO3	2	1	1	-	3	-	2	-	-	-	-	-	3	2	-
CO4	3	1	1	-	3	-	2	-	-	-	-	-	3	2	-
CO5	-	1	1	-	3	-	2	1	-	-	-	-	3	2	-
WT. AVG	**2**	**1**	**1**		**3**	**2**	**1.6**	**1**					**2.6**	**2**	

C. LIST OF EXPERIMENTS

1	Prepare and install infrastructure for setting up MongoDBlab. InstallMongoDBCommunityEdition Download MongoDBCommunityEdition Run the Mongo DBinstaller Follow theMongoDBCommunityEditioninstallation wizard RunMongoDB Community Edition as a Windows Service • RunMongoDB Community Edition from the Command InterpreterItis advised to follow below URL: https://docs.mongodb.com/manual/tutorial/install-mongodb-on-windows/
2	Perform/execute below sets of basic commands on Mongo DB lab environment. Loginto Lab Show all Databases, Select data base to work with Authenticate and Logout from databases List down Collections, Users, Roles Create Collection
3	Perform/execute below sets of basic commands on MongoDB lab environment. Insert Document, Save Document, Update Document, Display Collection Records Drop Function
4	Perform/execute below sets of advanced commands on Mongo DB lab environment. Administrative Commands Projection Limit Method, Skip Method, Sort Records, Indexing, Aggregation Interacting with cursors
5	Execute below steps by inserting some data which we can work with. Paste the following into your terminal to create a petshop with some pets in it use petshop db.pets.insert({name:"Mikey", species:"Gerbil"}) db.pets.insert({name:"DaveyBungooligan",species:"Piranha"}) db.pets.insert({name: "Suzy B", species: "Cat"}) db.pets.insert({name:"Mikey", species:"Hotdog"}) db.pets.insert({name: "Terrence", species: "Sausagedog"}) db.pets.insert({name:"PhilomenaJones",species: "Cat"})

	Add an other piranha,and an akedmolerat called Henry., Use find to listall the pets. Find the ID of Mikey the Gerbil. Use find to find Mikey by id., Use find to find all the gerbils. Find all the creatures named Mikey., Find all the creatures named Mikey who are gerbils. Find all the creatures with the string "dog" in their species.
6	AirPhone Corp is a famous telecom company. They have customers in all locations. Customers use AirPhone Corp's network to make calls. Government has brought in a regulation that all telecom companies should store call details of their customers. This is very important from asecurity point of view and all telecom companies have to retain this data for 15 years. AirPhone Corpal ready stores all customer details data,for their analytics team.But due to a surge in mobile users in recent years, their current data base cannot handle huge amounts of data. Current data base stores only six months of data. AirPhone Corp now wants to scale their database and wants to store15years of data. Datacontains following columns: Source: Phone number of caller Destination:Phone number of call receiver Source_location: Caller's city Destination_location : Call receiver's city Call_duration: phone call duration Roaming: Flag to check if caller is in roaming Call_charge: Money charged for call SampleData: { source:"+919612345670", destination: "+919612345671",source_location: "Delhi",destination_location:"Mumbai",call_duration:2. 03, roaming: false,call_charge:2.03 } After discussing the requirements with database and architecture team, it has been decided that they should use MongoDb. You have been given the task to Setup a distributed system(database)suchthat datafrom different locationsgo to differentnodes (to distributethe load) Import data to sharded collection Checkdataon eachshardfordistribution
7	Execute below sets of problem by taking reference of Experiment Number 06 and find out: Add additional node to existing system (to test if we can add nodes easily when data increases) Checkthe behavior of cluster (data movement) on adding ashard. Check the behavior of query for finding a document with source location Mumbai

8	Anand Corp is a leading corporate training provider. A lot of prestigious organizations send their employees to Anand Corp for training on different skills. As a distinct training provider, Anand Corp has decided to share analysis report with their clients. This report will help their clientsknow the employees who have completed training and evaluation exam, what are their strengths,and what are the areas where employees need improvement. This is going to be a unique selling feature for the Anand Corp. As Anand Corp is already doing great business and they give training to a large number of people every month, they have huge amount of data to deal with. They have hired you as an expert and want your help to solve this problem.
	Attributes of data:
	Id:idofthepersonwhowastrained
	Name: name of the person who was trained Evaluation: evaluation term
	Score: score achieved bythe personfor the specific term
	A person can undergo multiple evaluations. Each evaluation will have a unique result score. You can see the sample data below.
	Sample Data
	{
	"_id":0, "name":"Andy","results":[
	{"evaluation":"term1","score":1.463179736705023},
	{"evaluation":"term2","score":11.78273309957772},
	{"evaluation":"term3","score":6.676176060654615}
	]
	}
	PQR Corp has assigned the following tasks to you to analyze the results:
	Find count and percentage of employees who failed in term1, the passings core being 37.

9	Execute below sets of problem by taking reference of Experiment Number 08 and find out: Find employees who failed in aggregate(term1+term2+term3). Find the Average score of trainees for term1.
10	Execute below sets of problem by taking reference of Experiment Number 08 and find out: Find the Averages core of trainees for aggregate (term1+term2+term3). Find number of employees who failed in all the three (term1+term2 +term3). Find the number of employees who failed in any of the three (term1+term2+term3).
11	Case study on 5 different IT Companies who are working on MongoDB. Explain on the below parameters: Why moved to NoSQL Advantages over NOSQL Business BenefitsTechnology Adaptation
12	Execute below sets of problem by taking reference of Experiment Number 08 and find out: Find employees who failed in aggregate(term1+term2+term3). Find the Average score of trainees for term1.

D. RECOMMENDED STUDY MATERIAL:

S. No	**Text Books:**	**Author**	**Edition**	**Publication**
1.	PROFESSIONAL NoSQL	Shashank Tiwari		Wrox
2.	MongoDB in Action	KYLE BANKER PETER	Second Edition	Manning
Reference Book				
1.	NoSQLfor Dummies, AdamFowler, JohnWiley&Sons,Inc			
2.	NoSQLDistilled, PramodJ.Sadalage&Martin Fowler, PearsonEducation, Inc.			
3.	MakingSenseof NoSQL, DanMcCreary& Ann Kelly, ManningShelterIsland			
4.	NoSQLforMereMortals, DanSullivan, PearsonEducation			
Online Resources				
1.	https://www.javatpoint.com/nosql-databases			
2.	https://www.tutorialspoint.com/mongodb/index.htm			

Code : BAMECE6212	Reinforcement Learning Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Understand and explain the core idea behind various reinforcement learning algorithms
CO – 02	Understand	Develop practical skills in implementing SARAS, DQN
CO – 03	Apply	Gain insights of advanced reinforcement learning techniques.
CO – 04	Analyze	Apply reinforcement learning algorithms to real world problem
CO – 05	Evaluate	Critically evaluate and compare different RL methods

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	2	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	3	2	1	2	-	-	-	-	-	-	-	3	1	3
CO3	2	3	1	-	-	-	-	-	-	-	-	-	3	2	2
CO4	1	1	1	-	-	-	-	-	-	-	-	-	-	2	2
CO5	1	1	1	1	-	-	-	-	-	-	-	-	-	2	2
WT. AVG	1.6	2.0	1.2	1.0	2.0								3.0	1.8	2.3

C. LIST OF EXPERIMENTS

1	Implement a simple grid-world environment using pythonUse the gym library to create the environment
2	Implement ε-greedy, UCB (Upper Confidence Bound), and Thompson Samplingalgorithms using Python.
3	Write code to simulate different exploration strategies in a simple environment andanalyze their performance.
4	Write code to solve the Bandit problem using different strategies and compare theirperformance.
5	Define a small MDP problem in Python Implement a simple MDP environment and calculate policies and value functions.
6	Write code in python to solve a grid world problem using dynamic programmingtechniques.
7	Use Monte Carlo methods to solve the BlackJack problem and compare results withdynamic programming.
8	Apply TD(0) to a simple environment and compare its performance with previouslyimplemented methods.
9	Write code to solve a grid world problem using both SARSA and Q-learning, andanalyze the results. Write code to solve a simple RL problem using TD(λ) and compare different traces.
10	Implement a DQN using a deep learning framework and test it on a simple RLproblem. Write code to enhance a basic DQN with these techniques and analyze their impact onperformance.
11	Write code for an Actor-Critic algorithm and review the AlphaGo case study,summarizing key learnings.
12	Design and implement a custom environment for reinforcement learning experimentsin Python using the gym library. Train an agent in the custom environment and evaluate its performance.Visualize the learning process using Matplotlib

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Introduction to ReinforcementLearning, 2nd Edition	Richard S. Sutton and Andrew G. Barto	2nd Publication	MIT Press. 2017. ISBN-13 978-0262039246
2	Neuro Dynamic Programming	Dimitri Bertsekas and John G. Tsitsiklis	Athena Scientific	Athena Scientific.1996. ISBN-13: 978-1886529106
Reference Book				
1	V. S. Borkar, Stochastic Approximation: A Dynamical Systems Viewpoint, Hindustan Book Agency,2009. ISBN-13: 978-0521515924			
2.	Deep Learning. Ian Goodfellow and Yoshua Bengio and Aaron Courville. MIT Press. 2016.ISBN-13: 978-0262035613.			
Online Resources				
1	NPTEL Course in Reinforcement Learning: https://onlinecourses.nptel.ac.in/noc22_cs75/preview			
2	Reinforcement Learning Course (Stanford University):			

D: Ability Enhancement Courses (AEC)

Code : BUACHM6219	Leadership Communication Lab	1 Credits [LTP: 0-0-2]

E. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Apply	Show the skills for reporting educational qualifications, skills & expertise sections on social media
CO – 02	Understand	Identify the stages of the writing process and formatting process
CO – 03	Analyse	Analyze the presentation's weak spots and areas for improvement.
CO – 04	Apply	Use various strategies and personal skills to perform well in interviews and Group Discussions.
CO – 05	Apply	Practice negotiation and team management skills and their mutual dynamics.

F. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Personal Branding**	**2**
2.	**Professional Writing-II**	**8**
3.	**Presentation Skills: Professional Setting**	**2**
4.	**Job Interview & Group Discussion: Preparation by Mock Practice**	**4**
5.	**Negotiation Skills, Team Management & Professional Awareness**	**8**

G. List of Labs

LIST OF LABS	
1.	Introduction to Personal Branding and ways to create a brand. How to improve social Media visibility for career growth
2.	Learning Communication and Professional writing including formatting process
3.	Informal communication, non-verbal cues, Speakers vs communicators
4.	Leaders and Professional Appearance, ways to improve presentation skills. Planning your presentation, building, audio-visual aids, giving presentation, audience rapport, and overcoming stage fright.
5.	How leaders communicate, skills of a good communicator.
6.	40-question self-assessment, making sure your message matters, choosing your words, improving your skills
7.	Definition of defensive/non-defensive communication & 5 skills for communicating non-defensively.
8.	Definition of feedback, giving feedback, 10 common feedback mistakes, 5 tips on receiving feedback, 10 tips for dealing with difficult people.
9.	Barriers to listening, what is listening, listening vs hearing, bad listening habits, active listening
10.	Advanced Group Discussion-II: Analysis of professional GD Videos and Practices on Topics/Video/Article based topics
11.	When to use/not use e-mail, e-mail etiquette, when to use/not use telephone and video conferencing, 7 tips on telephone etiquette
12.	Team Management and Negotiation Skills. How to improve negotiation Skills

E: Skill Enhancement Courses (SEC)

Code : BULCSE6201	Skill Enhancement Generic Course-VI	1 Credits [LTP: 0-0-2]

1

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Apply	Understand basic problems based on arithmetic and soft skills area which are asked in aptitude test taken by companies
CO – 02	Understand	Effectively solve these problems by applying the knowledge earned.
CO – 03	Analyse	Actively participate in group discussion / meetings / interviews and prepare & deliver presentations
CO – 04	Apply	Become more effective individual through goal/target setting, self-motivation and practicing creative thinking
CO – 05	Apply	Function effectively in multi-disciplinary and heterogeneous teams through the knowledge of team work, Inter-personal relationships, conflict management and leadership quality

B. List of Labs

LIST OF LABS	
1.	Work & Wages, Ratio & Proportions
2.	Effective Communication and Managing Conflict, Story telling
3.	Heights & Distances, Probability
4.	Comprehension & Para Jumble,
5.	Written Communication, Behavioral Interview Skills
6.	Effective Presentation skills, How to become more approachable
7.	Odd one out, Order & Ranking
8.	Deductive Reasoning, Divergent Thinking
9.	How to brainstorm effectively, Mirror & Water images
10.	Mind Mapping, Closing deals
11.	Project Management, Team Management
12.	Emotion Management, Delivering constructive feedback

G: Summer Internship / Research Project / Dissertation

Code : BCECCE6401	Industrial Training Seminar-I	1 Credits [LTP: 0-0-2]

A. DETAILED SYLLABUS

1. Unit	Unit Details
1.	
	• **Introduction to Industrial Training Seminar-I** Overview and objectives Importance in the curriculum • **Module 1: [Topic Name]** Sub-topic 1 Sub-topic 2 ... • **Module 2: [Topic Name]** Sub-topic 1 Sub-topic 2 ... • **Module 3: [Topic Name]** Sub-topic 1 Sub-topic 2 ... • **Assessment Methods** Details on how each CO will be assessed (assignments, projects, exams, etc.) • **References** List of recommended textbooks and readings

Your Dreams Our Goal

POORNIMA UNIVERSITY

Member of Association of Indian Universities & Approved by UGC (Govt. of India) under 2(f) & 12(B)

B. TECH - CSE

FOURTH YEAR
SEMESTER: VII

COMMON SYLLABUS

For

Cyber Security
Cloud & Full Stack Development
Artificial Intelligence & Data Science
Artificial Intelligence & Machine Learning

BATCH: 2024-28

A: Major (Core Courses)
A.1: Theory

Code :BCECCE7101	Big Data Analytics	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Defining the Big Data technology and its popular platform Hadoop.
CO – 02	Understand	Explain the Distributed Storage Systems and its implementation in Hadoop HDFS framework
CO – 03	Apply	Apply Mapping and Reducing algorithms to solvethe analytical problems on Big Data using MapReduce
CO – 04	Analyze	Analyze the Big Data using Hive HQL and PIG scripting programming languages.
CO – 05	Evaluate	Evaluate the Big Data technology with Machine Learning using R programming language.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	2	1	-
CO2	3	2	3	2	-	-	-	-	-	-	--	-	-	2	-
CO3	3	1	3	3	-	-	-	-	-	-	-	-	1	1	-
CO4	3	-	2	1	-	-	-	-	-	-	-	-	-	-	-3
CO5	3	2	3	2	-	-	-	-	-	-	-	-	--	3	-
WT. AVG	**3**	**1.75**	**2.75**	**2**									**1.5**	**1.75**	**3**

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Big Data And Hadoop**	**07**
2.	**HDFS(Hadoop Distributed File System)**	**08**
3.	**Map Reduce**	**08**
4.	**Hadoop Eco System**	**07**
5.	**Data Analytics with R**	**07**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Big Data And Hadoop**
	Introduction of Unit The Design of HDFS, HDFS Concepts, Command Line Interface, Hadoop file system interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives, Hadoop I/O: Compression, Serialization, Avro and File-Based Data structures Conclusion of Unit
2.	**HDFS (Hadoop Distributed File System)**
	Introduction of Unit The Design of HDFS, HDFS Concepts, Command Line Interface, Hadoop file system interfaces, Data flow, Data Ingest with Flume and Scoop and Hadoop archives, Hadoop I/O: Compression, Serialization, Avro and File-Based Data structures Conclusion of Unit
3.	**Map Reduce**

	Introduction of Unit Anatomy of a Map Reduce Job Run, Failures, Job Scheduling, Shuffle and Sort, Task Execution, Map Reduce Types and Formats, Map Reduce Features. Conclusion of Unit
4.	**Hadoop Eco System**
.	Introduction of Unit Pig : Introduction to PIG, Execution Modes of Pig, Comparison of Pig with Databases, Grunt, Pig Latin, User Defined Functions, Data Processing operators. Hive : Hive Shell, Hive Services, Hive Metastore, Comparison with Traditional Databases, HiveQL, Tables, Querying Data and User Defined Functions. Hbase : HBasics, Concepts, Clients, Example, Hbase Versus RDBMS. Big SQL : Introduction Conclusion of Unit
5	**Data Analytics with R**
.	Introduction of Unit Machine Learning: Introduction, Supervised Learning, Unsupervised Learning, Collaborative Filtering, Big Data Analytics with BigR, Conclusion with R

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1.	Hadoop: The Definitive Guide	Tom White	Third Editon	O'reily
2.	Big Data Analytics	Seema Acharya, Subhasini Chellappan	2015	Wiley
Reference Book				
1	Michael Berthold, David J. Hand, "Intelligent Data Analysis", Springer, 2007.			
2.	Jay Liebowitz, "Big Data and Business Analytics" Auerbach Publications, CRC press (2013)			
3.	Tom Plunkett, Mark Hornick, "Using R to Unlock the Value of Big Data: Big Data Analytics with Oracle R			
Online Resources				
1	http://www.bdbanalytics.ir/media/1121/big-data-analytics_turning-big-data-into-big-money.pdf			
2	https://www.techtarget.com/searchbusinessanalytics/definition/big-data-analytics			
3	https://www.tutorialspoint.com/hadoop/hadoop_big_data_overview.htm			

A.2: Practical

Code :BCECCE7201	Big Data Analytics Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall the steps required to prepare the infrastructure for setting up a single-node Hadoop cluster.
CO – 02	Understand	Understand the purpose and significance of installing all necessary software components for a Hadoop cluster.
CO – 03	Apply	Apply configuration settings to set up a single-node Hadoop cluster, including modifying hdfs-site.xml, core-site.xml, and yarn-site.xml.
CO – 04	Analyze	Analyze the functionality and output of basic Hadoop commands performed on the single-node cluster.
CO – 05	Evaluate	Evaluate the efficiency and performance of different aspects of Hadoop cluster setup and job execution.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	1	-	-	-	-	-	-	-	1	1	-	-
CO2	2	-	1	1	-	-	-	-	-	-	-	1	1	-	-
CO3	2	-	2	1	-	-	-	-	-	-	-	1	1	1	-
CO4	2	1	2	1	-	-	-	-	-	-	-	1	1	1	-
CO5	2	2	2	1	-	-	-	-	-	-	-	1	1	1	-
WT. AVG	**2**	**1.5**	**1.75**	**1**								**1**	**1**	**1**	

C. LIST OF EXPERIMENTS

1	Hadoop Installation: Ubuntu & THEL 9 Operating System in stand-alone mode
2	File Management tasks in Hadoop
3	Implement the following Data structures in Java: Linked Lists, Stacks, Queues, Set, Map
4	Word Count Map Reduce program to understand Map Reduce
5	Implement the following file management tasks in Hadoop: Adding files and directories, Retrieving files, Deleting files
6	Implement Matrix Multiplication with Hadoop Map Reduce
7	Install and Run Pig then write Pig Latin scripts to sort, group, join, project, and filter your data.
8	Install and Run Hive then use Hive to create, alter, and drop databases, tables, views, functions, and indexes
9	Weather Report POC-Map Reduce Program to analyses time-temperature statistics and generate report with max/min temperature.
10	Implementing Matrix Multiplication with Hadoop Map Reduce
11	Pig Latin scripts to sort, group, join, project, and filter your data.
12	Hive Databases :Tables, Views, Functions and Indexes

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Hadoop in Practice	Alex	2014	
2	Big Data	Holmes	2016	Black Book
3	Big Data and Hadoop	V.K. Jain	2017	
Reference Book				
1	Hadoop Practice Guide,"Jisha Mariam Jose"			
2.	Hadoop: The Definitive Guide ,"Tom White",O'Relly			
Online Resources				
1	https://ia600201.us.archive.org/7/items/HadoopInPractice/Hadoop%20in%20Practice.pdf			

Code :BCECCE7202	IoT Lab/ Mazor Project/ Internship	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Acquire practical knowledge within the chosen area of technology for project development.
CO – 02	Understand	Identify, analyze, formulate and handle programming projects with a comprehensive and systematic approach.
CO – 03	Apply	Develop effective communication skills for presentation of project related activities.
CO – 04	Analyze	Contribute as an individual or in a team in development of technical projects.
CO – 05	Evaluate	Work effectively as part of a team to develop the major project, actively participating in discussions, brainstorming sessions, and decision-making processes

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2		2									3	2	
CO2		3	2		2								2	3	
CO3		2	3		2								3	2	
CO4				3		2				2				2	3
CO5			2							2				3	
WT. AVG	**3**	**2.3**	**2.3**	**2.5**	**2**	**2**				**2**			**2.6**	**2.4**	**3**

C. LIST OF EXPERIMENTS

1	LED Control via Smartphone App Use an Arduino or Raspberry Pi to control an LED using a smartphone app over Wi-Fi or Bluetooth
2	Temperature and Humidity Monitoring Build a temperature and humidity sensor using a microcontroller and display the data on an LCD or send it to a cloud platform
3	Home Automation System Create a smart home automation system using IoT devices to control lights, appliances, and security systems.
4	Smart Irrigation System Develop an IoT-based system that monitors soil moisture and weather conditions to automate the irrigation process.
5	Smart Health Monitoring Create a wearable IoT device to monitor vital signs like heart rate, blood pressure, and temperature, and transmit the data to a mobile app or cloud platform.
6	Smart Parking System Build a system that uses sensors to detect parking spot availability and notifies drivers via a mobile app or LED displays.
7	Air Quality Monitoring Design an IoT-based air quality monitoring system using gas sensors and display real-time data on a web dashboard.
8	Waste Management System Develop a smart waste management system that monitors garbage levels in bins and notifies waste collection teams when they need emptying.
9	Smart Energy Management Create an energy management system that monitors and controls energy consumption in buildings using IoT devices and data analytics.
10	Traffic Monitoring and Control System Develop a system that uses IoT devices to monitor traffic flow, control traffic signals, and provide real-time traffic updates to drivers.

B: Minor Stream Courses/ Department Electives
B.1: Theory - Professional Elective – VIII

Code : BCEECE7111	ASP.Net	3 Credits [LTP: 3-0-0]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	will know all root concept of asp.net and compiling their website.
CO – 02	Understand	Will be able to integrate all kind of server controls and ceating their own web application
CO – 03	Apply	Will be able to manage sessions and validations.
CO – 04	Analyze	will be able to create their own MVC projects.
CO – 05	Evaluate	will be able to create and integrate their own web API.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	-	2	-	-	-	-	-	-	-	-	1	-
CO2	3	-	-	2	-	-	-	-	-	-	-	-	-	-	-
CO3	2	1	1	-	1	-	-	-	-	-	-	-	-	1	3
CO4	2	-	2	-	-	-	-	-	-	-	-	-	-	-	1
CO5	2	1	2	-	2	-	-	-	-	-	-	-	-	-	3
WT. AVG	2.2	1.6	1.5	2	1.6									1	2.3

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	ASP.NET Overview and developing a web application	07
2.	Application structure and state's standard web forms	08
3.	Working with navigation and validation controls, Web parts Control	07
4.	Working with database controls and ADO.net	08
5.	ASP.net MVC and Web API	07

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**ASP.NET Overview and developing a web application**
	Introduction of Unit Exploring New Features of ASP.NET 4.5, ASP.NET Technologies, The ASP.NET Life Cycle, Exploring a Sample ASP.NET Web Application, Creating a Sample ASP.NET Website, Specifying a Location for a Web Application, File Types in ASP.NET Exploring ASP.NET Web Pages, ASP.NET Coding Model, Understanding ASP.NET Directives, Working with Server Controls, Understanding the Provider Model in ASP.NET, Implementing Code Sharing, Compilation in ASP.NET, Dynamic Compilation in ASP.NET Conclusion of Unit
2.	**Application structure and state's standard web forms**
	Introduction of Unit Structure of an Application, The Global.asax Application File, Using States, HTTP Handlers, Postback and Cross-Page Posting, Using the Global.asax File, Using Application State, Session State, View state. Creating an HTTP Handler Application, Working with Postback and Cross-Page Posting, Web Forms-Standard Controls Conclusion of Unit
3.	**Working with navigation and validation controls, Web parts Control**
	Introduction of Unit.

	Using the Tree View Class, The Tree View Control, Using the Menu Class, The Menu Control, Using the SiteMapPath Class, The SiteMapPath Control, Creating All Controls and Validation Controls. Creating web pages with Web PartManager Control, The Proxy Web ParManager Control, The Connections Zone Control, Creating all controls. Conclusion of Unit
4.	**Working with database controls and ADO.net**
.	Introduction of Unit The Grid View Control, The Data List Control, The Details View Control, The Form View Control, The List View Control, The Repeater Control, The DataPager Control, The Chart Control, The Query Extender Control The SQL Data Source Control, The AccessDataSource Control, The Link DataSource Control. The Object Data Source Control, The XmlDataSource Control The Exntity Data SourceControl, The SiteMap Data Source Control,developing application with ADO.Net. Conclusion of Unit
5	**ASP.net MVC and Web API**
.	Introduction to ASP.NET MVC First ASP.NET MVC application. Exploring with MVC Controllers Installing ASP.NET Core SDK and Runtime New Web API project with Visual Studio Default ASP.Net core project files Testing the Web API Project with Postman and Swagger Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	ASP.NET 4.5	Kogent	Fourth Edition	Learning Solutions Inc,
2	Programming ASP.NET Core	Dino Esposito	Professional Edition	Microsoft
Reference Book				
1	The Complete Reference ASP.NET Matthew MacDonald Indian Edition.			
Online Resources				
1	https://www.w3schools.com			
2	https://www.udemy.com/courses/search/?src=ukw&q=ASP.NET			
3	https://www.microsoft.com/en-in/search/explore?q=asp+net+notes			

Code :BADECE7111	Python for Time Series Data Analysis	3 Credits [LTP: 3-0-0]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Understand and Analyze	Recall and identify basic Python syntax, including data types, variables, and operators.
CO – 02	Understand and Evaluate	Demonstrate an understand the concept of Decision making in python, Control statements and loop manipulation and how it is implemented in Python.
CO – 03	Understand and Apply	Put the different Python Packages like pandas, numpy, matplot, scipy along with lists, tuple,set and dictionary and how it is imported in the python notebook.
CO – 04	Apply	Evaluate and Examine object-oriented design principles to create well-structured Python classes and objects.
CO – 05	Apply	How to Handle files in python and Also evaluate the efficiency and performance of different approaches to solving a programming problem in Python

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	1	1	-	-	-	-	-	-	-	-	-	2	1	1
CO2	2	2	1	-	-	-	-	-	-	-	-	-	2	1	1
CO3	3	2	2	1	-	-	-	-	-	-	-	-	3	2	1
CO4	3	3	2	2	-	-	-	-	-	-	-	-	3	3	2
CO5	3	3	3	3	-	-	-	-	-	-	-	-	3	3	3
WT. AVG	**2.6**		**1.3**	**2**									**2.75**	**2.6**	**1**

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Time Series Analysis**	**07**
2.	**Time Series Preprocessing and Feature Engineering**	**08**
3.	**Time Series Forecasting with Classical Methods**	**07**
4.	**Time Series Forecasting with Machine Learning**	**08**
5.	**Time Series Analysis with Deep Learning**	**07**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Time Series Analysis**
	Understanding the concept of time series data and its applications. Basic time series data visualization and exploration techniques. Handling datetime objects and time-indexed data in Python. Time series decomposition: trend, seasonality, and residual components. Introduction to popular time series libraries in Python (e.g., Pandas, NumPy).
2.	**Time Series Preprocessing and Feature Engineering**
	Handling missing values and outliers in time series data. Resampling and interpolation techniques for irregular time series. Feature engineering for time series data (lag features, rolling statistics, etc.). Time series differencing and transformation to achieve stationarity. Implementing time series preprocessing using Pandas and NumPy.
3.	**Time Series Forecasting with Classical Methods**
	Understanding classical time series forecasting methods (e.g., ARIMA, SARIMA). Parameter tuning for ARIMA models using ACF and PACF plots. Seasonal decomposition of time series for forecasting. Evaluating forecast accuracy using metrics like RMSE and MAE.

	Building and validating ARIMA models in Python.
4.	**Time Series Forecasting with Machine Learning**
.	Introduction to machine learning-based time series forecasting. Feature selection and engineering for machine learning models. Regression-based time series forecasting with linear and non-linear models. Time series forecasting with ensemble methods (e.g., Random Forest, XGBoost). Implementing machine learning-based time series forecasting in Python.
5	**Time Series Analysis with Deep Learning**
.	Introduction to deep learning-based time series forecasting. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks. Time series forecasting using RNNs and LSTMs in Python (using libraries like TensorFlow or Keras). Handling large-scale time series data with deep learning models. Advanced topics in time series analysis: multivariate time series, attention mechanisms, and sequence-to-sequence models.

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	"Practical Time Series Analysis: Master Time Series Data Processing, Visualization, and Modeling using Python"	Aileen Nielsen	Latest	Apress
2	"Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython"	Wes McKinney	Latest	O'Reilly Media
Reference Book				
1.	"Time Series Analysis and Its Applications: With R Examples" Robert H. Shumway, David S. Stoffer Publisher: Springer			
Online Resources				
1.	https://www.analyticsvidhya.com/blog/2021/07/time-series-forecasting-complete-tutorial-part-1/			
2.	https://www.datacamp.com/tutorial/tutorial-time-series-forecasting			
3.	https://www.kaggle.com/code/prashant111/complete-guide-on-time-series-analysis-in-python			

Code :BCSECE7111	Malware Analysis & Wireless Security	3 Credits [LTP: 3-0-0]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Identify various malwares and understand the behavior of malwares in real world applications.
CO – 02	Understand	Demonstrate different malware analysis techniques.
CO – 03	Apply	Analyze knowledge about the purpose of malware analysis
CO – 04	Analyze	Recognize the purpose of malware analysis.
CO – 05	Evaluate	Identify the various tools for wireless network analysis.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	-	-	-	-	-	-	-	1	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO5	-	2	-	-	-	-	-	-	-	-	-	-	-	-	-
WT. AVG	**3**	**2**											**1**		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction & Data Collection Methods**	**07**
2.	**Windows Basics & Dynamic Malware Analysis**	**08**
3.	**Basic Static Analysis & Advanced Static Analysis Reverse Engineering**	**08**
4.	**Android Malware Analysis**	**07**
5.	**Wireless Security**	**07**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Cloud Solution**
	Introduction of Unit Malware Analysis Goals of Malware Analysis, Techniques Static and Dynamic Analysis, Types of Malware Backdoor, Botnet, Downloader, Information Stealing malware, Launcher, Rootkit, Scareware, Worm or Virus. Data Collection Methods:Volatile Data Collection Methodology-Preservation of Volatile Data, Physical Memory Acquisition on a Live Windows System, Identifying Users Logged into the System, Non-Volatile Data Collection Inspect Prefetch Files, Examine the File System, Remote Registry Analysis, Examine Web Browsing Activities, Examine Cookie Files Conclusion of Unit
2.	**Cloud & Tools**
	Introduction of Unit Introduction to Windows Malware - Windows Basics Relevant to Malware Behavior-File System and Directory structure, Registry, Boot Sequence, Malware payloads. Dynamic Malware Analysis: Malware activities, Self-Start techniques, Essential setup for executing malware, Executing DLL files, Classifying Malware Based on their Behavior Conclusion of Unit
3.	**Basic Static Analysis & Advanced Static Analysis Reverse Engineering**
	Introduction of Unit Number System Static Analysis with File Attributes and PE Header Packet Identification

	Advanced Static Analysis Reverse Engineering Assembly level computing Standard x86 instructions, Introduction to IDA, OllyDbg, Advanced Malware Analysis Virus, Trojan. Parsing Basic Analysis of an APK. Conclusion of Unit
4.	**Android Malware Analysis**
.	Introduction of Unit APK File Structure Security Model Android Root Brief Description of Spreading and Dis- tribution Introduction to Android Debugging Tools and Their Usage Dex Structure Parsing Basic Analysis of an APK. Exploits Master Key Vulnerability File Name Length Vulnerability Introduction to Obfuscation DEX code obfuscation Conclusion of Unit
5	**Wireless Security**
.	Introduction of Unit Introduction , Overview of Cryptography and Wireless Networks Security in Wireless LANs , Security in Cellular Networks Bluetooth Security , Ad hoc and sensor network security Student presentations in other related areas that instructor did not cover (security and privacy in RFID systems, vehicular networks, wireless mesh networks, satellite networks, etc.) Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Malware Forensics Field Guide for Linux Systems	Cameron H. Malin	2014	Elsevier
2	**802.11 Wireless Networks**	Matthew Fast		O'relly
Reference Book				
1	Cameron H. Malin, Eoghan Casey, James M. Aquilina and Curtis W. Rose, Malware Forensics Field Guide for Windows Systems, Syngress, Elsevier, 2012			
2.	Christopher C. Elisan , Advanced Malware Analysis, Tata McGraw Hill, 2015 3.Cameron H. Malin, Eoghan Casey, James M. Aquilina and Curtis W. Rose, Malware			
3.	**Designing and Deploying 802.11 Wireless Networks**			
Online Resources				
1	https://www.cybrary.it/resources/study-guides/comptia-security-plus/			
2	https://www.tutorialspoint.com/wireless_security/index.htm			
3	https://github.com/topics/android-security?l=html&o=asc&s=updated			

Code :BCGECE7111	Docker Containers & Kubernetes Fundamentals	3 Credits [LTP: 3-0-0]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Remember Docker terminologies and basic commands.
CO – 02	Understand	Understand Docker architecture and its components.
CO – 03	Apply	Apply Docker commands to manage images and containers effectively.
CO – 04	Analyze	Analyze differences between Docker images and containers, Docker Swarm vs Kubernetes.
CO – 05	Evaluate	Evaluate advantages and use cases of Docker in virtualization and containerization.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1											1		
CO2	2	2	1											2	
CO3		2	1	2	1								2	1	
CO4		1		1					2					1	
CO5					2		1				2				1
WT. AVG	**2.5**	**1.5**	**1**	**1.5**	**1.5**		**1**		**2**		**2**		**1.5**	**1.333333**	**1**

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction of Docker**	**8**
2.	**Docker deployment orchestration**	**8**
3.	**Docker container**	**7**
4.	**Web services of docker container**	**8**
5.	**Mongo DB**	**8**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Dockers over the cloud computing**
	Introduction of Unit Introduction to Docker, features of Docker Important terminology in Docker Architecture of Docker : Docker Daemon, Dcoker client, host, Docker registry, Dcoker objects Docker images, Docker Containers Docker Storage , types, Docker Networking, types Conclusion of Unit
2.	**Docker Hub**
	Introduction of Docker hub Docker Hub features Advantages of docker hub Push and Pull images from Docker hub Docker cloud , terminologies of Docker Cloud Operation of Docker cloud, Dcoker cloud features Understand the containerization Container enhance the deployment process Unit conclusion
3.	**Docker Linux command environment**
	Docker instruction commands Running commands inside Docker Container

	Docker –user instruction Docker images Uses of Docker Images Difference between Docker images and Dcoker container Structure of Docker image Conclusion of unit
4.	**Working with Docker Images**
.	Docker images listing. Their names and tags Filters to list images , Pulling Docker images with specific tags Publshing images to Docker hub Docker customized images from container Conatinerization using Docker Containerization Docker architecture, components, compose,network Advantages of Docker Conclusion of unit
5	**Virtualization with Docker containers**
.	Introduction of virtualization with Docker container Provide the static IP to a Docker container Docker compose tool to run multi container application Introduction Docker swarm mode Working process of Docker Swarm Used of Docker swarm Different mode of Docker swarm Features of Docker swarm Difference between Docker Container and Docker swarm Difference between Docker swarm and Kubernetes Conclusion of unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Cloud Computing: Concepts, Technolo gy & Architecture	Zaigham Mahmood, Ricardo Puttini, and Thomas Erl	20013	Pearson Education
Reference Book				
1	"Cloud Computing For Dummies" by Judith Hurwitz			
Online Resources				
1	**https://www.geeksforgeeks.org/kubernetes-tutorial/**			
2	https://www.javatpoint.com/cloud-computing			

Code :BAMECE7111	Optimization Concepts for Data Science and Artificial Intelligence	3 Credits [LTP: 3-0-0]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Remember the basic concepts and terminologies of mathematical optimization, including linear programming, nonlinear programming, and integer/mixed-integer linear programming.
CO – 02	Understand	Understand the OPTMODEL procedure and its applications in various optimization problems
CO – 03	Apply	Apply the OPTMODEL procedure to formulate and solve linear programming problems using index sets and arrays.
CO – 04	Analyze	Analyze the results of optimization problems, including dual values and reduced costs in the simplex method.
CO – 05	Evaluate	Evaluate different optimization models and solutions, assessing their effectiveness and applicability to real-world problems.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO 2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	1											1		
CO2	3	1	2											1	
CO3			1	1	2								1	2	
CO4		3		2					2					1	
CO5					1		1				1				1
WT. AVG	2	1.6	1.5	1.5	1.5		1		2		1		1	1.3	1

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Introduction to Mathematical Optimization	07
2.	Linear Programming	08
3.	Nonlinear Programming	07
4.	Integer and Mixed-Integer Linear Programming	08
5.		

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Mathematical Optimization**
	Introduction. A simple example. The OPTMODEL procedure.
2.	**Linear Programming**
	Introduction to linear programming. Formulating and solving linear programming problems using the OPTMODEL procedure. Using index sets and arrays in the OPTMODEL procedure. Dual values and reduced costs in the simplex method (self-study). Applied data envelopment analysis (self-study) Reading SAS data sets (self-study).
3.	**Nonlinear Programming**
	Introduction to nonlinear programming. Solving nonlinear programming problems using the OPTMODEL procedure.

4.	Integer and Mixed-Integer Linear Programming
.	Introduction to integer and mixed-integer linear programming. Solving integer and mixed-integer linear programming problems using PROC OPTMODEL

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	The Complete Reference Linux		Seventh Edition	Mc graw hill
2	Linux Bible	Christopher Negus	Kindle Edition	Wiley
Reference Book				
1	Windows 10 Portable, Wiley, Paul Mac Fedrics.			
	Desktop OS for Expert, Sayan Banerjee and Swati Goel			
Online Resources				
1	https://www.bu.edu/tech/files/2018/05/2018-Summer-Tutorial-Intro-to-Linux.pdf			
2	https://www.youtube.com/watch?v=67keaaWOKzE			
3	https://www.coursera.org/os			

B.1: Theory - Professional Elective – IX

Code : BCEECE7112	Cloud Computing	3 Credits [LTP: 3-0-0]

F COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Learn the core concepts of the cloud computing paradigm
CO – 02	Understand	Understand the underlying principles of Cloud Technology and various types of cloud Computing architecture and types.
CO – 03	Apply	Learn to evaluate between different cloud solutions offered by various providers based on their merits and demerits.
CO – 04	Analyze	Integrating the fundamental concepts in cloud infrastructures to understand the tradeoffs in power, efficiency and cost.
CO – 05	Evaluate	Access various cloud programming models and apply them to solve problems on the cloud

G. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	2	1	-	-	-	-	-	-	-	-	2	1	1
CO2	3	2	2	1	-	-	-	-	-	-	-	-	2	1	1
CO3	2	2	1	1	-	-	-	-	-	-	-	-	1	1	1
CO4	2	1	1	1	-	-	-	-	-	-	-	-	1	1	-
CO5	2	1	1	-	-	-	-	-	-	-	-	-	-	-	-
WT. AVG	**2.4**	**1.6**	**1.4**	**1**									**1.5**	**1**	**1**

H. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Cloud Computing**	**06**
2.	**Cloud Computing Companies and Migrating to Cloud**	**07**
3.	**Cloud Cost Management and Selection of Cloud Provider**	**07**
4.	**Governance in the Cloud**	**08**
5.	**Cloud Deployment and Integration**	**08**

I. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Cloud Computing**
	Introduction to Unit Introduction to Cloud Computing, History and Evolution of Cloud Computing, Types of clouds, Private and Public clouds, Cloud Computing architecture, Cloud computing infrastructure, Merits of Cloud computing, Practical applications of cloud computing, Cloud computing delivery models and services (IaaS, PaaS, SaaS) Obstacles for cloud technology, Cloud vulnerabilities, Cloud challenges, Practical applications of cloud computing Conclusion of the Unit
2.	**Cloud Computing Companies and Migrating to Cloud**
	Introduction of Unit Web-based business services, Delivering Business Processes from the Cloud: Business process examples, Broad Approaches to Migrating into the Cloud, The Seven-Step Model of Migration into a Cloud, Efficient Steps for migrating to cloud Risks: Measuring and assessment of risks, Company concerns Risk Mitigation methodology for

	Cloud computing, Case Studies Conclusion of Unit
3.	**Cloud Cost Management and Selection of Cloud Provider**
	Introduction of Unit. Assessing the Cloud: software Evaluation, System Testing, Seasonal or peak loading, Cost cutting and cost- benefit analysis, selecting the right scalable application. Considerations for selecting cloud solution. Understanding Best Practices used in selection of Cloud service and providers, Clouding the Standards and Best Practices Issue: Interoperability, Portability, Integration, Security, Standards Organizations and Groups associated with Cloud Computing, Commercial and Business Consideration Conclusion of Unit
4.	**Governance in the Cloud**
.	Introduction of Unit Industry Standards Organizations and Groups associated with Cloud Computing, Need for IT governance in cloud computing Cloud Governance Solution: Access Controls, Financial Controls, Key Management and Encryption, Logging and Auditing, API integration Legal Issues: Data Privacy and Security Issues, Cloud Contracting models, Jurisdictional Issues Raised by Virtualization and Data Location, Legal issues in Commercial and Business Considerations Conclusion of Unit
5	**Cloud Deployment and Integration**
.	Introduction Explore cloud deployment models, including public, private, community, and hybrid clouds. Understand the process of migrating applications and data to the cloud. Learn about cloud integration techniques, including API management and data synchronization. Explore emerging trends and technologies in cloud computing, such as serverless computing and edge computing. Conclusion of Unit

RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg Andrzej M. Goscinski	Latest	John Wiley and Sons Publications
2	Brief Guide to Cloud Computing	Christopher Barnett	Latest	Constable & Robinson
Reference Book				
1	"Cloud Computing Theory and Practice" by Dan C Marinescu Publisher: Elsevier			
Online Resources				
1	https://cloud.google.com/learn			

Code :BADECE7112	Introduction to Gen AI & Prompt Engineering	3 Credits [LTP: 3-0-0]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Understand	To understand what Generative AI is and how it is being used in different domains.
CO – 02	Understand	To understand Large Language Models (LLMs) and everything around them.
CO – 03	Understand, Analyze and Apply	To understand what prompt engineering is and then apply the learned concepts to improve the prompts.
CO – 04	Understand and Apply	To understand how Retrieval Augmented Generation can be used to improve the responses of an LLM when it is used for querying from custom database
CO – 05	Understand, Remember, Analyze and Apply	To understand how to perform fine tuning and different fine-tuning techniques and applying them to tailor models and putting them into production

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	2	-	3	3	3	3	3	3	-	3
CO2	3	3	3	3	3	2	-	3	3	3	3	3	3	-	3
CO3	3	3	3	3	3	2	-	3	3	3	3	3	3	-	3
CO4	3	3	3	3	3	2	-	3	3	3	3	3	3	-	3
CO5	3	3	3	3	3	2	-	3	3	3	3	3	3	-	3
WT. AVG	**3**	**3**		**3**		**2**		**3**	**3**				**3**		**3**

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Generative AI**	**6**
2.	**Introduction to Large Language Models (LLMs)**	**6**
3.	**Guide to Prompt Engineering**	**8**
4.	**RAG and Chatting with your data**	**8**
5.	**Model Adaptation, Fine Tuning and Production**	**8**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Introduction to Generative AI**
	Introduction of Unit. What is Generative AI? What can we generate? Generating Text, Images, Code, Music and Videos. Enterprise Use case examples. How is Generative AI different from Traditional AI? What approach should Enterprises take? Architecture Considerations. Conclusion & Real-Life Applications.
2.	**Introduction to Large Language Models (LLMs)**
	Introduction of Unit. Foundational Models, LLMs, Transformer Architecture, Types of LLMs. Prompts, Tokens, Embeddings, Model Configuration. Context Window, Prompt Engineering, Model Adaptation and Emergent Behavior.

	Conclusion & Real-Life Applications.
3.	**Guide to Prompt Engineering**
	Introduction of Unit, what is Prompt Engineering? Basics of Prompt Engineering. Prompt Engineering Techniques, Zero-Shot, Few-Shot, and Many-Shot Learning Reasoning – Chain of Thought (CoT), Self-Consistency Sampling Image Prompting, Prompt Injection. Prompt Engineering Challenges and Best Practices. Conclusion of the Unit
4.	**Retrieval Augmented Generation (RAG) and Chatting with your data**
.	Introduction of Unit. What is RAG? Benefits of RAG, RAG Architecture, Retriever System. Understanding the Vector Database, what is Vector Index? Vector Search, Challenges with RAG Chunking Strategies Using Vector Database, Retrieving the data, an end-to-end Chat Implementation powered by RAG Conclusion of the Unit
5	**Model Adaptation, Fine Tuning and Production**
.	Introduction of Unit Key Stages of Fine Tuning LLMs, Preparing dataset for fine tuning Fine-tuning, Fine-tuning Training Metrics, Deployment of Fine-tuned models Model Adaptation – LoRA, RLHF Best Practices for Production Deployment, LLMOps and MLOps Checklist for Production Deployment Conclusion of the Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Generative AI in Action	Anit Behree	1st Edition	Manning Publications
2	LLMs in Production: From Language Models to Successful Products	Christopher Brousseau	1st Edition	Manning Publications
Reference Book				
1	Introduction to Generative AI, Numa Dhamani, Maggie Engler, 1st Edition, 2024, Manning Publications			
2.	Learn Generative AI with Pytorch, Mark Liu, 1st Edition, 2024, Manning Publications			
Online Resources				
1	https://www.tensorflow.org			
2	https://www.pytorch.org			
3	https://www.keras.io			
4	https://huggingface.co			

Code :BCSECE7112	Cyber Threat Intelligence & Bug Bounting	3 Credits [LTP: 3-0-0]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	To recognize why Intelligence and cyber threat intelligence is a useful for developers.
CO – 02	Understand	Identify the key issues in apply Intelligence, develop and experiment with Intelligence & Cyber.
CO – 03	Apply	Develop problem solving and critical thinking skills in fundamental enable techniques like SQL Injection conditionals and loops.
CO – 04	Analyze	Construct and explain with structure and concept of different Bug Bounty.
CO – 05	Evaluate	Implement Read and write data from/to SQL Injection Cross Site Script step-wise by defining functions.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	1	1	—	—	—	—	—	—	—	—	2	—	—
CO2	3	2	1	2	—	—	—	—	—	—	—	—	2	—	—
CO3	2	3	2	1	—	—	—	—	—	—	—	—	1	—	—
CO4	2	2	2	1	—	—	—	—	—	—	—	—	2	—	—
CO5	2	3	1	1	—	—	—	—	—	—	—	—	2	—	—
WT. AVG	**2.2**	**2.6**	**1.4**	**1.2**									**1.8**		

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Understanding Intelligence & Cyber Threat Intelligence**	**07**
2.	**Threat Intelligence Consumption**	**08**
3.	**The Bug Bunty**	**07**
4.	**SQL Injection**	**08**
5.	**Cross Site Script**	**07**

D. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Understanding Intelligence & Cyber Threat Intelligence**
	Introduction of Unit Intelligence Lexicon and Definitions, Traditional Intelligence Cycle, Structured Analytical Techniques, Defining Threats, Understanding Risk, Cyber Threat Intelligence and Its Role, Expectation of Organizations and Analysts, Diamond Model and Activity Groups, Four Types of Threat Detection The Threat Intelligence Lifecycle 1. Direction 2. Collection 3. Processing 4. Analysis 5. Dissemination 6. Feedback Conclusion of Unit
2.	**Threat Intelligence Consumption**
	Introduction of Unit Sliding Scale of Cyber security, Consuming Intelligence for Different Goals, Enabling Other Teams with Intelligence, Building an Intelligence Team, Positioning the Team in the Organization, Prerequisites for Intelligence Generation, Intelligence Requirements, Priority Intelligence Requirements, Beginning the Intelligence Lifecycle, Threat Modeling

	Conclusion of Unit
3.	**The Bug Bounty**
	Introduction of Unit What is a Bug Bounty Program?, Popular Bug Bounty Platforms Bug crowd (Demo) HackerOne(Demo),Benefits of Bug Bounty, Brief About Common Vulnerabilities, Hacking Terminologies, What is Information Gathering?, Concept of Digital Footprinting, What Information to gather?, What is Whois Information, Information gathering about People & Organization, Gathering Information about Websites, Google Dorking & GHDB, DVWA Introduction, bWAPP Introduction, Introduction to Burp Suite. Conclusion of Unit
4.	**SQL Injection**
.	Introduction of Unit Introduction to SQL, Writing Basic SQL Query, Different types of comments used in SQL, SQLi Introduction & Impact, Union Based SQLi (Demo), Boolean Based SQli, Time Based SQLi, Validation Bypass (Client and Server), IDOR Vulnerability, IDOR on bWAPP, Rate Limiting Flaw, File Upload Vulnerability, File Upload on DVWA, Live IDOR POC, Live Rate Limiting Flaw POC Conclusion of Unit
5	**Cross Site Script**
.	Introduction of Unit What Is Cross Site Scripting(XSS)?, Stored XSS, Stored XSS (DVWA), Reflected XSS Reflected XSS (DVWA), DOM based XSS, Blind XSS, Live XSS POC, Host Header Injection methods & URL redirection, Live Host Header Injection POC, Live URL Redirection POC, Understanding Session, Cookies & Session Fixation, Forced Browsing, Cross Site Request Forgery Introduction, CSRF Attack(DVWA), Open Redirections, Personally Identifiable Information (PII) Leakage, Sensitive, Information Disclosure, Live CSRF POC, Live Sensitive Information POC, Live Session Fixation POC Conclusion of Unit

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Cyber Threat Intelligence [The Beginner's Guide]	Kurt Baker	March 2022	Springdell
2	Bug Bounty Hunting: A Complete Guide	K. Vilith	Aug 2020	Krademy
Reference Book				
1	Bug bounty Hunting Essential, Caarlos A., Shahmeer Amir,Packt			
2.	The Hacker Play Book3:Practical Guide to Penetration Testing,peter Kim			
Online Resources				
1	https://krademy.com/bug-bounty-hunting-complete-guide			
2	https://owasp.org/www-pdf-archive/Getting_Started_with_Bug_Bounty.pdf			
3	https://www.codegrazer.com/conference/dc151_Talk_BugBounty_scene_nijagaw.pdf			

Code: BCGECE7112 WEB FRAMEWORKS USING PYTHON 3 Credits [LTP:3-0-0]

Course Outcomes: -

CO	Cognitive Abilities	Course Outcomes
CO – 01	Understand	To Understand the fundamentals of Web Frameworks
CO – 02	Understand	To Understand the basics of Flask Framework
CO – 03	Understand and apply	To understand and apply advanced Flask Features for web development
CO – 04	Understand	To Understand the basics of Django Framework
CO – 05	Understand and apply	To understand and apply advanced Django and RESTful APIs for web development.

A. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Introduction to Web Frameworks**	6
2.	**Introduction to Flask**	8
3.	**Advanced Flask Features**	8
4.	**Introduction to Django**	6
5.	**Advanced Django and RESTful APIs**	8

B. CO-PO-PSO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	1	1	-	-	1	-	-	-	-	-	-	1	-	1	-	-
CO2	2	2	1	-	2	-	-	-	-	-	2	2	2	2	-	-
CO3	3	3	2	-	3	-	-	-	-	-	3	2	3	2	2	-
CO4	3	3	3	-	3	-	-	-	-	-	3	3	3	3	2	-
CO5	3	3	3	3	2	-	-	-	-	1	2	3	3	3	3	-

C. DETAILED SYLLABUS

Unit	Unit Details
1.	**Introduction to Web Frameworks**
	Introduction of Frameworks Web Application Frameworks MVC Architectures Overview of Python frameworks and their importance Comparison of popular Python frameworks (e.g., Django, Flask, Pyramid) Setting up a Python environment for framework development Conclusion of the Unit
2.	**Introduction to Flask**
	Overview of Flask and its features Setting up the Flask development environment Creating and structuring a basic Flask application Understanding URL routing and dynamic URLs Implementing variable rules in Flask Handling HTTP methods like GET, POST, etc.
3.	**Advanced Flask Features**
	Working with Jinja2 templates for dynamic HTML Serving and managing static files Understanding and using the request object Processing and validating form data Implementing cookies and managing sessions Handling redirects, custom error pages, and message flashing

	Introduction to Flask extensions, focusing on SQLAlchemy for database integration
4.	**Introduction to Django**
	Introduction to Django and its role in back-end web development. Understanding the HTTP protocol and its importance in web development. Learning the MVC model and its application in Django (MTV model). Setting up and using virtual environments for Django projects. Exploring Django project structure and the built-in admin interface. Working with generic views for rapid development. Creating and managing HTML templates. Configuring URL dispatcher and routing in Django
5.	Advanced Django and RESTful APIs
	Building custom views and handling GET/POST methods. Implementing a URL shortener and managing the user model. Adding logic in templates and querying models. Serving static files and deploying Django applications. Introduction to Django REST framework for building APIs. Creating and customizing generic views and serializers in Django REST. Handling JSON data and building RESTful APIs. Filtering models, working with images, and API authentication with tokens. Using Postman for API testing and managing related models. Deploying web APIs and using API endpoints. Deploying a Django REST project.

I. RECOMMENDED STUDY MATERIAL

S. No	Text Books:	Author	Edition	Publication
1.	Python Web Frameworks	Carlos de la Guardia	7th Edition	O'Reilly Media, Inc
2.	Build a Frontend Web Framework (From Scratch	Ángel Sola Orbaiceta	1st Edition	Manning Publishing
3.	Python Flask and Django	Emenwa Global	1st Edition	CreateSpace Independent Publishing Platform
Reference Book				
3.	"**Flask Mega-Tutorial** " by Miguel Grinberg			
4.	"**Mastering Django: A Beginner's Guide (Mastering Computer Science)** " by Dan Appleman			
Online Resources				
5.	**URL**: https://www.geeksforgeeks.org/differences-between-django-vs-flask/			
6.	**URL**: https://www.simplilearn.com/flask-vs-django-article			

Code :BAMECE7112	Speech & Video Processing	3 Credits [LTP: 3-0-0]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Describe the mechanisms of human speech production systems and methods for speech feature extraction.
CO – 02	Understand	Understand basic algorithms of speech analysis and speech recognition.
CO – 03	Apply	Explain basic techniques in digital video processing, including imaging characteristics and sensors.
CO – 04	Analyze	
CO – 05	Evaluate	Apply motion estimation and object tracking algorithms on video sequence.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	3	1	-	-	-	-	-	-	-	2	-	-
CO2	1	2	3	1	2	-	-	-	-	-	-	-	-	2	-
CO3	2	1	1	2	3	-	-	-	-	-	--	-	-	-	3
CO4	1	1	2	2	3	-	-	-	-	-	-	-	-	1	-
CO5	-	-	3	1	2	-	-	-	-	-	-	-	-	-	1
WT. AVG	1.3	2.0		2.0	2.0									1.5	2.0

C. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	Speech processing concepts	6
2.	Speech recognition	8
3.	Basics of Video Processing	8
4.	Motion estimation Techniques	6
5.	object tracking and segmentation	8

D. DETAILED SYLLABUS

Unit	Unit Details
1.	**Speech processing concepts**
	The speech production mechanism, Discrete time speech signals, Pole-Zero modeling of speech, relevant properties of the fast Fourier transform for speech recognition, convolution, linear and nonlinear filter banks, spectral estimation of speech using DFT. Linear Prediction analysis of speech.
2.	**Speech recognition**
	Real and Complex Cepstrum, application of cepstral analysis to speech signal, feature extraction for speech, static and dynamic feature for speech recognition, robustness issues, discrimination in the feature space, feature selection, MFCC, LPCC, Distance measures, vector quantization models. Gaussian Mixture model, HMM.
3.	**Basics of Video Processing**
	Video formation, perception and representation: Principle of color video, video cameras, video display, pinhole model, CAHV model, Camera motion, Shape model, motion model, Scene model, two-dimensional motion models. Three-Dimensional Rigid Motion, Approximation of projective mapping.
4.	**Motion estimation Techniques**
.	Optical flow, motion representation, motion estimation criteria, optimization methods, pixel-based motion estimation, Block matching algorithm, gradient Based, Intensity matching, feature matching, frequency domain motion estimation, Depth from motion. Motion analysis applications: Video

	Summarization, video surveillance.
5	**Object tracking and segmentation**
.	2D and 3D video tracking, blob tracking, kernel based counter tracking, feature matching, filtering Mosaicing, video segmentation, mean shift based, active shape model, video shot boundary detection. Interframe compression, Motion compensation

E. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Fundamentals of Speech recognition	L. Rabiner and B. Juang	3rd	Prentice Hall signal processing series.
2	Digital Video processing	A Murat Tekalp	2nd	Prentice Hall.
Reference Book				
1	1. "Speech and Audio Signal Processing", B.Gold and N. Morgan, Wiley. 2. "Digital image sequence processing, Compression, and analysis", Todd R. Reed, CRC Press 3. "Handbook of Image and Video processing", Al Bovik, Academic press, second Edition			
Online Resources				
1	**URL**: https://www.geeksforgeeks.org			

B.2: Practical - Professional Elective Lab- VI

Code :BCEECE7211	ASP.Net Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Remember basic ASP.Net concepts, terminologies, and development environment setup.
CO – 02	Understand	Understand the structure and components of ASP.Net applications.
CO – 03	Apply	Implement form handling, validation, and state management in ASP.Net applications.
CO – 04	Analyze	Analyze the functionality and performance of ASP.Net applications.
CO – 05	Evaluate	Evaluate the security aspects and best practices in ASP.Net applications.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3												3		
CO2	2	1											3	1	
CO3		2	2	1	2								3	2	
CO4		2	1	2	2				1				1	2	
CO5			2	1	1		1		1					1	2
WT. AVG	**2.5**	**1.66**	**1.66**	**1.33**	**1.6**		**1**		**1**				**2.5**	**1.5**	**2**

C. LIST OF EXPERIMENTS

1	Write a program to display the addition using the windows application
2	Write a program to perform all string operations using windows application.
3	Write a program forsimple calculator using windows application.
4	Write a program working with page using ASP.NET
5	Write a program working with forms using ASP.NET.
6	Write a program working with validation controls.
7	Write a programfor connectivity with Oracle database
8	Write a programto perform WebPartManager Control.
9	Write a programto access data source through ADO.NET.
10	Write a programto manage the session
11	Write a program to perform a GridView Control.
12	Develop an applet that displays a simple message in center of the screen
13	Design and develop a first MVC application.
14	Design and develop the new web API project.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	ASP.NET 4.5	Kogent	Fourth	Learning Solutions
2	Programming ASP.NET Core	Dino Esposito	Professional Edition	Microsoft
Reference Book				
1	The Complete Reference ASP.NET Mattew MacDonaldIndian Edition			
Online Resources				
1	https://www.w3schools.com			
2	https://www.udemy.com/courses/search/?src=ukw&q=ASP.NET			

Code :BADECE7211	Python for Time Series Data Analysis Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Discuss file handling concept and apply them to solve python program
CO – 02	Understand	Demonstrate proficiency in handling Strings and File Systems
CO – 03	Apply	Implement programs in Python using conditionals and loops for solving problems.
CO – 04	Analyze	Analyze various built-in python functions and formulate user-defined functions.
CO – 05	Evaluate	Create, run and manipulate Python Programs using core data structures like Lists, Dictionaries and use Regular Expressions.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	-	2	1	-
CO2	2	2	3	2	1	-	-	-	-	-	-	-	-	2	-
CO3	-	1	3	1	2	-	-	-	-	-	-	-	1	1	-
CO4	-	-	2	1	2	-	-	-	-	-	-	-	-	-	-
CO5	1	2	3	2	3	-	-	-	-	-	-	-	-	2	-
WT. AVG	**2**	**1.75**	**2.75**	**1.5**	**2**								**1.5**	**1.5**	

C. LIST OF EXPERIMENTS

1	Explore time series data using Pandas and NumPy.
2	Time Series Preprocessing - Handle missing values, resample, and interpolate time series data..
3	Feature Engineering - Generate lag features and transform data for forecasting.
4	ARIMA Modeling - Implement ARIMA and SARIMA models for time series forecasting.
5	Seasonal Decomposition - Decompose time series into trend, seasonality, and residual components.
6	Evaluating Forecast Accuracy - Evaluate forecasting performance using metrics like RMSE and MAE.
7	Regression-Based Forecasting - Apply linear regression for time series prediction.
8	Ensemble Methods - Use Random Forest and XGBoost for time series forecasting.
9	LSTM Implementation - Build LSTM networks for deep learning-based time series forecasting.
10	Multivariate Time Series - Handle and forecast multivariate time series data.
11	Attention Mechanisms - Implement attention mechanisms for improved forecasting.
12	Sequence-to-Sequence Models - Create sequence-to-sequence models for time series analysis.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	"Practical Time Series Analysis: Master Time Series Data Processing, Visualization, and Modeling using Python"	Aileen Nielsen	Latest	Apress
2	"Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython"	Wes McKinney	Latest	O'Reilly Media
Reference Book				
1	"Time Series Analysis and Its Applications: With R Examples" Robert H. Shumway, David S. Stoffer Publisher: Springer			
Online Resources				
1	https://www.analyticsvidhya.com/blog/2021/07/time-series-forecasting-complete-tutorial-part-1/ https://www.datacamp.com/tutorial/tutorial-time-series-forecasting			
2	https://www.kaggle.com/code/prashant111/complete-guide-on-time-series-analysis-in-python			
3	https://www.analyticsvidhya.com/blog/2021/07/time-series-forecasting-complete-tutorial-part-1/ https://www.datacamp.com/tutorial/tutorial-time-series-forecasting			

Code :BCSECE7211	Malware Analysis & Wireless Security Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Identify various malwares
CO – 02	Understand	understand the behavior of malwares in real world
CO – 03	Apply	applications.
CO – 04	Analyze	Apply different malware analysis techniques.
CO – 05	Evaluate	Analyze the malware behavior in windows and android.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2	PSO1	PSO2	PSO3
CO1	-	1	1	-	3	2	1	-	1	-	-	1	2	2	-
CO2	-	1	1	-	3	2	1	-	1	-	-	1	2	2	-
CO3															
CO4	-	1	1	-	3	2	2	1	1	1	-	1	3	2	-
CO5	-	1	1	-	3	2	2	1	1	1	-	1	3	2	-
WT. AVG	-	1	1	-	3	2	2	1	1	2	1	1	3	2	-

C. LIST OF EXPERIMENTS

1	Implementation of MITM- attack using wireshark/ network sniffers.
2	Implementation of IT Audit, malware analysis and Vulnerability assessment and generate the report.
3	Implementation of Windows security using firewall and other tools
4	Implementation to identify web vulnerabilities, using OWASP project
5	Implementation of Symmetric and Asymmetric cryptography.
6	Implementation to gather information from any PC's connected to the LAN using whois, port scanners, network scanning, Angry IP scanners etc.
7	Android malware analysis
8	Implementation of IT Audit, malware analysis and Vulnerability assessment and generate the report.
9	Comparative study of various malware analysis tools
10	Tools available in Antivirus Application
11	Analysis of different types of vulnerabilities for hacking a websites / Web Applications.
12	Analysis Unauthorized Access in Wireless Networks

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Malware Forensics Field Guide for Linux Systems	Cameron H. Malin	2014	Elsevier
2	Mobile Malware Attacks and Defense	Ken Dunham, Saeed Abu-Nimeh	2009	Elsevier
3	**802.11 Wireless Networks**	Matthew Fast		O'relly
Reference Book				
1	ErciFiliol, Computer Viruses: from theory to applications, Springer, 2005.			
2.	Designing and Deploying 802.11 Wireless Networks			
Online Resources				
1	https://www.cybrary.it/resources/study-guides/comptia-security-plus/			
2	https://www.tutorialspoint.com/wireless_security/index.htm			

Code :BCGECE7211	Docker Containers & Kubernetes Fundamentals Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Remember Docker terminologies and basic commands.
CO – 02	Understand	Understand Docker architecture and its components.
CO – 03	Apply	Apply Docker commands to manage images and containers effectively.
CO – 04	Analyze	Analyze differences between Docker images and containers, Docker Swarm vs Kubernetes.
CO – 05	Evaluate	Evaluate advantages and use cases of Docker in virtualization and containerization.

F. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	1											1		
CO2	2	2	1											2	
CO3		2	1	2	1								2	1	
CO4		1		1					2					1	
CO5					2		1				2				1
WT. AVG	**2.5**	**1.5**	**1**	**1.5**	**1.5**		**1**		**2**		**2**		**1.5**	**1.333333**	**1**

B. LIST OF EXPERIMENTS

1	**Installation Docker on Ubuntu**
2	**Create an application in Docker**
3	**Push an image to Docker Hub**
4	**Fetch and run the image from Docker Hub**
5	**Creating first repository in Docker Hub using GUI**
6	**How to Push and Pull images from Docker Hub**
7	**Deploy the Containerization Process**
8	**Docker installation process in windows**
9	**Create a Docker image and run it as container**
10	**Create a repository on Docker Hub**
11	**How to provide the static IP to a Docker container**
12	**Migration of containers**

C. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Cloud Computing: Concepts, Technology & Architecture	Zaigham Mahmood, Ricardo Puttini, and Thomas Erl	20013	Pearson Education
Reference Book				
1	"Cloud Computing For Dummies" by Judith Hurwitz			
Online Resources				
1	**https://www.geeksforgeeks.org/kubernetes-tutorial/**			
2	https://www.javatpoint.com/cloud-computing			

Code :BAMECE7211	Data Exploration & Visualization Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Use concepts in data visualization, such as types of charts, visualization techniques, and principles of effective visualization design.
CO – 02	Understand	Understand the significance of data visualization in exploring and communicating insights from data, including its role in identifying patterns, trends, and relationships.
CO – 03	Apply	Apply data visualization techniques using visualization tools like Tableau, matplotlib, ggplot, or D3.js to create visual representations of datasets.
CO – 04	Analyze	Analyze and interpret visualizations to extract meaningful insights from data, including identifying outliers, trends, and correlations.
CO – 05	Evaluate	Evaluate the effectiveness of different visualization types (e.g., bar charts, line charts, scatter plots) in representing specific types of data and conveying information to the audience.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	1	2	-	3	-	1	-	1	-	1	1	3	2	2
CO2	-	1	2	-	3	-	1	-	1	-	1	1	3	2	2
CO3	-	1	2	-	2	-	1	-	1	-	1	1	3	2	2
CO4	-	1	2	-	2	-	1	-	1	-	1	1	3	2	2
CO5	-	1	2	-	2	-	1	-	1	-	1	1	3	2	2
WT. AVG		**1**	**2**		**2.4**		**1**		**1**		**1**	**1**	**3**	**2**	**2**

C. LIST OF EXPERIMENTS

1	Perform the Visualization of Spread sheet Models according to given requirement. For the given data set that contains immigration details to Canada from 1980 to 2013, Create an area plot for top 6 immigrant countries from 1990 to 2013 Create and year-wise immigrant bar chart from India to Canada during the period of 1980 to 2013. Create a box plot for Indian, Phillip in and China immigrants. Show the total no. of immigrants from India and France countries using Area Chart and Pie chart. Create a scatter Histogram for the immigrants from Fiji and Singapore in the year 2013. LinkforDataSet- https://www.un.org/en/development/desa/population/migration/data/empirical2/migrationflows.asp Visualize the given Placement Data Full Class dataset that contains details about Campus Recruitment using the below techniques for appropriate dimensions and differentiate between the two techniques: Histogram and Bar Chart [For histogram let no. of bins=10] Facet Plot and Pair Plot Area Chart and Pie Chart [For yes or no data]
2	RDBMS Connectivity using Python Find out output of the joint operation applied to the company database. Apply inner join type to the following queries; apart from this apply other joins type to the first question. List the name of all employees who works for the research department. For every project located at 'Stafford' list the project number, the controlling Departmentt number and Departmentt manages last name. Find the name of all employees who works on the projects controlled by Dno=4. Make the list of project numbers for projects that involve an employee whose last name is' Jennifer' as a worker or as a manager of the dept that controls the project. List the name of the employees who have no dependents. List the name of manager that have at least one dependent.

3 Visualization of Semi-Structured Data

Create a dictionary for the below data and convert the data into JSON.

S.NO	Name	Department	GPA	Future Preference
1	Amy	CSE	8.7	Placements
2	Rebekah	ECE	9.2	Higher Education
3	David	CSE	5.6	Higher Education
4	Sophia	CSE	6.8	Placements
5	Lucas	ECE	7.5	Placements
6	Andrew	CSE	8.9	Higher Education
7	Evan	CSE	7.9	Placements
8	Rose	CSE	8.7	Higher Education
9	Luis	ECE	7.2	Higher Education
10	Blake	ECE	6.8	Higher Education
11	Finn	CSE	7.2	Placements
12	Alan	ECE	8.7	Placements
13	Olivia	ECE	6.8	Higher Education
14	Isabella	CSE	8.7	Placements
15	Scarlett	ECE	6.8	Higher Education

using dump()method from JSON package.

Plot a graph showing the difference in Future Preferences of the students.
Visualize the student's statistics based on the feature "Department".
Plot a pie chart for the feature GPA.

For the below given data set which contains world population in json format:
https://query.data.world/s/uvvfp4usm2q4mlapbqtoi2stgunwda

Read the data using pandas in column orient.
Using appropriate plotting technique visualize the given data on the basis of population feature.

5 Introduction to Tableau and Aggregation Methods in Any Data Visualization tool of your choice. Connect the given Bus Safety dataset to Tableau and perform the below tasks on separate sheets.
Go to meta-data of the data set and change the column name form 'Date Of Incident' to 'Date' and 'Bus Garage' to 'Garage'.
Visualize the no. of Incidents by different Operators and explore various possible charts.
Show a pie chart depicting the age categories as Adult, Child, Elderly and Unknown and no. of incidents in each category.
Show the statistics of Route No.'s in purple color Bar Chart.
Create a chart for 'Borough' feature depicting the total count of each and then sort it in ascending order.
Depict the no. of incidents under the eight Incident Event Types for each of the Boroughs in the form of horizontal bar chart.

6 Visual Encodings and Basic Dash boards in Any Data Visualization tool of your choice

For the given dataset FIFA.csv that contains data about various football players, perform the following tasks on separate sheets:
After connecting the data use the data interpreter and clean the data.
Create a horizontal bar chart to depict the International Reputation of various nations on an avg.
Check if there is any relation between wage and position(left/right). If yes, describe the relation.
Plot a bar chart against Avg. Heading Accuracy and Body Type. Find out which body type has highest and least accuracy.
Create a yellow colored Tree Chart to depict the total penalties of each nation and thus determine the highest and lowest.
Using the above sheets create a dashboard and write an analysis report of what insights can be drawn from this.

7 Interactive Plots in Python
Using the in-built "Car Crashes" dataset from seaborn library perform the below tasks in order to depict interactive plots.
. Create a sub-dataset df that contains 'total', 'speeding', 'alcohol' columns only.

i. Visualize an interactive bar plot for df.
ii. Using bar iplot, display the mean of all columns in the original dataset.

	v. Visualize a scatter matrix plot for the dataset. (The scatter matrix plot is basically a set of all the scatter plots for numeric columns in your dataset) vi. Depict an interactive box plot for df. vii. Show a histogram plot for df interactively. viii. Visualize 3D iplot for the data and give your insights so as to why and when should 3D visualization be used.
8	Hierarchical and Topographical Data Visualizations in Any Data Visualization tool of your choice. Using the in-built data set from following link : https://www.bls.gov/cpi/tables/relative-importance/home.htm#Archived%20Relative%20Importance%20Data Find the suitable answer of following Develop a sunburst pie chart to visualize all items. Create a tree map graph to display data in rectangular box Display the data in hierarchical format using shankey diagram.
9	Calendar Heat maps Data Visualizations in Python Write a Pandas program to create a heat map (rectangular data as a color-encoded matrix) for comparison of the top 10 years in which the UFO was sighted vs each Month. Load the dataset from "flight_dealy.csv" and create a heat map to show relationship between various fields of dataset.
10	Time Series Data Visualization in Python Collect the dataset from link https://github.com/Neelu-Tiwari/dataset/blob/main/stock_data.csv and perform the following task. Plot the changes that occurred in data over time. Create a bar plot of month data for 2016 and 2017. Perform the more practices from https://learnche.org/pid/data-visualization/data-visualization-exercises https://www.r-exercises.com/2017/04/10/forecasting-time-series-exploration-exercises-part-1/
11	Imagine that you work at one location of a retail department store chain. You're curious to see how the proportion of sales by product category at your particular store differs from the average of sales distribution numbers across all locations. Download the dataset from https://docs.google.com/spreadsheets/d/1VDG- ZpkkRaAituejvzIJ1Ky24LMpQfTN/edit#gid=416232713.
12	Select a member of the MIT aesthetics and computation research group (http://acg.media.mit.edu/). Briefly discuss that person's work and provide a review of the potential for that technique to help in information visualization (amount of information communicated vs. amount of aesthetics).

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Effective Data Storytelling: How to Drive Change with Data, Narrative, and Visuals	Brent Dykes	2018	
2	Effective Data Visualization: The Right Chart for the Right Data	Stephanie D.H. Evergreen	2021	Evergreen Data & Evaluation, LLC
Reference Book				
1	"Information Dashboard Design: Displaying Data for At-a-glance Monitoring" by Stephen Few,O'Rellay			
2.	"The Accidental Analyst: Show Your Data Who's Boss" by Eileen and Stephen McDaniel, O'Rellay			
Online Resources				
1	https://towardsdatascience.com/visualize-hierarchical-data-using-plotly-and-datapane-7e5abe2686e1			
2	https://www.idvbook.com/index.html%3Fp=44.html			

Professional Elective Lab- VII

Code :BCEECE7212	Cloud Computing Lab	1 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Articulate the main concepts, key technologies, strengths and limitations of cloud computing.
CO – 02	Understand	Explain the key and enabling technologies that help in the development of cloud.
CO – 03	Apply	Make use of NIST cloud computing architecture to solve architecture design challenges
CO – 04	Analyze	Explain the core issues of cloud computing such as resource management and security.
CO – 05	Evaluate	Install and use current cloud technologies. llustrate and choose the appropriate technologies, algorithms and approaches for implementation and use of cloud.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	2	1	1	-	-	-	-	1	2	2	-
CO2	-	-	-	-	2	1	1	-	-	-	-	1	2	2	-
CO3	-	-	-	-	3	2	2	1	-	-	-	1	2	3	-
CO4	-	-	-	-	2	1	1	-	-	-	-	1	2	2	-
CO5	-	-	-	-	3	2	2	1	-	-	-	1	2	3	-
WT. AV G					**2.4**	**1.4**	**1.4**	**1**				**1**	**2**	**2.4**	

C. LIST OF EXPERIMENTS

1	Introduction to cloud computing
2	Creating a Warehouse Application in SalesForce.com.
3	Creating an Application in SalesForce.com using Apex programming Language.
4	Implementation of SOAP Web services in C#/JAVA Applications.
5	Implementation of Para-Virtualization using VM Ware‘s Workstation/ Oracle‘s Virtual Box and Guest O.S.
6	Installation and Configuration of Hadoop.
7	Create an application (Ex: Word Count) using Hadoop Map/Reduce.
8	Case Study: PAAS(Facebook, Google App Engine)
9	Case Study: Amazon Web Services.
10	Introduction to cloud computing
11	Creating a Warehouse Application in SalesForce.com.
12	Creating an Application in SalesForce.com using Apex programming Language.

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg, Andrzej M. Goscinski	Latest	John Wiley and Sons Publications
2	Brief Guide to Cloud Computing	Christopher Barnett	Latest	Constable & Robinson
Reference Book				
1	"Cloud Computing Theory and Practice" by Dan C Marinescu Publisher: Elsevier			
Online Resources				
1	https://cloud.google.com/learn			

Code :BADECE7212	Data Handling & Visualization lab	1 Credits [LTP: 0-0-2]	

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Use concepts in data visualization, such as types of charts, visualization techniques, and principles of effective visualization design.
CO – 02	Understand	Understand the significance of data visualization in exploring and communicating insights from data, including its role in identifying patterns, trends, and relationships.
CO – 03	Apply	Apply data visualization techniques using visualization tools like Tableau, matplotlib, ggplot, or D3.js to create visual representations of datasets.
CO – 04	Analyze	Analyze and interpret visualizations to extract meaningful insights from data, including identifying outliers, trends, and correlations.
CO – 05	Evaluate	Evaluate the effectiveness of different visualization types (e.g., bar charts, line charts, scatter plots) in representing specific types of data and conveying information to the audience.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	1	2	-	3	-	1	-	1	-	1	1	3	2	2
CO2	-	1	2	-	3	-	1	-	1	-	1	1	3	2	2
CO3	-	1	2	-	2	-	1	-	1	-	1	1	3	2	2
CO4	-	1	2	-	2	-	1	-	1	-	1	1	3	2	2
CO5	-	1	2	-	2	-	1	-	1	-	1	1	3	2	2
WT. AVG		**1**	**2**		**2.4**		**1**		**1**		**1**	**1**	**3**	**2**	**2**

C. LIST OF EXPERIMENTS

1	Perform the Visualization of Spread sheet Models according to given requirement. For the given data set that contains immigration details to Canada from 1980 to 2013, Create an area plot for top 6 immigrant countries from 1990 to 2013 Create and year-wise immigrant bar chart from India to Canada during the period of 1980 to 2013. Create a box plot for Indian, Phillip in and China immigrants. Show the total no. of immigrants from India and France countries using Area Chart and Pie chart. Create a scatter Histogram for the immigrants from Fiji and Singapore in the year 2013. LinkforDataSet- https://www.un.org/en/development/desa/population/migration/data/empirical2/migrationflows.asp Visualize the given Placement Data Full Class dataset that contains details about Campus Recruitment using the below techniques for appropriate dimensions and differentiate between the two techniques: Histogram and Bar Chart [For histogram let no. of bins=10] Facet Plot and Pair Plot Area Chart and Pie Chart [For yes or no data]
2	RDBMS Connectivity using Python Find out output of the joint operation applied to the company database. Apply inner join type to the following queries; apart from this apply other joins type to the first question. List the name of all employees who works for the research department. For every project located at 'Stafford' list the project number, the controlling Departmentt number and Departmentt manages last name. Find the name of all employees who works on the projects controlled by Dno=4. Make the list of project numbers for projects that involve an employee whose last name is' Jennifer' as a worker or as a manager of the dept that controls the project. List the name of the employees who have no dependents. List the name of manager that have at least one dependent.
3	Visualization of Semi-Structured Data

Create a dictionary for the below data and convert the data into JSON.

S.NO	Name	Department	GPA	Future Preference
1	Amy	CSE	8.7	Placements
2	Rebekah	ECE	9.2	Higher Education
3	David	CSE	5.6	Higher Education
4	Sophia	CSE	6.8	Placements
5	Lucas	ECE	7.5	Placements
6	Andrew	CSE	8.9	Higher Education
7	Evan	CSE	7.9	Placements
8	Rose	CSE	8.7	Higher Education
9	Luis	ECE	7.2	Higher Education
10	Blake	ECE	6.8	Higher Education
11	Finn	CSE	7.2	Placements
12	Alan	ECE	8.7	Placements
13	Olivia	ECE	6.8	Higher Education
14	Isabella	CSE	8.7	Placements
15	Scarlett	ECE	6.8	Higher Education

using dump()method from JSON package.

Plot a graph showing the difference in Future Preferences of the students.
Visualize the student's statistics based on the feature "Department".
Plot a pie chart for the feature GPA.

For the below given data set which contains world population in json format:
https://query.data.world/s/uvvfp4usm2q4mlapbqtoi2stgunwda

Read the data using pandas in column orient.
Using appropriate plotting technique visualize the given data on the basis of population feature.

5 Introduction to Tableau and Aggregation Methods in Any Data Visualization tool of your choice. Connect the given Bus Safety dataset to Tableau and perform the below tasks on separate sheets.

Go to meta-data of the data set and change the column name form 'Date Of Incident' to 'Date' and 'Bus Garage' to 'Garage'.

i. Visualize the no. of Incidents by different Operators and explore various possible charts.

Show a pie chart depicting the age categories as Adult, Child, Elderly and Unknown and no. of incidents in each category.

Show the statistics of Route No.'s in purple color Bar Chart.

Create a chart for 'Borough' feature depicting the total count of each and then sort it in ascending order.

Depict the no. of incidents under the eight Incident Event Types for each of the Boroughs in the form of horizontal bar chart.

6 Visual Encodings and Basic Dash boards in Any Data Visualization tool of your choice

For the given dataset FIFA.csv that contains data about various football players, perform the following tasks on separate sheets:

i. After connecting the data use the data interpreter and clean the data.

i. Create a horizontal bar chart to depict the International Reputation of various nations on an avg.

Check if there is any relation between wage and position(left/right). If yes, describe the relation.

Plot a bar chart against Avg. Heading Accuracy and Body Type. Find out which body type has highest and least accuracy.

Create a yellow colored Tree Chart to depict the total penalties of each nation and thus determine the highest and lowest.

Using the above sheets create a dashboard and write an analysis report of what insights can be drawn from this.

7 Interactive Plots in Python

Using the in-built "Car Crashes" dataset from seaborn library perform the below tasks in order to depict interactive plots.

viii. Create a sub-dataset df that contains 'total', 'speeding', 'alcohol' columns only.

x. Visualize an interactive bar plot for df.

x.Using bar iplot, display the mean of all columns in the original dataset.

xi. Visualize a scatter matrix plot for the dataset. (The scatter matrix plot is basically a set of all the scatter

	plots for numeric columns in your dataset) xii. Depict an interactive box plot for df. xiii. Show a histogram plot for df interactively. xiv. Visualize 3D iplot for the data and give your insights so as to why and when should 3D visualization be used.
8	Hierarchical and Topographical Data Visualizations in Any Data Visualization tool of your choice. Using the in-built data set from following link : https://www.bls.gov/cpi/tables/relative-importance/home.htm#Archived%20Relative%20Importance%20Data Find the suitable answer of following Develop a sunburst pie chart to visualize all items. Create a tree map graph to display data in rectangular box Display the data in hierarchical format using shankey diagram.
9	Calendar Heat maps Data Visualizations in Python Write a Pandas program to create a heat map (rectangular data as a color-encoded matrix) for comparison of the top 10 years in which the UFO was sighted vs each Month. Load the dataset from "flight_dealy.csv" and create a heat map to show relationship between various fields of dataset.
10	Time Series Data Visualization in Python Collect the dataset from link https://github.com/Neelu-Tiwari/dataset/blob/main/stock_data.csv and perform the following task. Plot the changes that occurred in data over time. Create a bar plot of month data for 2016 and 2017. Perform the more practices from https://learnche.org/pid/data-visualization/data-visualization-exercises https://www.r-exercises.com/2017/04/10/forecasting-time-series-exploration-exercises-part-1/
11	Imagine that you work at one location of a retail department store chain. You're curious to see how the proportion of sales by product category at your particular store differs from the average of sales distribution numbers across all locations. Download the dataset from https://docs.google.com/spreadsheets/d/1VDG- ZpkkRaAituejvzIJ1Ky24LMpQfTN/edit#gid=416232713.
12	Select a member of the MIT aesthetics and computation research group (http://acg.media.mit.edu/). Briefly discuss that person's work and provide a review of the potential for that technique to help in information visualization (amount of information communicated vs. amount of aesthetics).

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Effective Data Storytelling: How to Drive Change with Data, Narrative, and Visuals	Brent Dykes	2018	
2	Effective Data Visualization: The Right Chart for the Right Data	Stephanie D.H. Evergreen	2021	Evergreen Data & Evaluation, LLC
Reference Book				
1	"Information Dashboard Design: Displaying Data for At-a-glance Monitoring" by Stephen Few,O'Rellay			
2.	"The Accidental Analyst: Show Your Data Who's Boss" by Eileen and Stephen McDaniel, O'Rellay			
Online Resources				
1	https://towardsdatascience.com/visualize-hierarchical-data-using-plotly-and-datapane-7e5abe2686e1			
2	https://www.idvbook.com/index.html%3Fp=44.html			

Code : BCSECE7212	Cyber Threat intelligence & Bug Bounty Lab	1 Credits [LTP: 0-0-2]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	To recognize why Intelligence and cyber threat intelligence is a useful for developers.
CO – 02	Understand	Identify the key issues in apply Intelligence, develop and experiment with Intelligence & Cyber.
CO – 03	Apply	Develop problem solving and critical thinking skills in fundamental enable techniques like SQL Injection conditionals and loops.
CO – 04	Analyze	Construct and explain with structure and concept of different Bug Bounty.
CO – 05	Evaluate	Implement Read and write data from/to SQL Injection Cross Site Script step-wise by defining functions.

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	2	1	1	-	-	-	-	-	-	-	-	1	2	-
CO2	2	2	1	1	-	-	-	-	-	-	-	-	1	3	-
CO3	2	2	1	1	-	-	-	-	-	-	-	-	1	3	-
CO4	2	2	1	1	-	-	-	-	-	-	-	-	-	2	-
CO5	2	2	1	1	-	-	-	-	-	-	-	-	-	3	-
WT. AVG	2.0	2.0	1.0	1.0									1.0	2.6	

C. LIST OF EXPERIMENTS

1	Performing Vulnerabilities Analysis and the OWASP
2	Using Web Proxies, DaaS, and Remote Systems
3	Prepare to Attack Our Fake Company with Bug Hunter Skills
4	Performing Web Hunting Edition, Active & Passive Reconnaissance
5	Performing Digital Footprint, Google Dorking & GHDB
6	Using Purposely Buggy Frameworks like BWapp, DVWA, Webgoat
7	Using RegEx: Basics, Tools, and Bypassing
8	Checking Out Auth with Basic Login and Password Bypass
9	Understanding Fuzzing Tools like FFuF, Libfuzz, Wfuzz
10	ExploreWeb Sockets, Session ID, Scripts and CSWH
11	Explore JavaScript, Credentials, De-obfuscation
12	What is SSRF Forgery, Bind, Time-Based
13	IDOR: Insecure Direct Object Reference, Identifiers
14	Validation Bypass (Client/Server)
15	Metadata in Websites: How to Find It and Its Value
16	SQLMap Essentials and Intermediate Capabilities
17	SQL Injection Fundamentals – Understanding SQL Formats
18	Time-Based SQLi, Writing Basic SQL Queries, Boolean
19	Wrap-Up: Day 1 Review and Homework Assignments
20	Command Injections: Payloads, Forms, Cookies
21	Host Header Injection, URL Redirection
22	Server-Side Attacks: SSRF and More SSRF, Session Security
23	Login Brute Forcing, Broken Authentication, 2FA Bypass
24	Web Attacks and File Inclusion, DNS, Subdomains, Domain Hunting

D. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	Cyber Threat Intelligence[The Beginner'sGuide]	KurtBaker	March2022	Springdell
2	BugBountyHunting:A Complete Guide	K. Vilith	Aug2020	Krademy
Reference Book				
1	BugbountyHuntingEssential,CaarlosA.,ShahmeerAmir,Packt			
2	TheHackerPlayBook3:PracticalGuidetoPenetrationTesting,peter Kim			
Online Resources				
1	https://krademy.com/bug-bounty-hunting-complete-guide			
2	https://owasp.org/www-pdf-archive/Getting_Started_with_Bug_Bounty..pdf			
3	https://www.codegrazer.com/conference/dc151_Talk_BugBounty_scene_nijagaw.pdf			

Code: BCGECE7212	Web Framework Lab	1 Credit	[LTP: 0-0-2]

A. Course Outcome:

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Understand web framework for Web Development
CO – 02	Understand	Reduce the amount of code you write to build rich user interface applications
CO – 03	Apply	Increase the reliability and maintainability of UI by using Flask and Django
CO – 04	Analyze	Modularize your code with the custom services and directives.
CO – 05	Evaluate	Develop a complete module

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	2	1	-	-	-	-	-	-	-	-	-	1	2	-
CO2	-	-	-	1	-	-	-	-	-	-	-	-	1	1	-
CO3	1	2	-	1	-	-	-	-	-	-	-	-	1	2	-
CO4	2	2	1	1	-	-	-	-	-	-	-	-	-	1	-
CO5	-	2	1	-	-	-	-	-	-	-	-	-	-	3	-
WT. AVG	1.3	2.0	1.0	1.0									1.0	1.8	

C. LIST OF EXPERIMENTS:

1	Build a responsive web application for shopping cart with registration, login, catalog and cart pages using CSS3 features, flex and grid.
2	Develop a Django app that displays current date and time in server
3	Develop a simple Django app that displays an unordered list of fruits and ordered list of selected students for an event
4	Develop a layout.html with a suitable header (containing navigation menu) and footer with copyright and developer information. Inherit this layout.html and create 3 additional pages: contact us, About Us and Home page of any website.
5	Develop a model form for student that contains his topic chosen for project, languages used and duration with a model called project
6	Develop example Django app that performs CSV and PDF generation for any models created in previous laboratory component.
7	Create a simple Flask application that displays "Hello, World!" on the homepage. Add a route that takes a user's name as a parameter and greets them personally.
8	Develop a Flask application with a login and registration system. Implement form validation and password hashing for security.
9	Create a RESTful API using Flask that allows clients to perform CRUD operations on a collection of products. The API should return data in JSON format.
10	Build a Flask web application that integrates Bootstrap for responsive design. Create a blog site with features like adding posts, editing posts, and viewing a list of all posts.
11	Write a program Create a Form to Edit Your Data
12	Develop a Flask application that allows users to send emails. Implement a contact form that users can fill out, and send the form data to an email address using Flask-Mail.

D. RECOMMENDED STUDY MATERIAL

S. No	Text Books:	Author	Edition	Publication
1.	Mastering Django: A Beginner's Guide (Mastering Computer Science)	Sufyan bin Uzayr	2022	
2.	Flask Web Development 2e: Developing Web Applications with Python	Miquel Grinberg		
3.	Angular 6 for Enterprise Ready-Web Applications			
Reference Book				
4.	Mastering Django : George, Nigel			
5.	Two Scoops of Django: Audrey Roy Greenfeld			
Online Resources				
6.	https://kinsta.com/blog/flask-vs-django/			
7.	https://www.sitepoint.com/premium/books/python-web-development-with-django-and-flask-getting-started/read/1/			
8.	https://www.interviewbit.com/blog/flask-vs-django/			

Code :BAMECE7212	Time Series Data Analysis Lab	1 Credits [LTP: 0-0-2]

A. COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Remember	Recall and describe the fundamental concepts of time series data, including the distinction between time series and cross-sectional data
CO – 02	Understand	Explain and interpret the behavior of different time series models (e.g., AR, MA, ARMA, ARIMA) and the significance of stationarity, unit roots, and spectral analysis in time series forecasting.
CO – 03	Apply	Implement various time series models (e.g., AR, MA, ARIMA) and techniques (e.g., differencing, ADF test) using Python to analyze and forecast time series data
CO – 04	Analyze	Critically examine and analyze time series data to identify and decompose its components, evaluate model performance using statistical metrics (e.g., MSE, ACF, PACF)
CO – 05	Evaluate	Evaluate and compare the effectiveness of different time series models and techniques for forecasting, assessing the significance of model parameters

B. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	1	0	2	0	0	0	0	0	0	0	2	1	1
CO2	3	3	2	2	3	0	0	0	0	0	0	0	3	2	2
CO3	2	3	3	3	3	1	1	0	2	1	0	0	3	3	2
CO4	2	3	3	3	3	2	1	1	2	2	0	1	3	3	3
CO5	2	3	3	3	3	3	2	1	3	2	2	1	3	3	3
WT. AVG	2.4	2.8	2.4	2.2	2.6	1.2	0.8	0.4	1.4	1	0.4	0.4	2.8	2.4	2.2

C. LIST OF EXPERIMENTS

Experiment 1:	**Introduction to Time Series Data and Plotting**
	Objective: Understand and visualise time series data, and differentiate it from cross-sectional data. **Tasks:** **Load Time Series Data:** Use a built-in dataset in Python (e.g., AirPassengers dataset from seaborn or statsmodels). Plot the time series data using matplotlib or seaborn. **Differentiate Between Data Types:** Create a simple cross-sectional dataset (e.g., annual revenue data of different companies) and plot it. Compare the visual difference between time series and cross-sectional data. **Explore Components of Time Series:** Identify and visualise the components of the time series: trend, seasonal, and irregular components using seasonal_decompose from statsmodels. **Expected Outcome:** Students will understand how time series data differs from cross-sectional data. They will be able to plot and interpret the different components of time series data.
Experiment 2:	**arity and Its Importance in Time Series Analysis**
	Objective: Learn the concept of stationarity, its importance, and how to test for stationarity in a time series. **Tasks:** **Understand Stationarity:** Generate synthetic time series data with and without a trend and seasonal component. Visualise the data to understand the concept of stationarity. **Test for Stationarity:**

	Apply the Augmented Dickey-Fuller (ADF) test using statsmodels to check for stationarity. Use differencing techniques to make a non-stationary series stationary. **Interpretation:** Analyse the results of the ADF test and interpret whether the series is stationary or not. Discuss why stationarity is crucial for time series forecasting. **Expected Outcome:** Students will understand stationarity and how to test for it. They will gain hands-on experience in making a time series stationary.
Experiment 3:	**tary Time Series Models and Model Evaluation**
	Objective: Implement simple time series models and evaluate their performance using various metrics. **Tasks:** **White Noise and Random Walk Models:** Generate and plot a white noise process. Create a random walk model and plot it to understand its characteristics. **Simple Time Series Models:** Implement a moving average (MA) model and an autoregressive (AR) model using Python. Use the ARIMA model in statsmodels for a simple forecasting task. **Model Evaluation:** Evaluate the models using Bias, Mean Absolute Deviation (MAD), Mean Squared Error (MSE), and Mean Absolute Percentage Error (MAPE). Compare the performance of different models using these metrics. **Expected Outcome:** Students will understand basic time series models and their implementation. **ill learn how to evaluate and compare time series models using different performance metrics**
Experiment 4:	**ing Stationary Data with AR and MA Models**
	Objective: Understand and implement Auto-Regressive (AR) and Moving Average (MA) models for stationary time series data. **Tasks:** **Generate Stationary Data:** Create a stationary time series using Python's numpy and pandas libraries. Visualise the data to confirm its stationarity. **Implement AR and MA Models:** Implement an Auto-Regressive (AR) model using the ARIMA class in statsmodels. Implement a Moving Average (MA) model using the same ARIMA class, setting the appropriate parameters. Fit these models to the generated stationary data. **Model Analysis:** Plot the predictions and residuals from both models. Analyse the residuals to ensure they are white noise, indicating a good fit. **Expected Outcome:** Students will gain hands-on experience in implementing AR and MA models on stationary data. They will learn how to assess the model fit by analysing residuals.
Experiment 5:	**Understanding and Testing for Unit Roots**
	Objective: Explore the concept of unit roots, their impact on time series analysis, and perform unit root tests. **Tasks:** **Concept of Unit Roots:** Generate synthetic time series data with and without a unit root. Visualise the difference between a series with a unit root and one without. **Dickey-Fuller Test:** Perform the Dickey-Fuller test on both synthetic and real-world time series data using statsmodels. Interpret the results to determine whether a unit root is present. **Augmented Dickey-Fuller (ADF) and KPSS Tests:** Apply the Augmented Dickey-Fuller test and the KPSS test on the same datasets. Compare the results of these tests and discuss any differences. **Expected Outcome:** Students will understand the concept of unit roots and their impact on model estimation. They will learn how to perform and interpret the Dickey-Fuller, ADF, and KPSS tests.
Experiment 6:	**Seasonal Unit Roots and Periodic Integration**

	Objective: Investigate the presence of seasonal unit roots in time series data and understand periodic integration. **Tasks:** **Identify Seasonality and Stationarity:** Use real-world time series data that exhibits strong seasonality (e.g., monthly sales data). Test for stationarity in the presence of seasonality using differencing techniques. **Testing for Seasonal Unit Roots:** Apply the seasonal differencing method to remove seasonal unit roots. Perform seasonal unit root tests using statsmodels and interpret the results. **Periodic Integration:** Introduce the concept of periodic integration and its relevance in time series analysis. Implement a simple periodic integration model and test it on the seasonal data. **Expected Outcome:** Students will learn how to detect and handle seasonal unit roots in time series data. They will gain an understanding of periodic integration and its application in time series analysis.
Experiment 7:	**Understanding ARMA (p, q) Models and ACF/PACF**
	Objective: Explore the ARMA (p, q) process and analyse the Auto-Correlation Function (ACF) and Partial Auto-Correlation Function (PACF). **Tasks:** **Generate ARMA (p, q) Data:** Generate synthetic time series data following an ARMA (p, q) process using Python's statsmodels. Visualise the generated data. **Analyse ACF and PACF:** Plot the ACF and PACF of the generated ARMA process using matplotlib and seaborn. Identify the order of the ARMA model by analysing the ACF and PACF plots. **Model Fitting:** Fit an ARMA (p, q) model to the generated data using ARIMA in statsmodels. Compare the model parameters with the true parameters used to generate the data. **Expected Outcome:** Students will understand the ARMA (p, q) process and how to generate and analyse it. They will learn how to use ACF and PACF plots to determine the appropriate order of an ARMA model.
Experiment 8:	**Forecasting with ARMA and ARIMA Models**
	Objective: Implement forecasting using ARMA and ARIMA models, and learn about integration for non-stationary data. **Tasks:** **ARMA Model Forecasting:** Fit an ARMA model to real-world time series data. Use the model to forecast future values and visualise the forecasted data along with confidence intervals. **Handling Non-Stationary Data:** Introduce non-stationarity by generating or using real-world time series data with a trend. Apply first-order and second-order differencing to achieve stationarity. **ARIMA Model Implementation:** Fit an ARIMA (p, i, q) model to the differenced data. Forecast future values using the ARIMA model and compare the results with the ARMA model. **Expected Outcome:** Students will gain experience in forecasting with ARMA and ARIMA models. They will understand the importance of differencing in handling non-stationary data
Experiment 9:	**Parameter Estimation in ARIMA Models and Wald Test**
	Objective: Estimate the parameters of an ARIMA model and perform the Wald Test to evaluate the significance of coefficients. **Tasks:** **Fit an ARIMA Model:** Use real-world or synthetic time series data to fit an ARIMA model. Estimate the model parameters using statsmodels and interpret the output. **Wald Test for Coefficient Significance:** Perform the Wald Test on the estimated ARIMA model coefficients to determine their statistical significance. Interpret the test results and discuss the importance of each coefficient in the model. **Model Validation:**

	Validate the model by checking residuals for white noise using ACF and PACF plots. Refine the model if necessary and re-evaluate the significance of coefficients. **Expected Outcome:** Students will learn how to estimate the parameters of an ARIMA model and assess their significance using the Wald Test. They will understand how to validate and refine ARIMA models to improve forecasting accuracy.
Experiment 10:	**Spectral Densities and Periodogram Analysis**
	Objective: Understand spectral densities and periodograms, and explore their applications in time series analysis. **Tasks:** **Generate Time Series Data:** Generate synthetic time series data using Python, such as a sinusoidal signal combined with random noise. Visualise the time series data. **Compute and Plot the Periodogram:** Calculate the periodogram of the time series data using scipy.signal.periodogram. Plot the periodogram and analyse the dominant frequencies present in the data. **Spectral Density Estimation:** Estimate the spectral density using the periodogram and other methods such as Welch's method (scipy.signal.welch). Compare the spectral density estimates and discuss the impact of windowing and segmenting on the spectral estimates. **Expected Outcome:** Students will understand the concept of spectral densities and periodograms. They will gain hands-on experience in estimating and analysing the spectral density of time series data.
Experiment 11:	**Spectral Density of ARMA Models and Smoothing Techniques**
	Objective: Explore the spectral density of ARMA models and learn about smoothing techniques and their impact on spectral estimates. **Tasks:** **Spectral Density of ARMA Model:** Generate time series data from an ARMA model using statsmodels. Compute the spectral density of the ARMA model using the theoretical formula and verify it with a periodogram. **Smoothing the Spectral Density:** Apply smoothing techniques to the spectral density estimate, such as kernel smoothing or using a moving average. Analyze the bias-variance trade-off introduced by smoothing. **Bandwidth and Confidence Intervals:** Discuss the concept of bandwidth in the context of spectral density estimation. Calculate and interpret confidence intervals for the spectrum. **Expected Outcome:** Students will understand the spectral properties of ARMA models and how to estimate their spectral densities. They will learn how to apply smoothing techniques and interpret the resulting spectral estimates, considering bias, variance, and confidence intervals.
Experiment 12:	**VAR Estimation and VECM in Multivariate Time Series Analysis**
	Objective: Implement Vector AutoRegression (VAR) estimation, explore Vector Error Correction Models (VECM), and understand the basics of GARCH models for multivariate time series data. **Tasks:** **Introduction to VAR Estimation:** Load a multivariate time series dataset (e.g., economic indicators like GDP, inflation rate, and unemployment rate). Implement a Vector AutoRegression (VAR) model using Python's statsmodels. Estimate the parameters using multivariate least squares and discuss the asymptotic properties. **Cointegration and Vector Error Correction Model (VECM):** Test for cointegration among the variables using the Johansen cointegration technique. If cointegration is found, implement a Vector Error Correction Model (VECM). Analyse the long-term relationships between the variables and the short-term adjustments. **Introduction to GARCH Models:** Introduce the concept of GARCH models and discuss their application in modelling volatility. Implement a univariate GARCH model using Python's arch library.

	Discuss the extension to multivariate GARCH models and their application in finance. **Expected Outcome:** Students will gain practical experience in estimating and interpreting VAR models. They will learn how to test for and model cointegrated processes using VECM. Students will also be introduced to GARCH models, understanding their significance in volatility modelling.

D. RECOMMENDED STUDY MATERIAL:

S. No.	Text Books	Author	Edition	Publication
1	Time Series Analysis: Forecasting and Control	George E. P. Box, Gwilym M. Jenkins, Gregory C. Reinsel, and Greta M. Ljung	5th Edition	Wiley
2	Introduction to Time Series and Forecasting	Peter J. Brockwell and Richard A. Davis	3rd Edition	Springer
3	Applied Time Series Analysis with R	Wayne A. Woodward, Henry L. Gray, and Alan C. Elliott	2nd Edition	CRC Press
Reference Books:				
S. No.	**Reference Books**	**Author**	**Edition**	**Publication**
1	Time Series Analysis and Its Applications	Robert H. Shumway and David S. Stoffer	4th Edition	Springer
2	Practical Time Series Forecasting with R	Galit Shmueli	3rd Edition	Axelrod Schnall Publishers
Online Resources:				
S. No.	**Online Resources**	**Link or Author**		
1	Time Series Analysis in Python with statsmodels	statsmodels Documentation		
2	Forecasting: Principles and Practice (online book)	Rob J Hyndman and George Athanasopoulos		

D: Ability Enhancement Courses (AEC)

Code : BUACHM7116	INTERPERSONAL COMMUNICATION & GROOMING	2 Credits [LTP: 0-0-2]

A COURSE OUTCOMES

CO	Cognitive Abilities	Course Outcomes
CO – 01	Apply	Implement various tips for grooming the personality after doing a Swot analysis.
CO – 02	Apply	Use the soft skills set for cohesive working with the team as a team player.
CO – 03	Apply	Practice the techniques to build Emotional Intelligence.
CO – 04	Analyse	Differentiate between attitude and motivation while learning self-motivation.
CO – 05	Apply	Present positive self-esteem in personal and professional spaces.

B. OUTLINE OF THE COURSE

Unit No.	Title of the unit	Time required for the Unit (Hours)
1.	**Personality Grooming**	**5**
2.	**Personality Traits & Interpersonal Skills**	**5**
3.	**Emotional Intelligence**	**5**
4.	**Attitude & Motivation**	**5**
5.	**Self Esteem**	**4**

C. DETAILED SYLLABUS

1. Unit	Unit Details
1.	**Personality Grooming**
	Introduction to the Unit Definition & Basics of Personality The concept of success and failure Personal Grooming & Dressing Sense Causes of failure. SWOT analyses. Team Building Activities Conclusion & Real-Life Application
2.	**Personality Traits & Interpersonal Skills**
	Introduction to the Unit Introduction of the topic Personality Traits (OCEAN): Big-Five Personality characteristics such as Openness, Conscientiousness, Extroversion, agreeableness, and Neuroticism. Psychometric Assessment (Open Source) Skills Building Sessions Elements of FIRO-B "Inclusion, Affection & Control" in both Wanted and Expressed Dimensions. Group Feedback Prior to the session Suggestions on the categorized information Rapport Building Establishing Complementary Transactions

	Tips for Effective Interpersonal Skills Skills Building Sessions Conclusion & Real-Life Application
3.	**Emotional Intelligence**
	Introduction to the Unit Emotional Intelligence: Awareness of the Basic Emotions such as Fear, Anger, Jealousy, Happiness, Affection, Sentiments, Disgust, Sadness & Surprise Identifying Personal Levels of Emotional Labours Experiencing Emotional Authenticity & Emotional Sensibility by application of Sensitivity Processes Skill Building for Strengthening the Elements of Self-awareness, Self-regulation, Internal motivation, Empathy, Social skills Conclusion & Real-Life Application
4.	**Attitude & Motivation**
.	Introduction to the topic Listening Skills activities Social Problem Solving Managing Conflicts Being a part of the group and expression of feelings Conclusion & Real-Life Application
5	**Self Esteem**
.	Introduction to the Unit Term self-esteem Symptoms - Advantages Do's and Don'ts to develop positive self-esteem Low self-esteem - Symptoms - Personality having low self-esteem Conclusion and Real-Life Application

C. RECOMMENDED STUDY MATERIAL:

S. No	Text Books:	Author	Edition	Publication
1	English for Engineers and Technologists		(Combined edition, Vol. 1 and 2)	Orient Blackswan 2010.
2	The Elements of Style	William Strunk Jr. & E.B. White	4th Edition	Pearson, 1999.
3	Technical Communications	Raman Sharma	London, 2004	Oxford Publication
	Success in Interview	Anand Ganguly	5 Edition, 2016	RPH

E: Skill Enhancement Courses (SEC)

Code : BULCSE7201	Skill Enhancement Technical Courses-I	3 Credits [LTP: 3-0-0]

A. LIST OF EXPERIMENTS:

LIST OF LABS	
1.	Introduction to Competitive Programming Overview of competitive programming and its significance in the IT industry Understanding the common programming contest platforms and their features (e.g., Codeforces, Topcoder, HackerRank) Introduction to algorithmic problem-solving techniques and strategies Familiarization with programming languages commonly used in competitive programming (e.g., C++, Java) Basic input/output operations and handling test cases
2.	Data Structures Review of fundamental data structures (arrays, linked lists, stacks, queues) Advanced data structures: trees (binary trees, binary search trees, heaps), graphs (representation, traversal algorithms), and hash tables Analyzing time and space complexities of data structures and their operations Implementing data structures from scratch and utilizing built-in libraries
3.	Algorithms and Problem Solving Understanding algorithmic paradigms: greedy, divide and conquer, dynamic programming. Sorting and searching algorithms (e.g., quicksort, mergesort, binary search) Graph algorithms: breadth-first search (BFS), depth-first search (DFS), shortest paths, minimum spanning trees. Dynamic programming: principles, memoization, tabulation Solving practice problems to reinforce algorithmic thinking
4.	Advanced Topics String manipulation algorithms (pattern matching, suffix trees, tries) Advanced graph algorithms: topological sorting, strongly connected components, network flow. Computational geometry: convex hull, line intersection, closest pair of points. Number theory: prime numbers, modular arithmetic, Euclidean algorithm. Utilizing advanced algorithms to solve competitive programming problems

Note: This syllabus provides a broad outline for a competitive programming course. It is recommended to allocate sufficient time for each unit and incorporate regular practice sessions, coding exercises, and mock contests to enhance problem-solving skills and familiarity with the competitive programming environment.

A. RECOMMENDED STUDY MATERIAL:

Reference Book	
1	"Competitive Programming 3" by Steven Halim and Felix Halim. "Introduction to the Design and Analysis of Algorithms" by Anany Levitin
2	"Data Structures and Algorithms Made Easy: Data Structures and Algorithmic Puzzles" by Narasimha Karumanchi
3	"Computational Geometry: Algorithms and Applications" by Mark de Berg, Otfried Cheong, Marc van Kreveld, and Mark Overmars
4	"Introduction to Algorithms" by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein

5	"Algorithms on Strings, Trees, and Sequences: Computer Science and Computational Biology" by Dan Gusfield
6	"Algorithms on Strings, Trees, and Sequences: Computer Science and Computational Biology" by Dan Gusfield
7	"Computational Geometry: Algorithms and Applications" by Mark de Berg, Otfried Cheong, Marc van Kreveld, and Mark Overmars
8	"Competitive Programmer's Handbook" by Antti Laaksonen "Competitive Programming: The New Lower Bound of Programming Contests" by Hasan Jaddouh
Online Resources	
1	Codeforces: https://codeforces.com/ Topcoder: https://www.topcoder.com/ HackerRank: https://www.hackerrank.com/ LeetCode: https://leetcode.com/ Codeforces: https://codeforces.com/
2	GeeksforGeeks: https://www.geeksforgeeks.org/ Brilliant: https://brilliant.org/ AtCoder: https://atcoder.jp/
3	Google Code Jam: https://codingcompetitions.withgoogle.com/codejam
4	"Algorithms, Part I" by Robert Sedgewick and Kevin Wayne (online course on Coursera)

G: Summer Internship / Research Project / Dissertation

Code : BCECCE7301	Minor Project	3 Credits [LTP: 0-0-4]

B COURSE OUTCOMES AND THEIR RESPECTIVE MAPPING

CO	Cognitive Abilities	Course Outcomes
CO – 01	Apply	Apply theoretical knowledge to solve practical problems in a professional or research environment.
CO – 02	Analyze	Plan, execute, and manage a project from inception to completion.
CO – 03	Apply	Conduct research, analyze data, and draw meaningful conclusions.
CO – 04	Analyze	Prepare and present technical reports, research papers, or project documentation.
CO – 05	Evaluate	Work effectively as a member of a team and communicate results clearly to both technical and non-technical audiences.

C. MAPPING MATRIX OF CO, PO, & PSO

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	3	2	1	0	1	2	1	3	3	2	1
CO2	3	3	2	3	2	3	2	1	2	2	2	2	3	3	2
CO3	3	3	3	2	3	3	2	1	2	3	2	2	3	3	3
CO4	3	2	2	3	3	2	2	0	2	3	2	2	2	2	3
CO5	3	3	3	3	3	3	3	2	3	3	2	3	3	3	3
WT. AVG	3	2.8	2.4	2.6	2.8	2.6	2	0.8	2	2.6	1.8	2.4	2.8	2.6	2.4

D. DETAILED SYLLABUS

1	Data Collection and Analysis:
2	Collection of data as per the project requirements.
3	Use of appropriate analytical methods and tools to process and analyze the data.
4	Interpretation of results in the context of the research objectives.
5	Documentation and Reporting:
6	Preparation of a detailed project report or dissertation following the prescribed format.
7	Inclusion of all relevant sections such as introduction, methodology, results, discussion, conclusion, and references.
8	Regular submission of progress reports and drafts to the faculty supervisor.
9	Presentation of the project findings to a panel of faculty members.
10	Defense of the project methodology, findings, and conclusions.
11	Addressing questions and feedback from the panel.
12	Final Submission:
13	Submission of the final project report or dissertation.
14	Submission of any supplementary materials such as code, data sets, or models developed during the project

Code : BCECCE7401	Industrial Training Seminar-II	1 Credits [LTP: 0-0-2]

A. DETAILED SYLLABUS

1. Unit	Unit Details
1.	
	• **Introduction to Industrial Training Seminar-II** Overview and objectives Importance in the curriculum • **Module 1: [Topic Name]** Sub-topic 1 Sub-topic 2 ... • **Module 2: [Topic Name]** Sub-topic 1 Sub-topic 2 ... • **Module 3: [Topic Name]** Sub-topic 1 Sub-topic 2 ... • **Assessment Methods** Details on how each CO will be assessed (assignments, projects, exams, etc.) • **References** List of recommended textbooks and readings

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Member of Association of Indian Universities & Approved by UGC (Govt. of India) under 2(f) & 12(B)

B. TECH - CSE

FOURTH YEAR
SEMESTER: VIII

COMMON SYLLABUS

For

Cyber Security
Cloud & Full Stack Development
Artificial Intelligence & Data Science
Artificial Intelligence & Machine Learning

BATCH: 2024-28

Code : BCECCE8301	Major Project/Dissertation	10 Credits [LTP: 0-0-20]

A	Details
	The students will undertake a project as part of their final semester. The students can do independent projects or can take up projects in groups of two or more depending on the complexity of the project. The maximum group sizewill be four and in case of team projects there should be a clear delineation of the responsibilities and work done by each project member. The projects must be approved by the mentor assigned to the student. The mentors will counsel the students for choosing the topic for the projects and together they will come up with the objectives and the process of the project. From there, the student takes over and works on the project. If the student chooses to undertake an industry project, then the topic should be informed to the mentor, and the student should appear for intermediate valuations. Prior to undertaking this project the students undergo a bridge course. **Bridge Course:** The bridge course ensures that all the students have the correct prerequisite knowledge before their industry interface. The purpose of a bridge course is to prepare for a healthy interaction with industry and to meet their expectations. It would be difficult to establish standards without appropriate backgrounds and therefore to bridge this gap, students are put through a week mandatory classroom participation where faculty and other experts will give adequate inputs in application based subjects, IT and soft skills. **The Project:** Each student will be allotted a Faculty Guide and an Industry Guide during the internship/project work. Students need to maintain a Project Diary and update the project progress, work reports in the project diary. Every student must submit a detailed project report as per the provided template. In the case of team projects, a single copy of these items must be submitted but each team member will be required to submit an individual report detailing their own contribution to the project. Each student/group should be allotted a supervisor and periodic internal review shall be conducted which isevaluated by panel of examiners. **Project Evaluation Guidelines:** The Project evaluator(s) verify and validate the information presented in the project report.The break-up of marks would be as follows: Internal Evaluation External Assessment Viva Voce **Internal Evaluation:** Internal Evaluator of project needs to evaluate Internal Project work based on the following criteria: Project Scope , Objectives and Deliverables Research Work, Understanding of concepts Output of Results and Proper Documentation Interim Reports and Presentations– Twice during the course of the project **External Evaluation:** The Project evaluator(s) perform the External Assessment based on the following criteria. Understanding of the Project Concept Delivery Skill The Final Project Report Originality and Novelty

	The Final Project Report Details: The report should have an excel sheet that documents the work of every project member **Viva Voce** Handling questions Clarity and Communication Skill **Marking Scheme:** **Internal Evaluation:** 35% of Total Marks **External Evaluation:** 50% of Total Marks **Viva Voce:** 15 % of Total Marks **For e.g.,** If the total mark for the project is 100, then Internal Evaluation = 35 marks The break-up of marks is shown below:- Interim Evaluation 1: 10 marks Interim Evaluation 2: 10 marks Understanding of concepts: 5 marks Programming technique: 5 marks Execution of code : 5 marks • External Evaluation = 50 marks The break-up of marks is shown below:- Project Report: 15 marks Explanation of project working: 10 marks Execution of code: 10 marks – (if done in industry, a stand-alone module can be reprogrammed andsubmitted. Error rectification etc. can be included by the evaluator) Participation in coding: 15 marks Viva Voce = 15 marks The break-up of marks is shown below: - Questions related to project: 10 marks Questions related to technology: 5 marks The Project evaluator(s) verifies and validates the information presented in the project report

********** HAPPY LEARNING **********